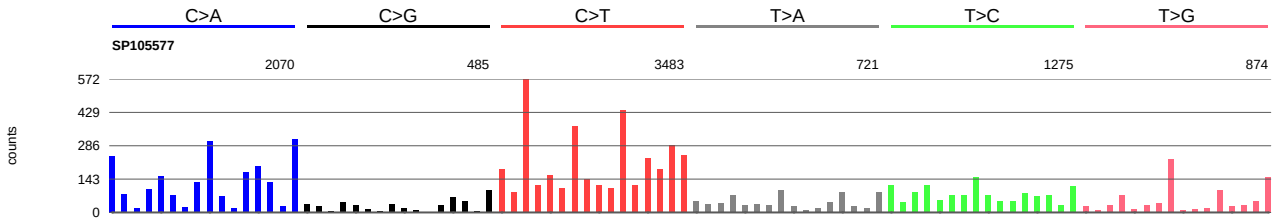
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2070
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874
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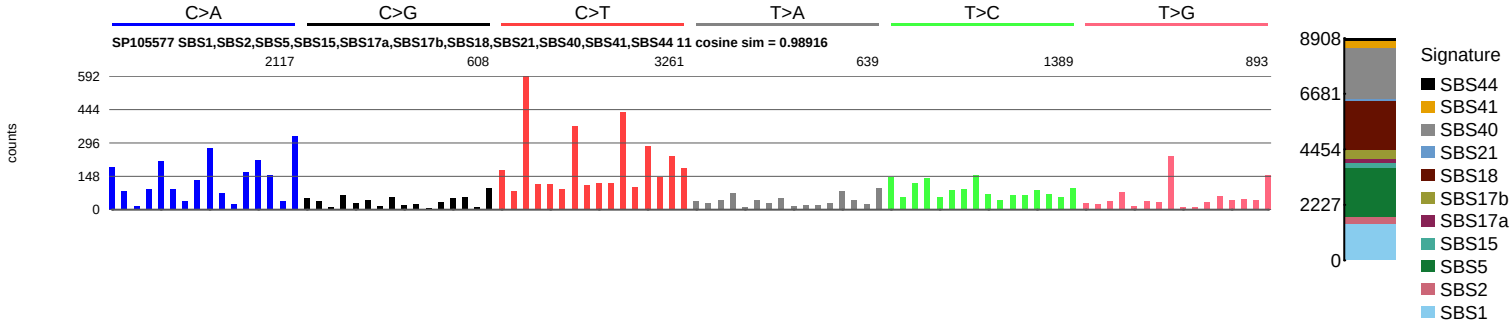
counts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2117
608
3261
639
1389
893
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
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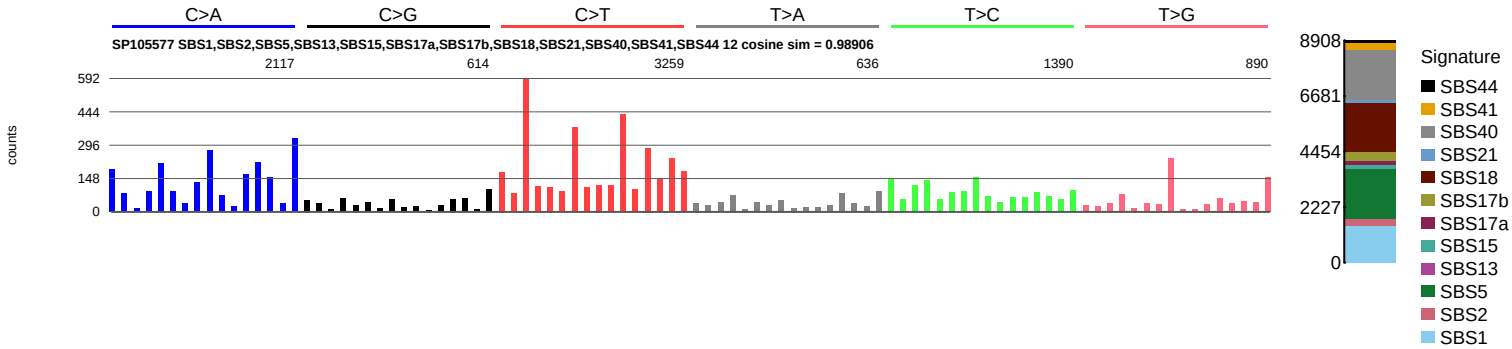
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98906
2117
614
3259
636
1390
890
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
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SBS2
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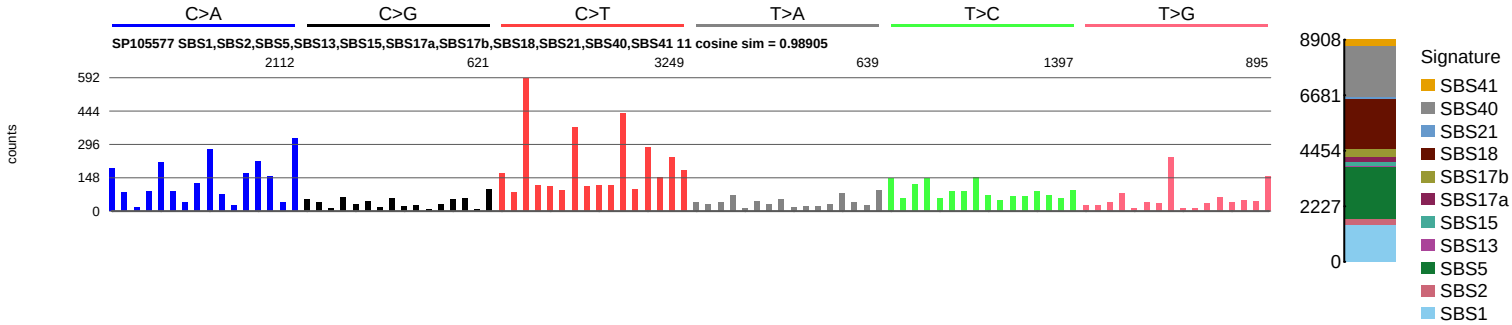
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2112
621
3249
639
1397
895
592
444
296
148
0
counts
8908
6681
4454
2227
0
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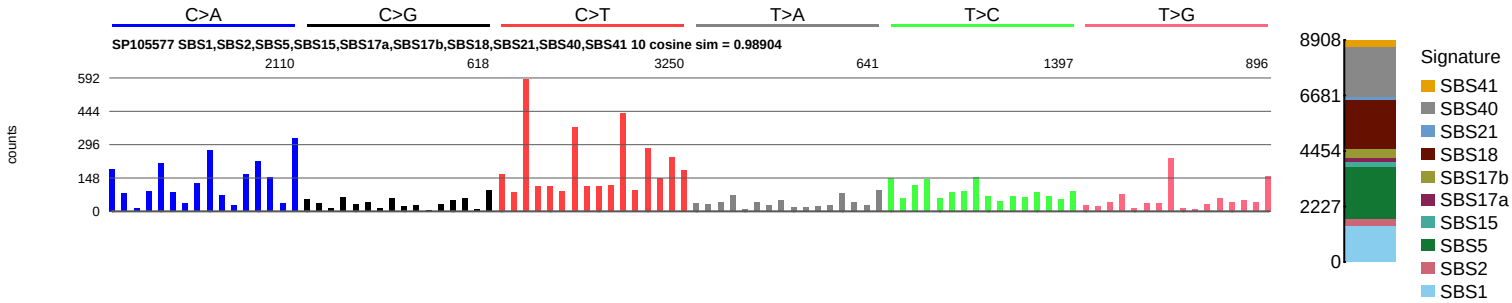
SBS1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2110
618
3250
641
1397
896
592
444
296
148
0
counts
8908
6681
4454
2227
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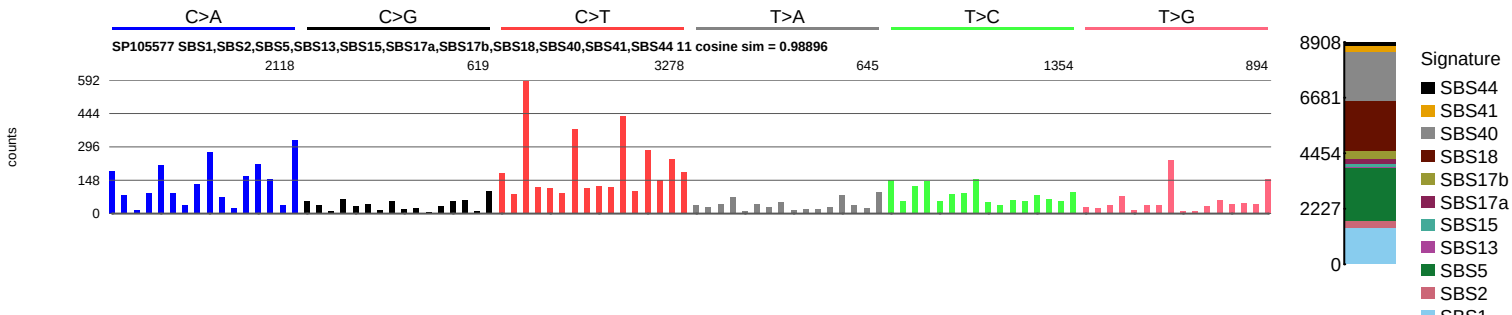
SBS15
SBS5
SBS2
SBS1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SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98896
2118
619
3278
645
1354
894
592
444
296
148
0
counts
8908
6681
4454
2227
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SBS40
SBS18
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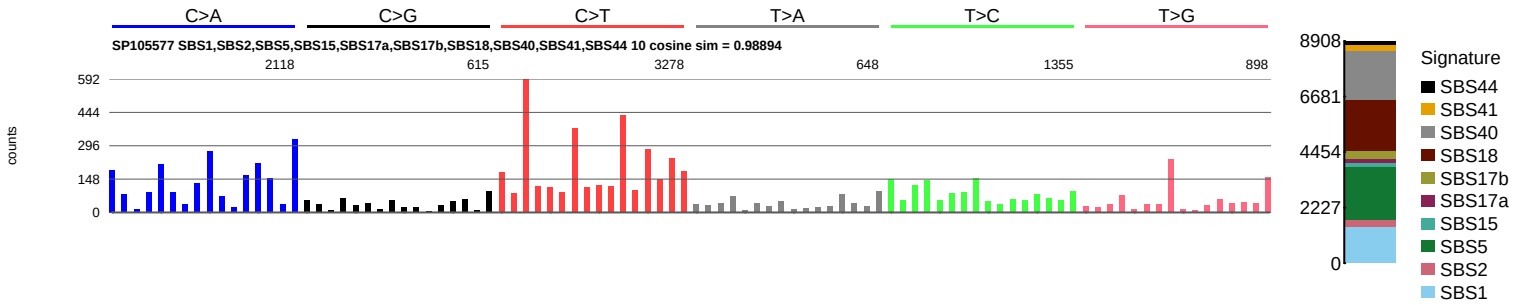
SBS17a
SBS15
SBS13
SBS5
SBS2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2118
615
3278
648
1355
898
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
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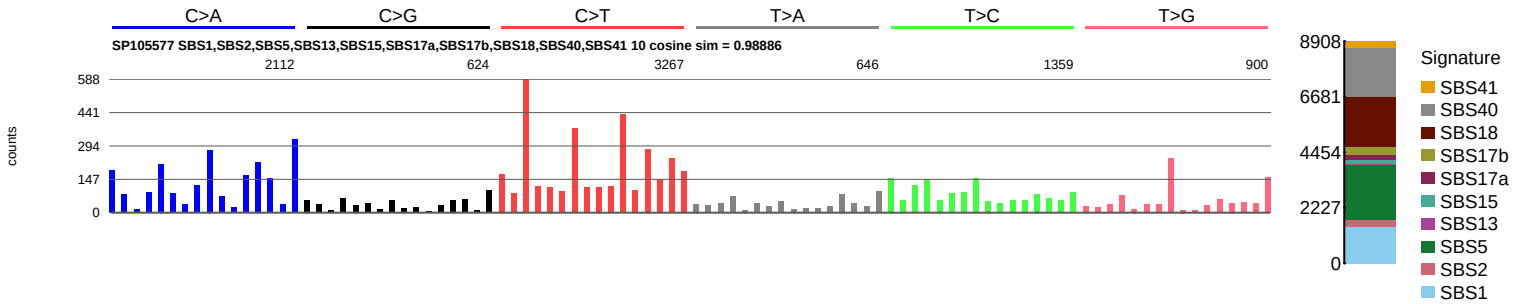
SBS2
SBS1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T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98886
2112
624
3267
646
1359
900
counts
588
441
294
147
0
8908
6681
4454
2227
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Signature
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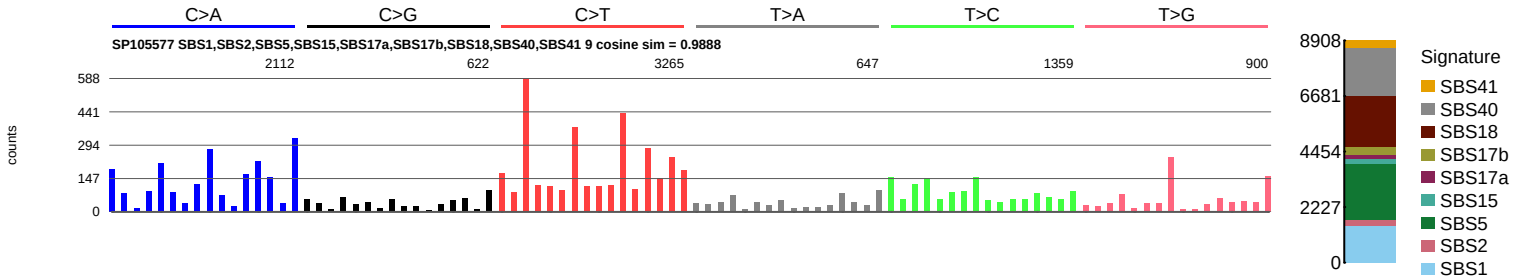
SBS2
SBS1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C>T
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T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.9888
2112
622
3265
647
1359
900
counts
588
441
294
147
0
8908
6681
4454
2227
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Signature
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SBS40
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SBS17b
SBS17a
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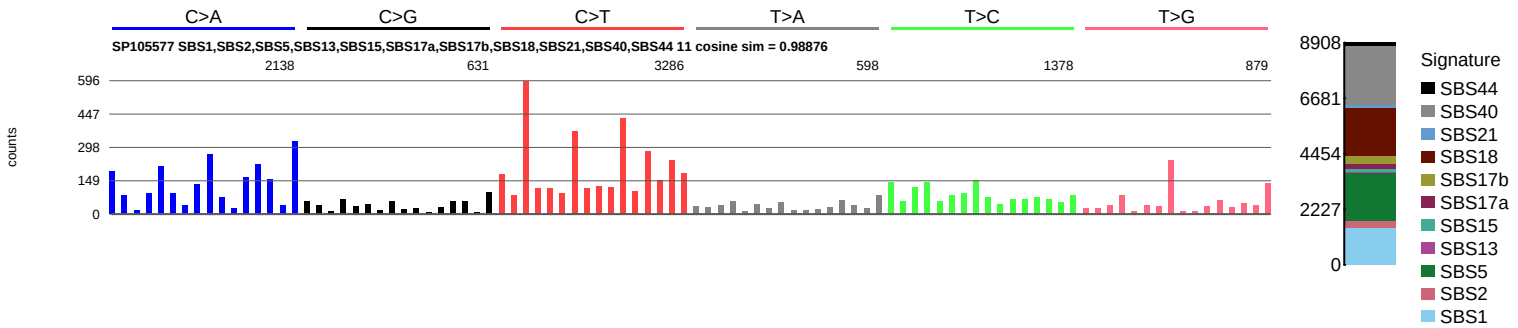
SBS2
SBS1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T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98876
2138
631
3286
598
1378
879
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
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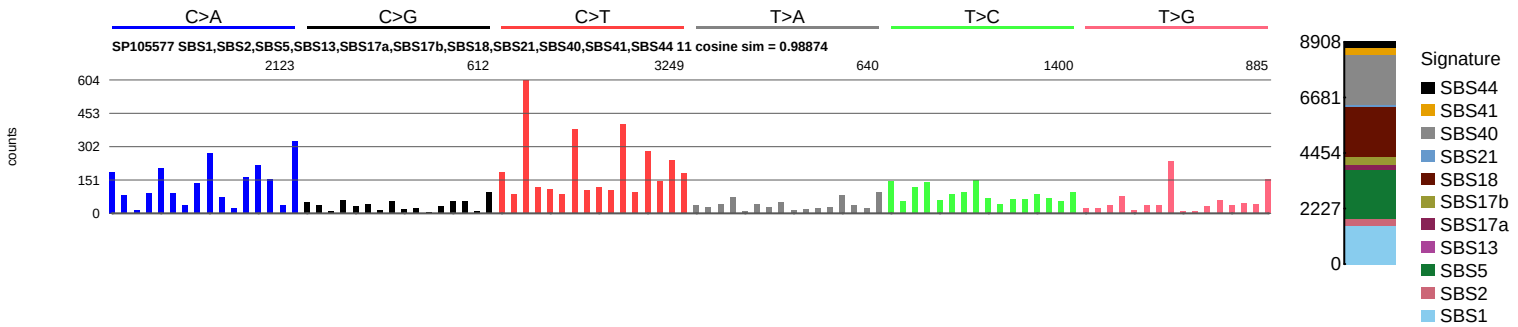
SBS2
SBS1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C>T
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T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98874
2123
612
3249
640
1400
885
counts
604
453
302
151
0
8908
6681
4454
2227
0
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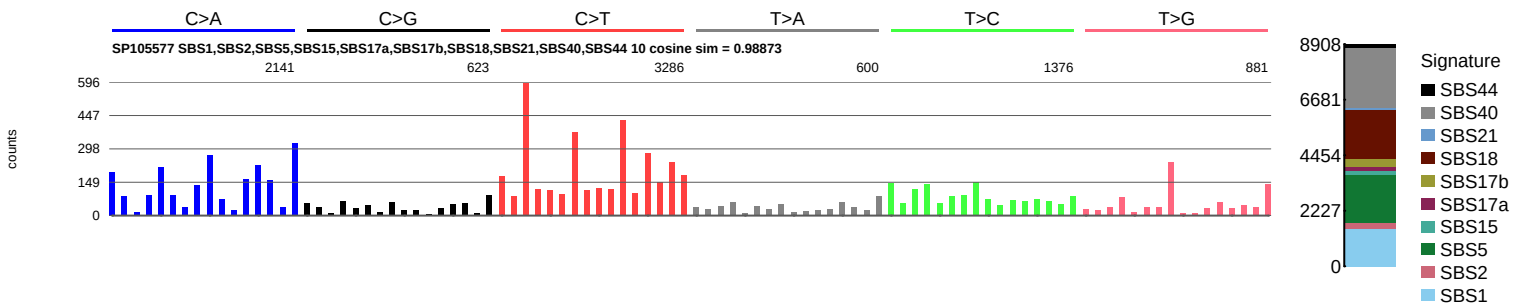
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98873
2141
623
3286
600
1376
881
counts
596
447
298
149
0
8908
6681
4454
2227
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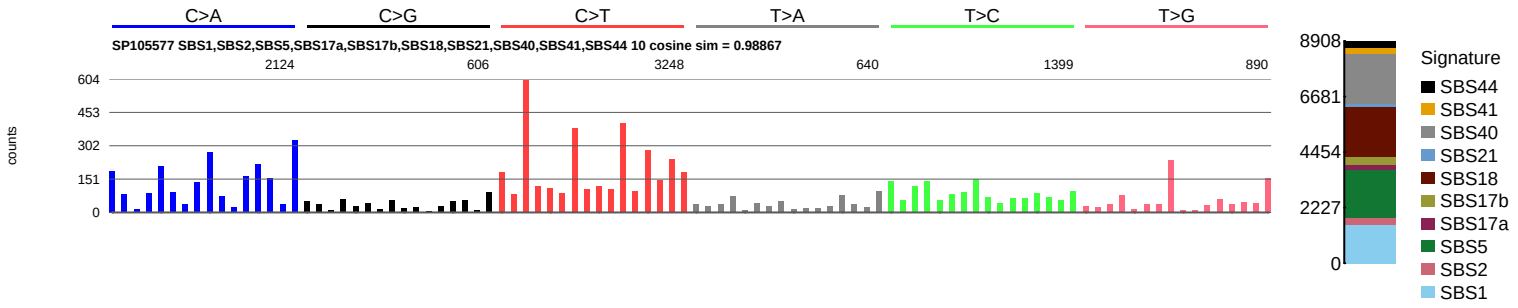
SBS2
SBS1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SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98867
2124
606
3248
640
1399
890
counts
604
453
302
151
0
8908
6681
4454
2227
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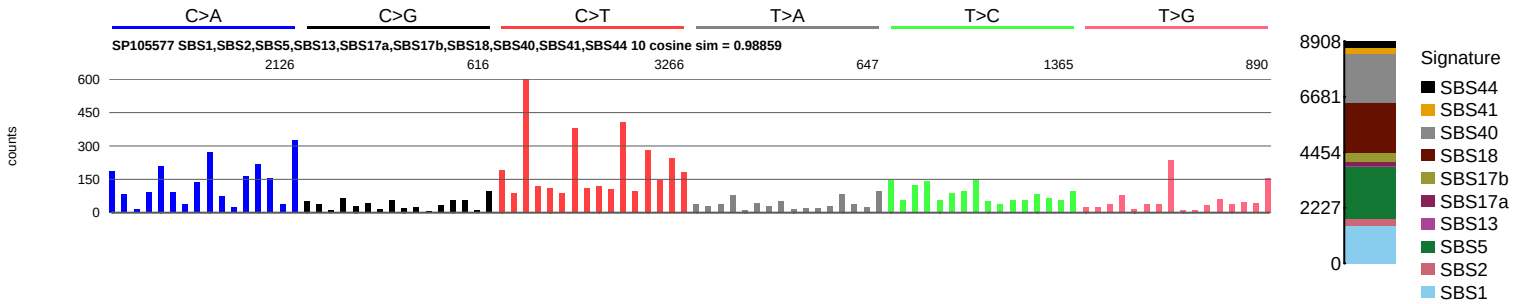
SBS5
SBS2
SBS1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T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98859
2126
616
3266
647
1365
890
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
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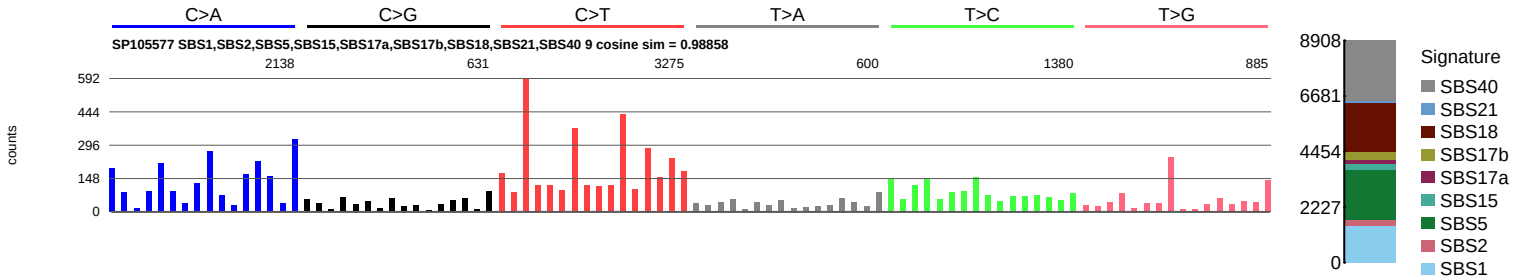
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98858
2138
631
3275
600
1380
885
counts
592
444
296
148
0
8908
6681
4454
2227
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Signature
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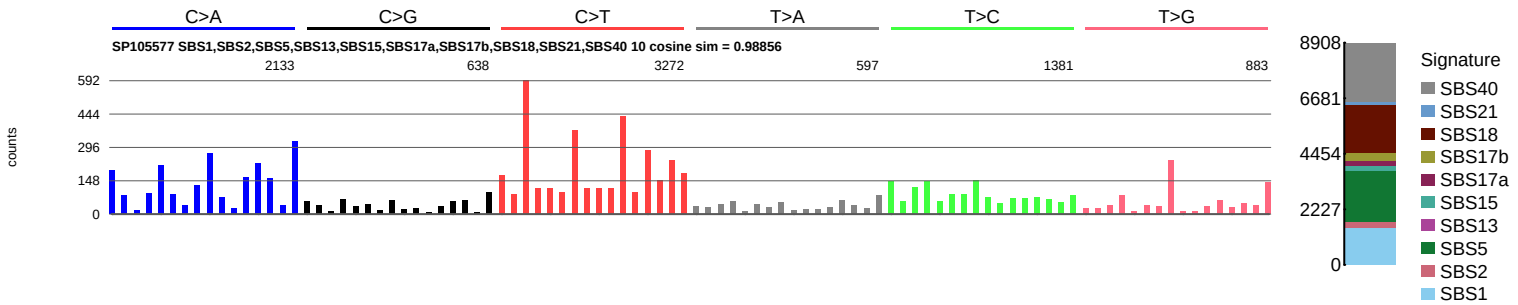
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98856
2133
638
3272
597
1381
883
counts
592
444
296
148
0
8908
6681
4454
2227
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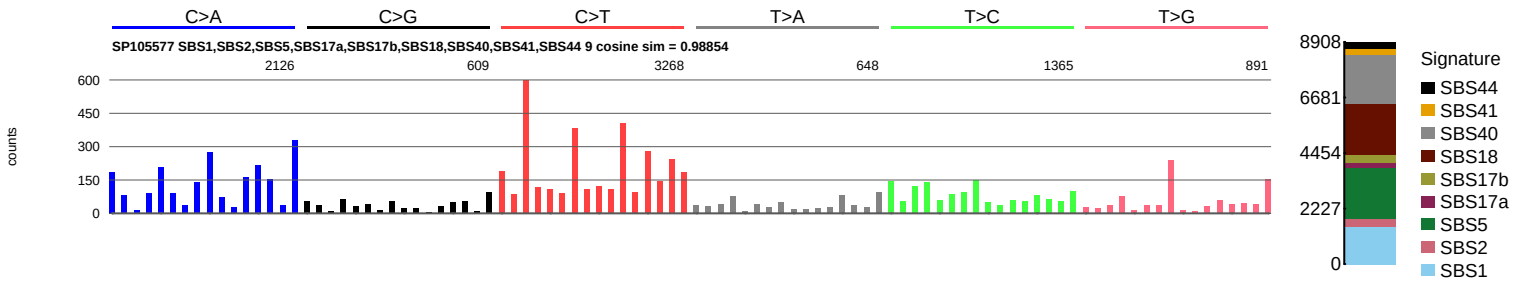
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98854
2126
609
3268
648
1365
891
counts
600
450
300
150
0
8908
6681
4454
2227
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SBS41
SBS40
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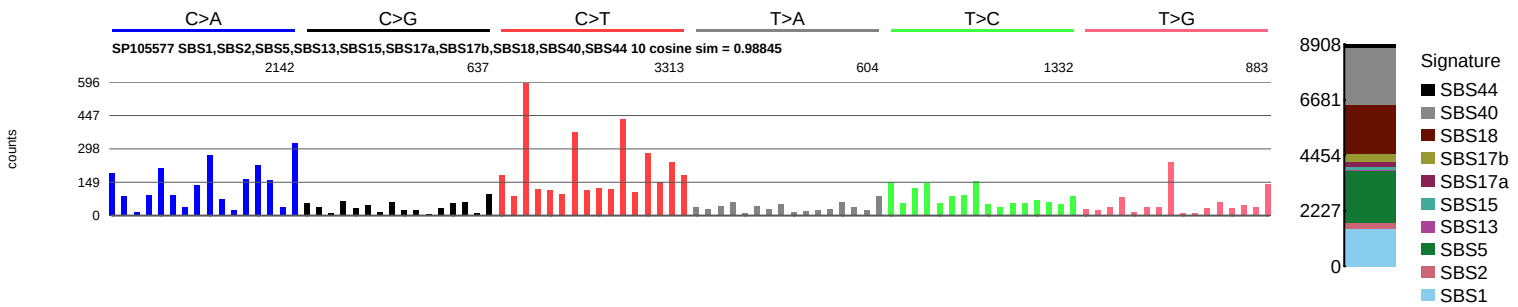
SBS17a
SBS5
SBS2
SBS1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T>C
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SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98845
2142
637
3313
604
1332
883
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
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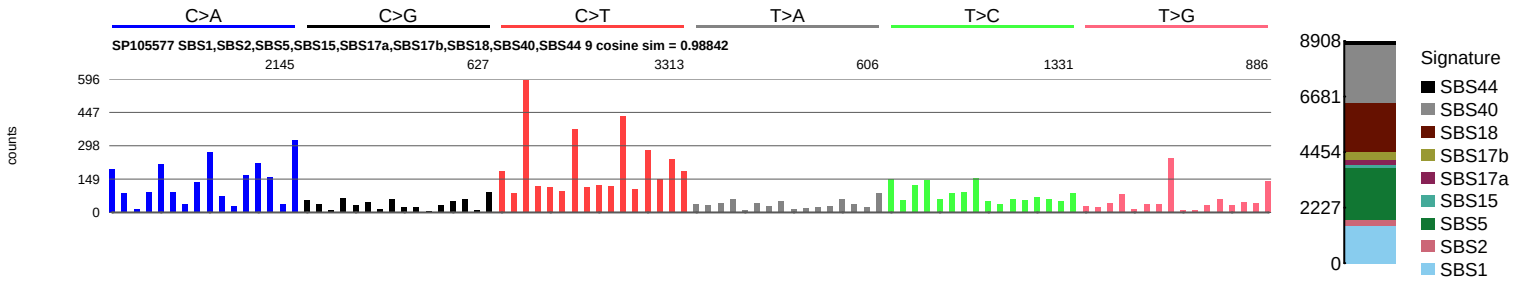
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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SP105577 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98842
2145
627
3313
606
1331
886
counts
596
447
298
149
0
8908
6681
4454
2227
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Signature
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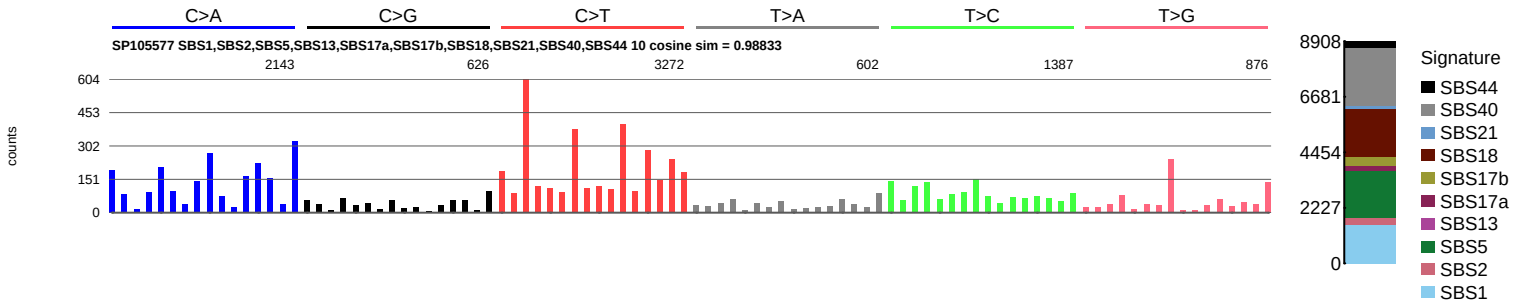
SBS2
SBS1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SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98833
2143
626
3272
602
1387
876
counts
604
453
302
151
0
8908
6681
4454
2227
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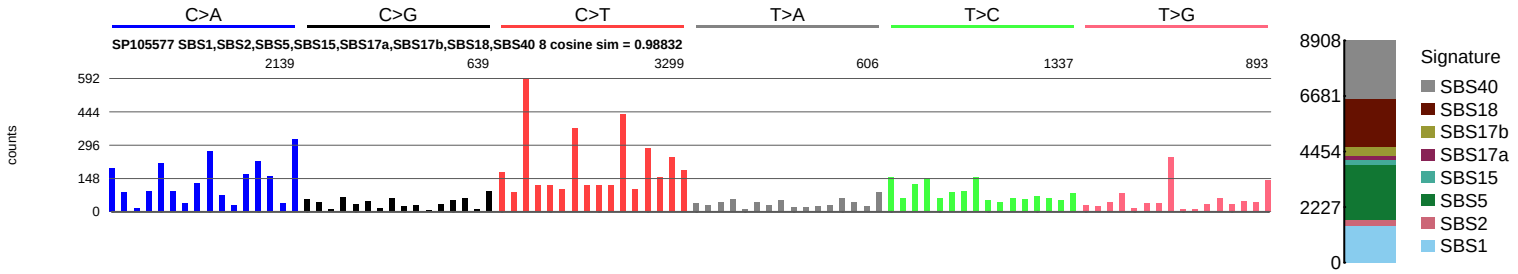
SBS2
SBS1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2139
639
3299
606
1337
893
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
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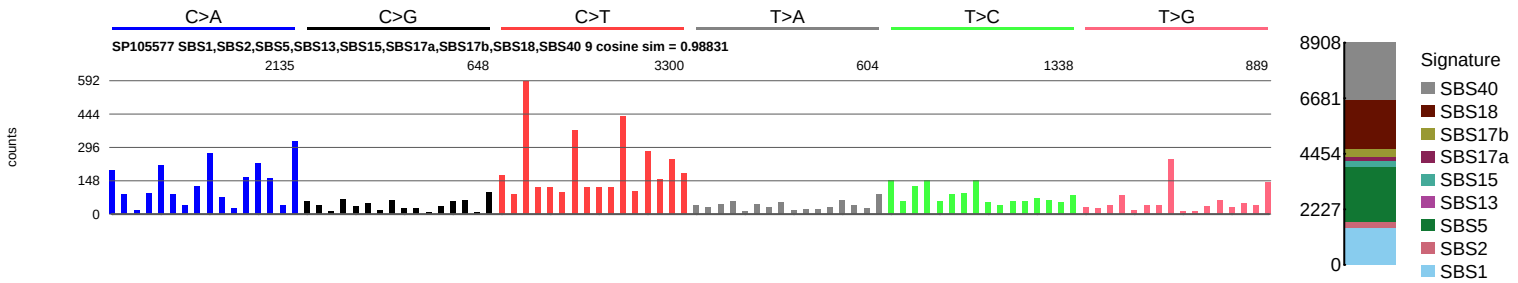
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98831
2135
648
3300
604
1338
889
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
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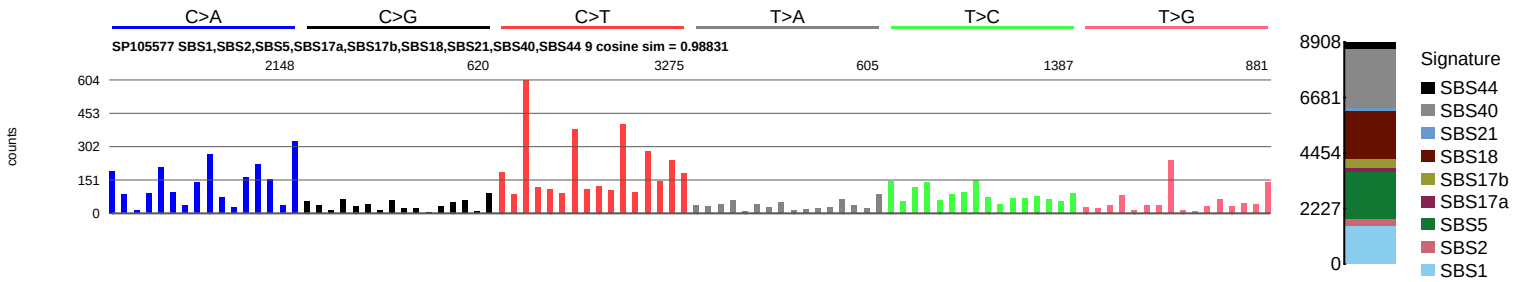
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98831
2148
620
3275
605
1387
881
counts
604
453
302
151
0
8908
6681
4454
2227
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Signature
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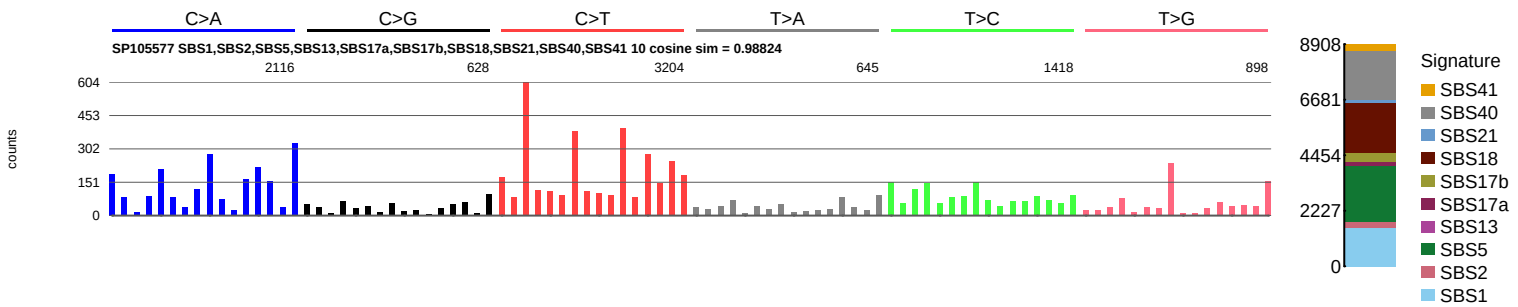
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
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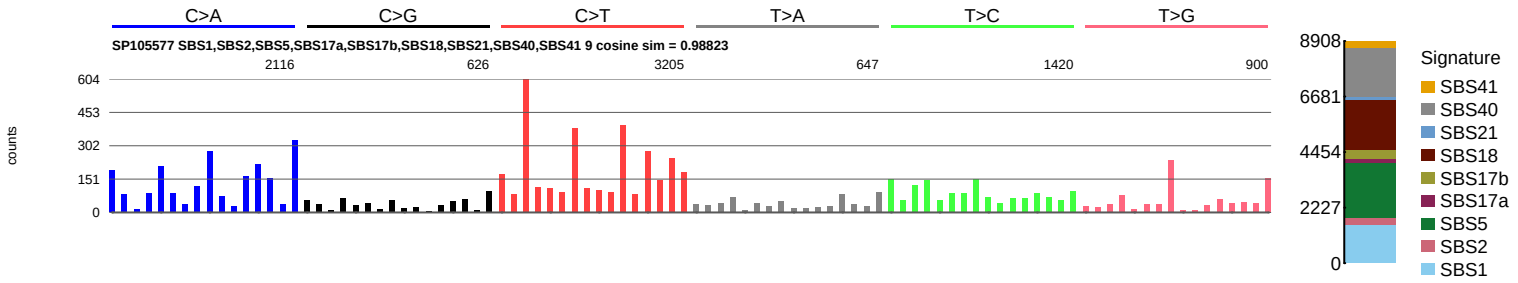
2116
628
3204
645
1418
898
counts
604
453
302
151
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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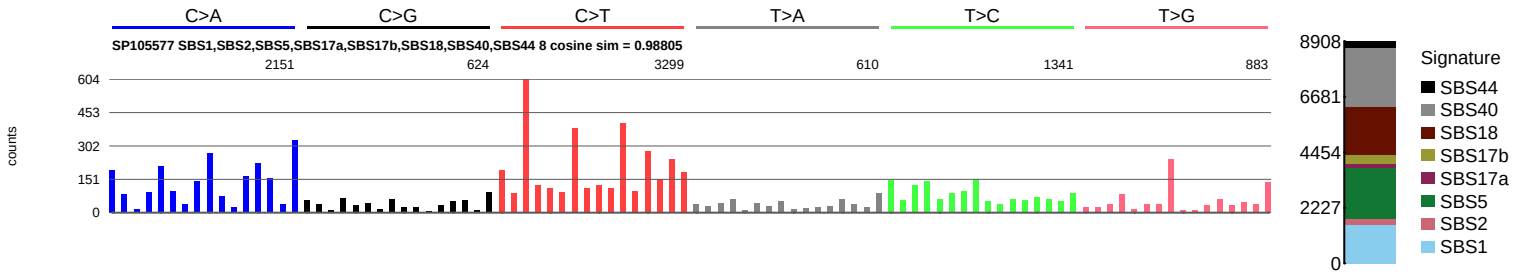
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98823
2116
626
3205
647
1420
900
counts
604
453
302
151
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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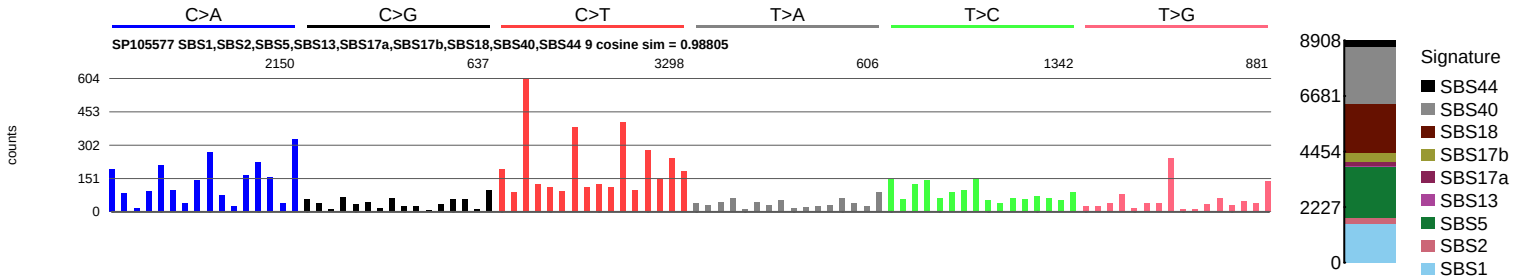
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98805
2151
624
3299
610
1341
883
counts
604
453
302
151
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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T>C
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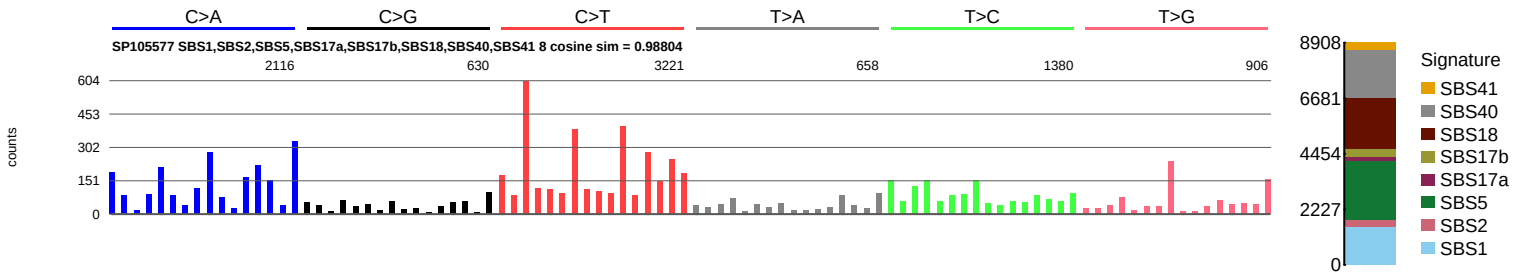
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98805
2150
637
3298
606
1342
881
counts
604
453
302
151
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
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SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98804
2116
630
3221
658
1380
906
counts
604
453
302
151
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
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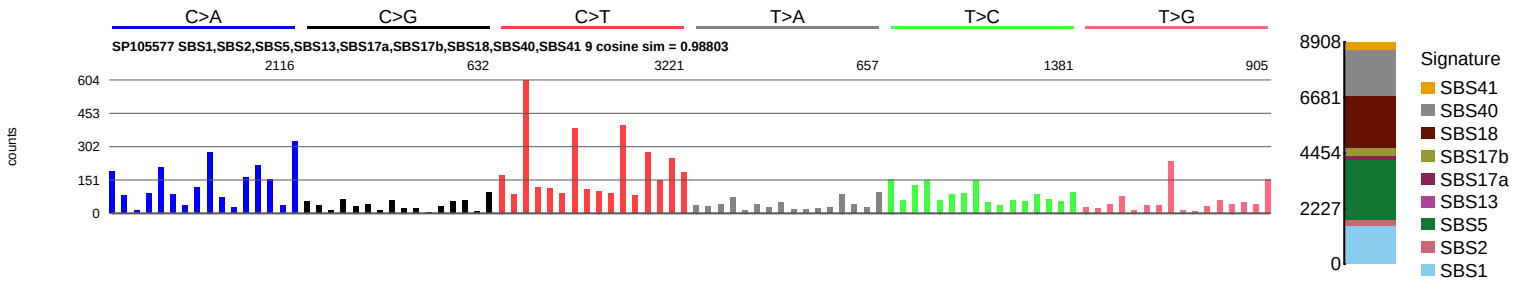
SBS2
SBS1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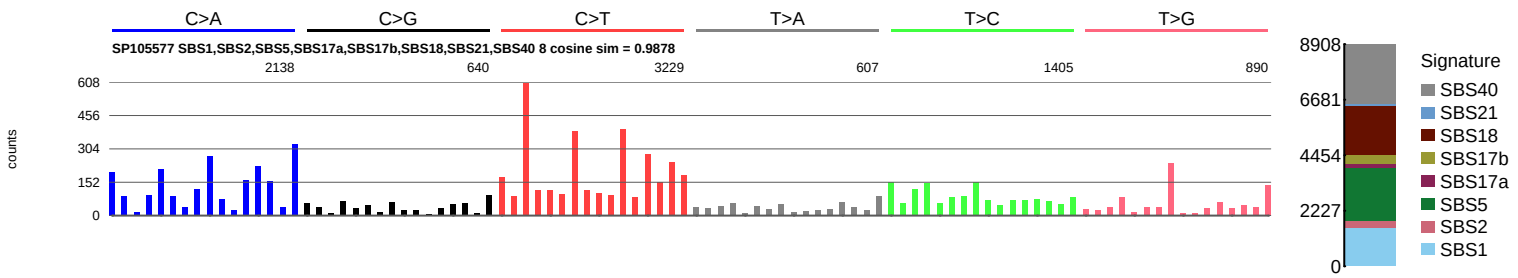
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98803
2116
632
3221
657
1381
905
counts
604
453
302
151
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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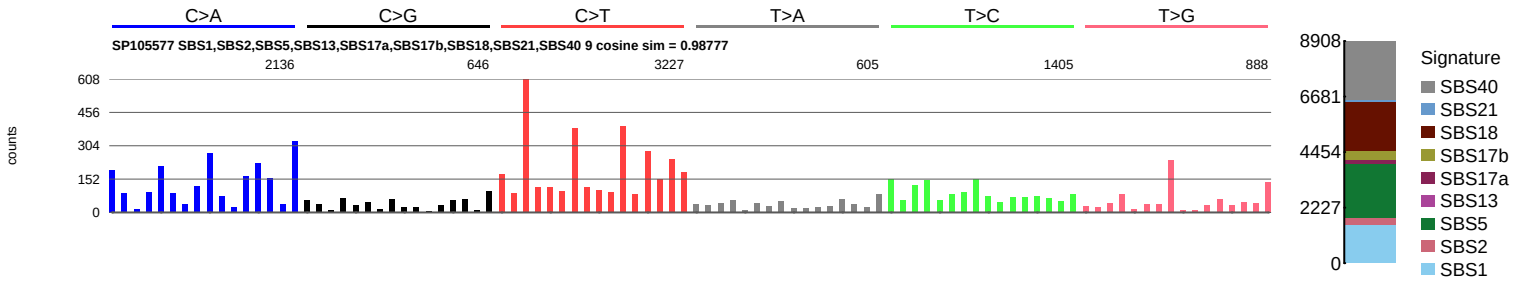
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.9878
2138
640
3229
607
1405
890
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98777
2136
646
3227
605
1405
888
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS40
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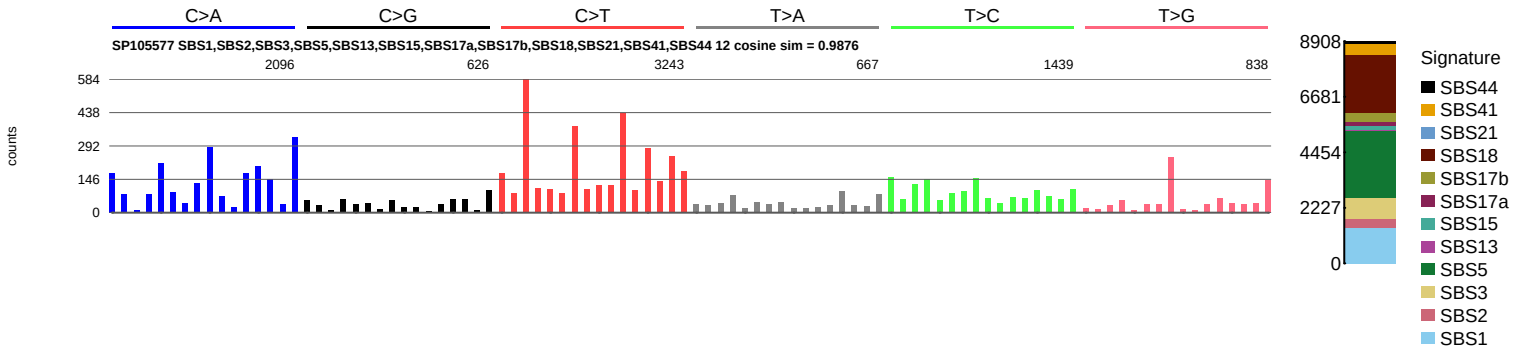
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 12 cosine sim = 0.9876
2096
626
3243
667
1439
838
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
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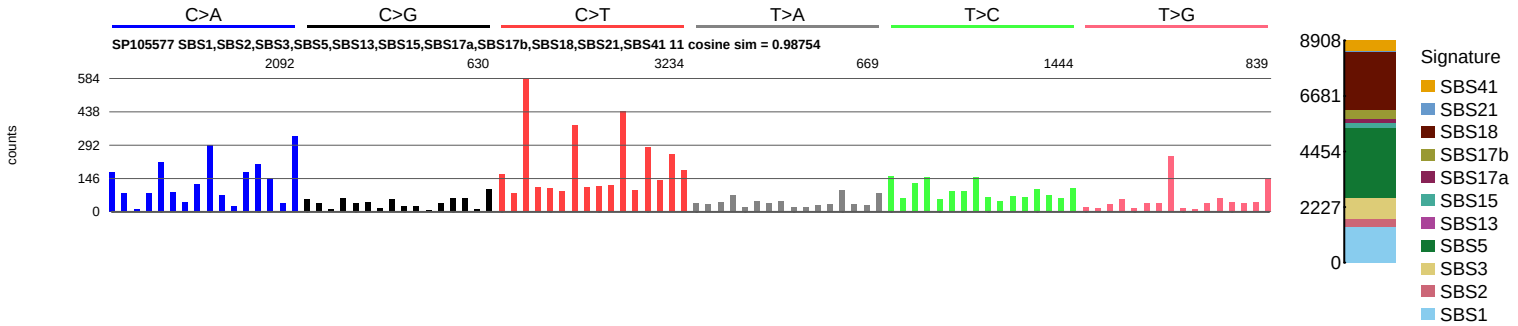
SBS2
SBS1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C>G
C>T
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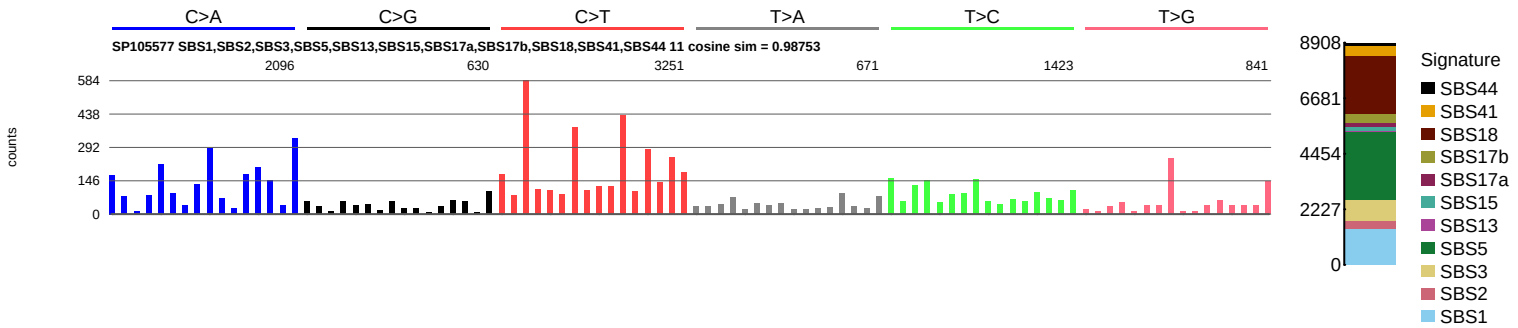
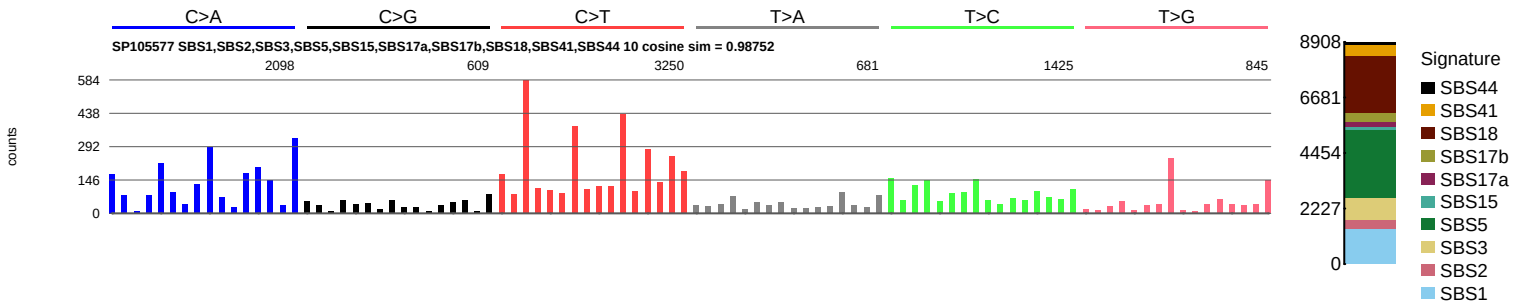
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.98754
2092
630
3234
669
1444
839
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 11 cosine sim = 0.98753
2096
630
3251
671
1423
841
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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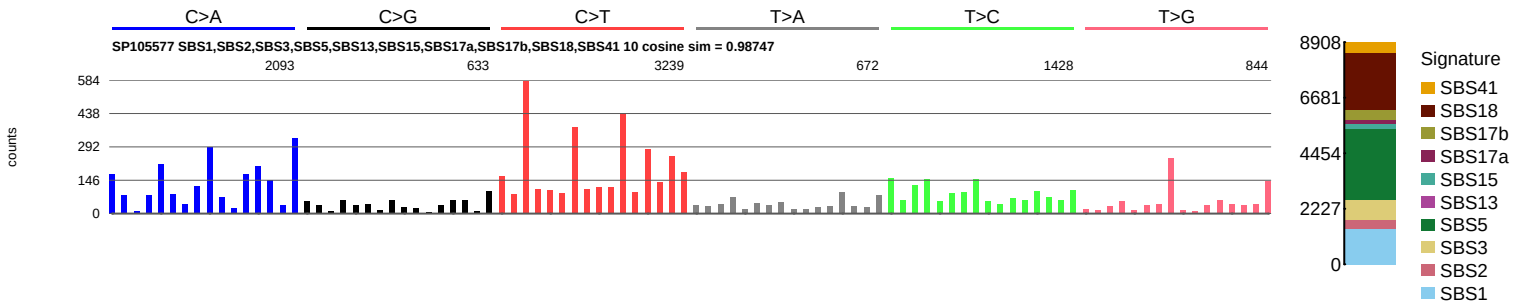
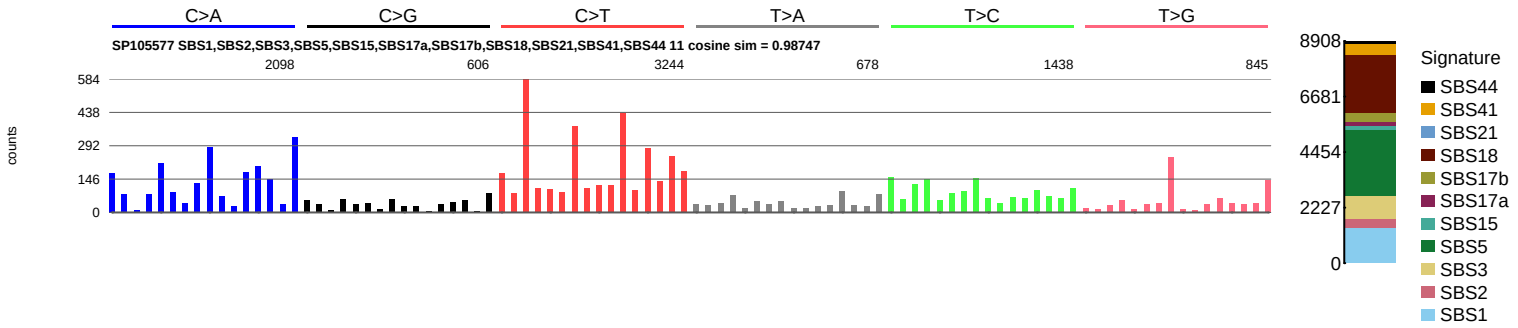
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98752
2098
609
3250
681
1425
845
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.98747
2093
633
3239
672
1428
844
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98747
2098
606
3244
678
1438
845
584
438
292
146
0
counts
8908
6681
4454
2227
0
Signature
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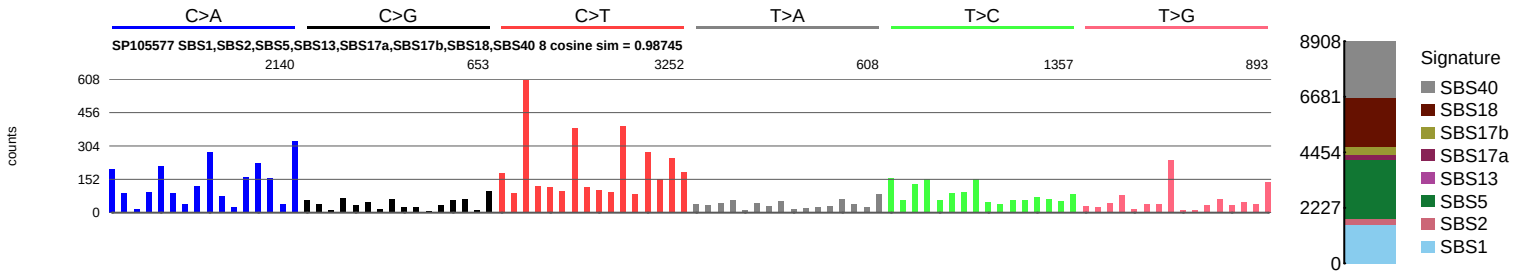
SBS2
SBS1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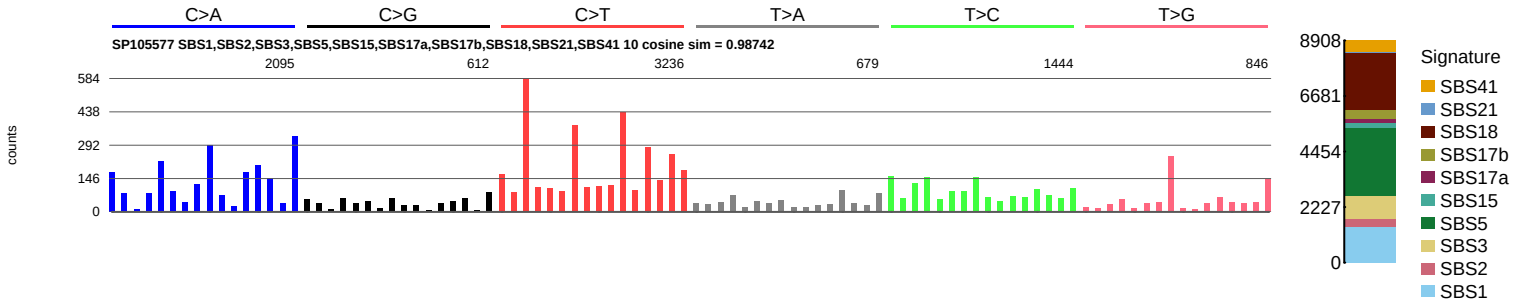
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98745
2140
653
3252
608
1357
893
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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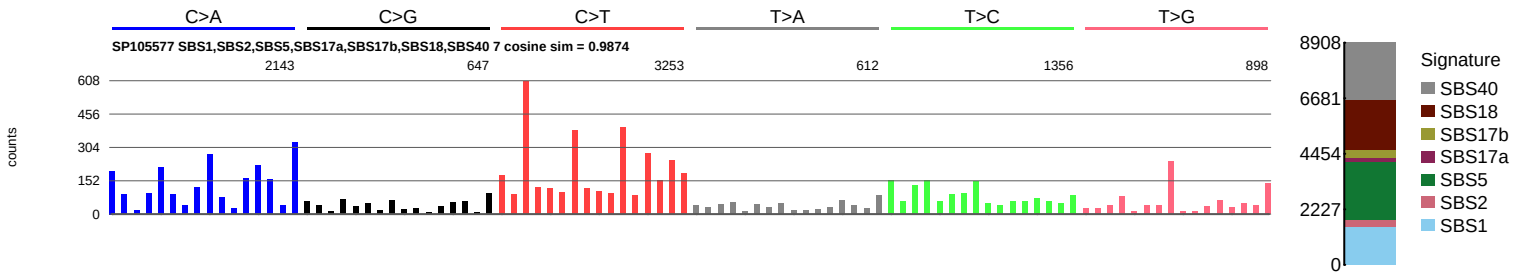
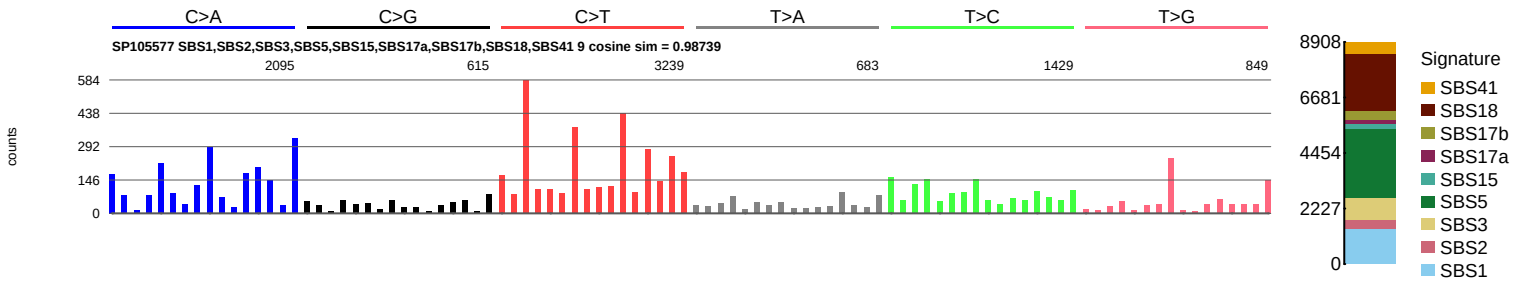
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98742
2095
612
3236
679
1444
846
584
438
292
146
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.9874
2143
647
3253
612
1356
898
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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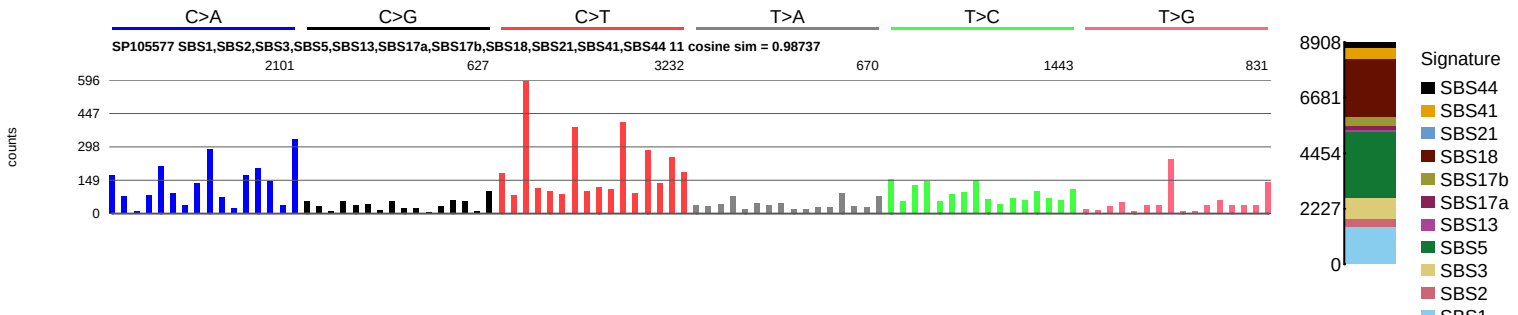
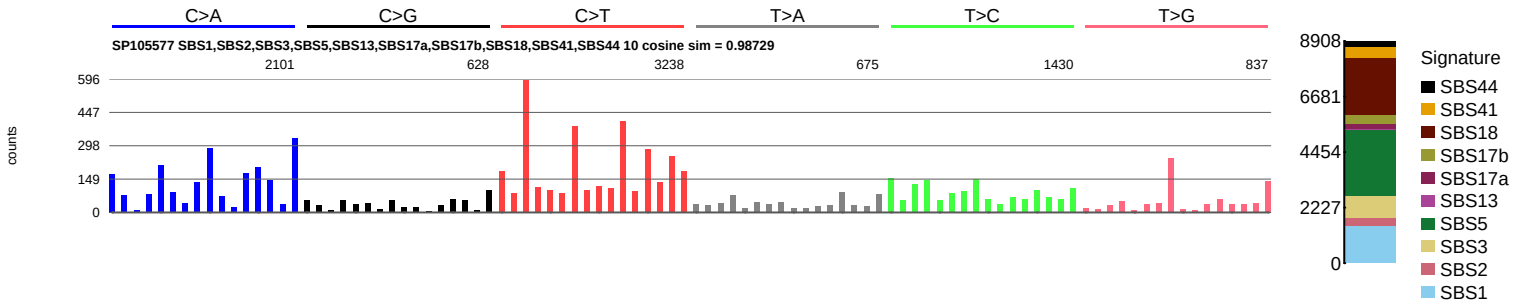
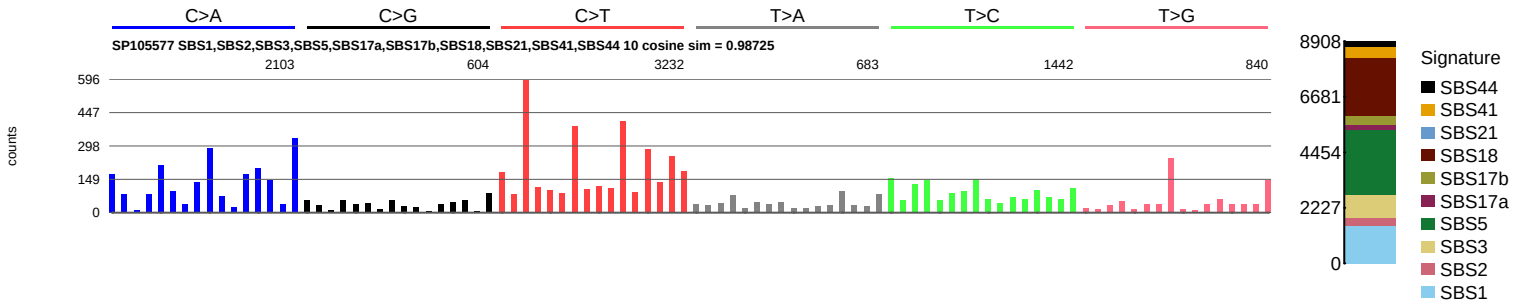
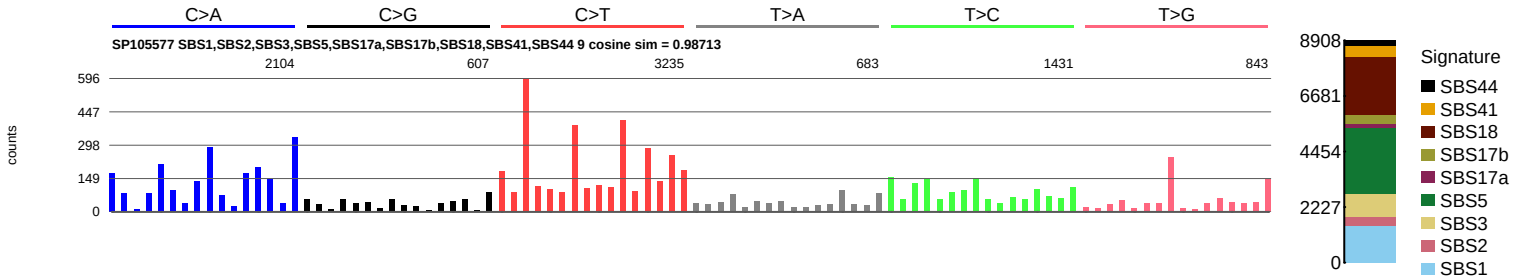
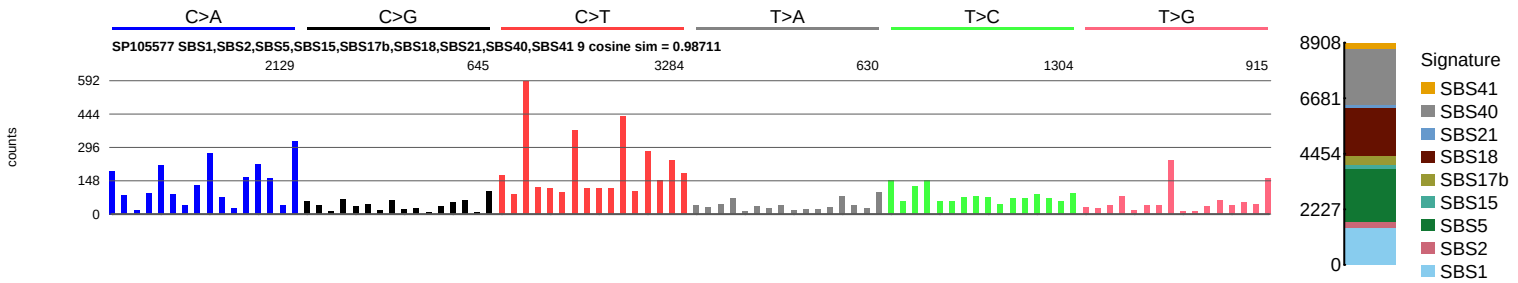
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98739
2095
615
3239
683
1429
849
584
438
292
146
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98737
2101
627
3232
670
1443
831
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98729
2101
628
3238
675
1430
837
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98725
2103
604
3232
683
1442
840
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98713
2104
607
3235
683
1431
843
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98711
2129
645
3284
630
1304
915
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
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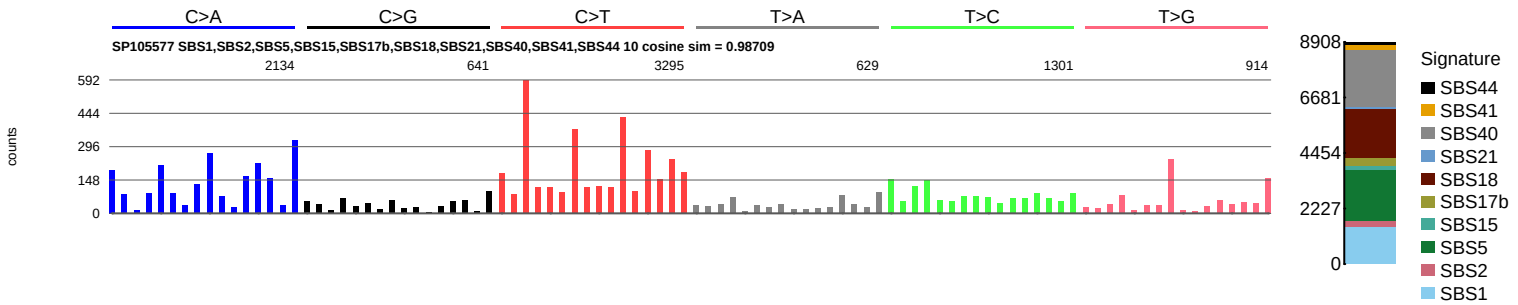
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98709
2134
641
3295
629
1301
914
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
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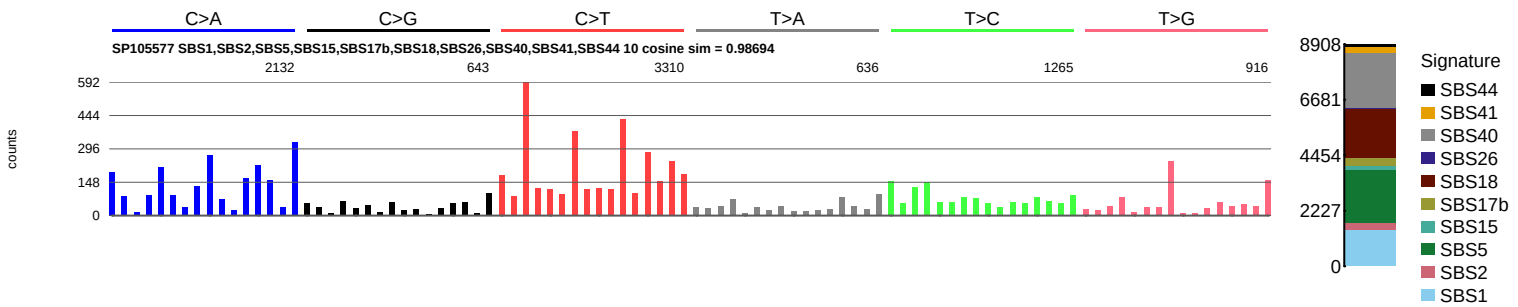
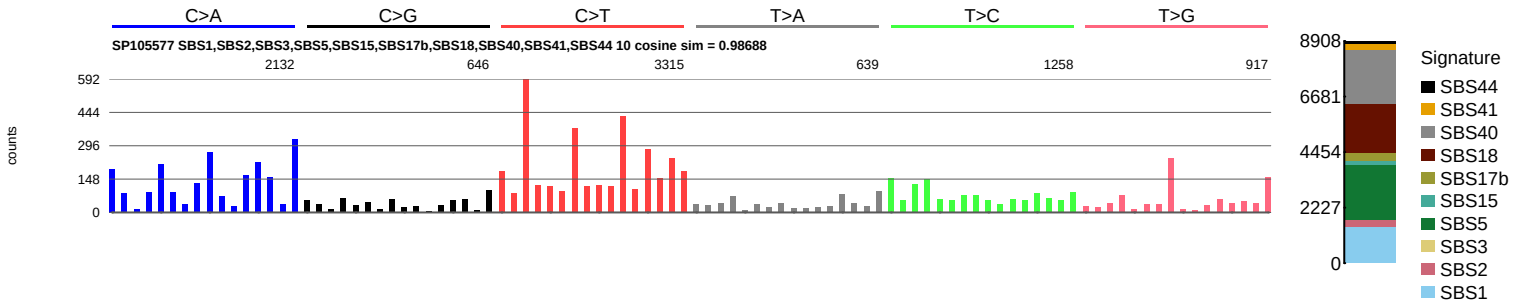
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98694
2132
643
3310
636
1265
916
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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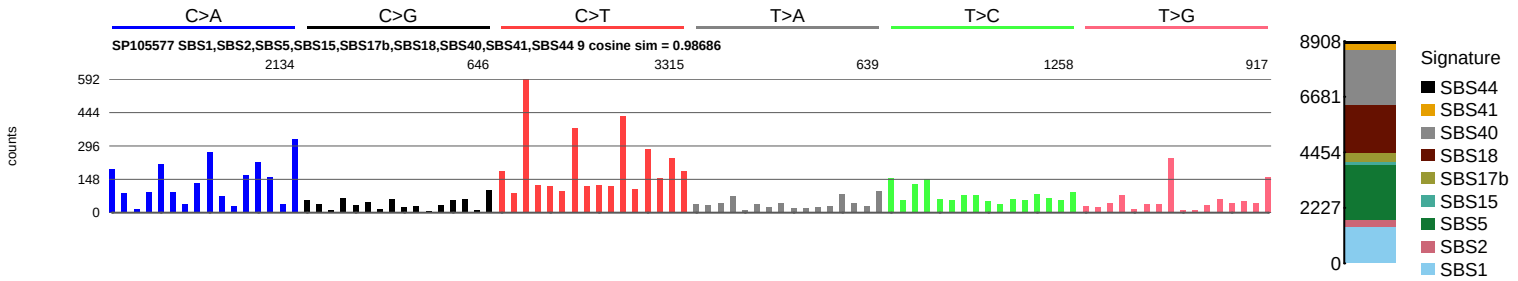
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98688
2132
646
3315
639
1258
917
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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C>T
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T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98686
2134
646
3315
639
1258
917
counts
592
444
296
148
0
8908
6681
4454
2227
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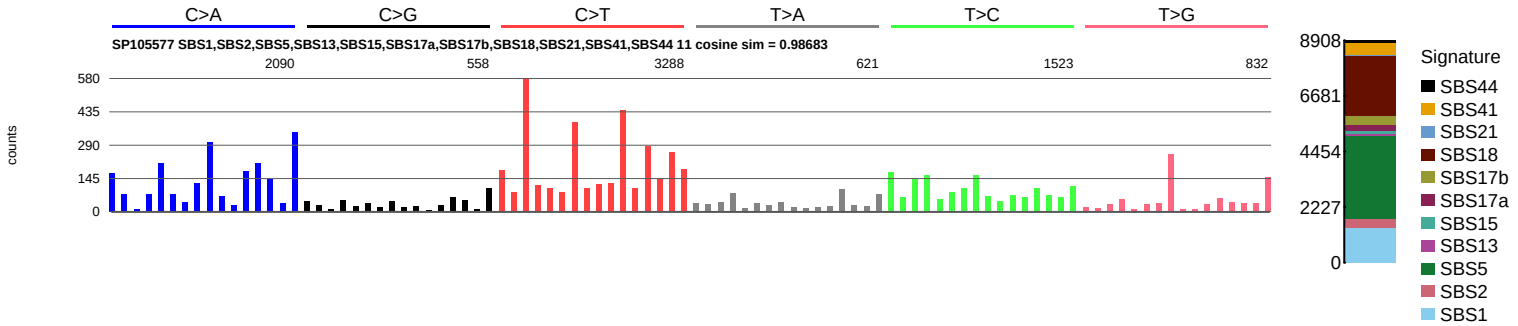
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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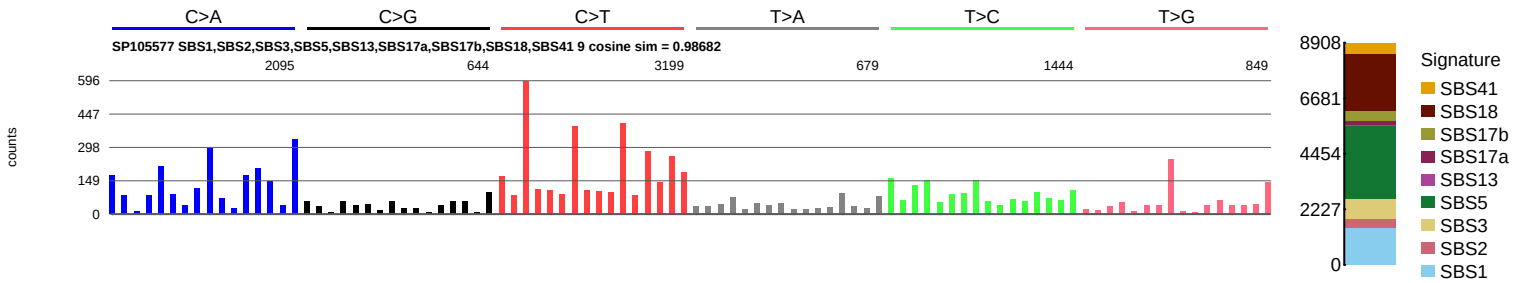
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98683
2090
558
3288
621
1523
832
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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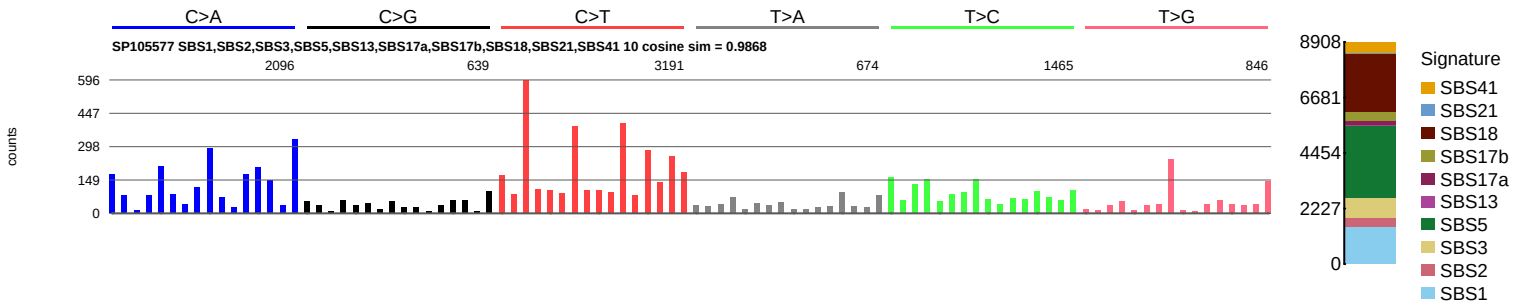
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98682
2095
644
3199
679
1444
849
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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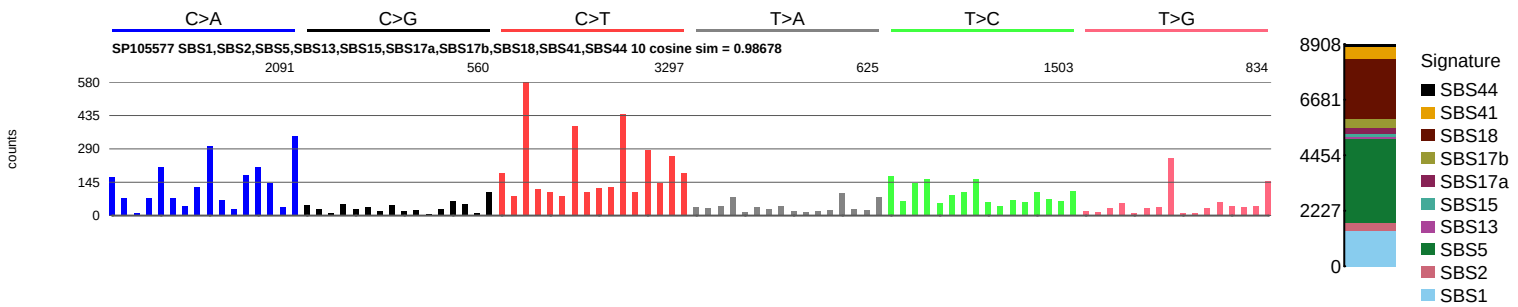
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.9868
2096
639
3191
674
1465
846
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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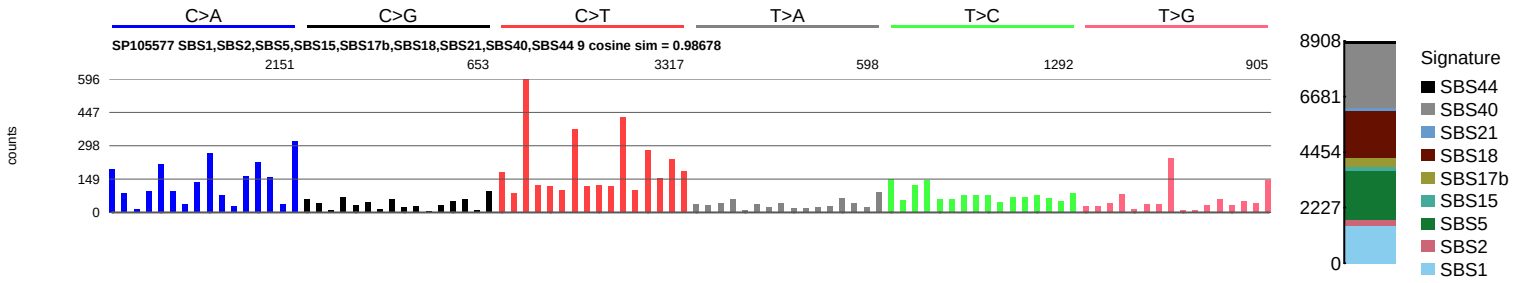
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98678
2091
560
3297
625
1503
834
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98678
2151
653
3317
598
1292
905
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
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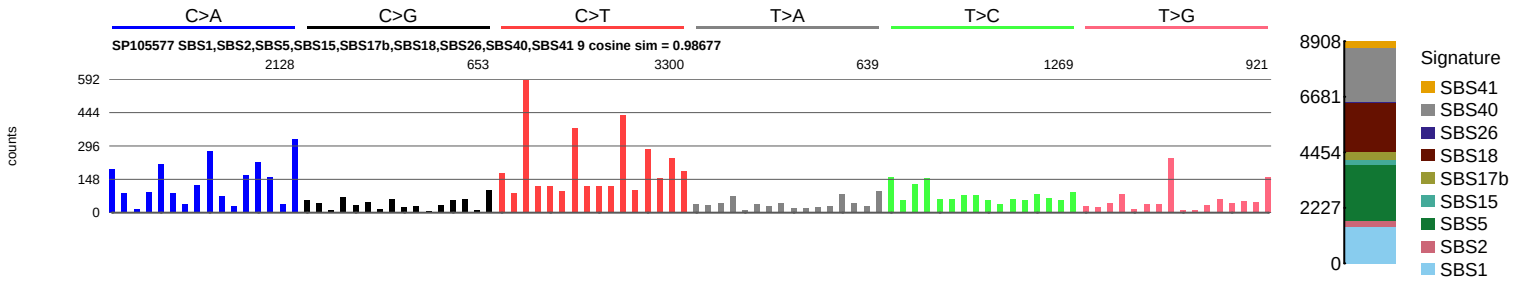
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98677
2128
653
3300
639
1269
921
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
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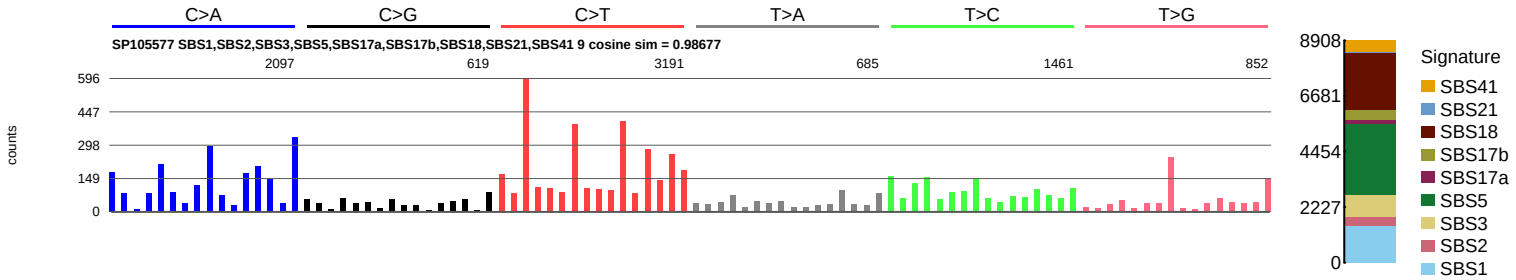
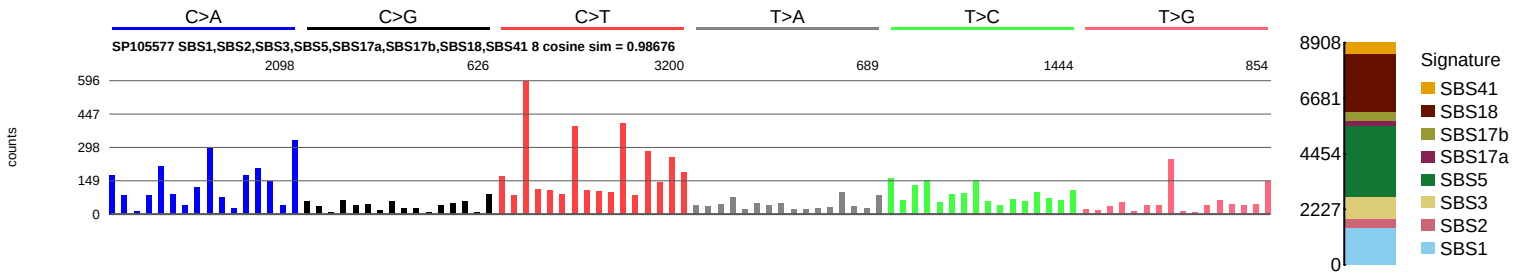
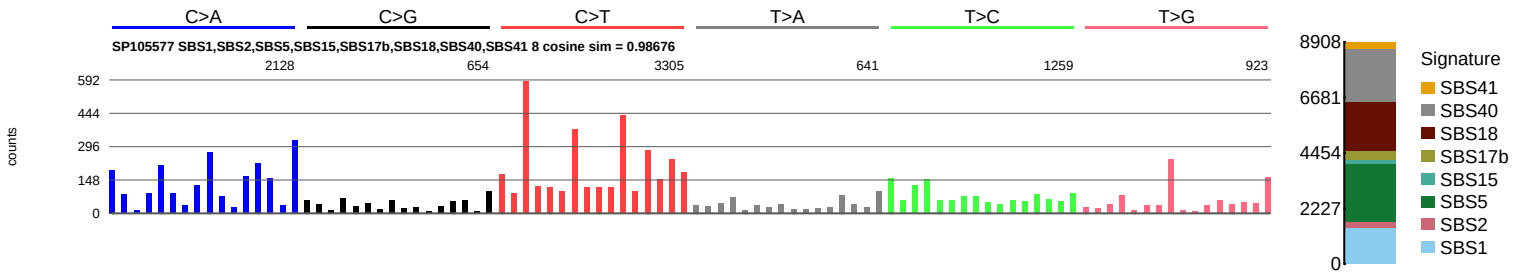
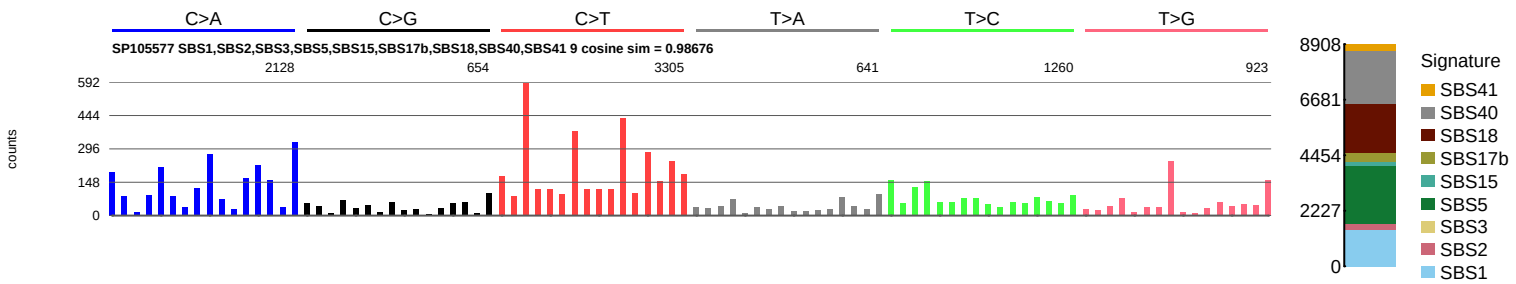
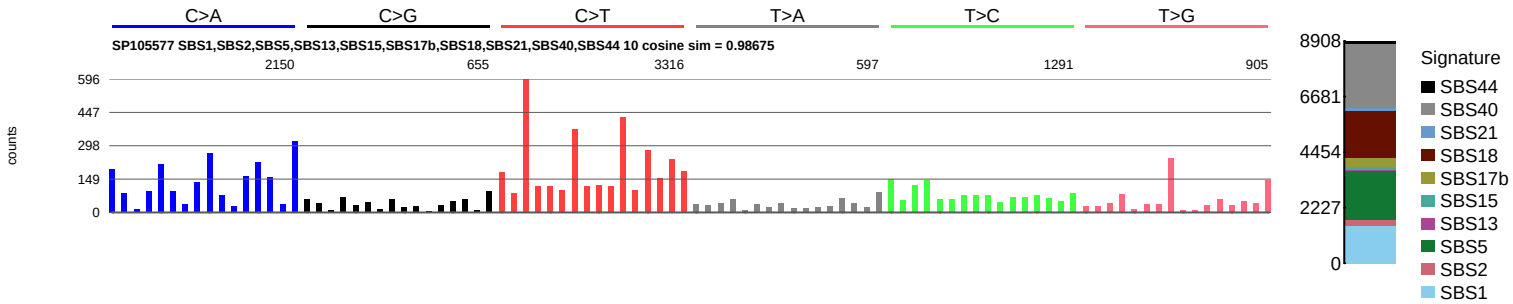
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98677
2097
619
3191
685
1461
852
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98676
2098
626
3200
689
1444
854
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98676
2128
654
3305
641
1259
923
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98676
2128
654
3305
641
1260
923
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98675
2150
655
3316
597
1291
905
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
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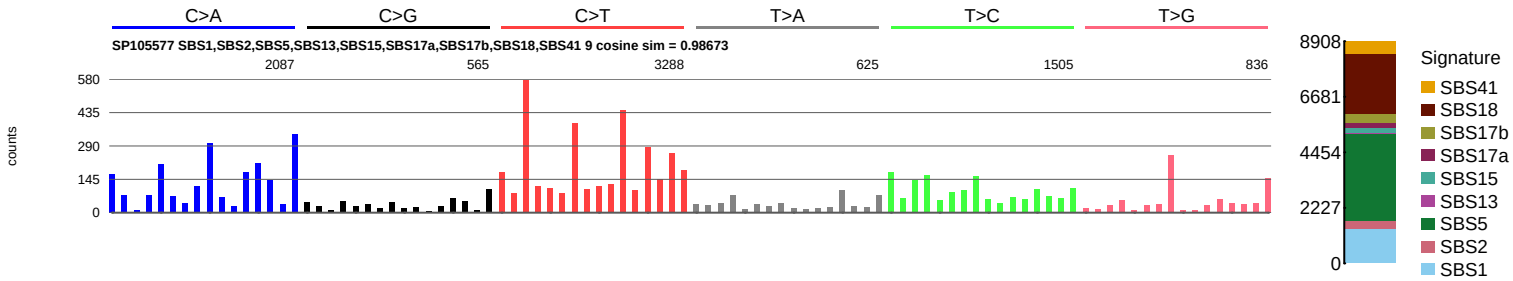
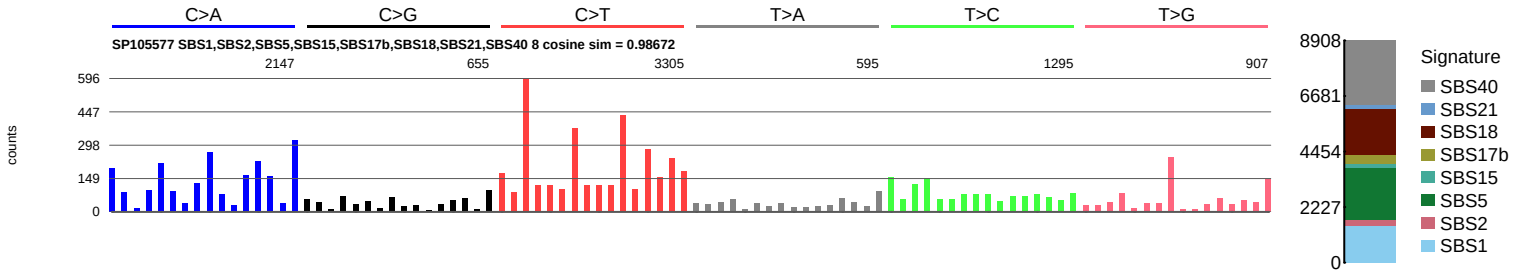
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98673
2087
565
3288
625
1505
836
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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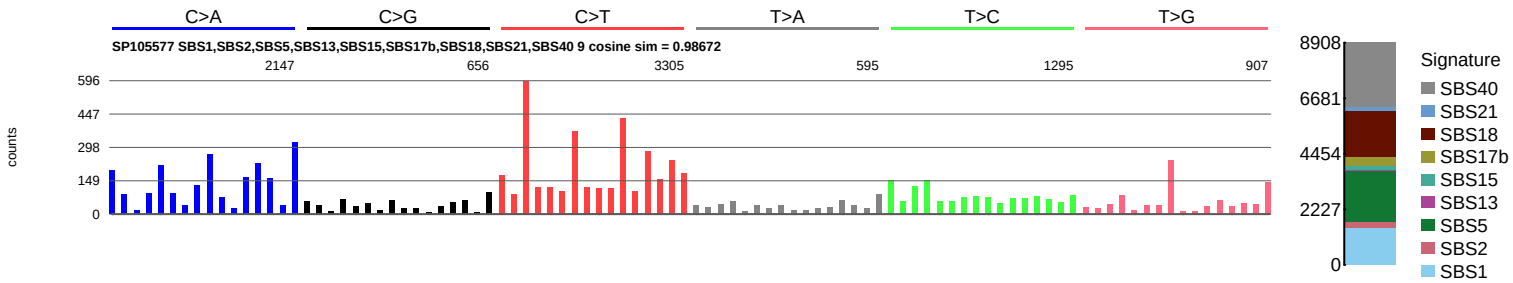
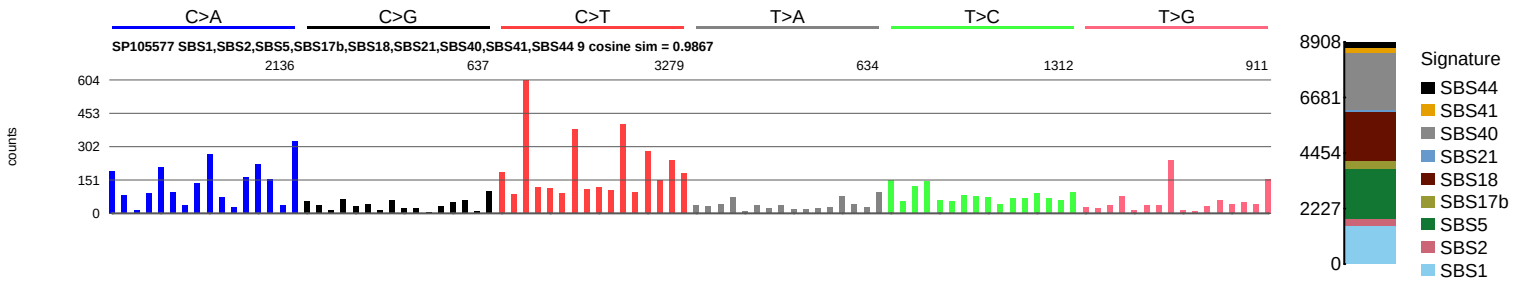
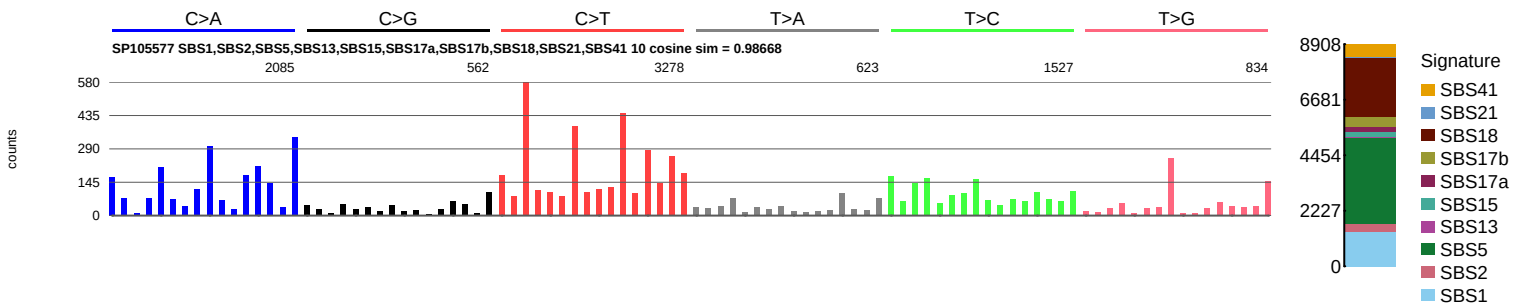
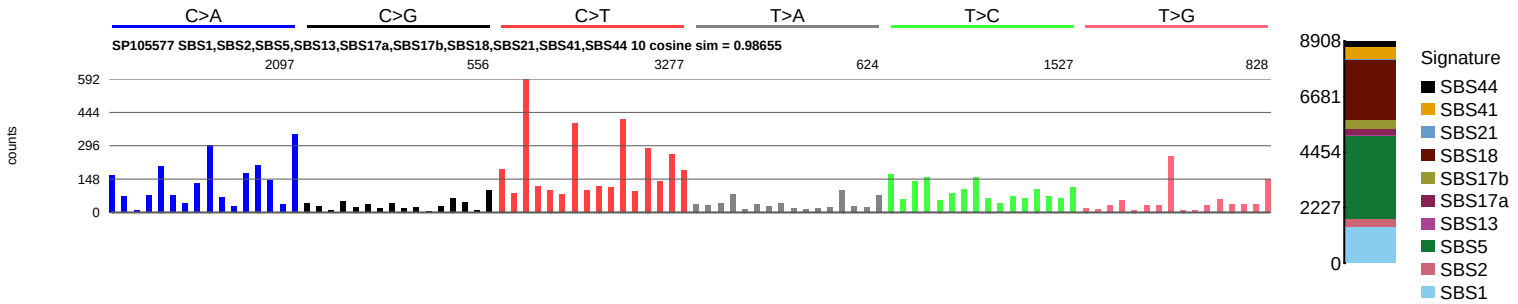
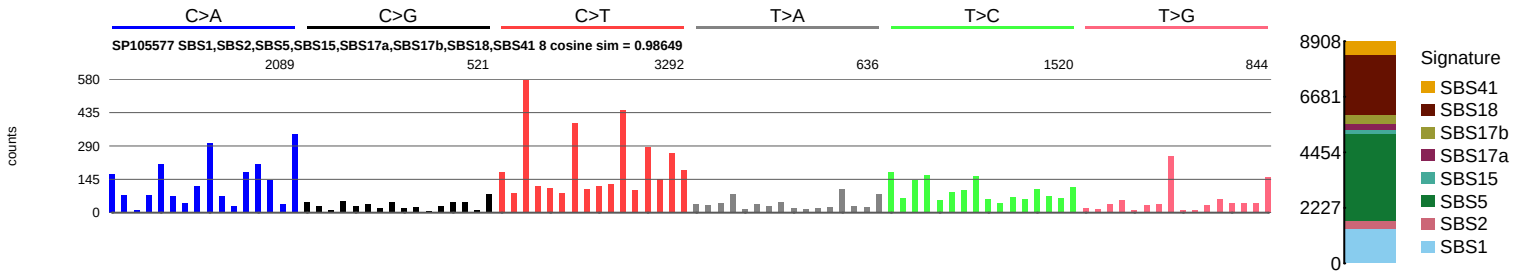
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98672
2147
655
3305
595
1295
907
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98672
2147
656
3305
595
1295
907
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9867
2136
637
3279
634
1312
911
counts
604
453
302
151
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98668
2085
562
3278
623
1527
834
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98655
2097
556
3277
624
1527
828
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98649
2089
521
3292
636
1520
844
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
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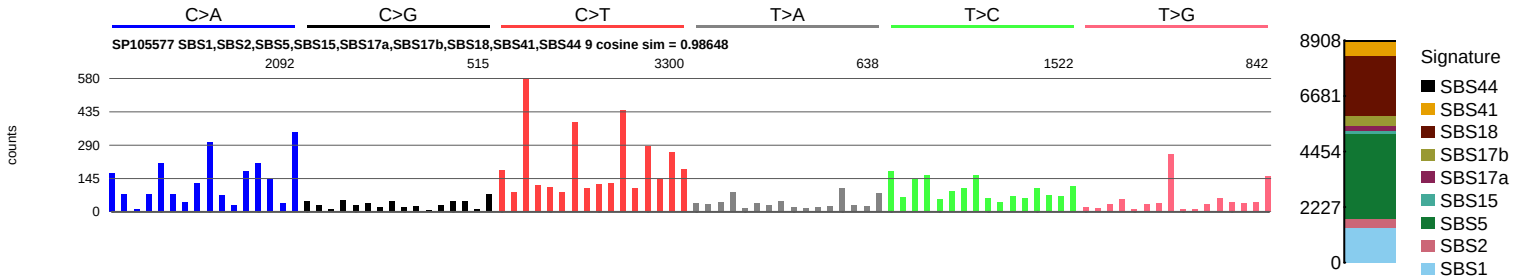
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98648
2092
515
3300
638
1522
842
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
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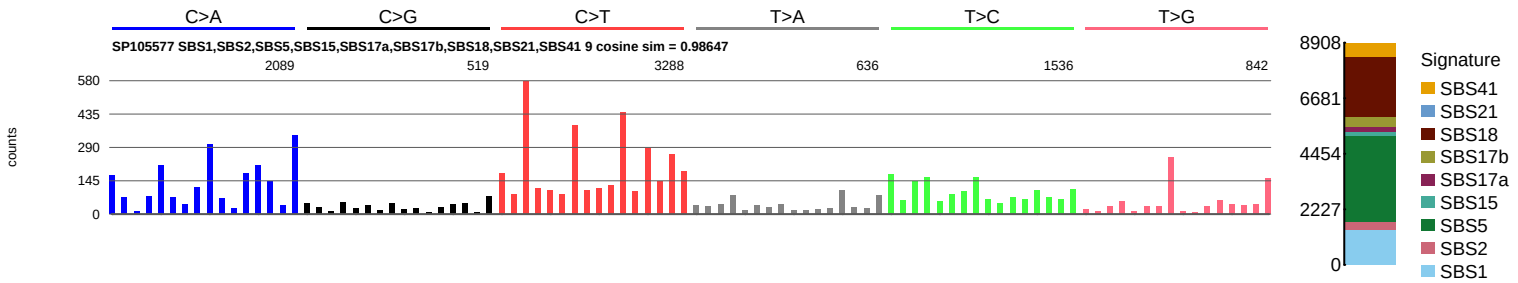
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98647
2089
519
3288
636
1536
842
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS41
SBS21
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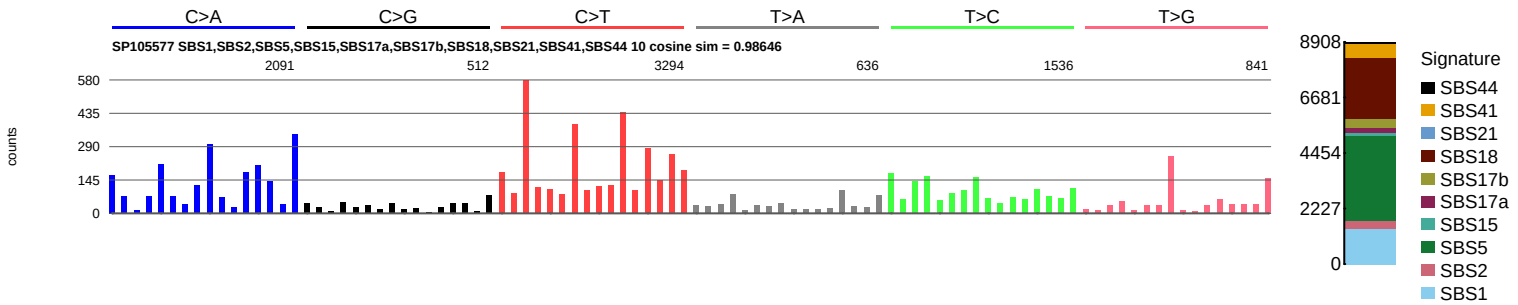
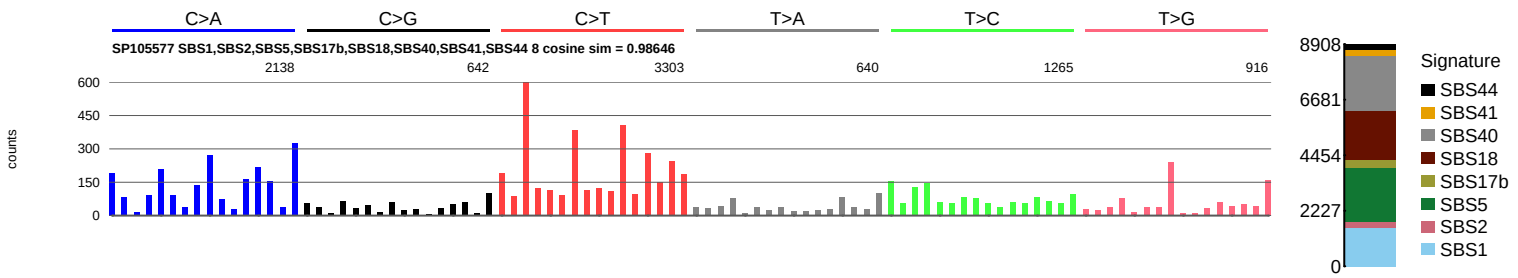
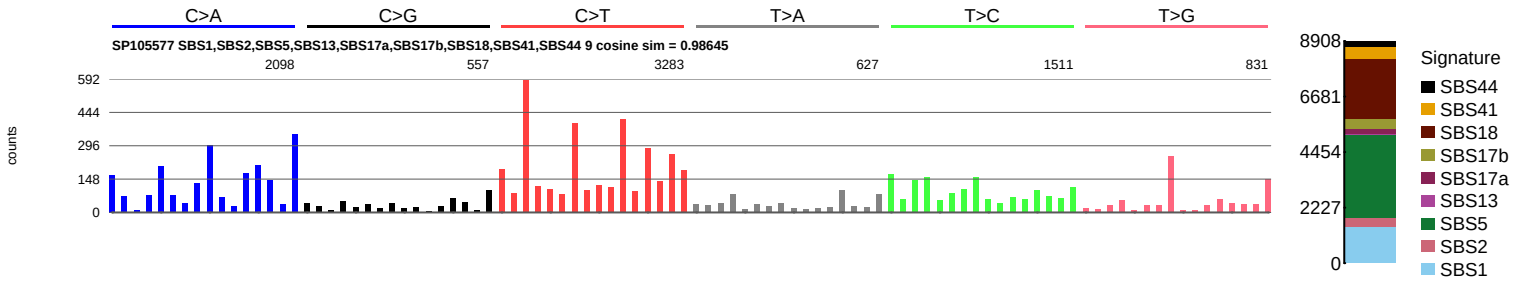
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98646
2091
512
3294
636
1536
841
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98646
2138
642
3303
640
1265
916
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1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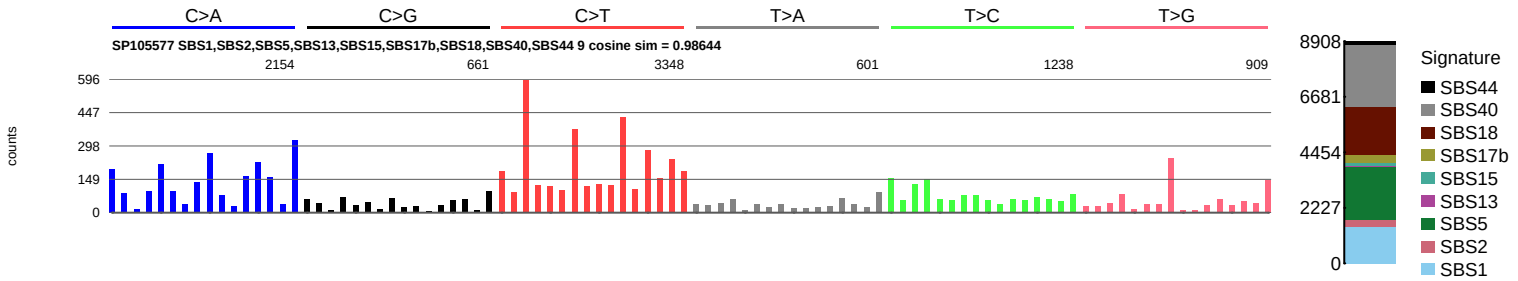
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98645
2098
557
3283
627
1511
831
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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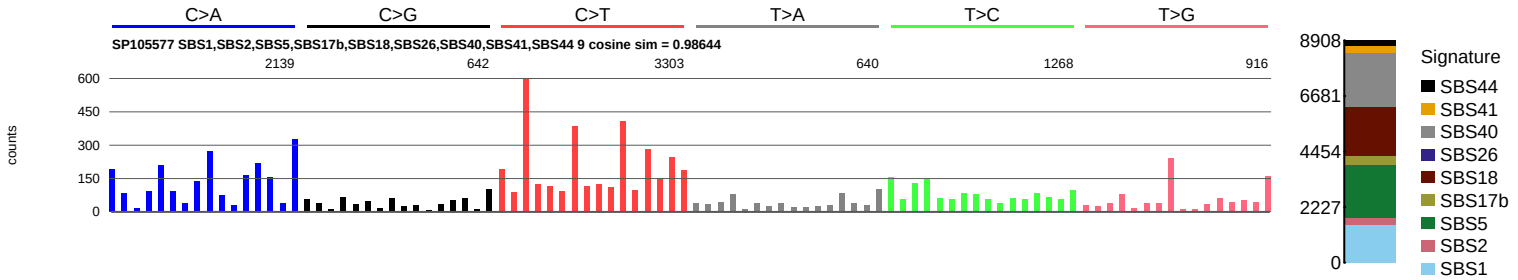
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98644
2154
661
3348
601
1238
909
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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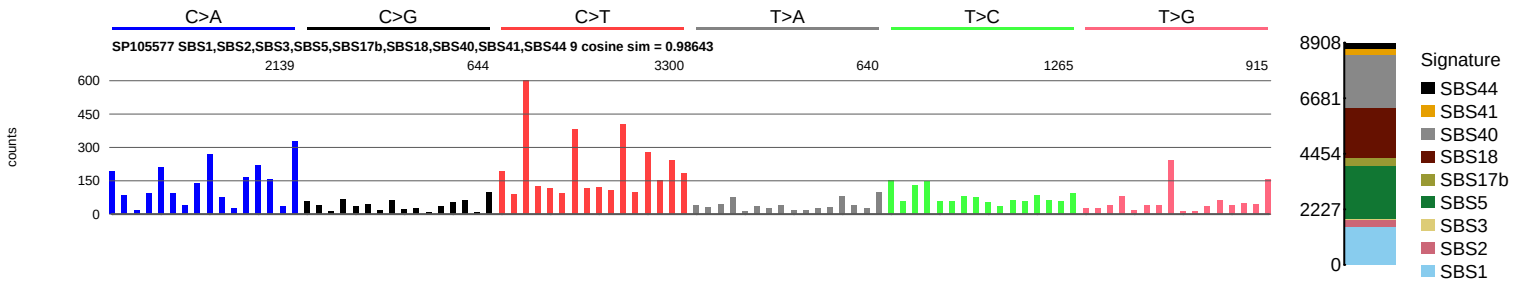
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98644
2139
642
3303
640
1268
916
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1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C>T
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T>C
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SP105577 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98643
2139
644
3300
640
1265
915
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
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SBS17b
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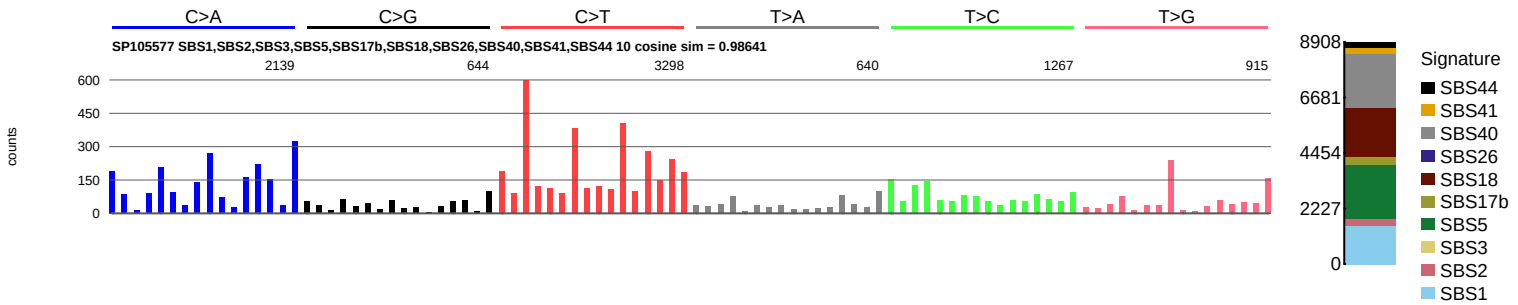
SBS2
SBS1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C>G
C>T
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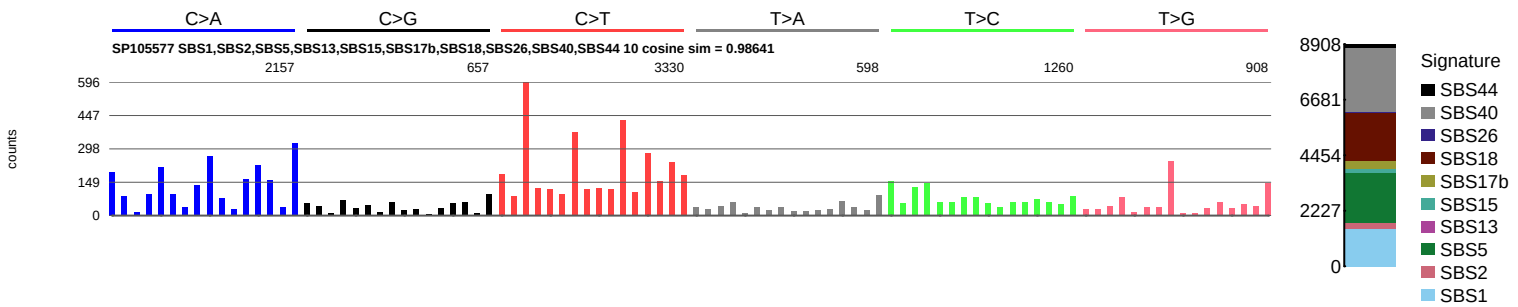
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98641
2139
644
3298
640
1267
915
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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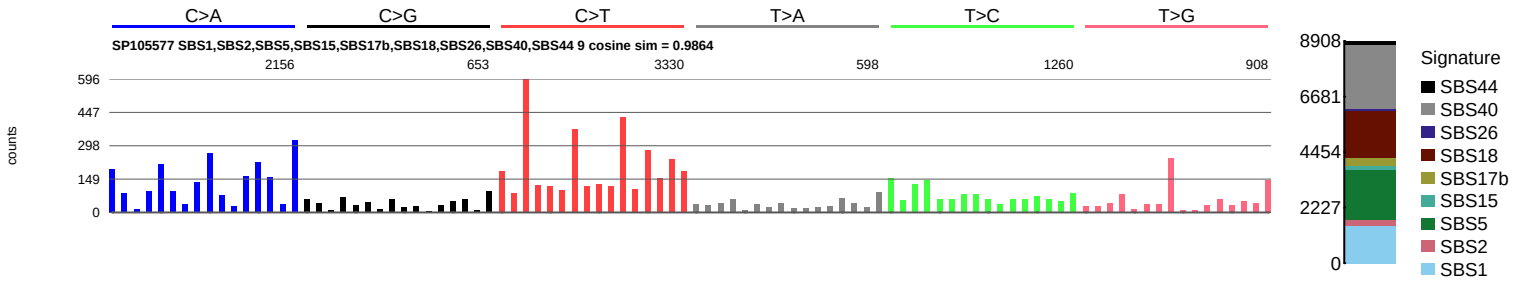
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98641
2157
657
3330
598
1260
908
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.9864
2156
653
3330
598
1260
908
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
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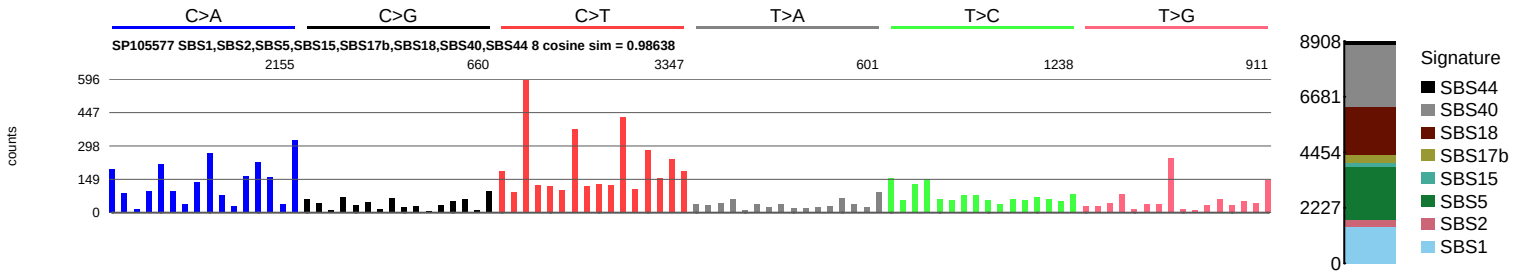
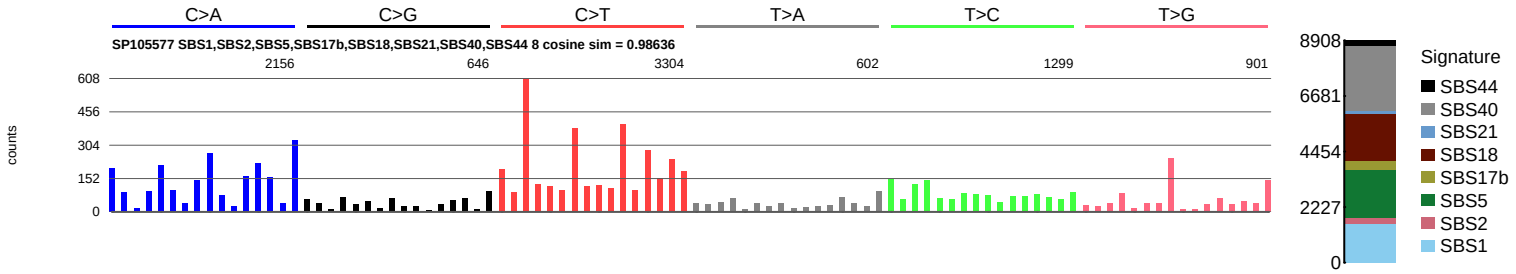
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98638
2155
660
3347
601
1238
911
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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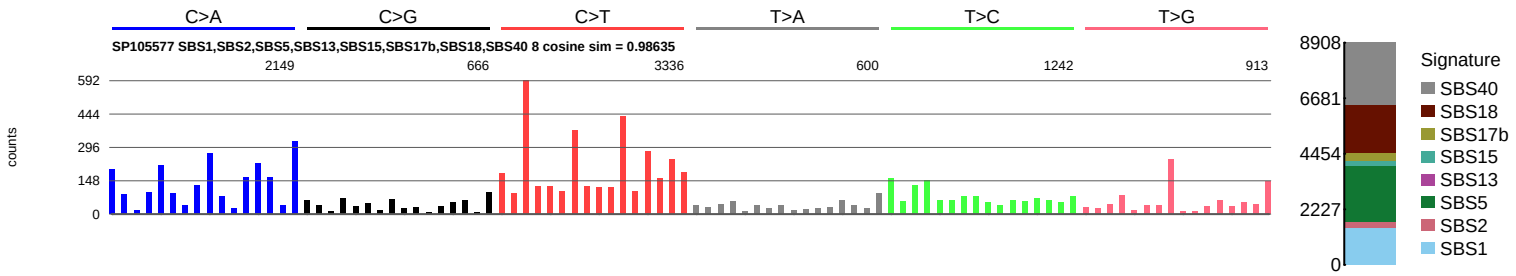
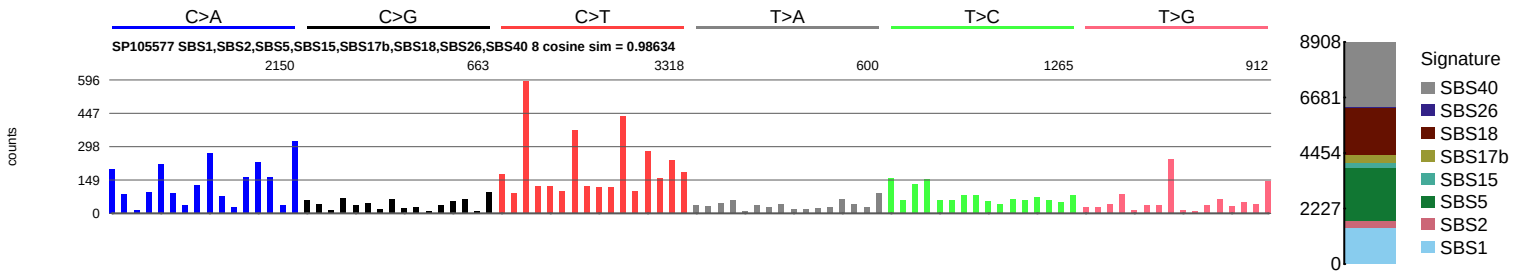
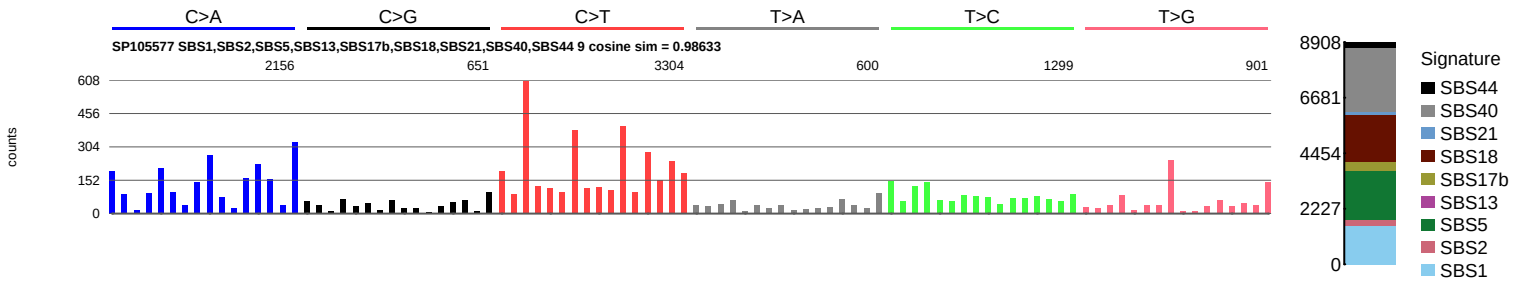
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98636
2156
646
3304
602
1299
901
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98635
2149
666
3336
600
1242
913
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98634
2150
663
3318
600
1265
912
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98633
2156
651
3304
600
1299
901
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
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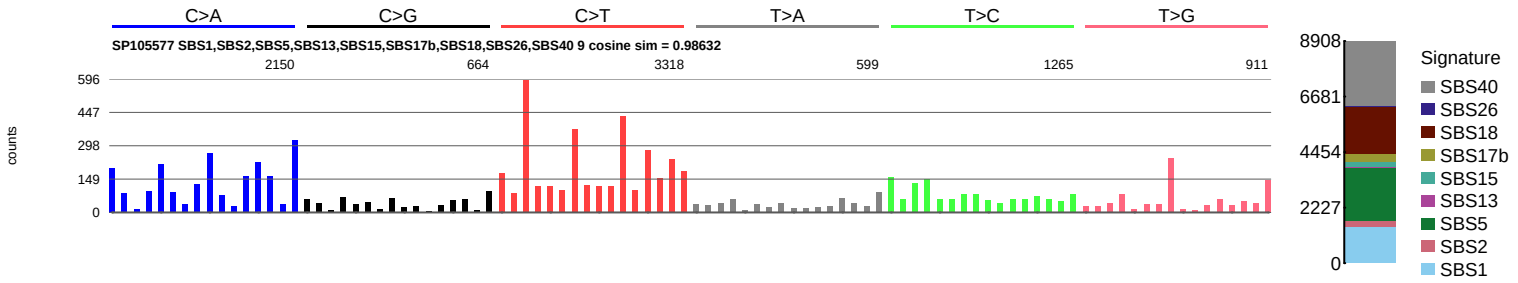
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98632
2150
664
3318
599
1265
911
596
447
298
149
0
counts
8908
6681
4454
2227
0
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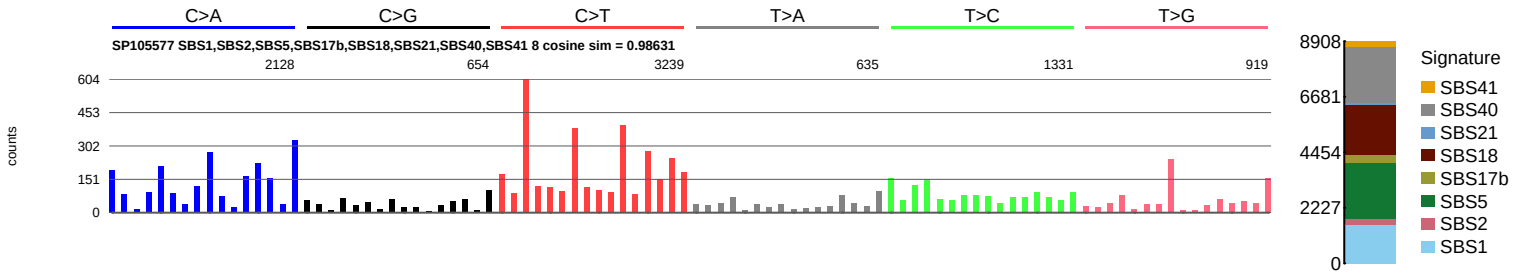
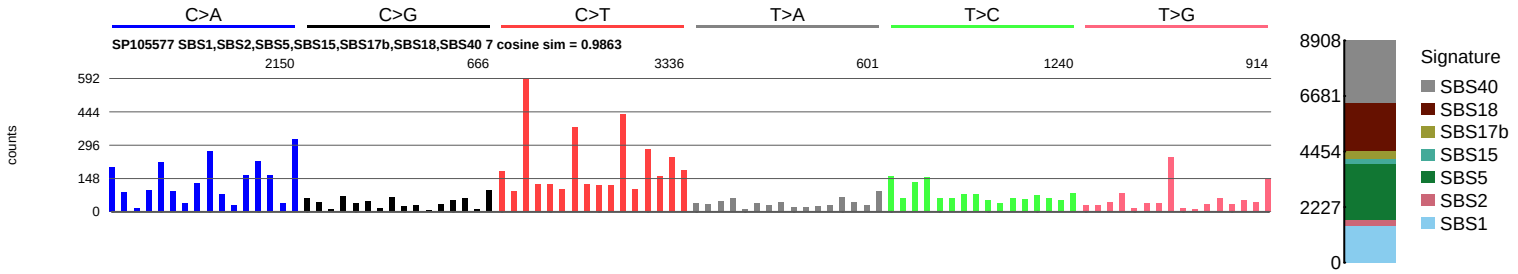
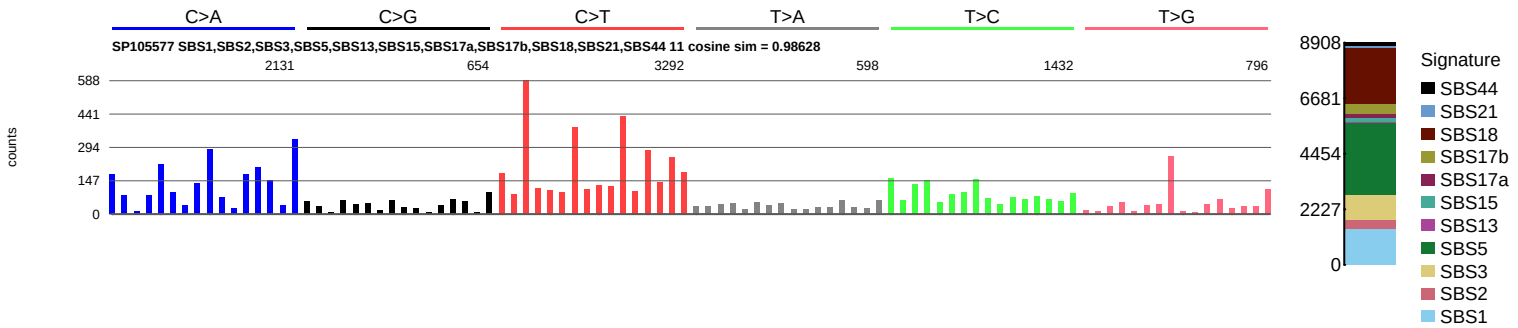
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98631
2128
654
3239
635
1331
919
604
453
302
151
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.9863
2150
666
3336
601
1240
914
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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C>G
C>T
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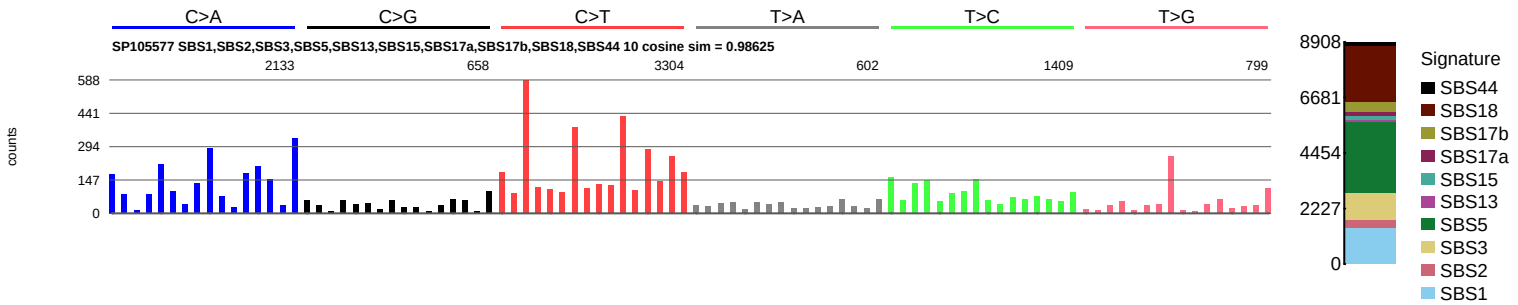
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98628
2131
654
3292
598
1432
796
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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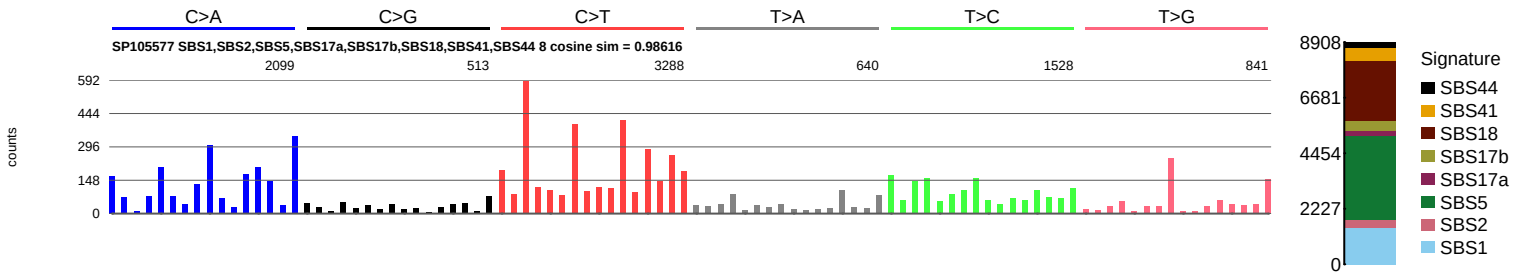
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98625
2133
658
3304
602
1409
799
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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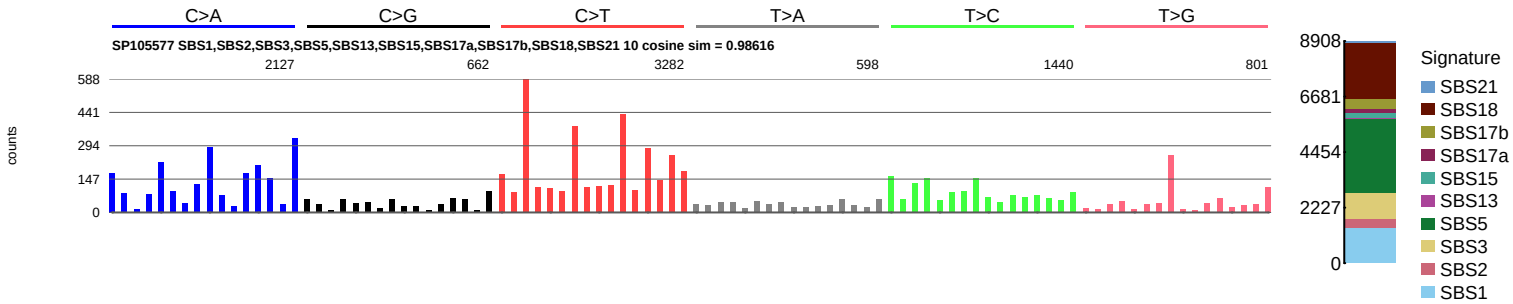
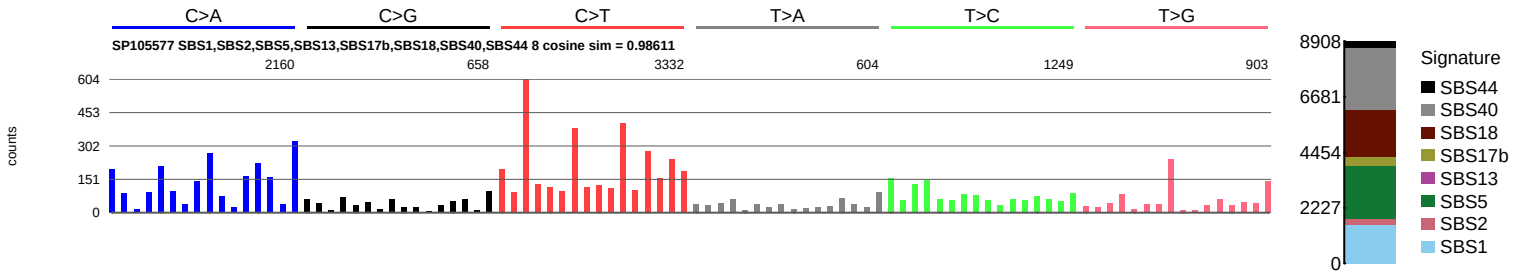
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98616
2099
513
3288
640
1528
841
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98616
2127
662
3282
598
1440
801
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98611
2160
658
3332
604
1249
903
604
453
302
151
0
counts
8908
6681
4454
2227
0
Signature
SBS44
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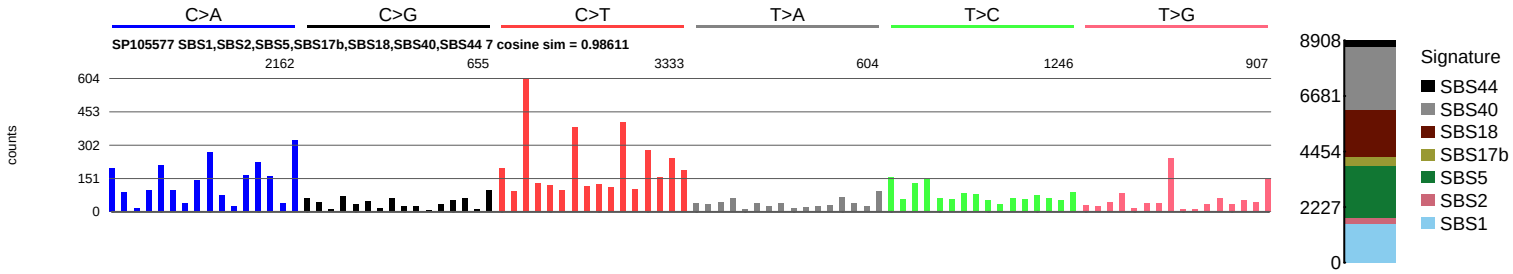
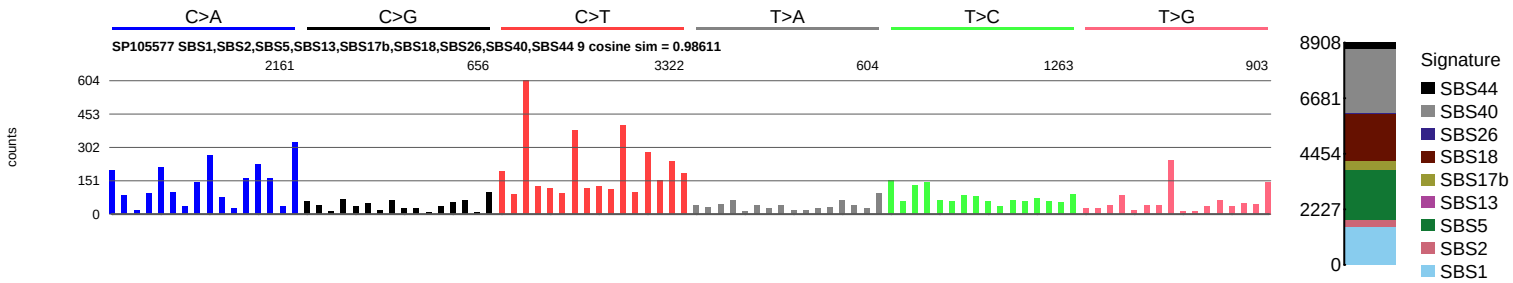
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98611
2162
655
3333
604
1246
907
604
453
302
151
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98611
2161
656
3322
604
1263
903
604
453
302
151
0
counts
8908
6681
4454
2227
0
Signature
SBS44
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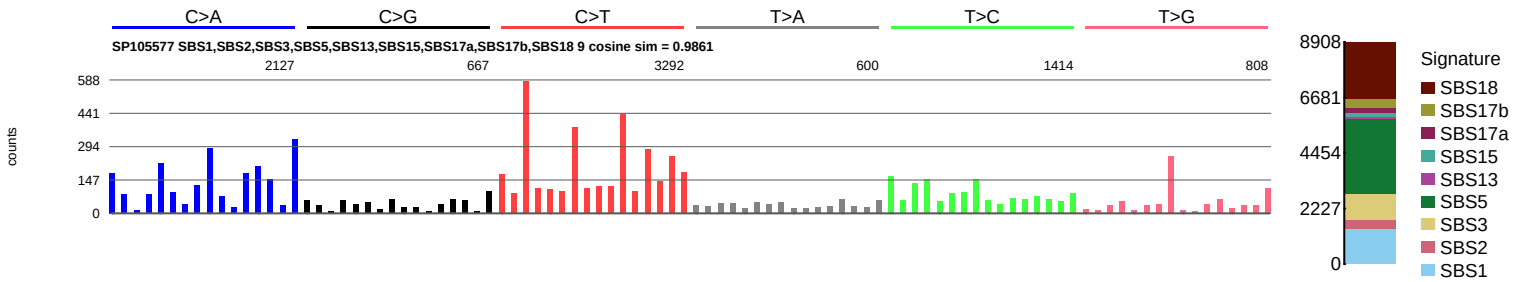
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.9861
2127
667
3292
600
1414
808
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS18
SBS17b
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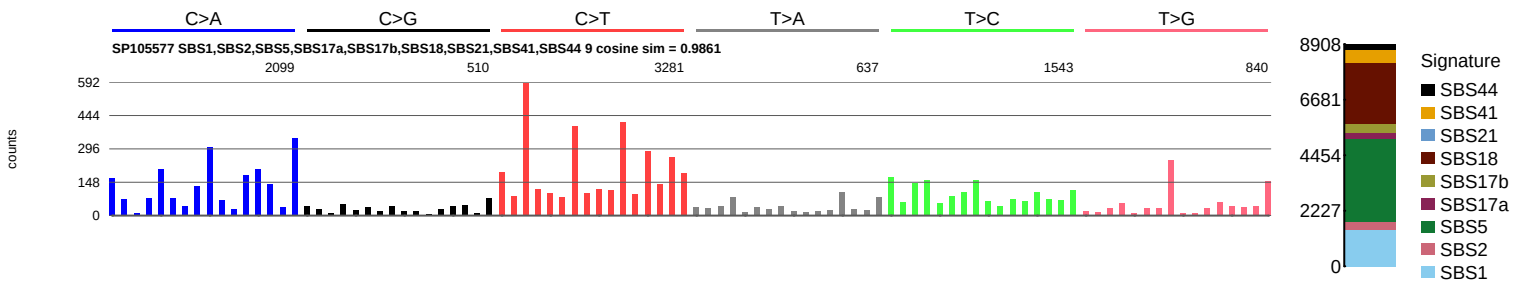
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.9861
2099
510
3281
637
1543
840
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
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SBS18
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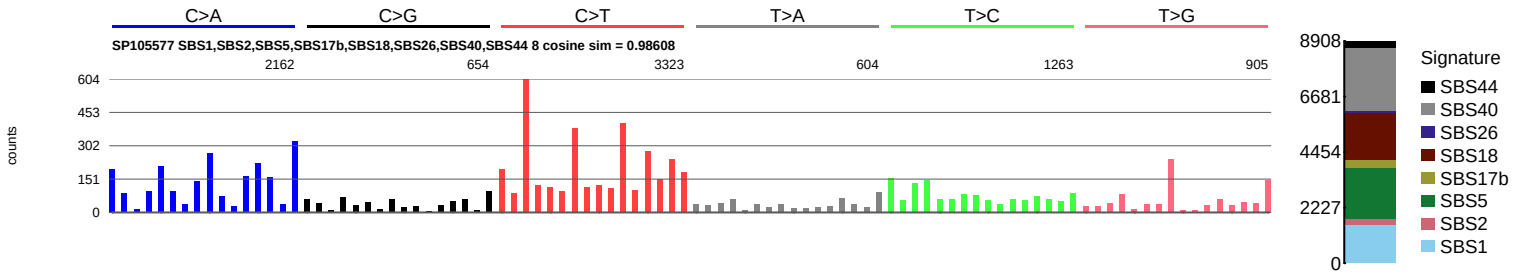
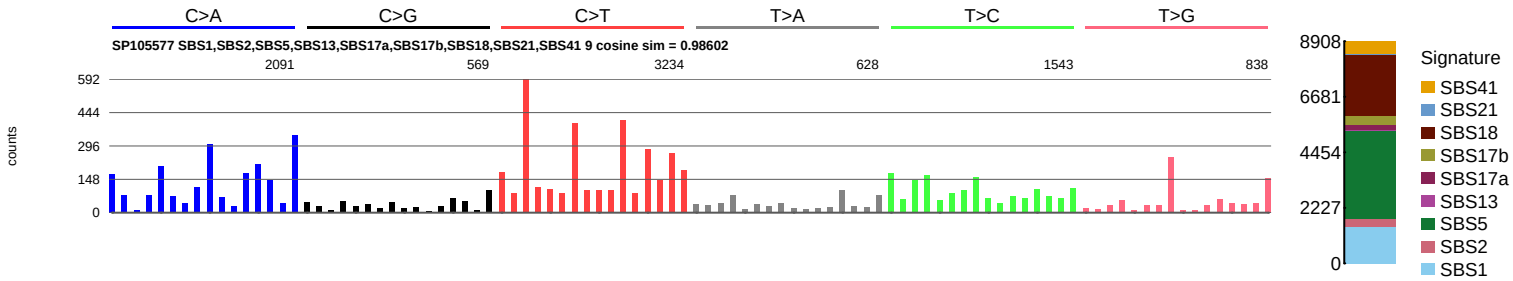
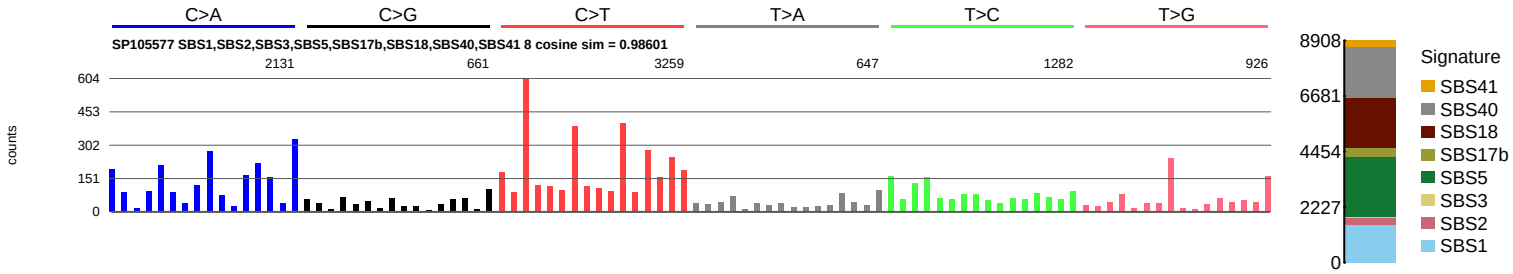
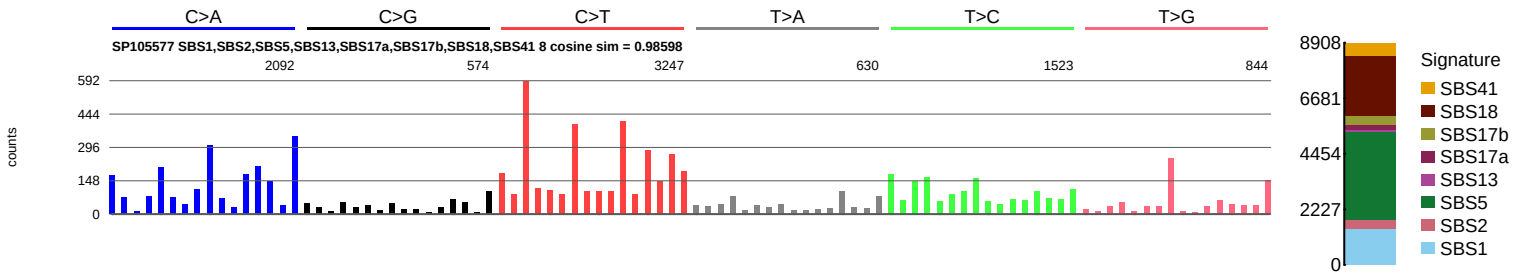
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98608
2162
654
3323
604
1263
905
counts
604
453
302
151
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98602
2091
569
3234
628
1543
838
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98601
2131
661
3259
647
1282
926
counts
604
453
302
151
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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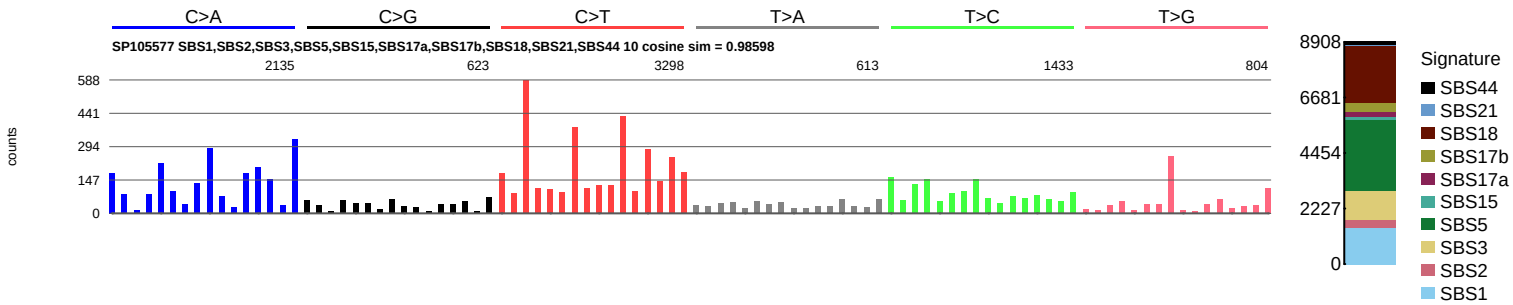
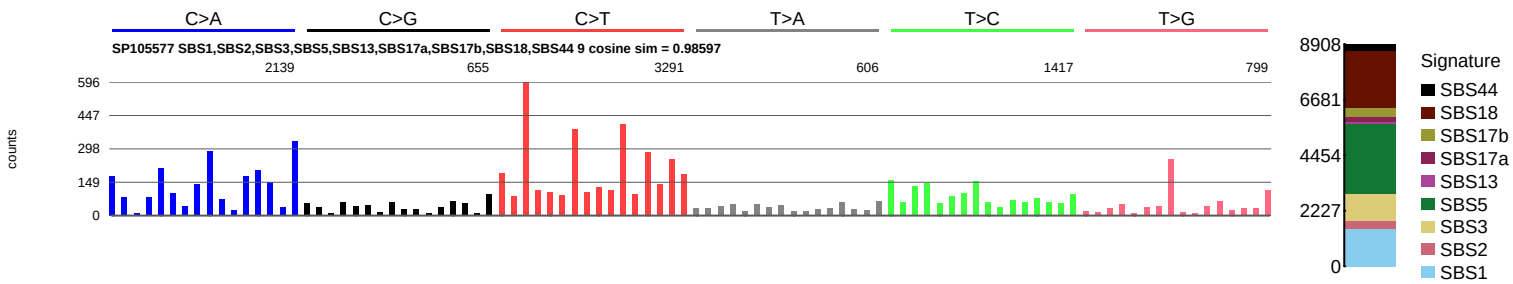
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98598
2092
574
3247
630
1523
844
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98598
2135
623
3298
613
1433
804
counts
588
441
294
147
0
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98597
2139
655
3291
606
1417
799
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS44
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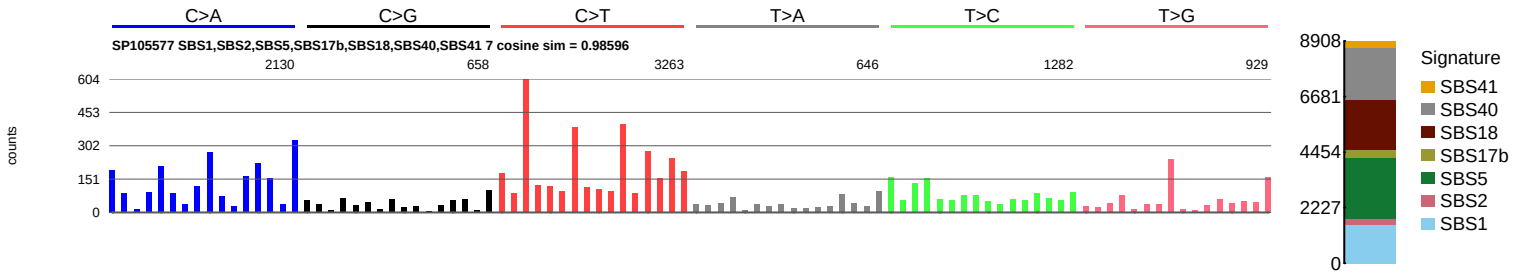
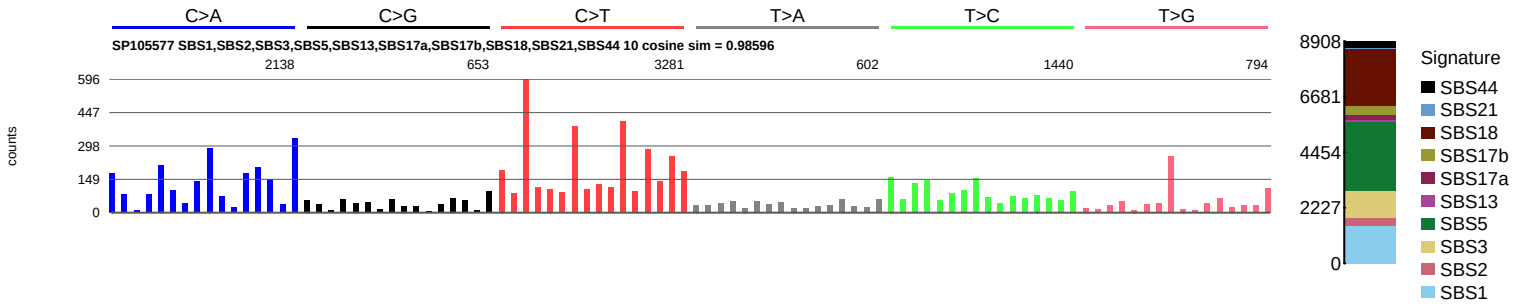
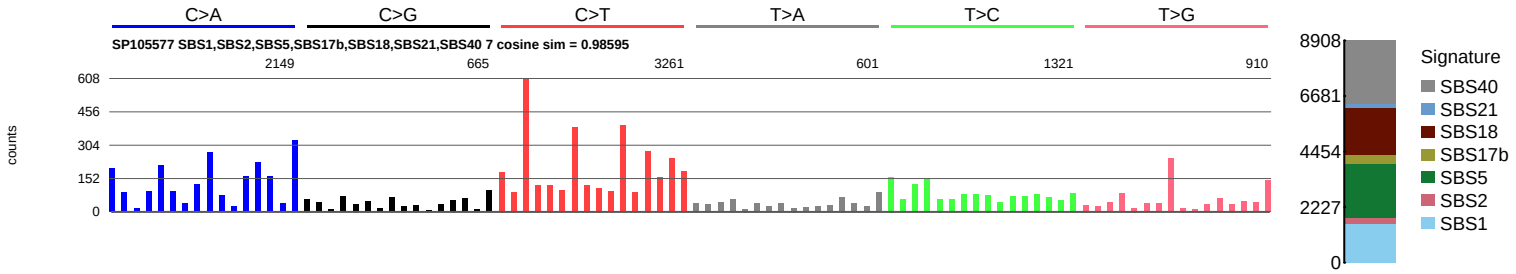
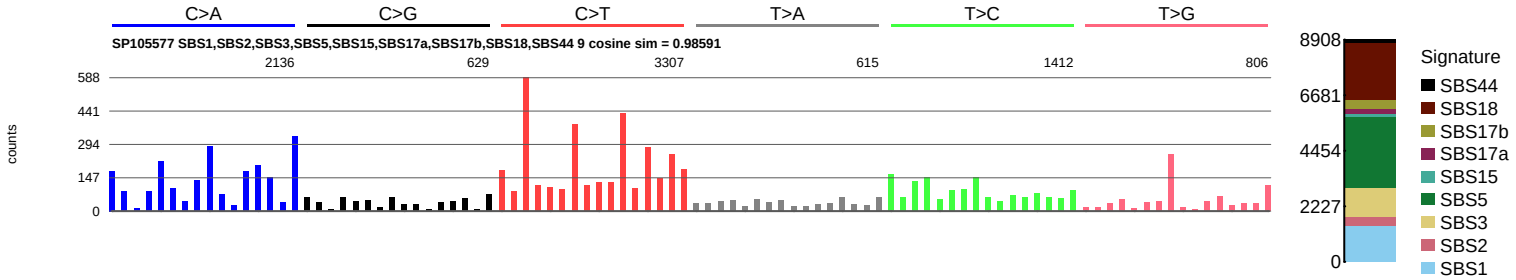
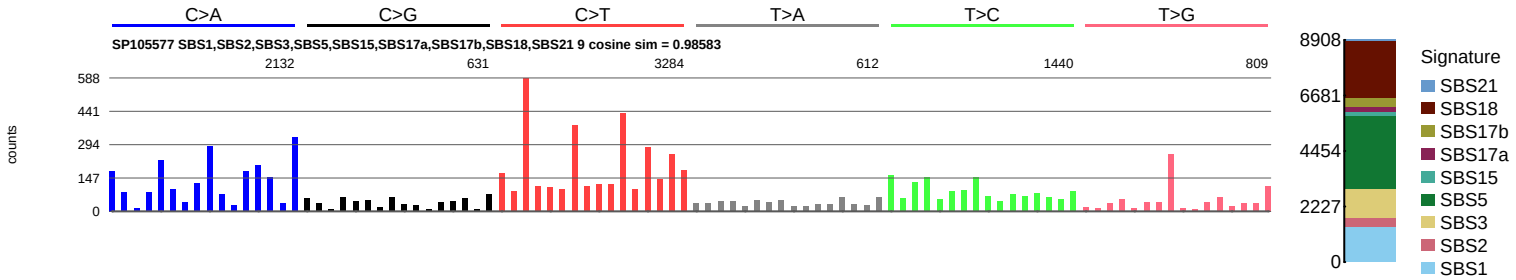
SBS3
SBS2
SBS1

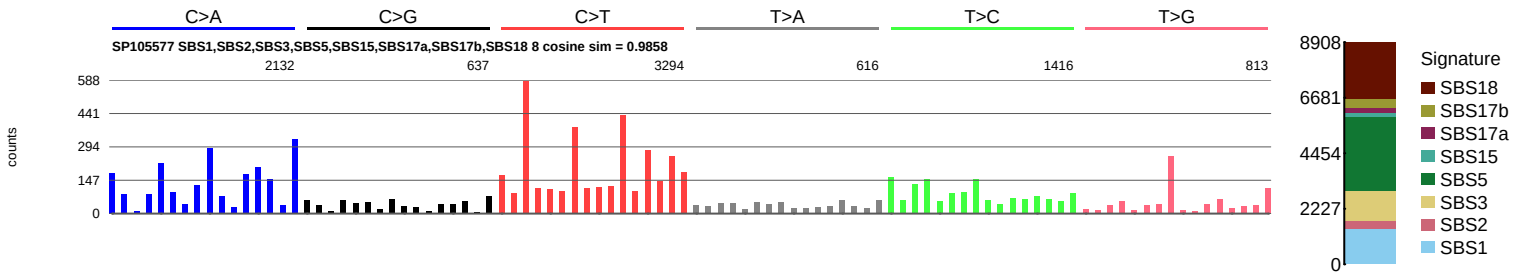

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98596
2130
658
3263
646
1282
929
604
453
302
151
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1

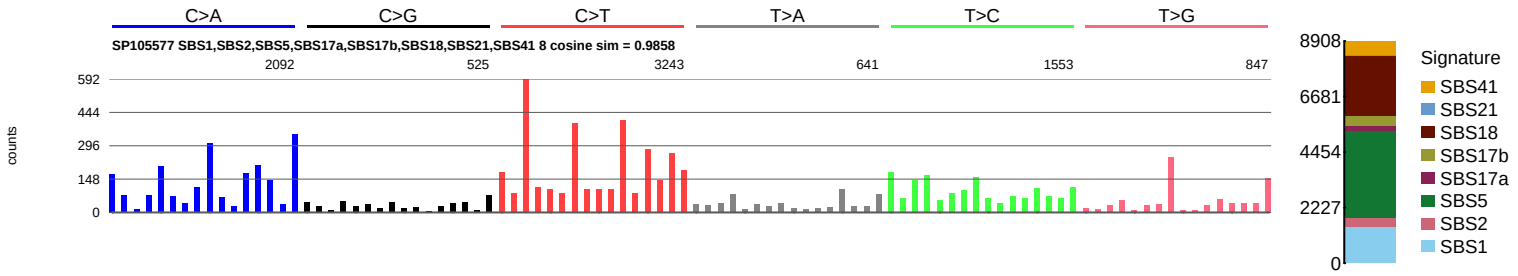

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98596
2138
653
3281
602
1440
794
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

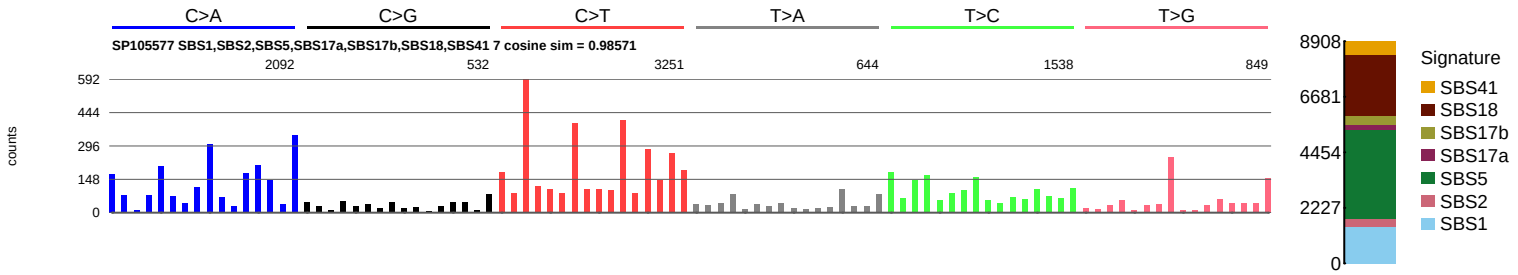

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98595
2149
665
3261
601
1321
910
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

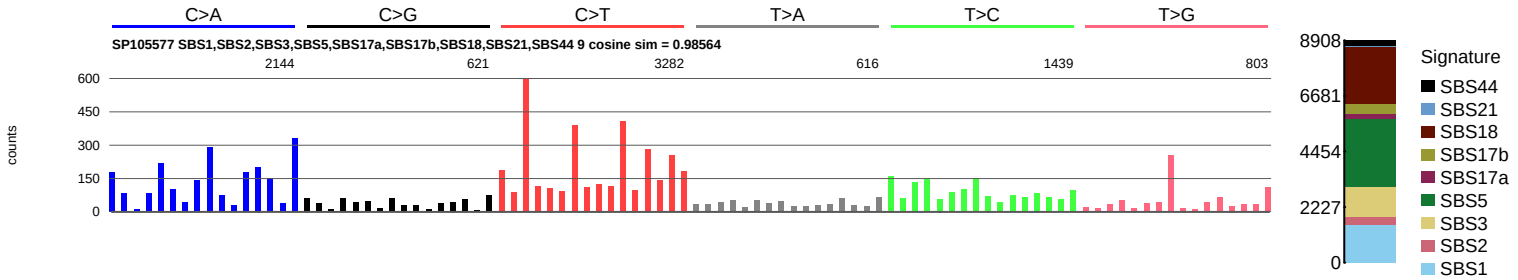

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98591
2136
629
3307
615
1412
806
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

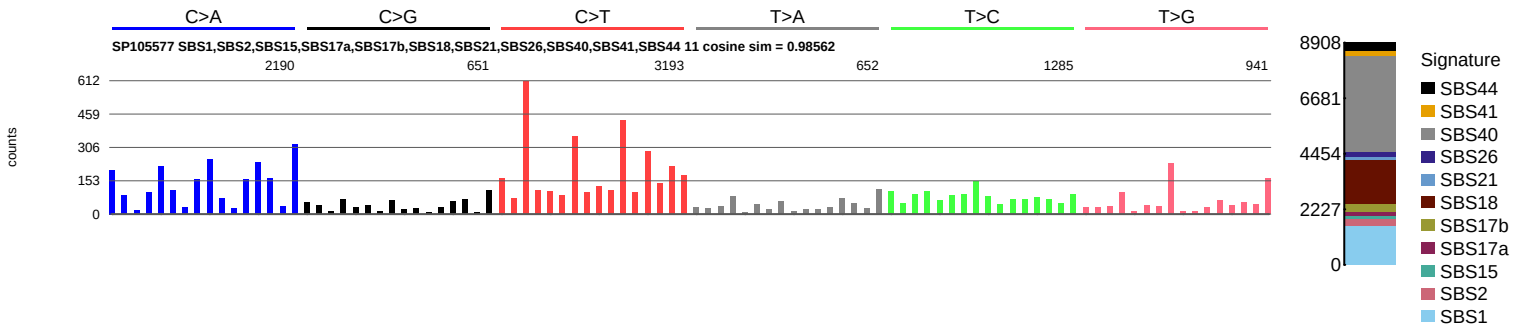

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98583
2132
631
3284
612
1440
809
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

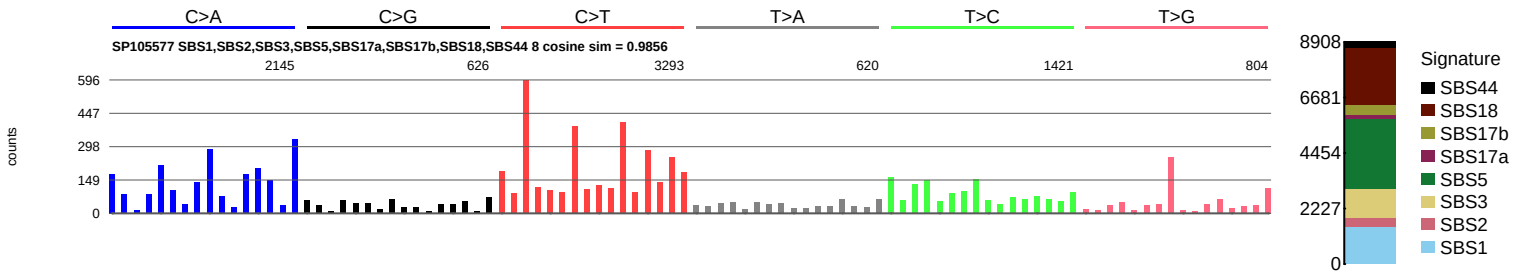

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.9858
2132
637
3294
616
1416
813
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

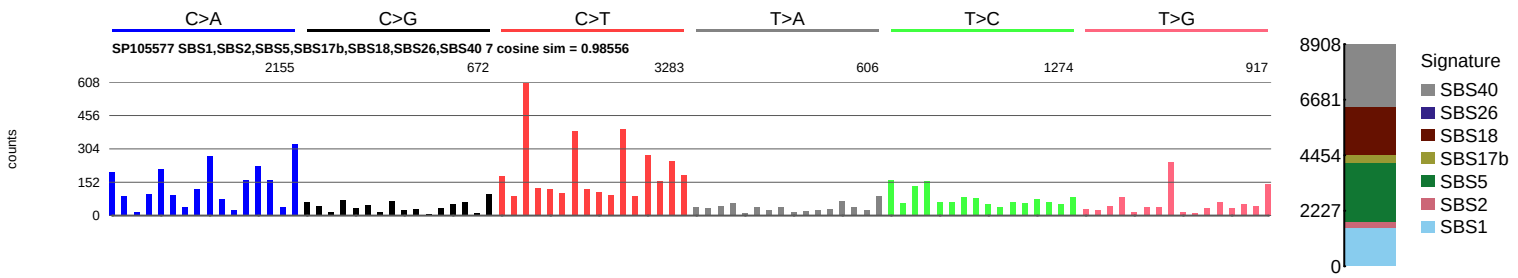

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.9858
2092
525
3243
641
1553
847
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

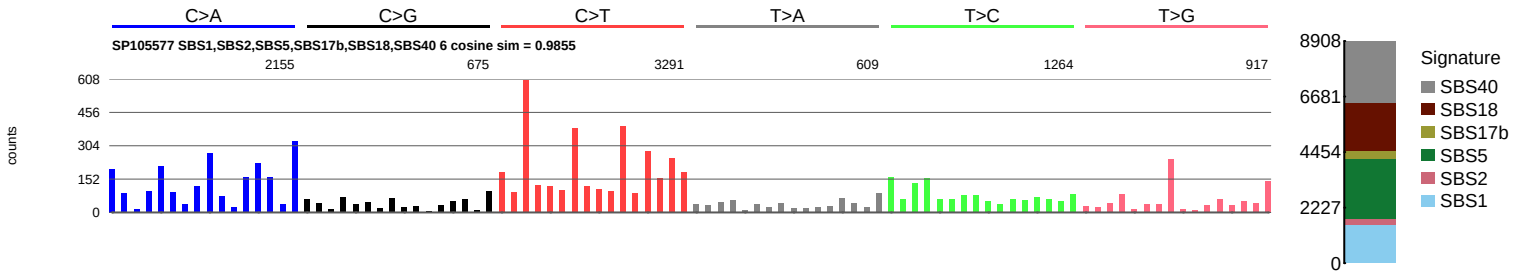

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98571
2092
532
3251
644
1538
849
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

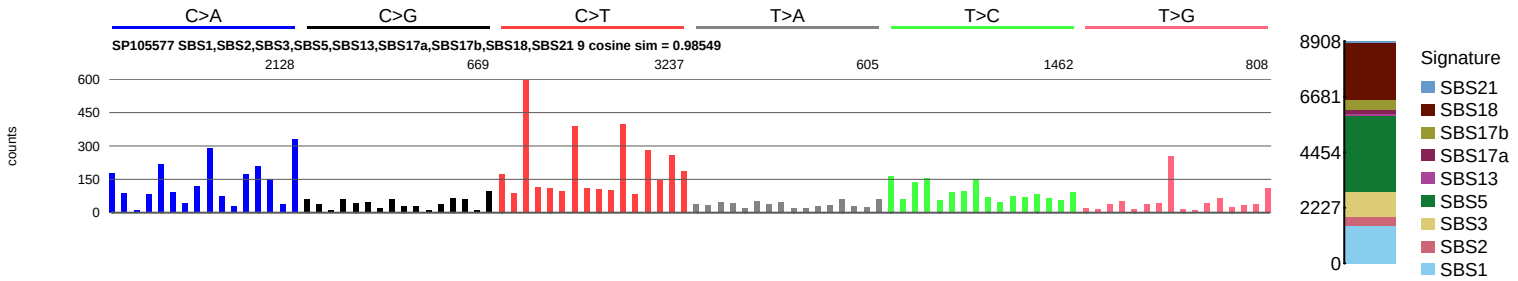

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98564
2144
621
3282
616
1439
803
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

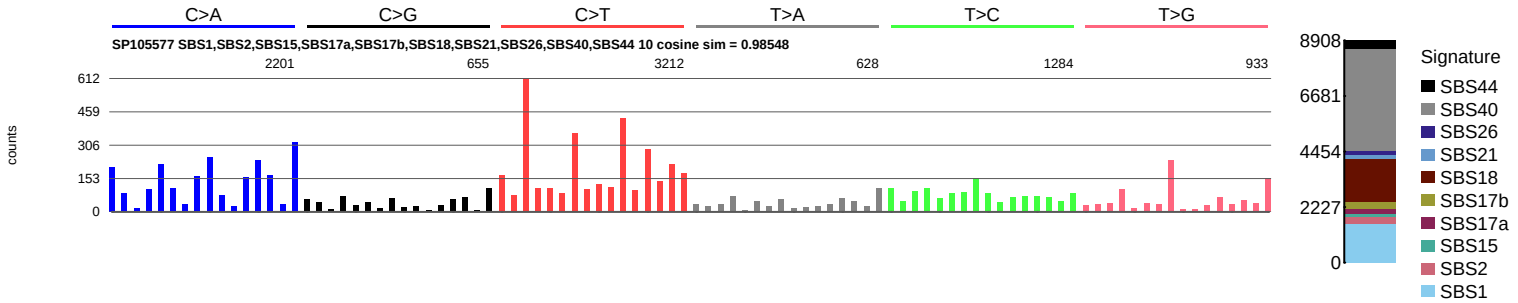

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98562
2190
651
3193
652
1285
941
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

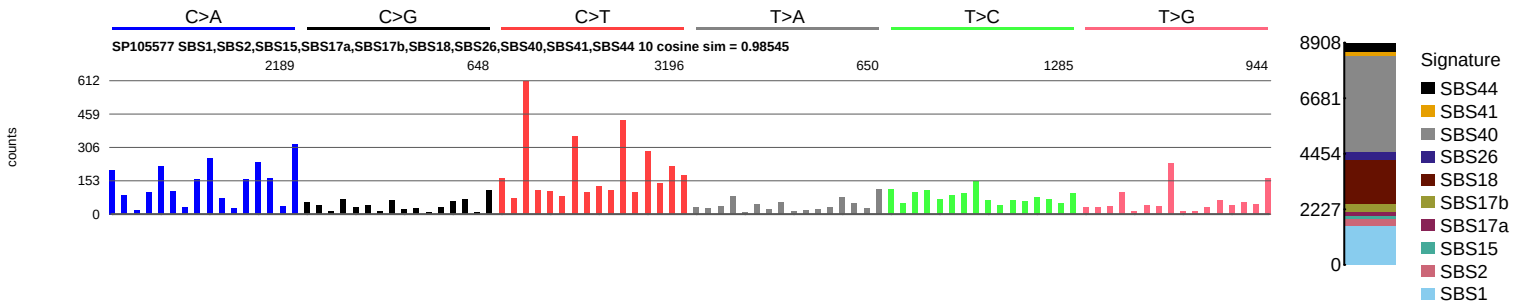

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.9856
2145
626
3293
620
1421
804
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

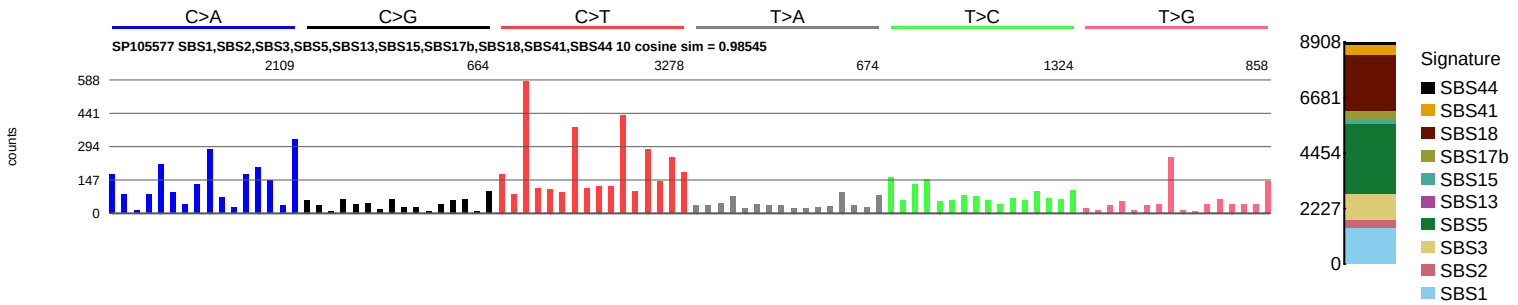

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98556
2155
672
3283
606
1274
917
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1

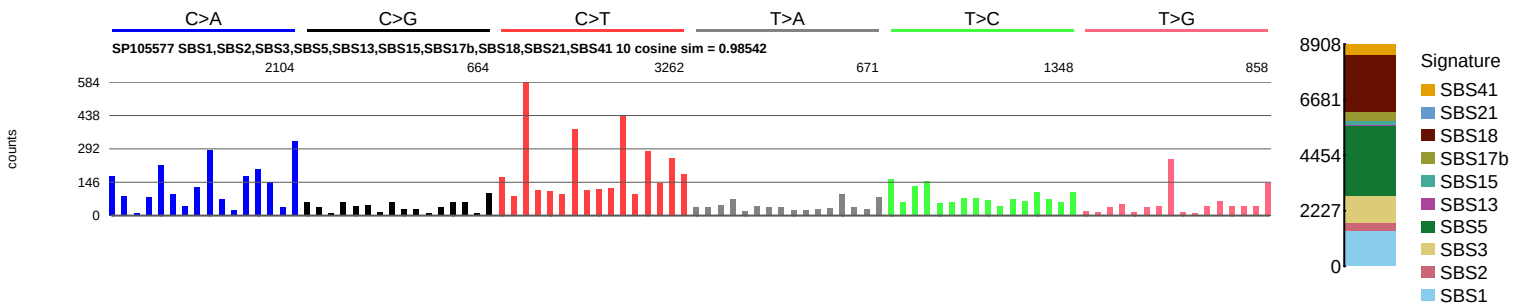

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40 6 cosine sim = 0.9855
2155
675
3291
609
1264
917
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1

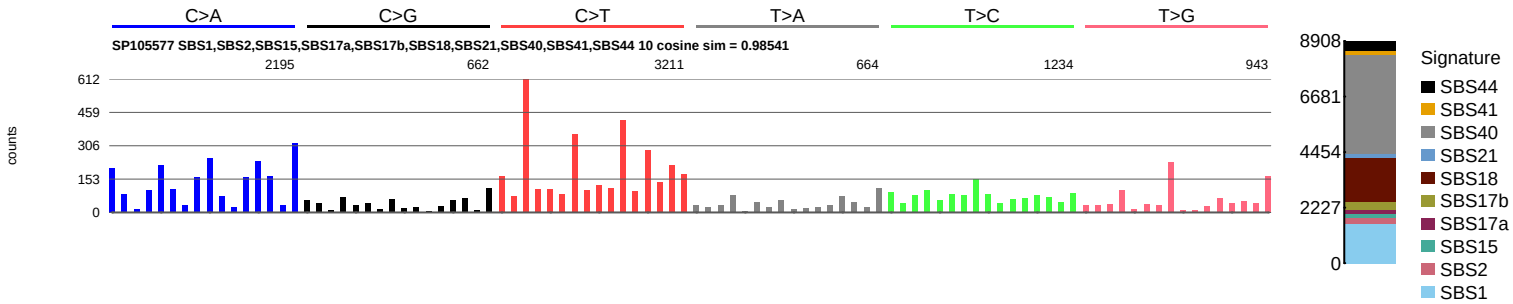

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98549
2128
669
3237
605
1462
808
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

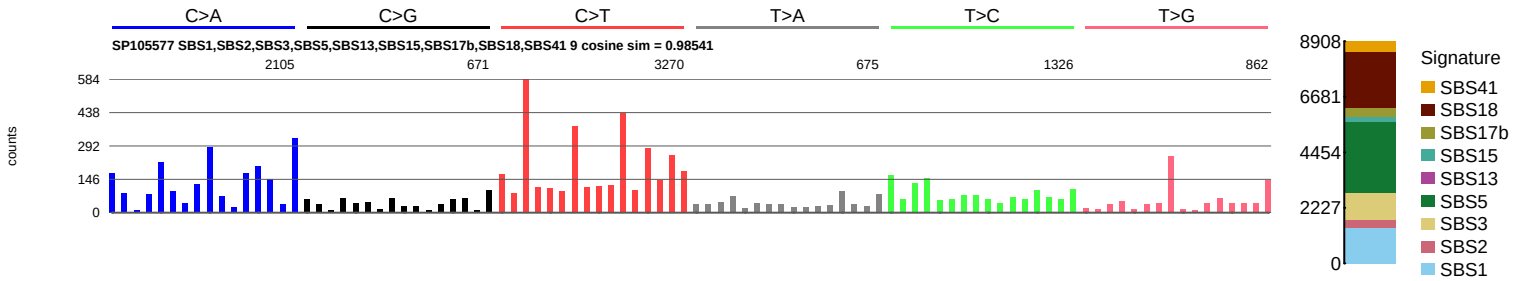

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98548
2201
655
3212
628
1284
933
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98545
2189
648
3196
650
1285
944
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98545
2109
664
3278
674
1324
858
counts
588
441
294
147
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98542
2104
664
3262
671
1348
858
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98541
2195
662
3211
664
1234
943
612
459
306
153
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 9 cosine sim = 0.98541
2105
671
3270
675
1326
862
584
438
292
146
0
counts
8908
6681
4454
2227
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Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
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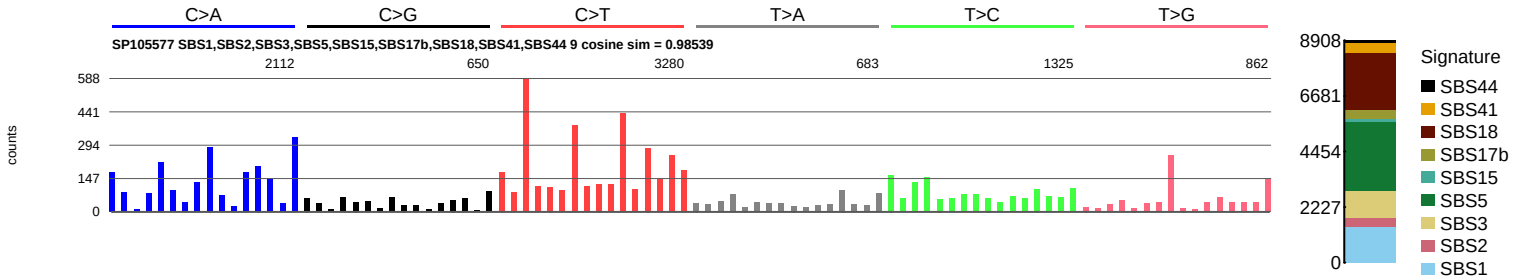
SBS2
SBS1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SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98539
2112
650
3280
683
1325
862
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
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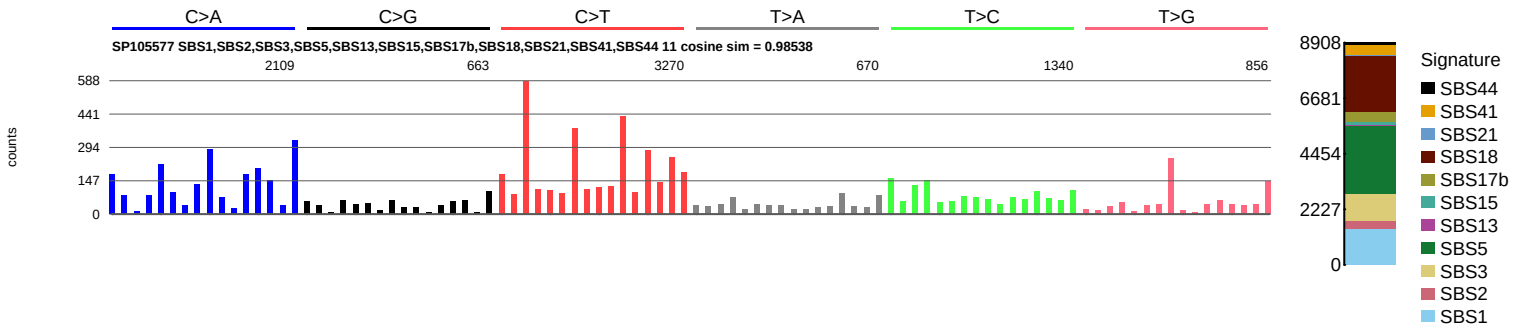
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98538
2109
663
3270
670
1340
856
588
441
294
147
0
counts
8908
6681
4454
2227
0
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SBS44
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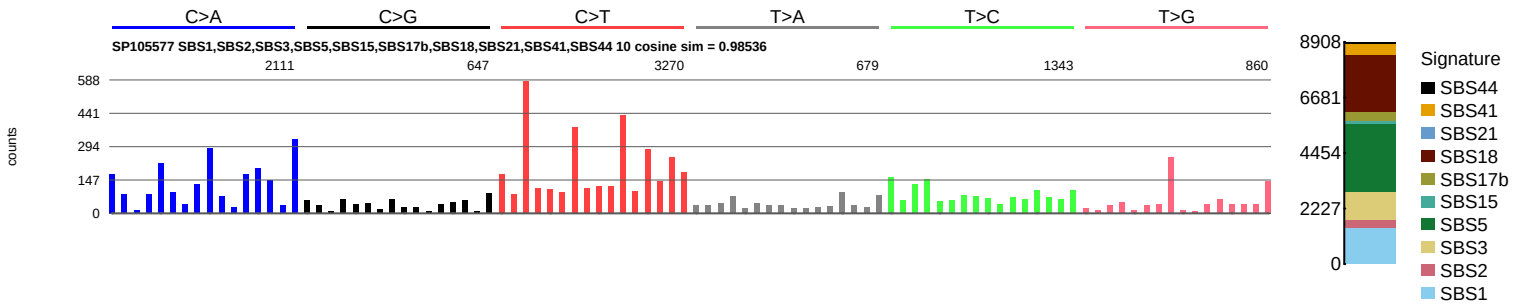
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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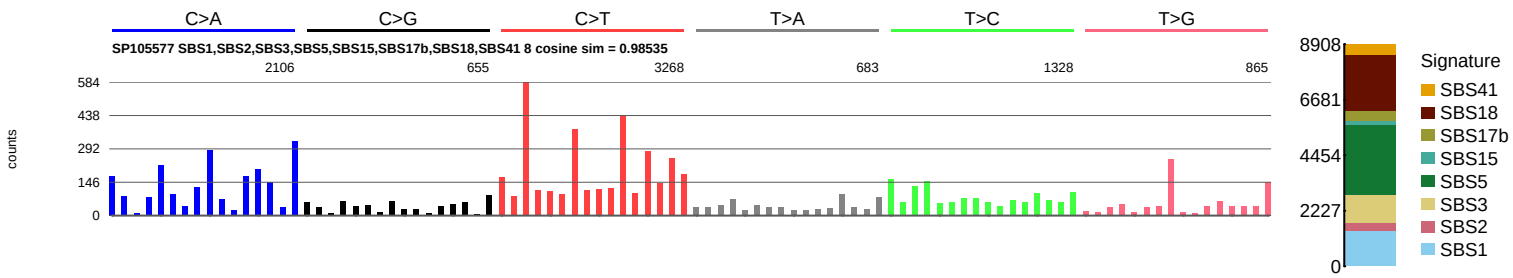
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98536
2111
647
3270
679
1343
860
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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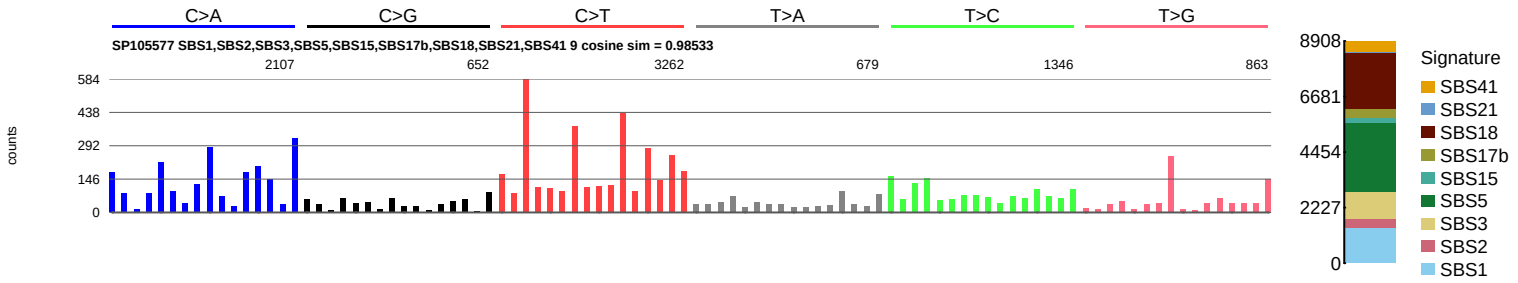
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98535
2106
655
3268
683
1328
865
584
438
292
146
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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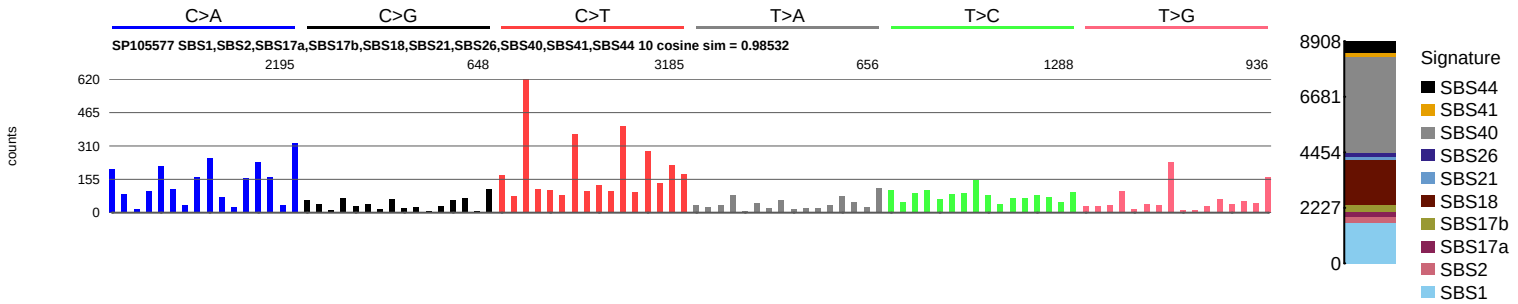
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98533
2107
652
3262
679
1346
863
584
438
292
146
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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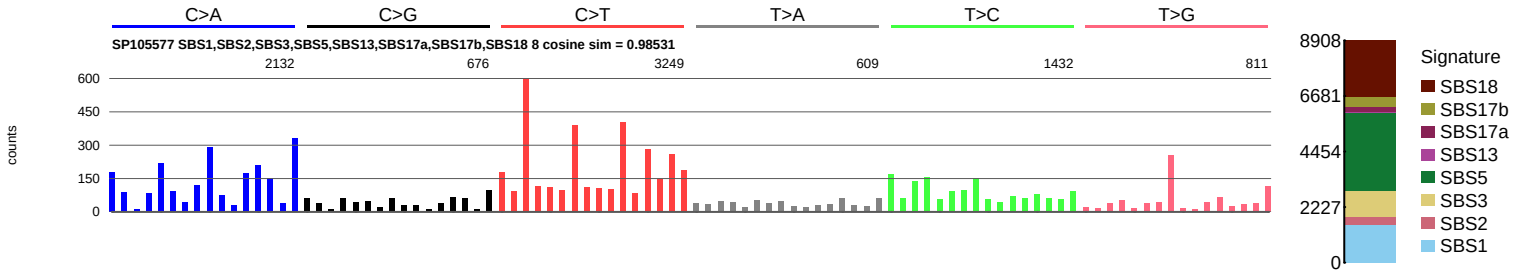
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98532
2195
648
3185
656
1288
936
620
465
310
155
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98531
2132
676
3249
609
1432
811
600
450
300
150
0
counts
8908
6681
4454
2227
0
Signature
SBS18
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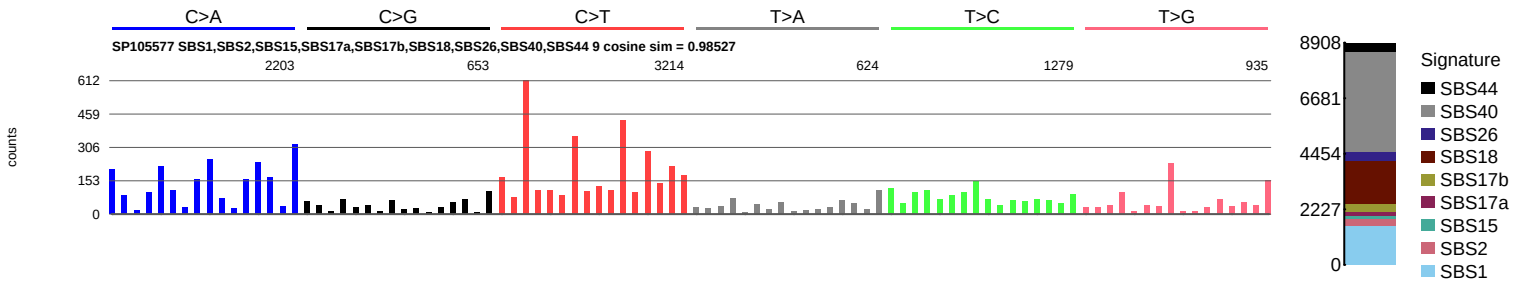
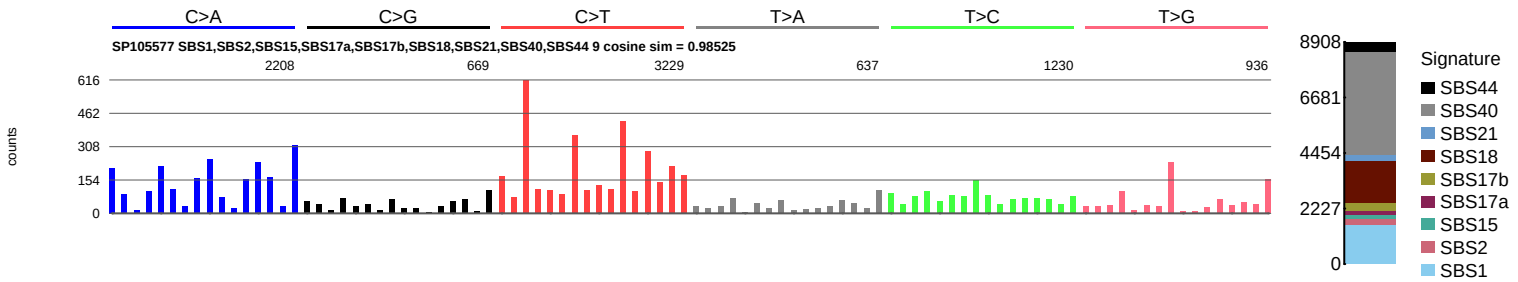
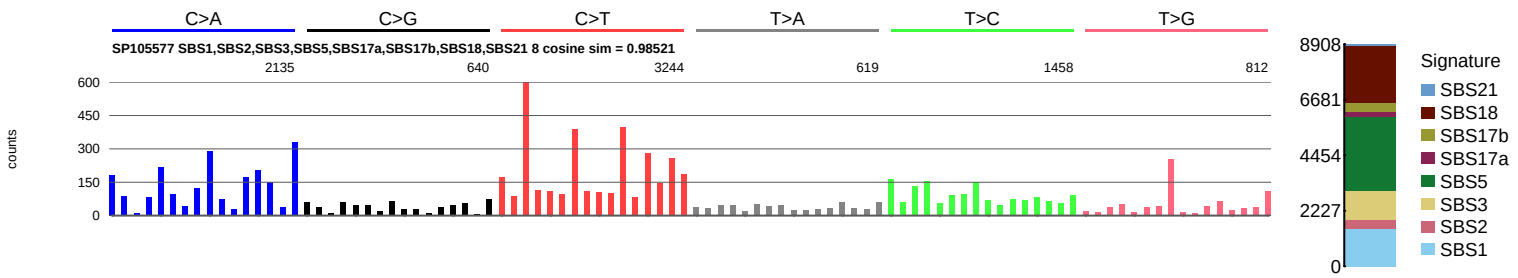
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98527
2203
653
3214
624
1279
935
612
459
306
153
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98525
2208
669
3229
637
1230
936
616
462
308
154
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98521
2135
640
3244
619
1458
812
600
450
300
150
0
counts
8908
6681
4454
2227
0
Signature
SBS21
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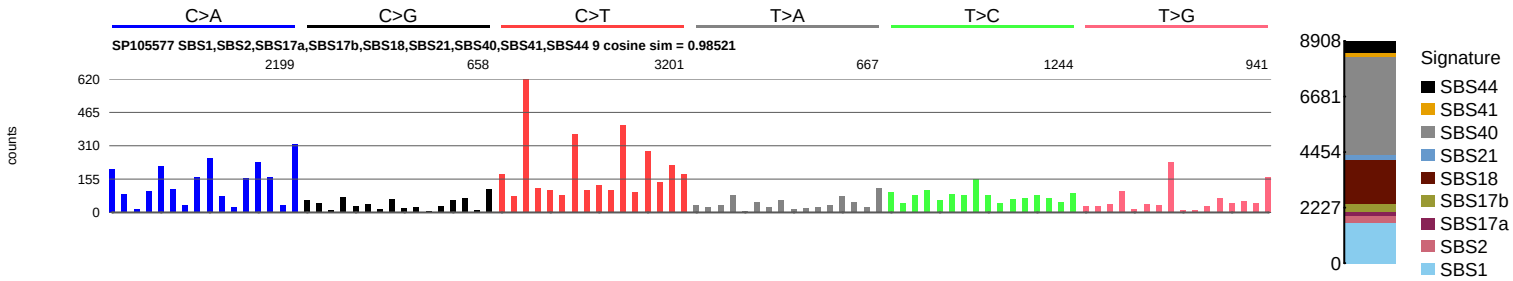
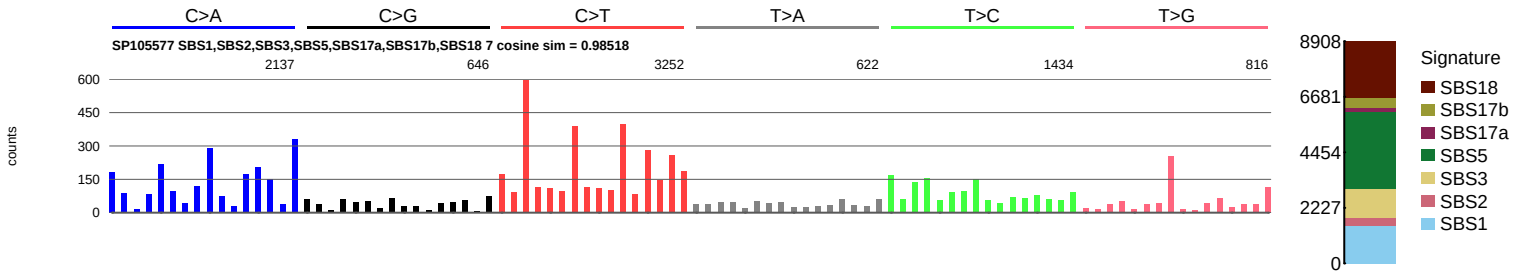
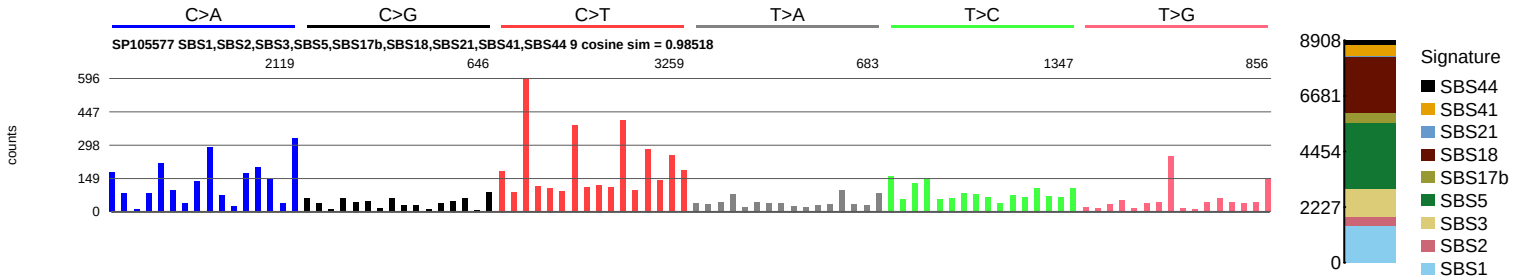
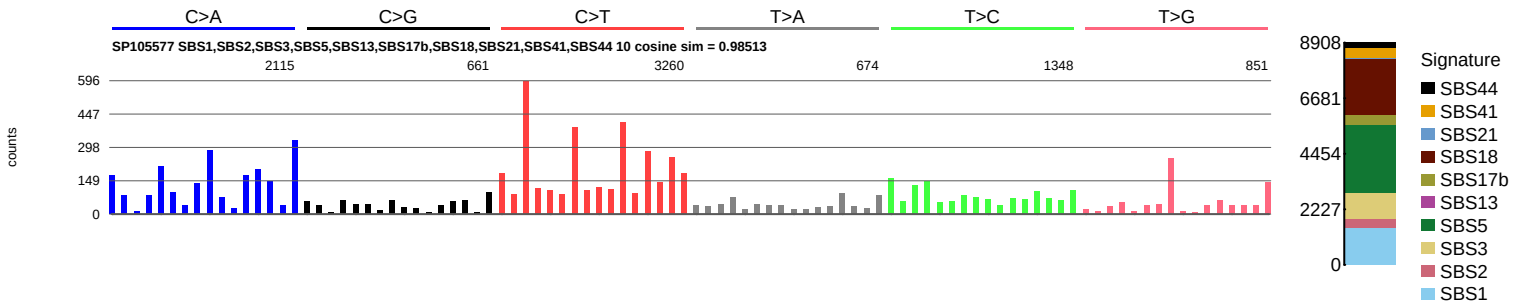
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98521
2199
658
3201
667
1244
941
counts
620
465
310
155
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98518
2137
646
3252
622
1434
816
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98518
2119
646
3259
683
1347
856
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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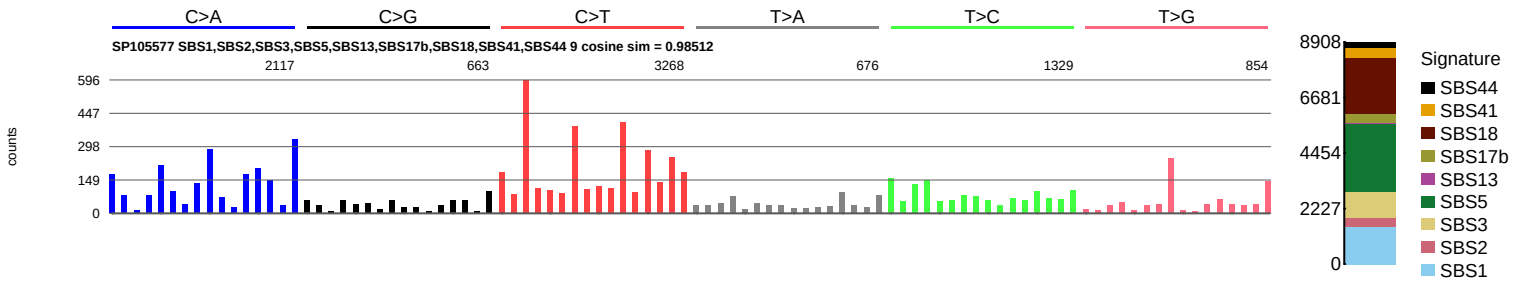
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98513
2115
661
3260
674
1348
851
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98512
2117
663
3268
676
1329
854
counts
596
447
298
149
0
8908
6681
4454
2227
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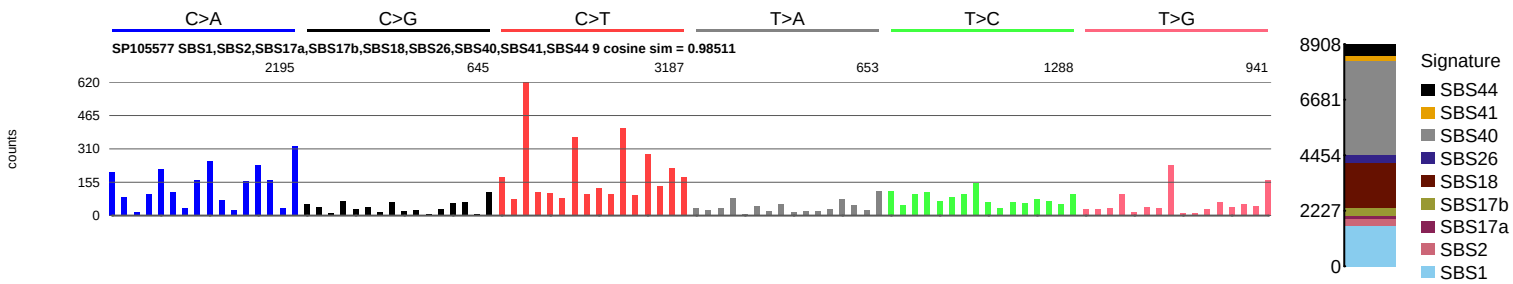
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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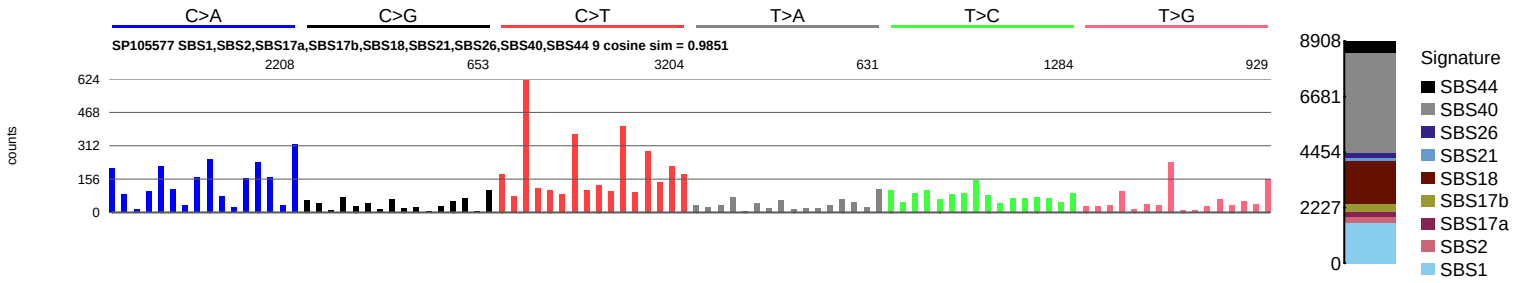
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98511
2195
645
3187
653
1288
941
counts
620
465
310
155
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
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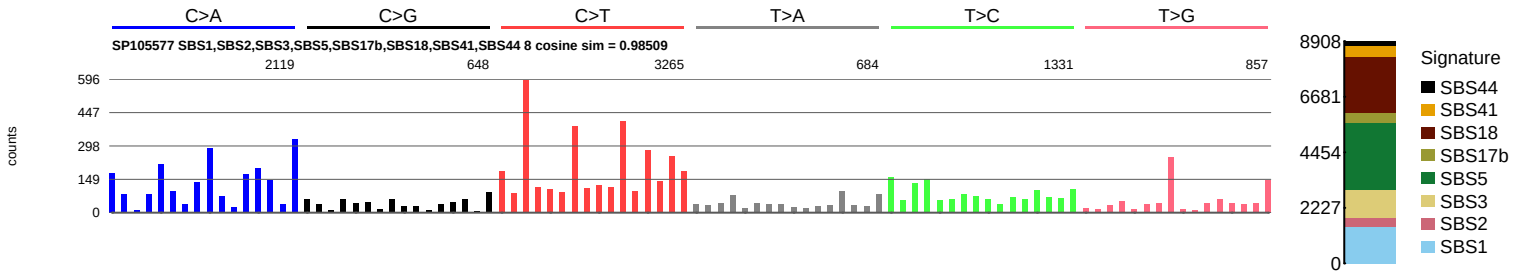
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.9851
2208
653
3204
631
1284
929
counts
624
468
312
156
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98509
2119
648
3265
684
1331
857
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
SBS17b
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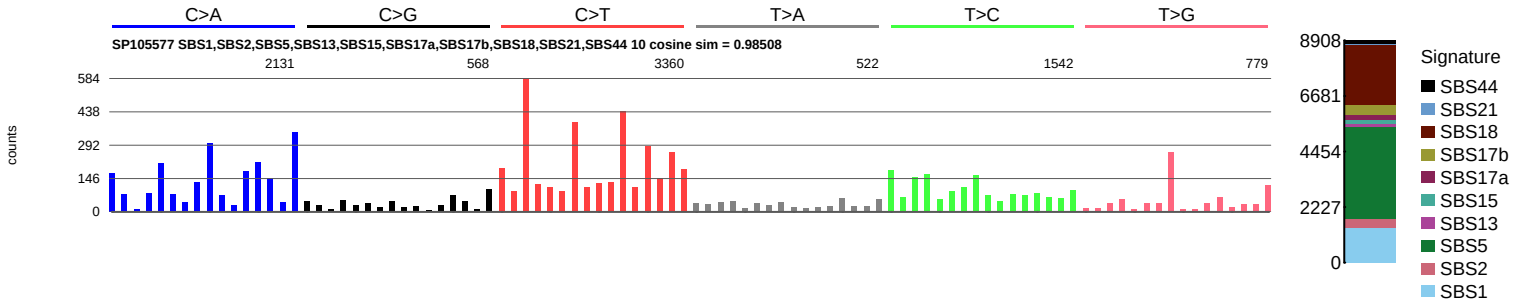
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98508
2131
568
3360
522
1542
779
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
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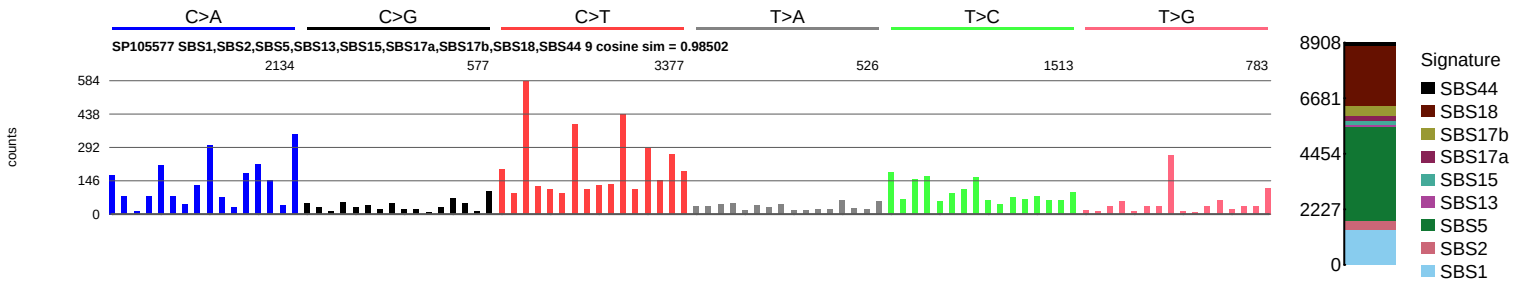
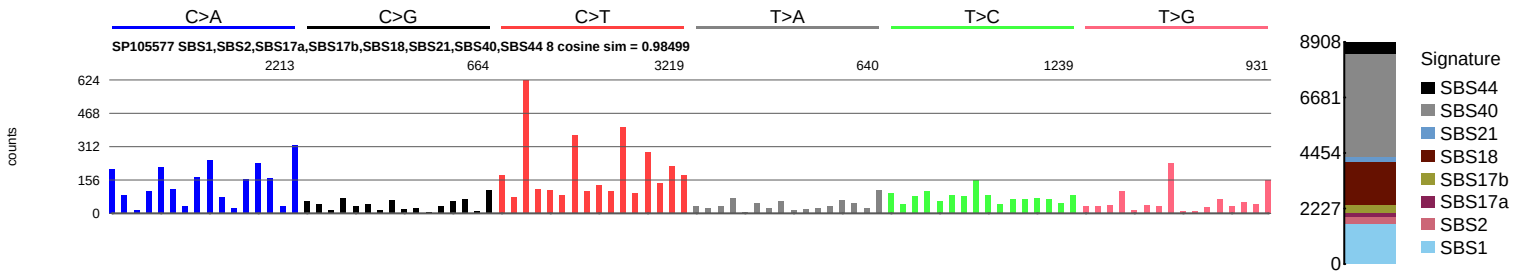
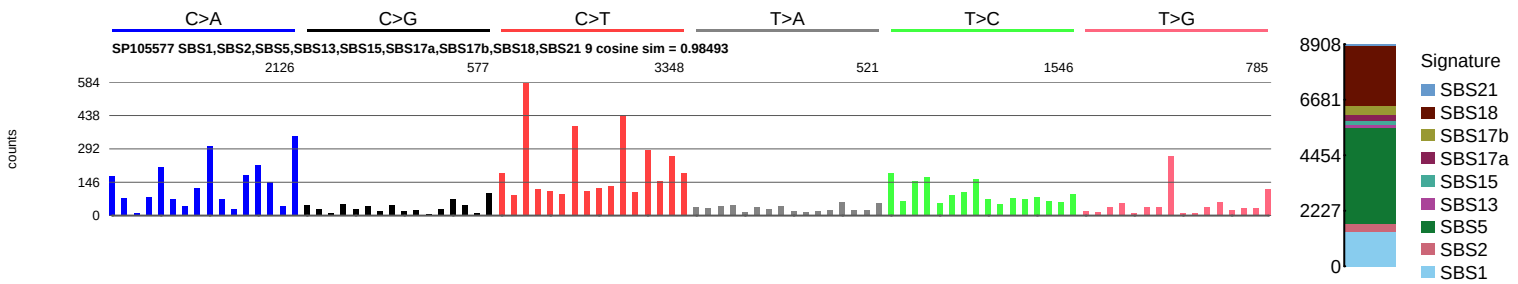
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98502
2134
577
3377
526
1513
783
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98499
2213
664
3219
640
1239
931
counts
624
468
312
156
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98493
2126
577
3348
521
1546
785
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS21
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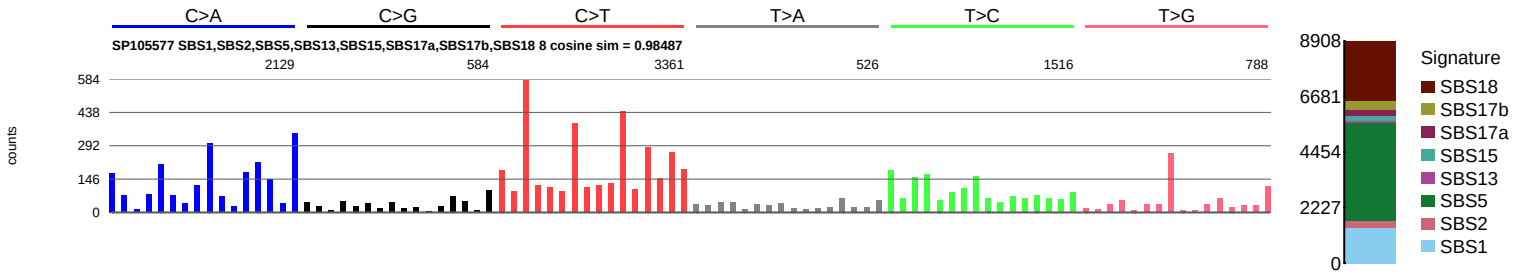
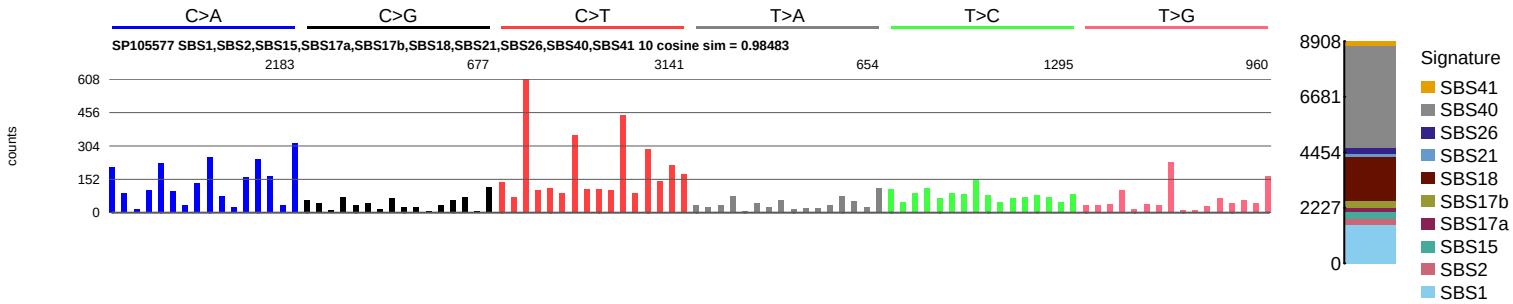
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98487
2129
584
3361
526
1516
788
584
438
292
146
0
counts
8908
6681
4454
2227
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>T
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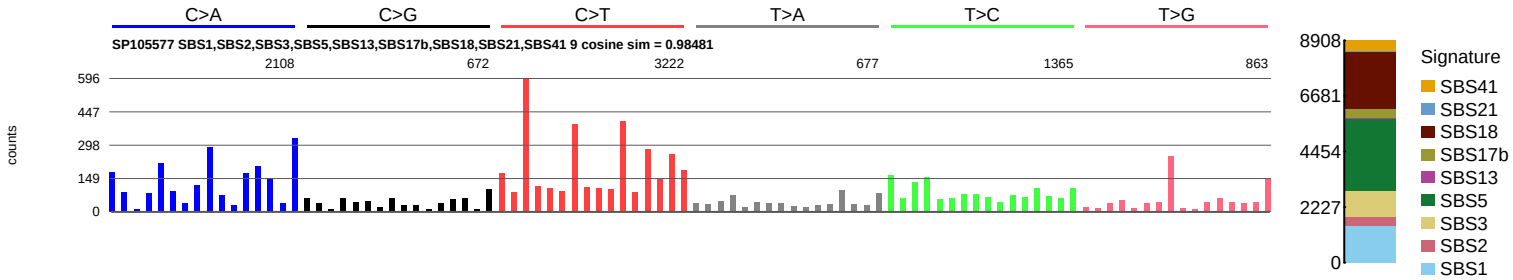
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98483
2183
677
3141
654
1295
960
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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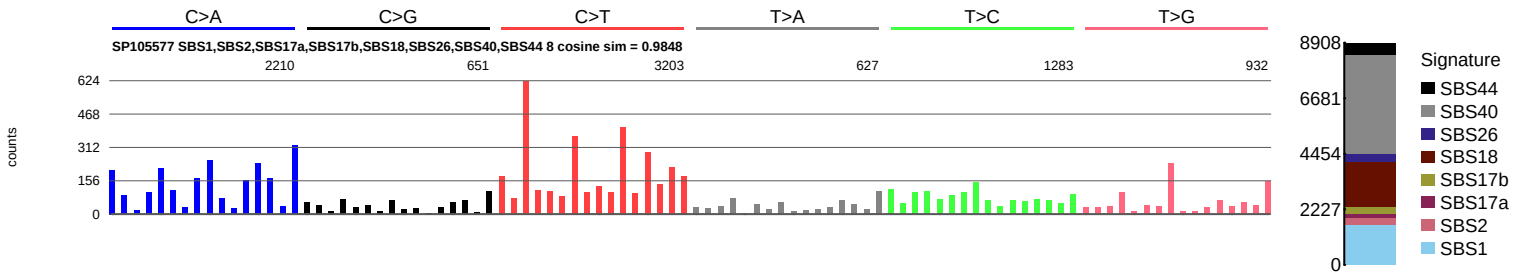
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98481
2108
672
3222
677
1365
863
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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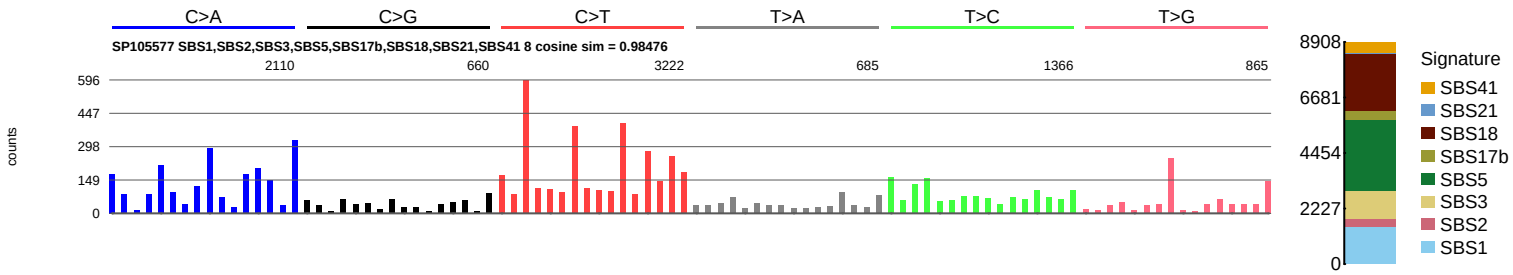
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.9848
2210
651
3203
627
1283
932
624
468
312
156
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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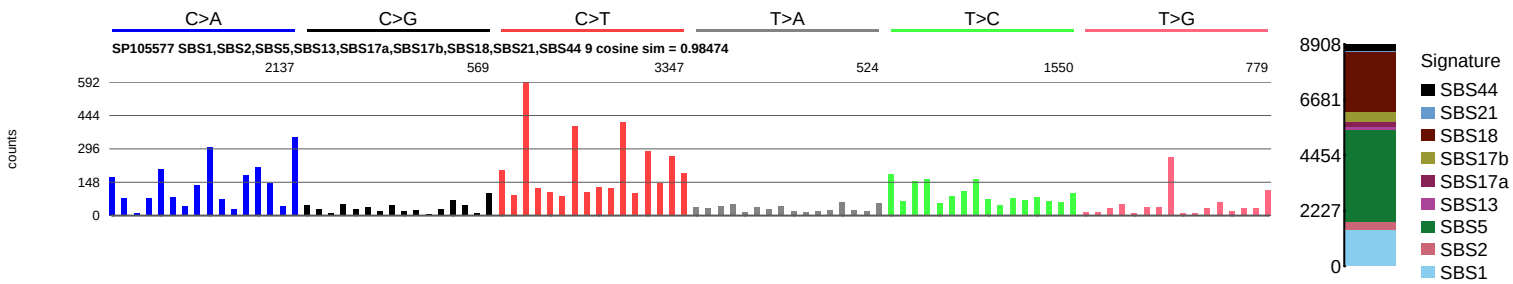
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98476
2110
660
3222
685
1366
865
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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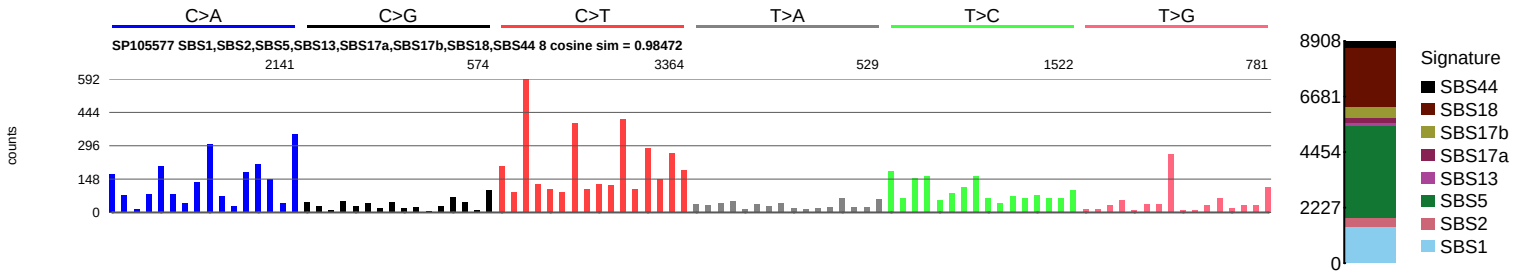
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98474
2137
569
3347
524
1550
779
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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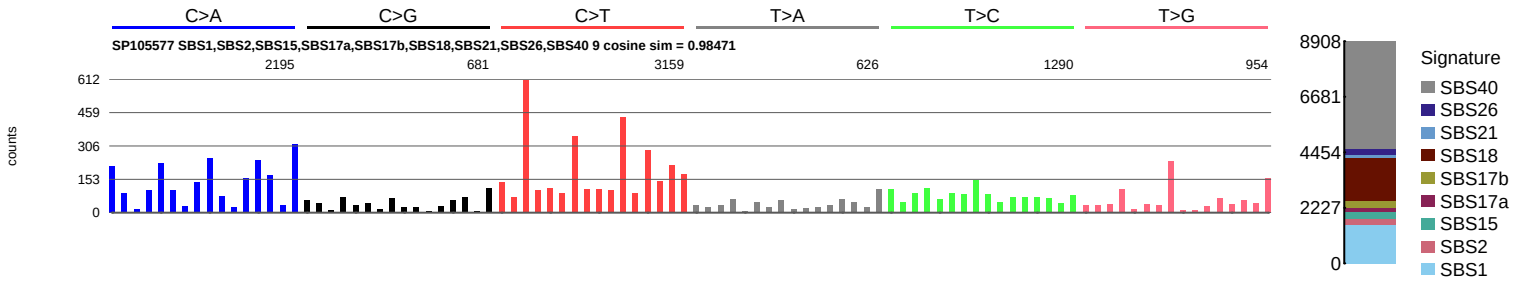
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98472
2141
574
3364
529
1522
781
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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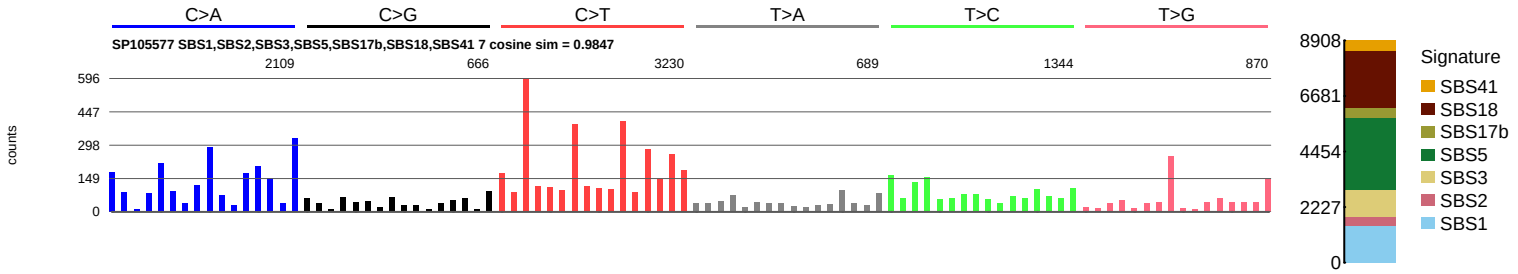
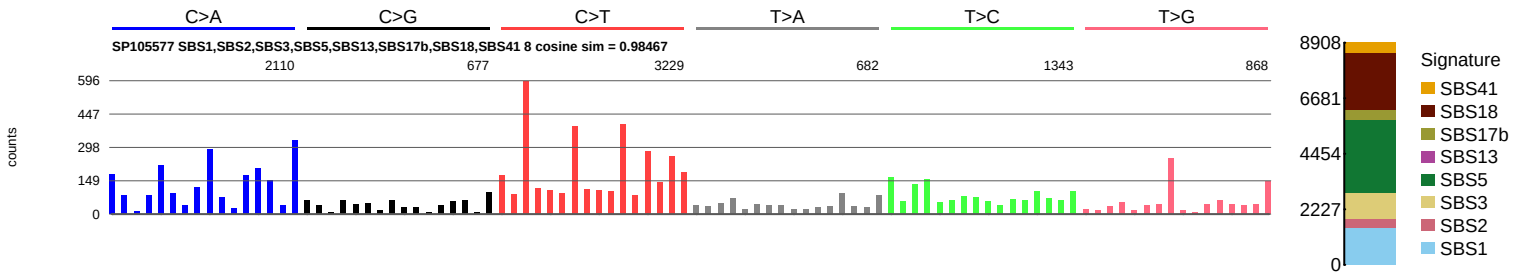
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98471
2195
681
3159
626
1290
954
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41 7 cosine sim = 0.9847
2109
666
3230
689
1344
870
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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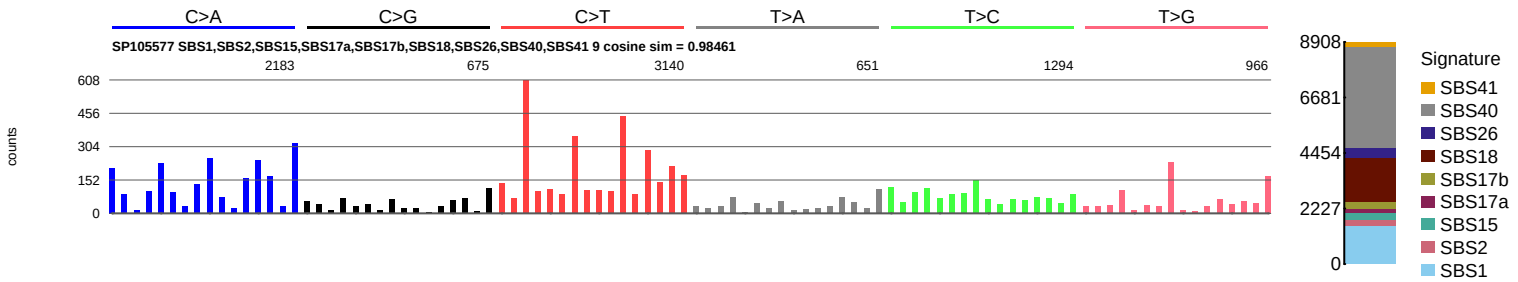
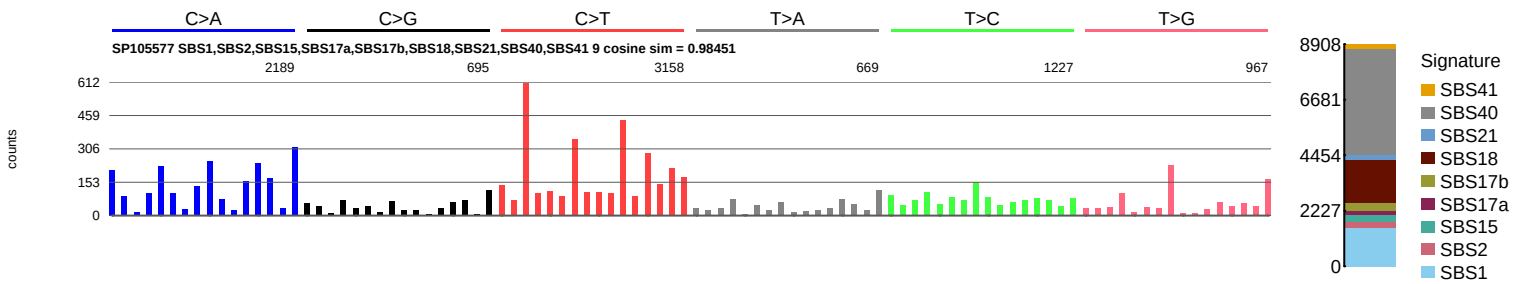
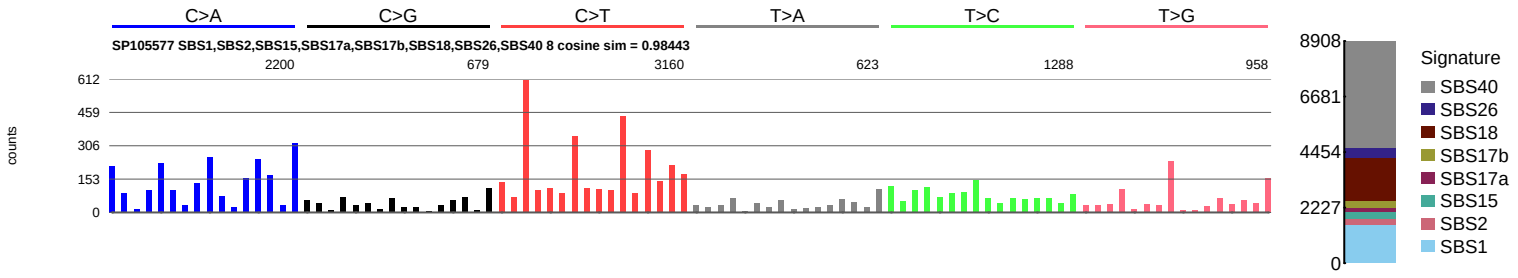
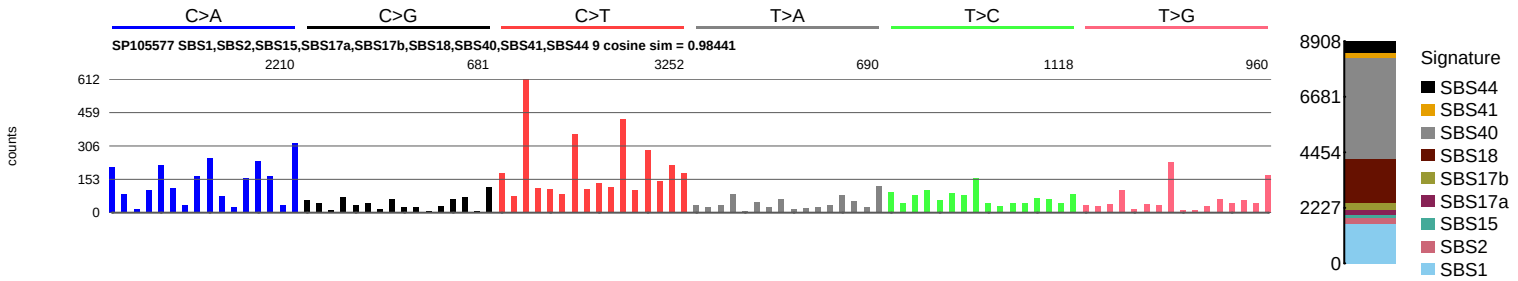
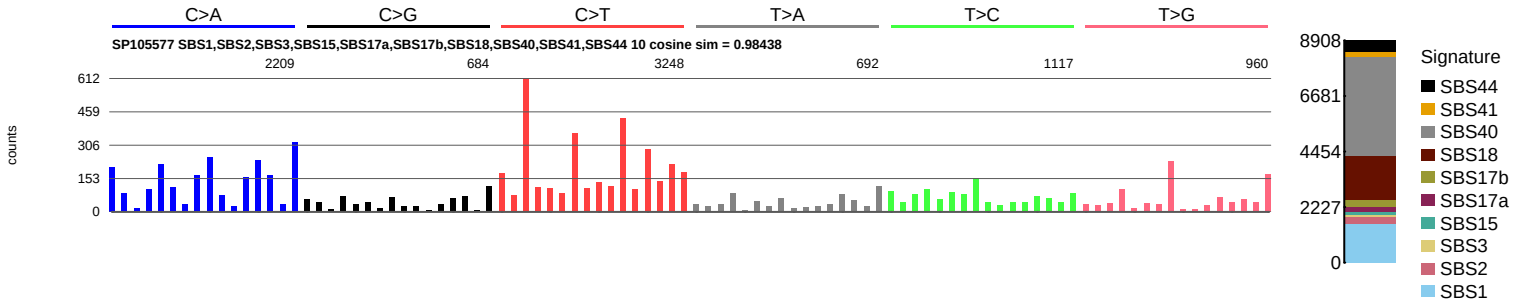
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41 8 cosine sim = 0.98467
2110
677
3229
682
1343
868
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

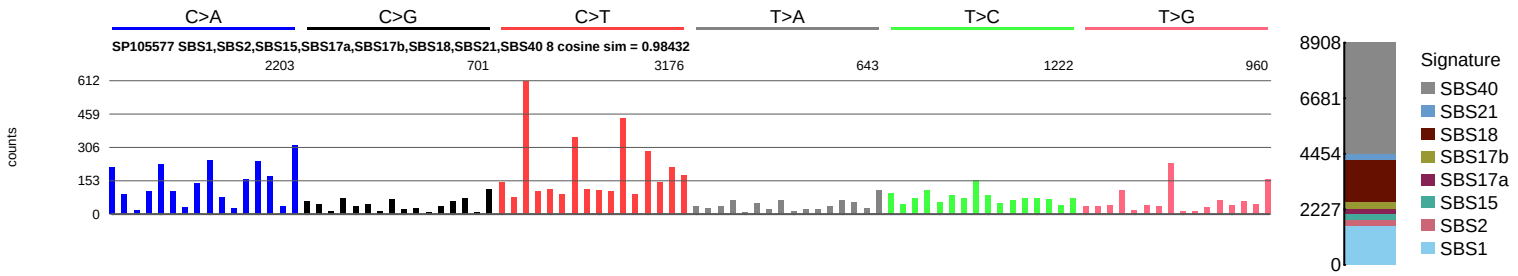

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98461
2183
675
3140
651
1294
966
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

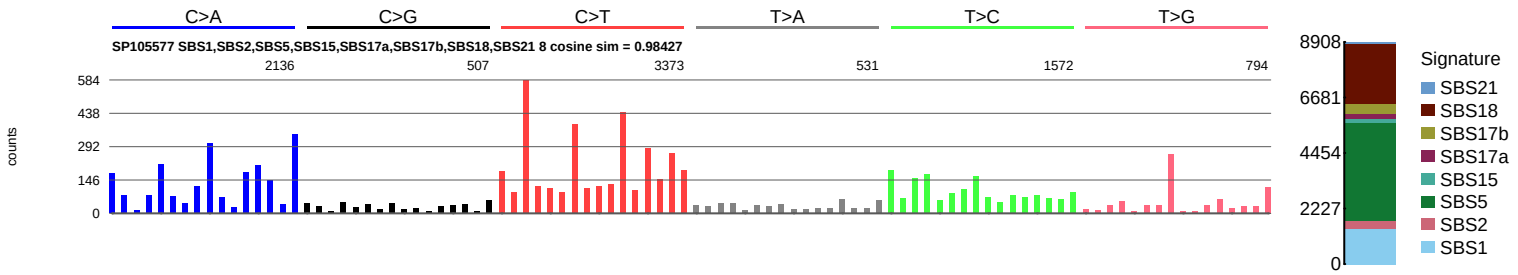

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98451
2189
695
3158
669
1227
967
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98443
2200
679
3160
623
1288
958
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98441
2210
681
3252
690
1118
960
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98438
2209
684
3248
692
1117
960
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98432
2203
701
3176
643
1222
960
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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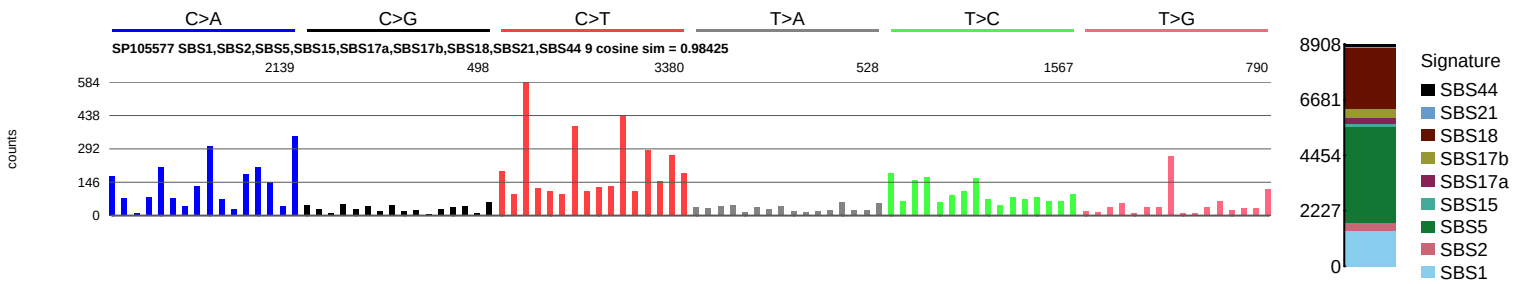
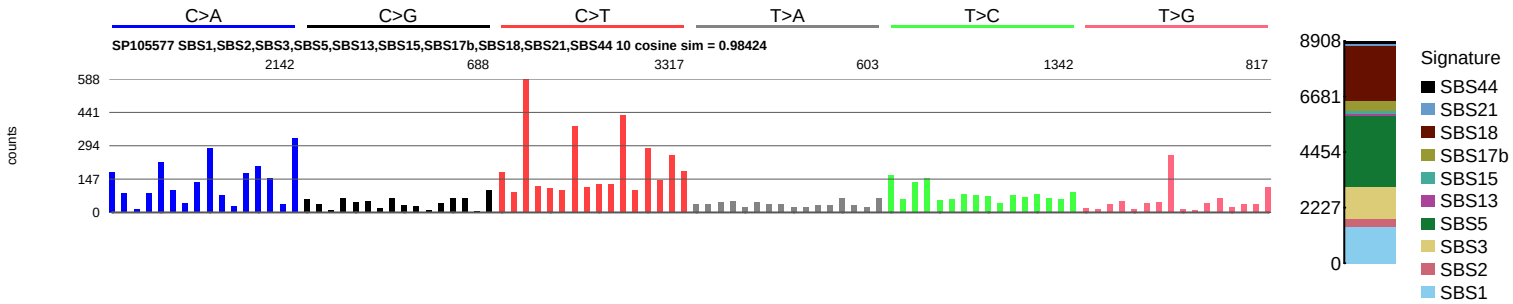
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98427
2136
507
3373
531
1572
794
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98425
2139
498
3380
528
1567
790
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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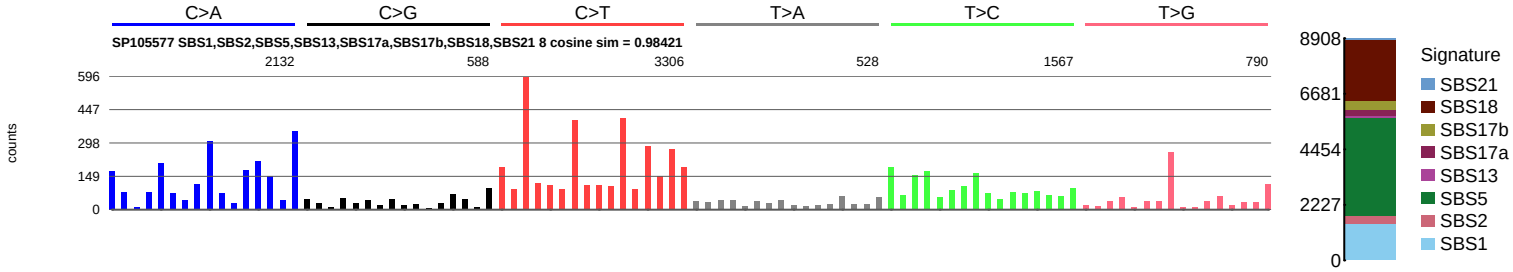
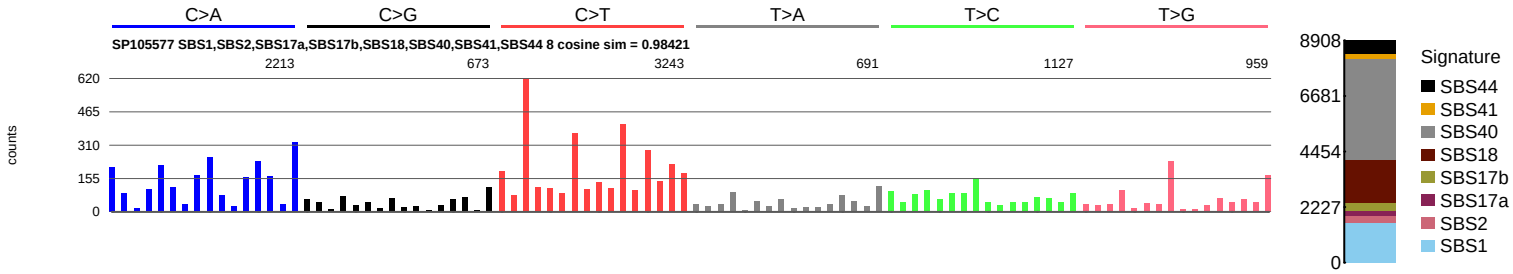
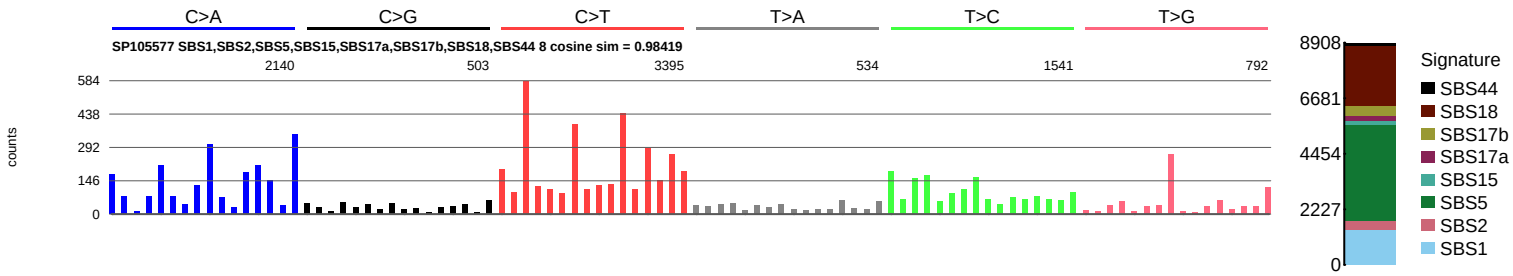
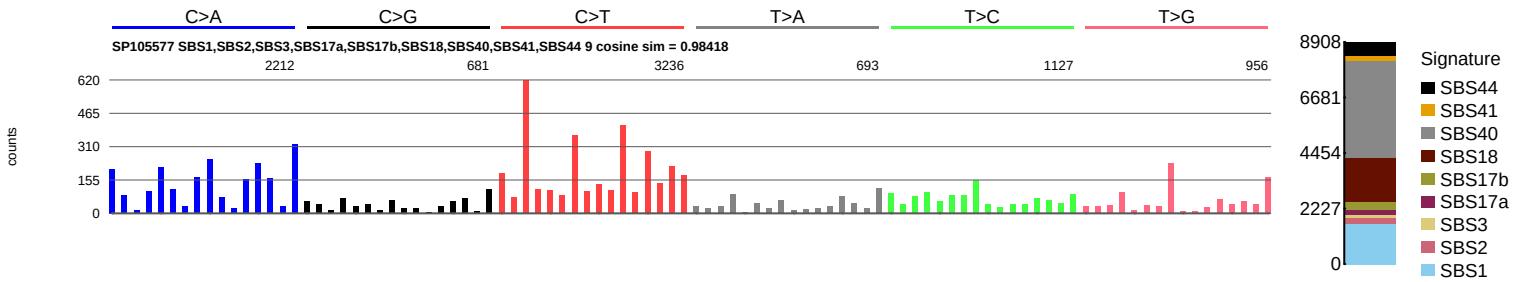
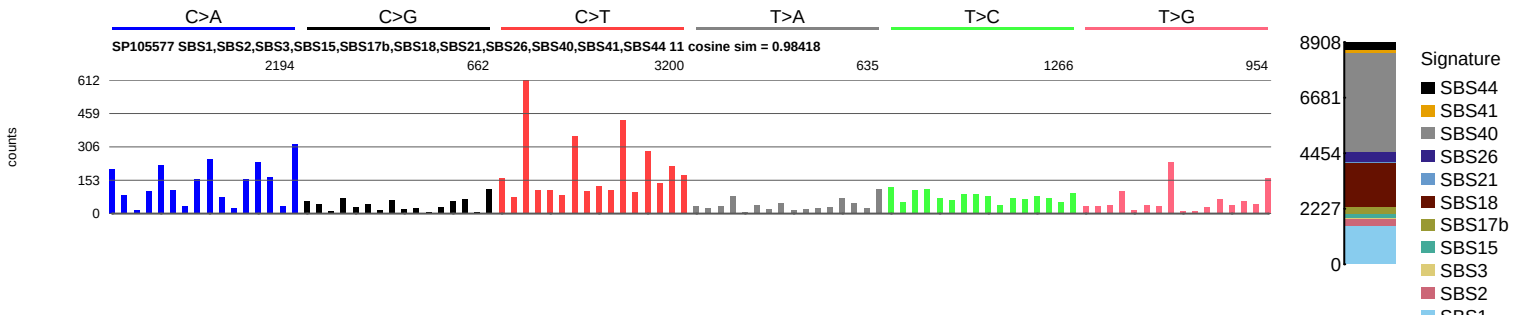
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98424
2142
688
3317
603
1342
817
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

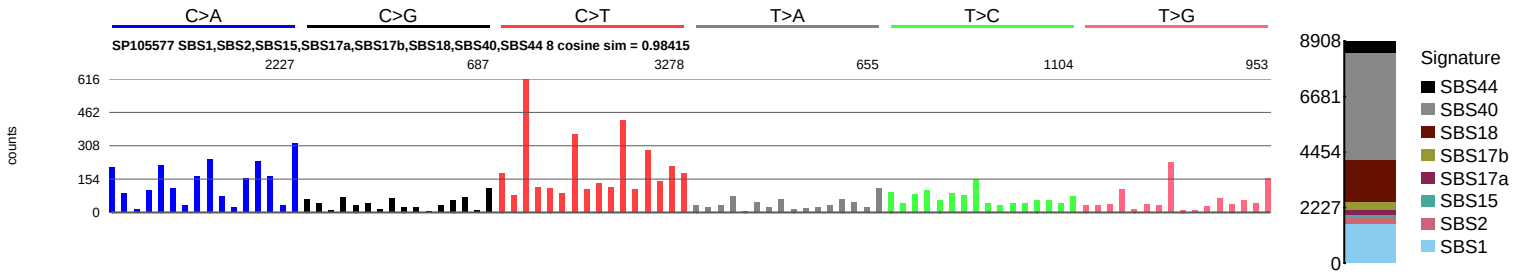

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98421
2132
588
3306
528
1567
790
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

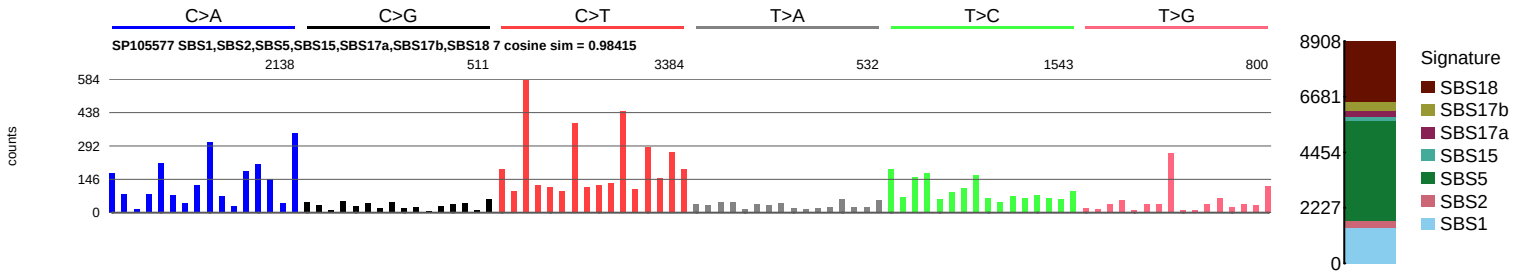

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98421
2213
673
3243
691
1127
959
620
465
310
155
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

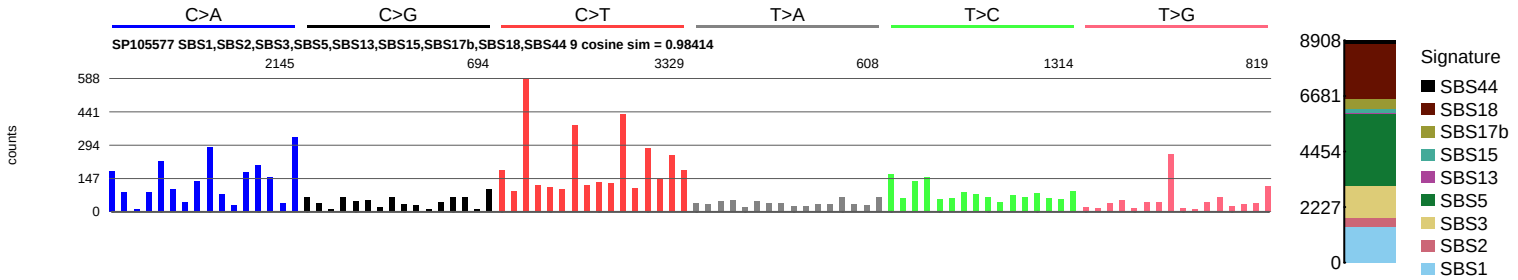

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98419
2140
503
3395
534
1541
792
584
438
292
146
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

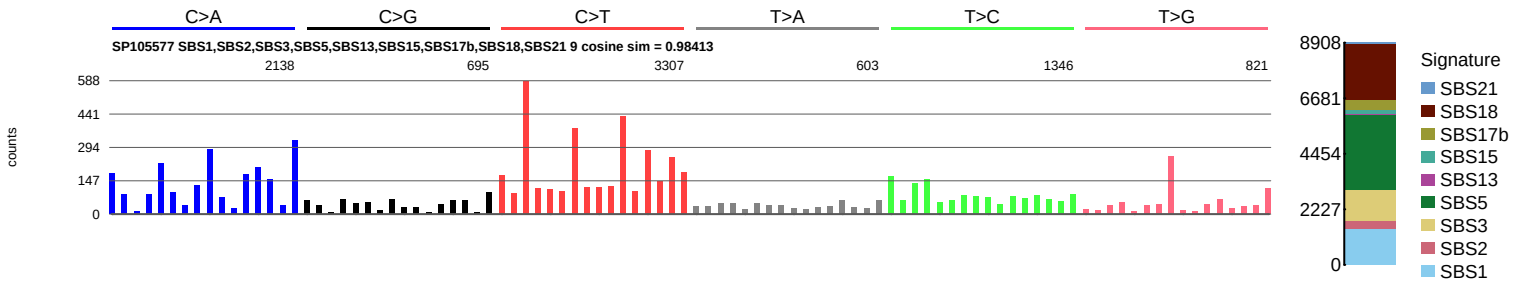

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98418
2212
681
3236
693
1127
956
620
465
310
155
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

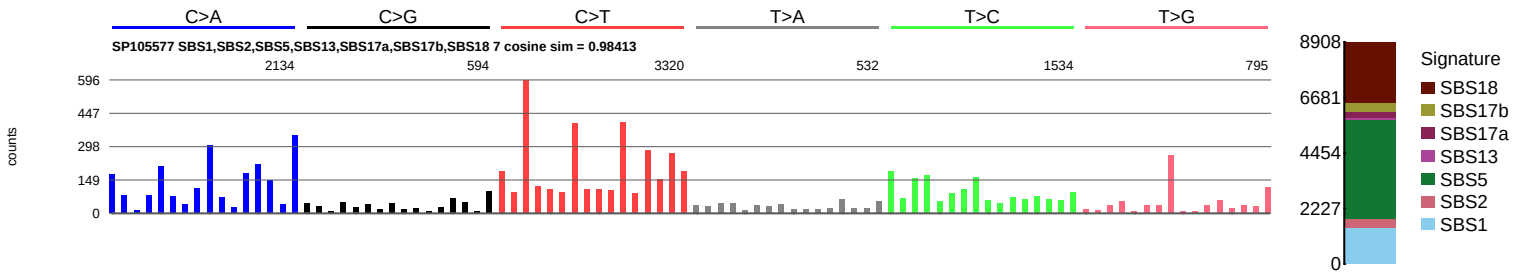

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98418
2194
662
3200
635
1266
954
612
459
306
153
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2

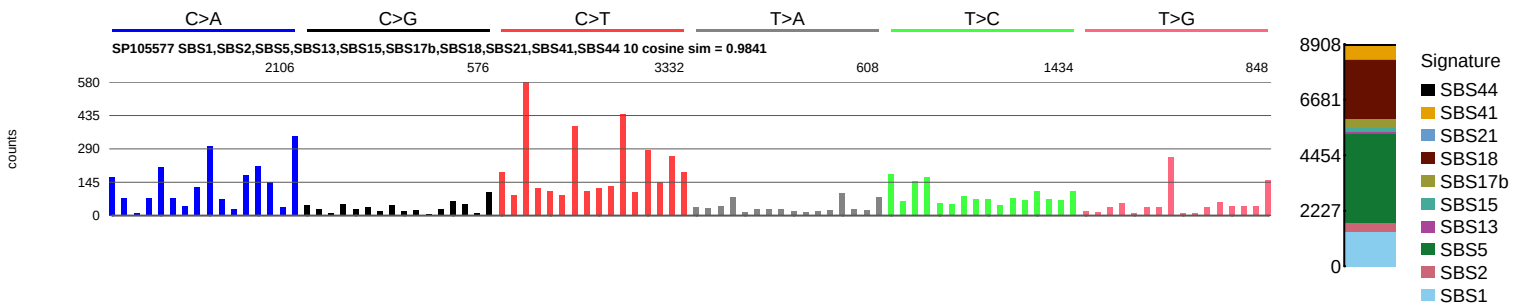

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98415
2227
687
3278
655
1104
953
616
462
308
154
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

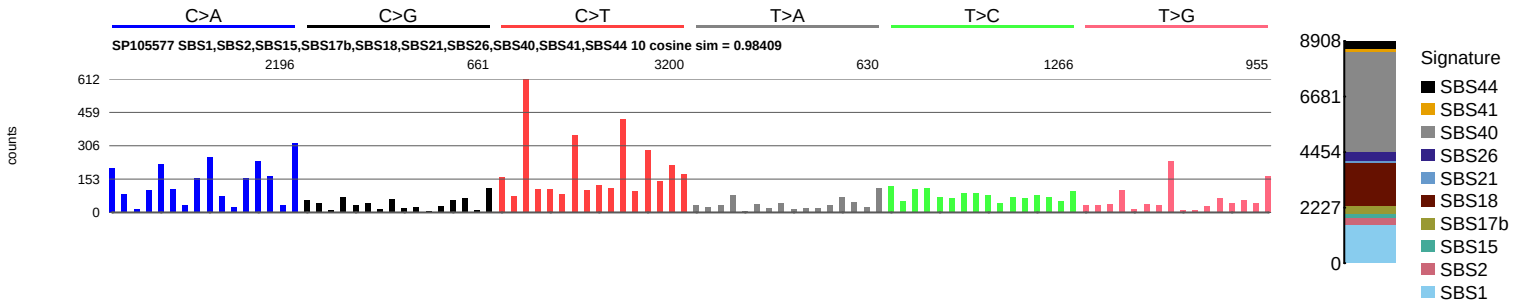

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98415
2138
511
3384
532
1543
800
584
438
292
146
0
counts
8908
6681
4454
2227
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

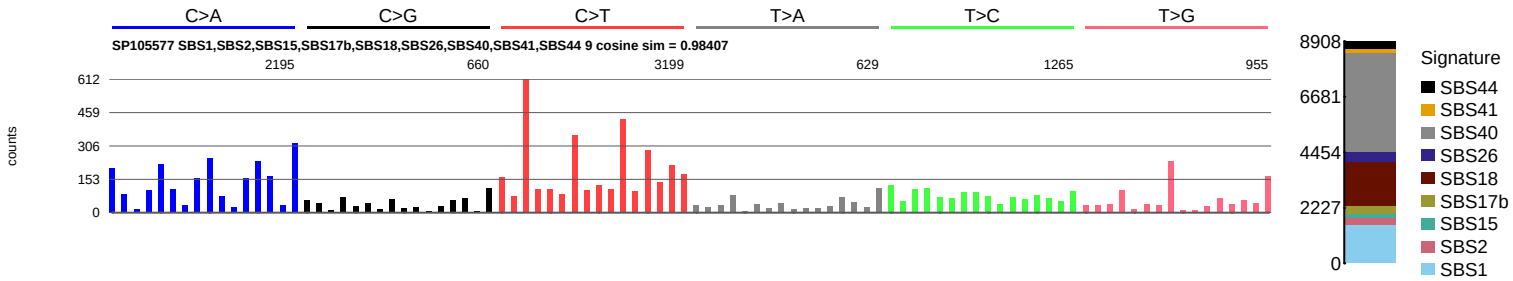

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 9 cosine sim = 0.98414
2145
694
3329
608
1314
819
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

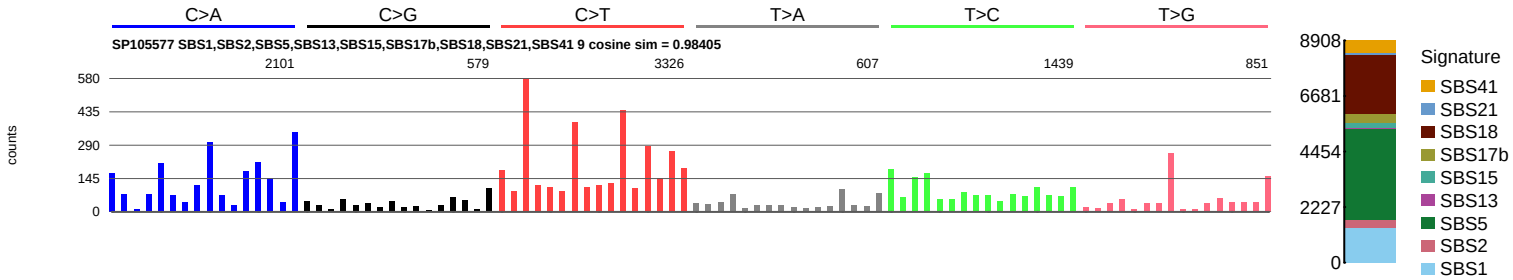

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 9 cosine sim = 0.98413
2138
695
3307
603
1346
821
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

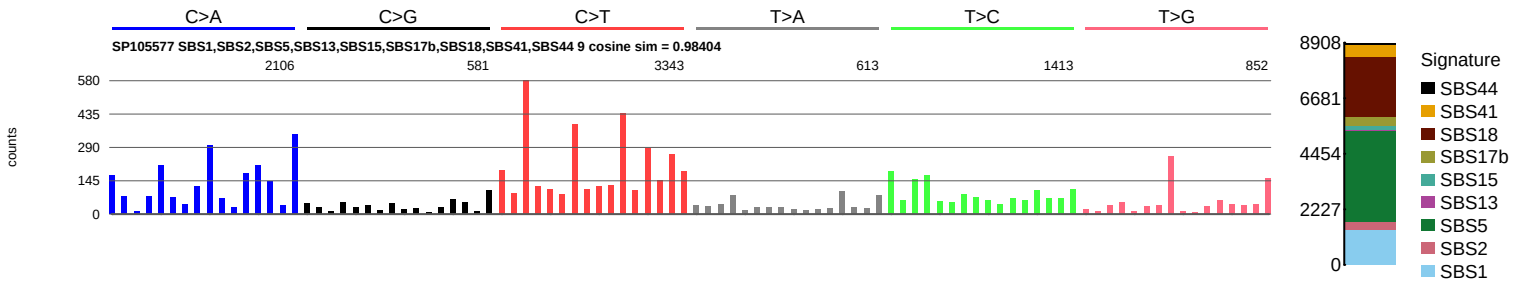

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98413
2134
594
3320
532
1534
795
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.9841
2106
576
3332
608
1434
848
580
435
290
145
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98409
2196
661
3200
630
1266
955
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98407
2195
660
3199
629
1265
955
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98405
2101
579
3326
607
1439
851
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98404
2106
581
3343
613
1413
852
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
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SBS41
SBS18
SBS17b
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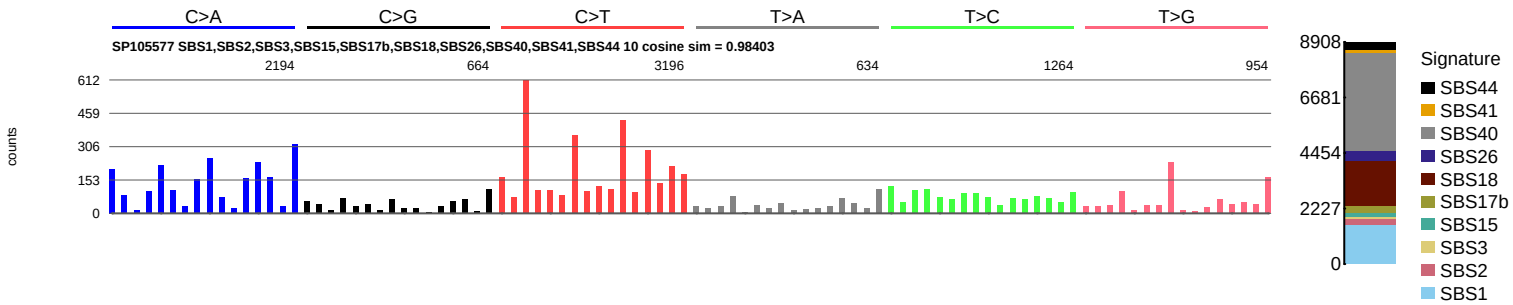
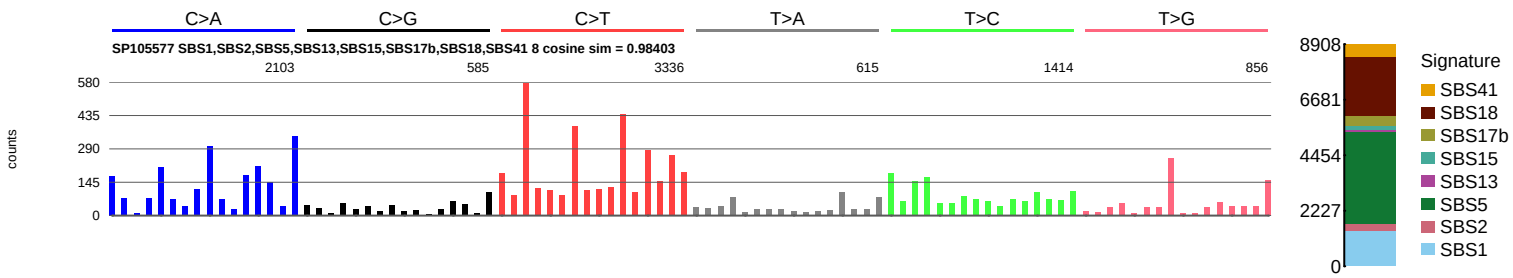
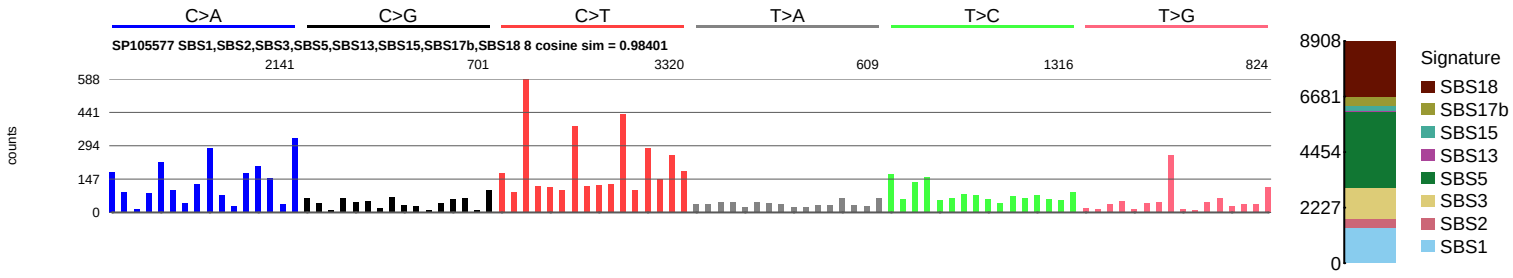
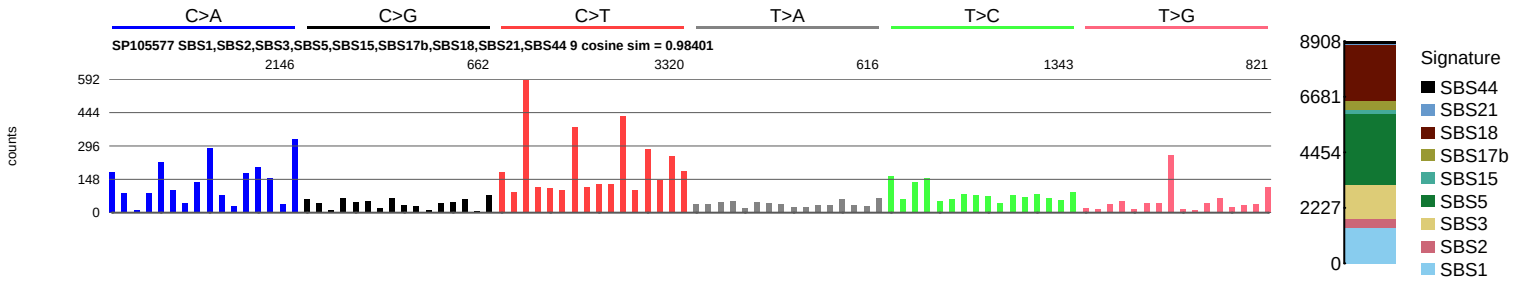
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98403
2194
664
3196
634
1264
954
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98403
2103
585
3336
615
1414
856
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18 8 cosine sim = 0.98401
2141
701
3320
609
1316
824
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98401
2146
662
3320
616
1343
821
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS44
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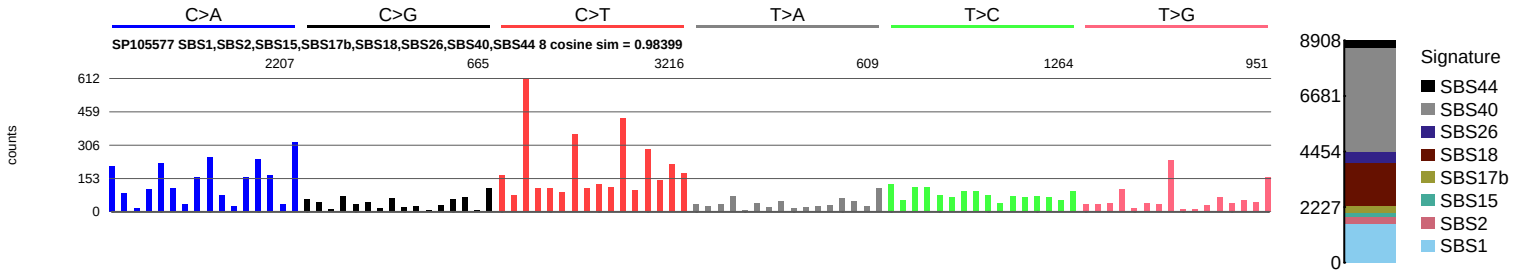
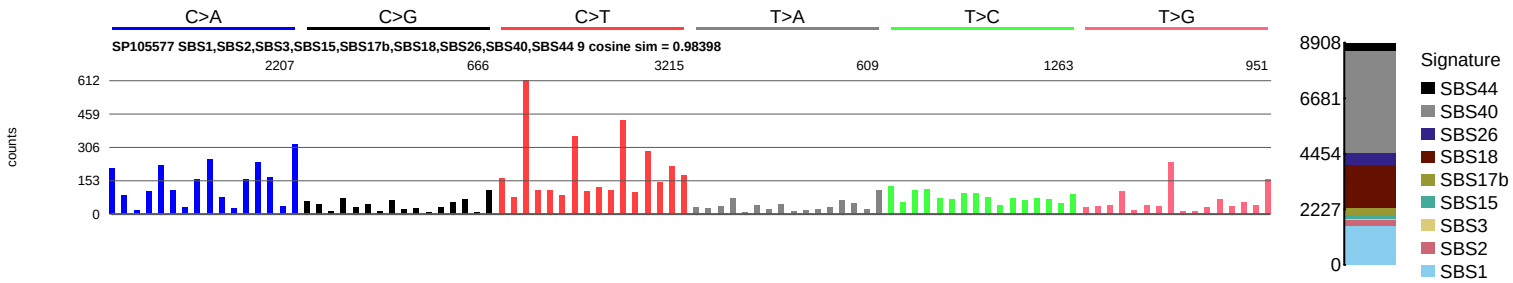
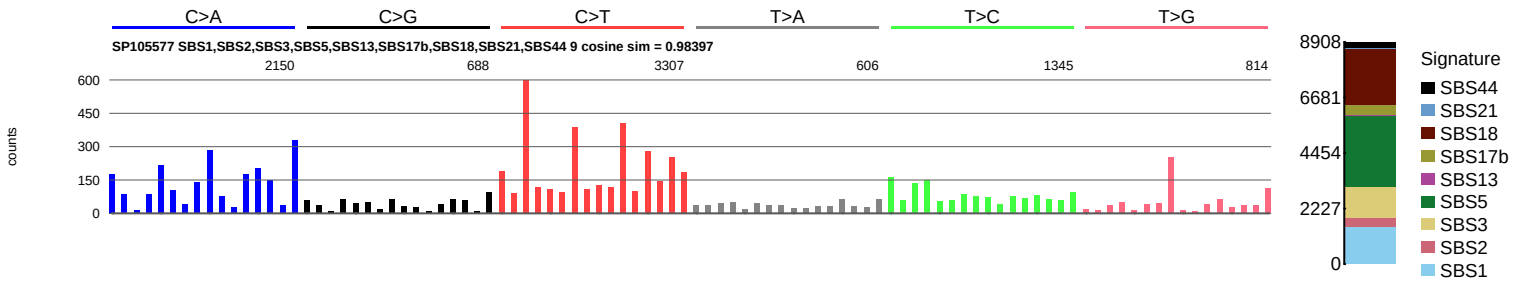
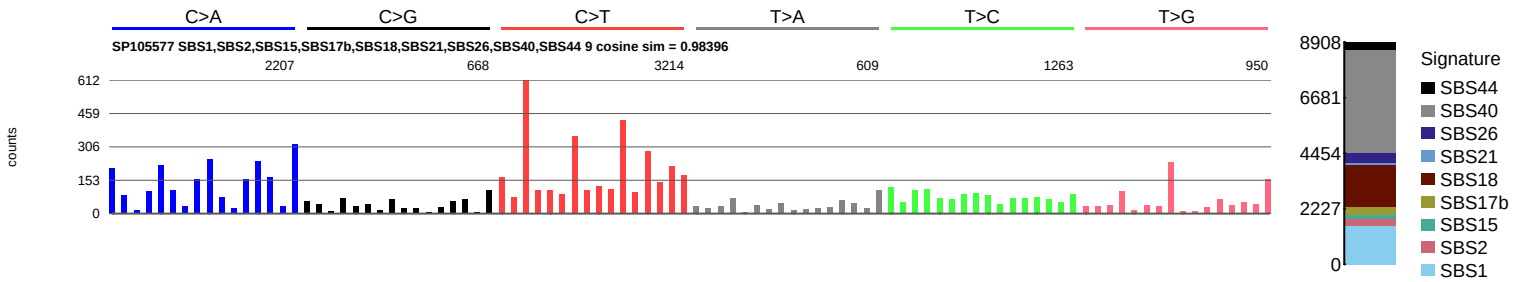
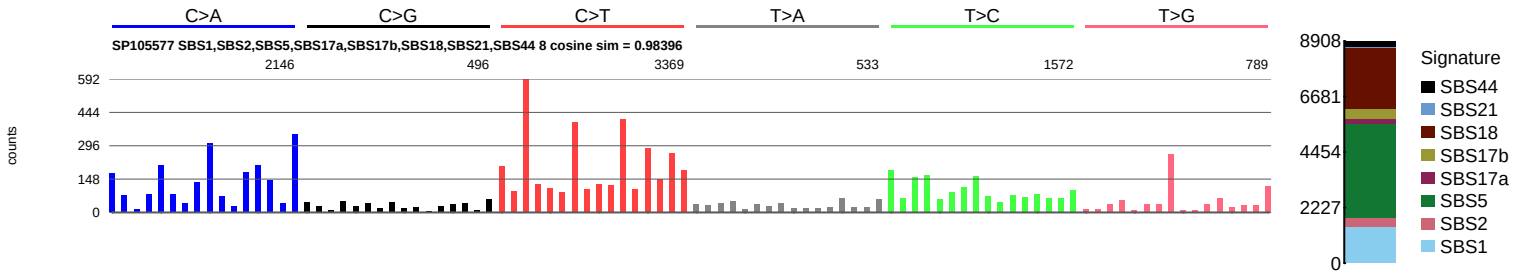
SBS2
SBS1

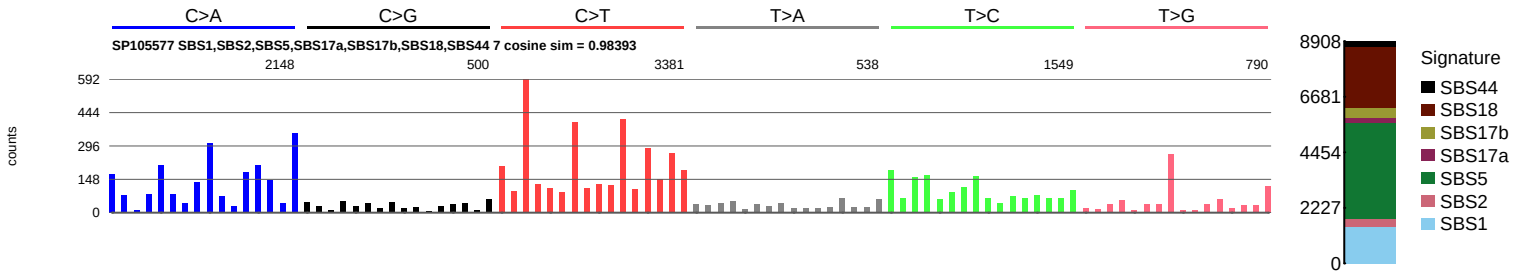

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98399
2207
665
3216
609
1264
951
612
459
306
153
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1

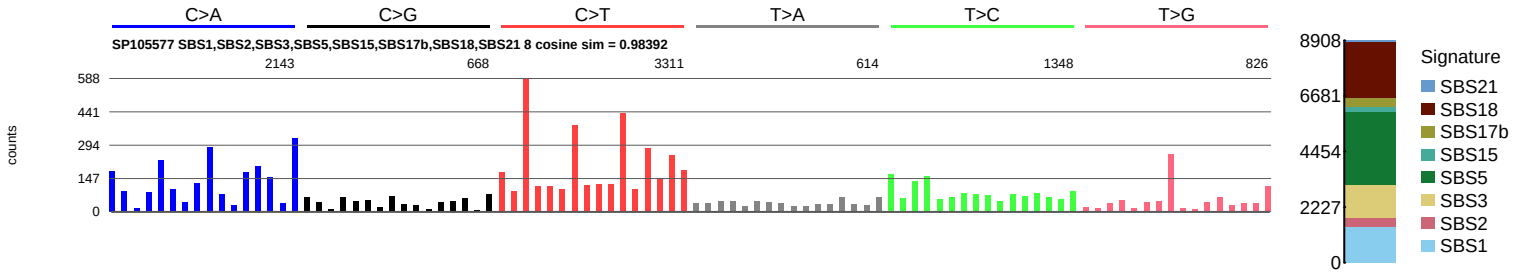

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98398
2207
666
3215
609
1263
951
612
459
306
153
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

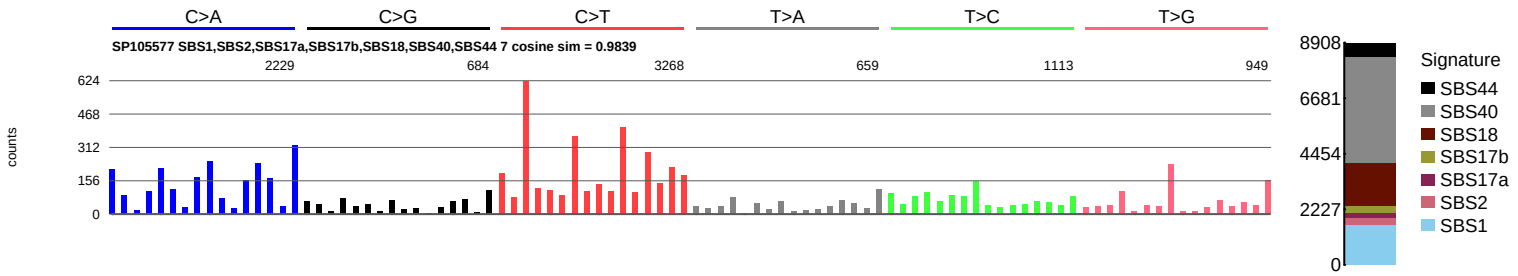

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98397
2150
688
3307
606
1345
814
600
450
300
150
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

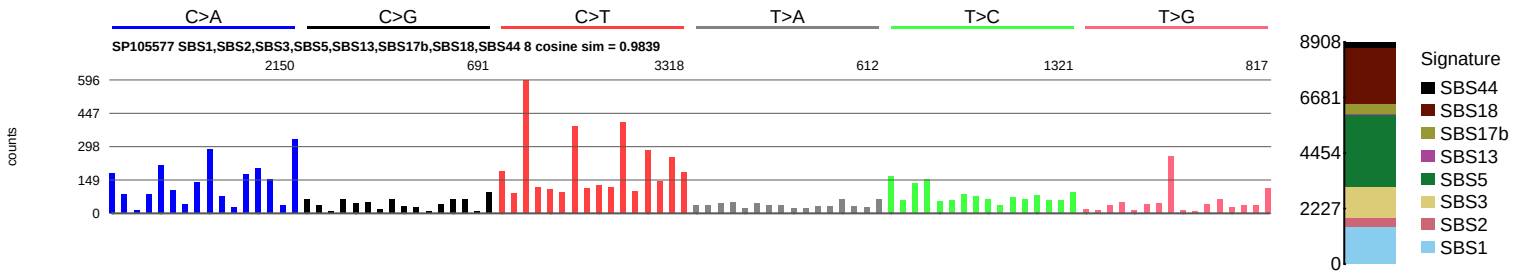

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98396
2207
668
3214
609
1263
950
612
459
306
153
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98396
2146
496
3369
533
1572
789
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98393
2148
500
3381
538
1549
790
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98392
2143
668
3311
614
1348
826
counts
588
441
294
147
0
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.9839
2229
684
3268
659
1113
949
counts
624
468
312
156
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS44 8 cosine sim = 0.9839
2150
691
3318
612
1321
817
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
SBS13
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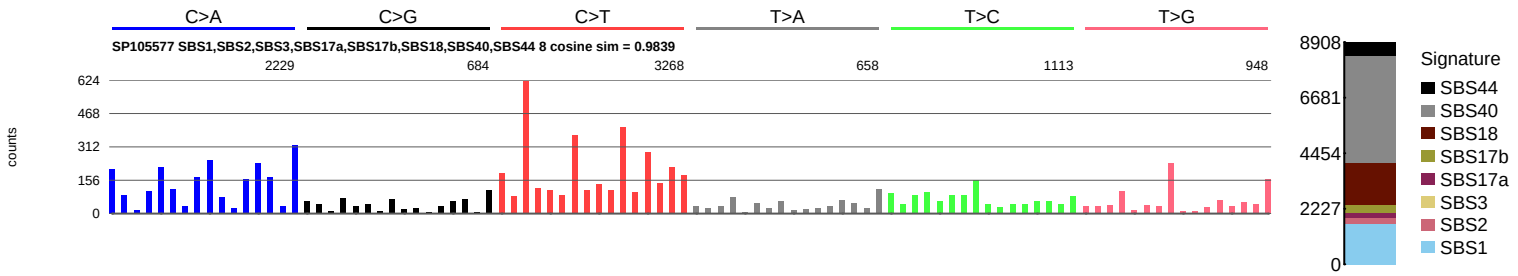
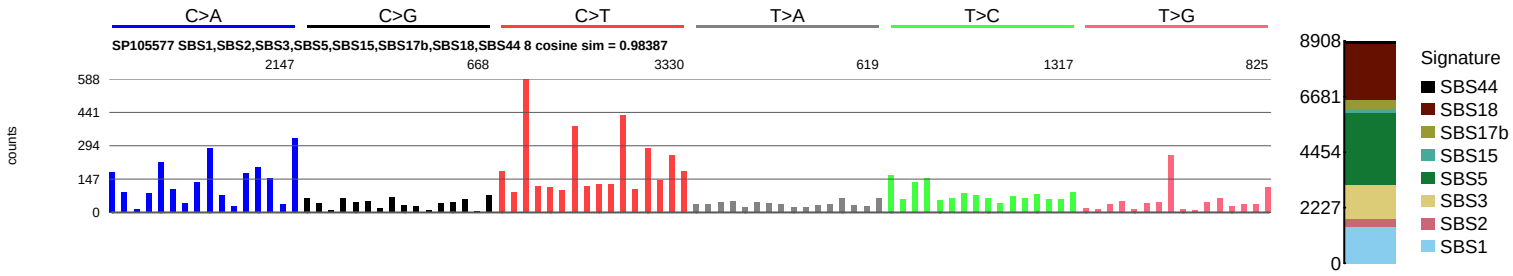
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.9839
2229
684
3268
658
1113
948
counts
624
468
312
156
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS44 8 cosine sim = 0.98387
2147
668
3330
619
1317
825
counts
588
441
294
147
0
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
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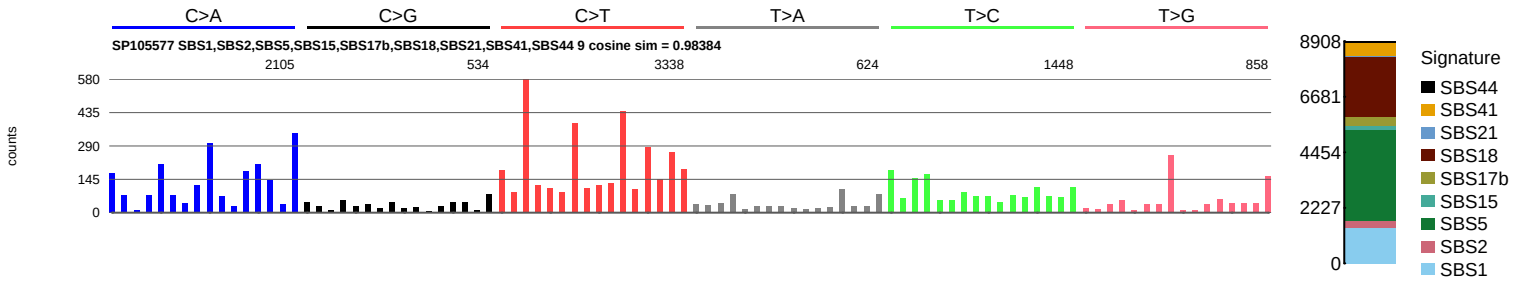
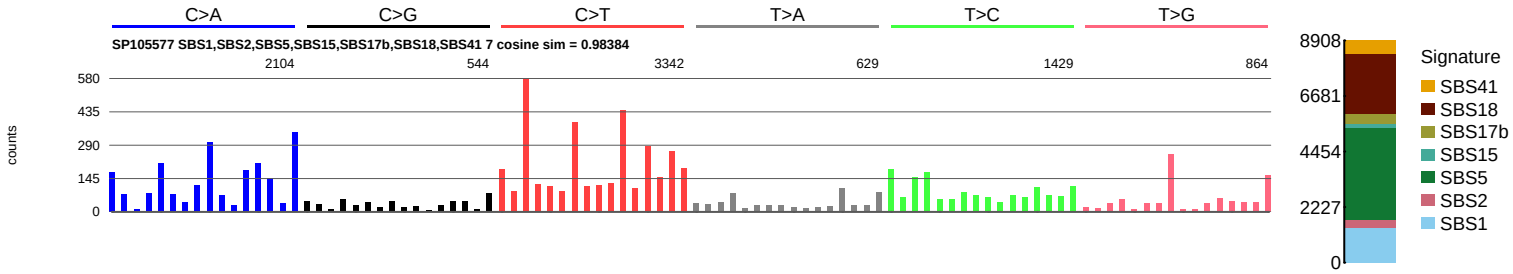
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98384
2105
534
3338
624
1448
858
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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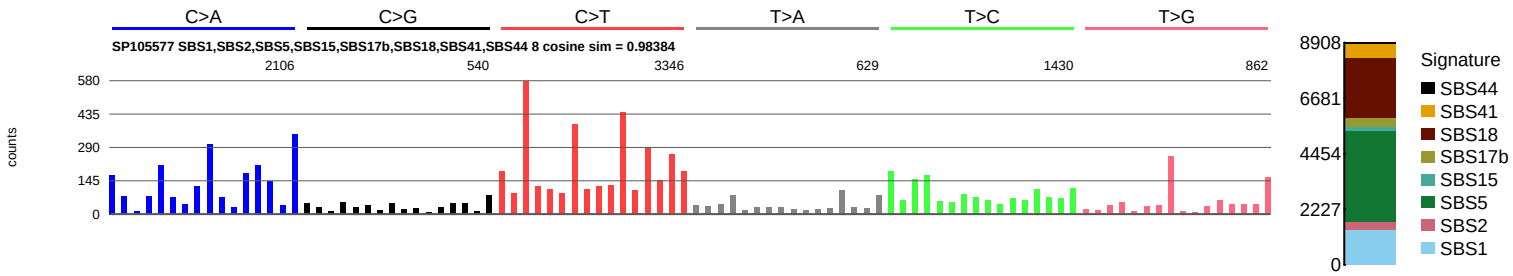
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS41 7 cosine sim = 0.98384
2104
544
3342
629
1429
864
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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C>T
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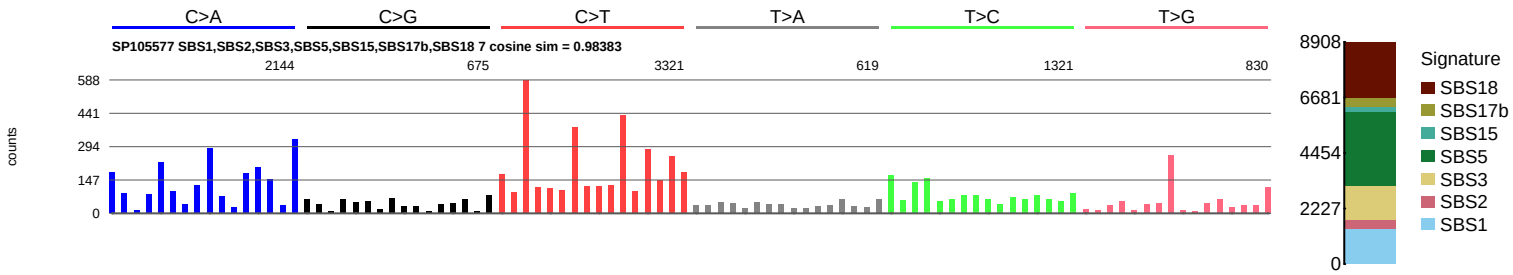
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98384
2106
540
3346
629
1430
862
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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C>G
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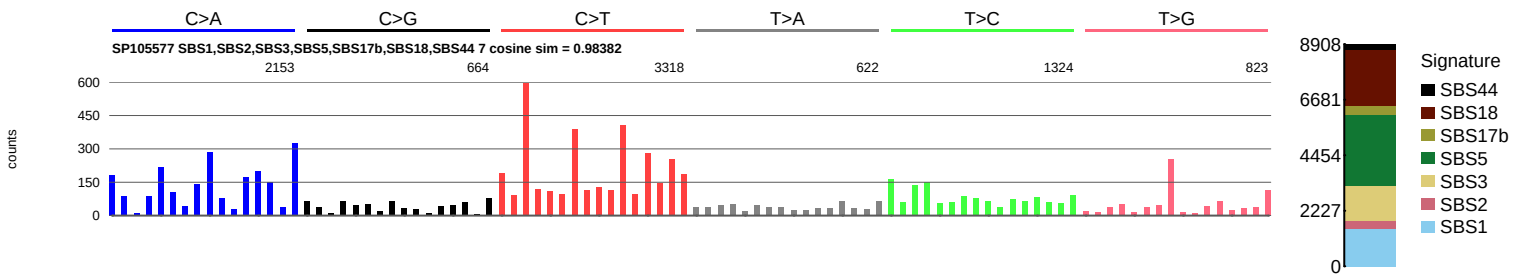
SP105577 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18 7 cosine sim = 0.98383
2144
675
3321
619
1321
830
counts
588
441
294
147
0
8908
6681
4454
2227
0
Signature
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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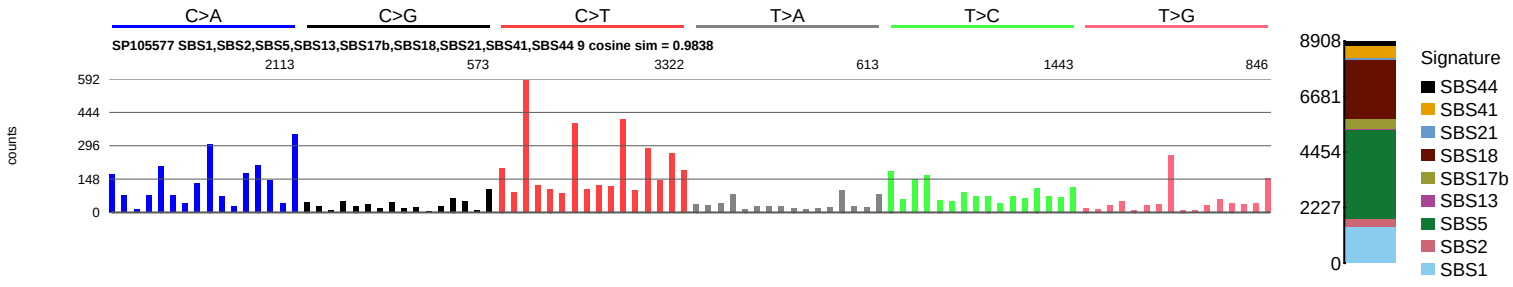
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS44 7 cosine sim = 0.98382
2153
664
3318
622
1324
823
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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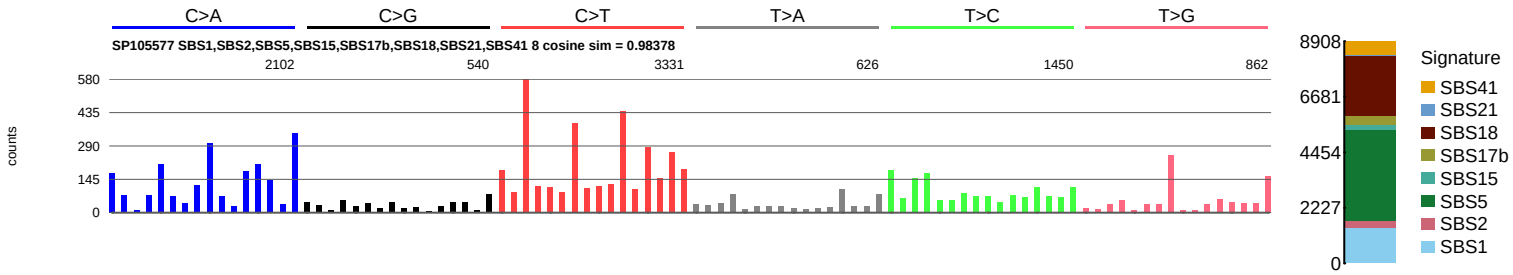
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.9838
2113
573
3322
613
1443
846
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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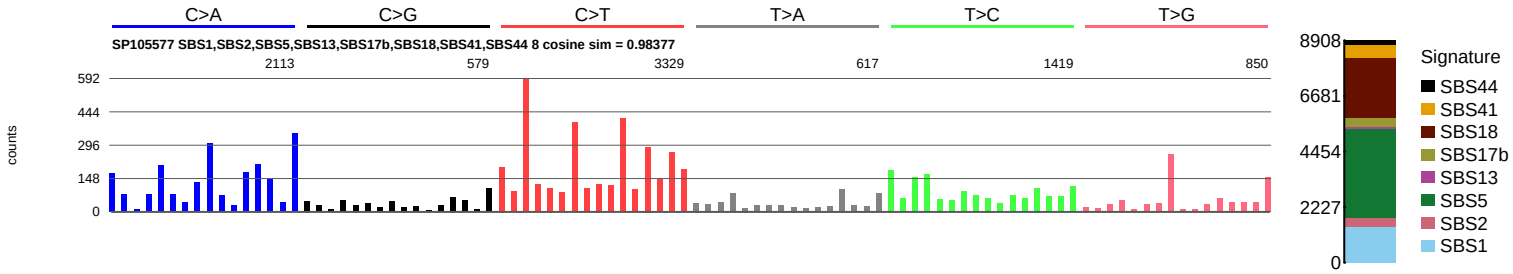
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98378
2102
540
3331
626
1450
862
counts
580
435
290
145
0
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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C>G
C>T
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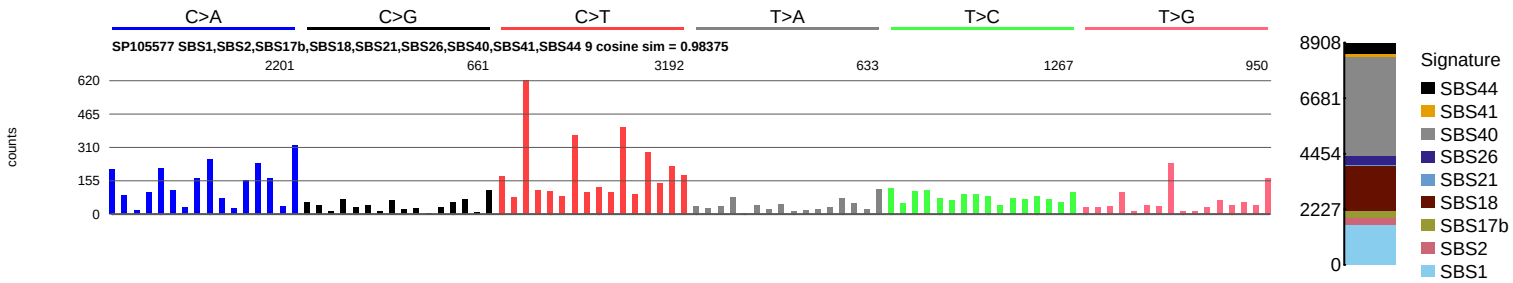
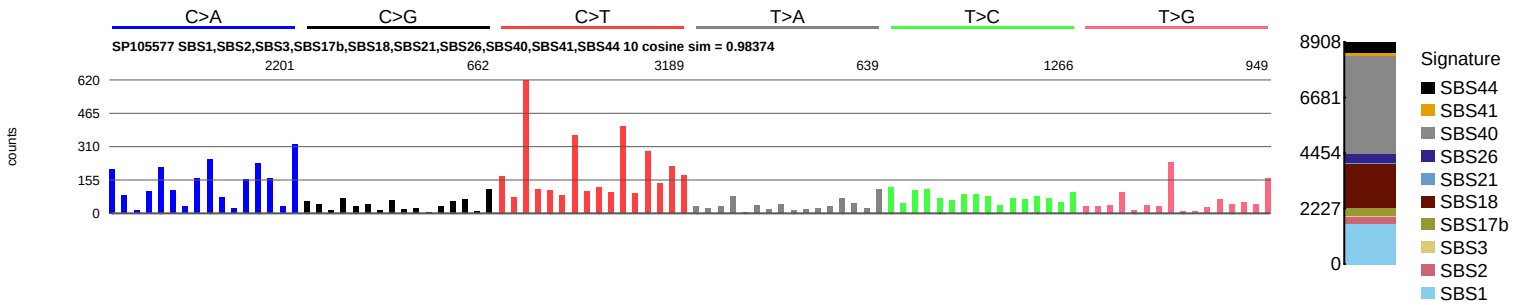
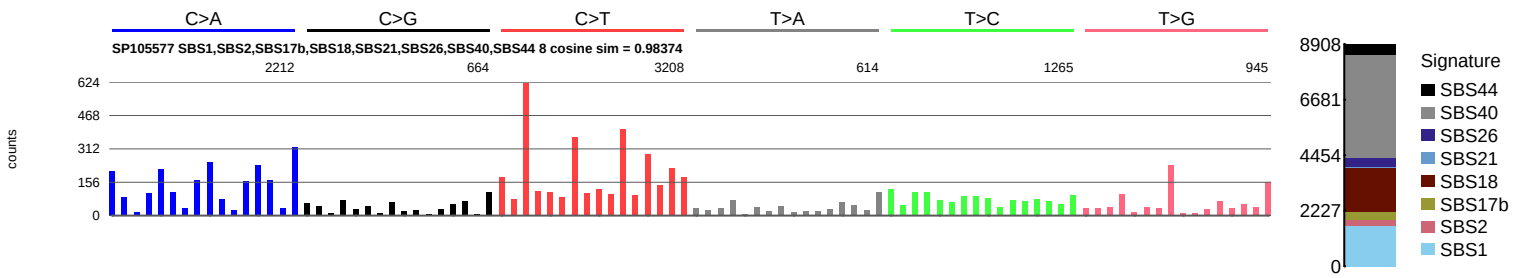
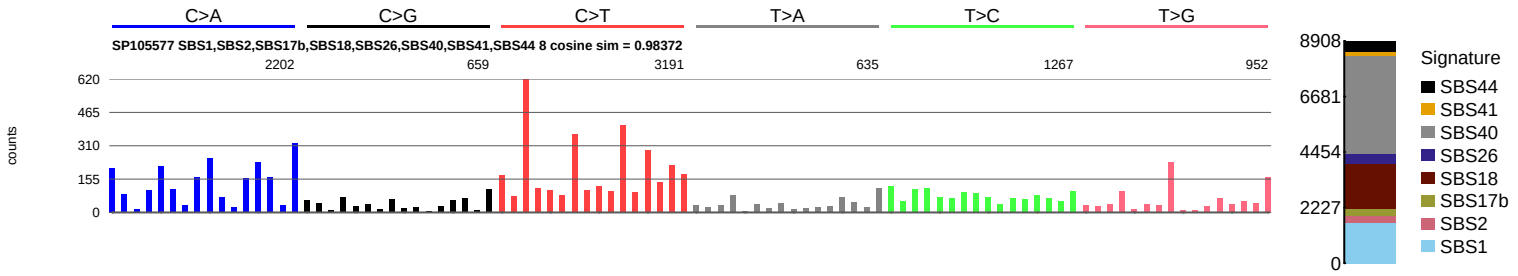
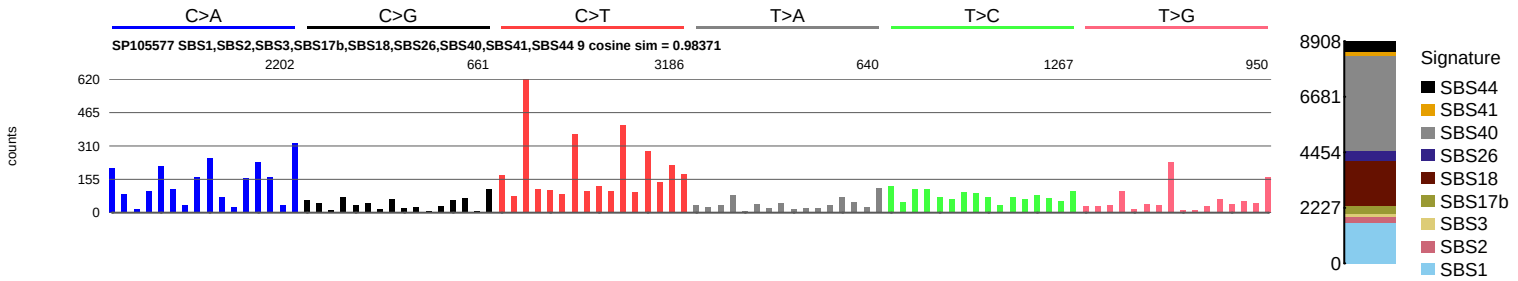
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98377
2113
579
3329
617
1419
850
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

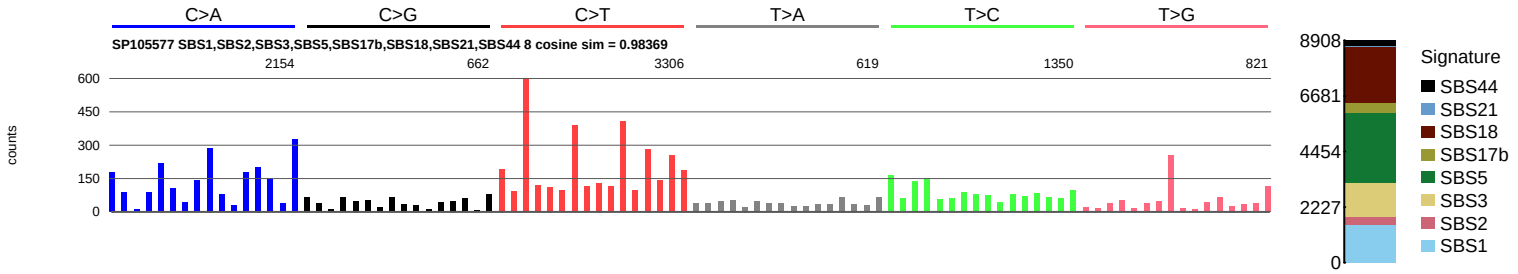

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98375
2201
661
3192
633
1267
950
counts
620
465
310
155
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1

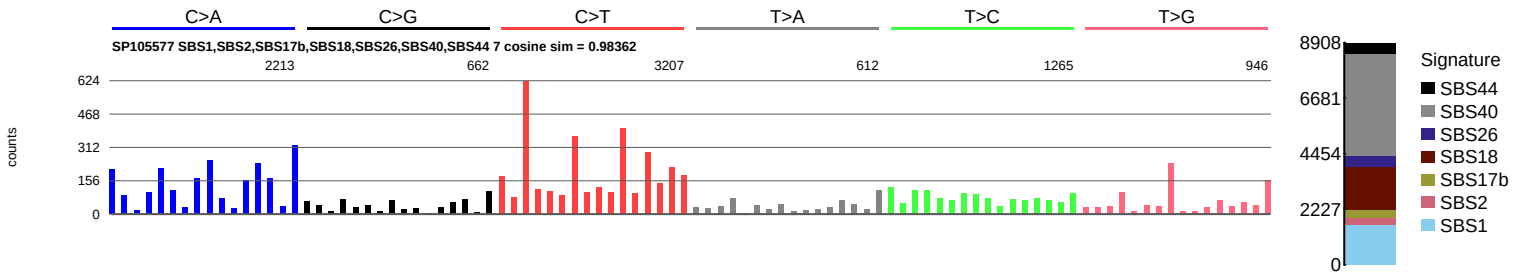

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98374
2201
662
3189
639
1266
949
counts
620
465
310
155
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1

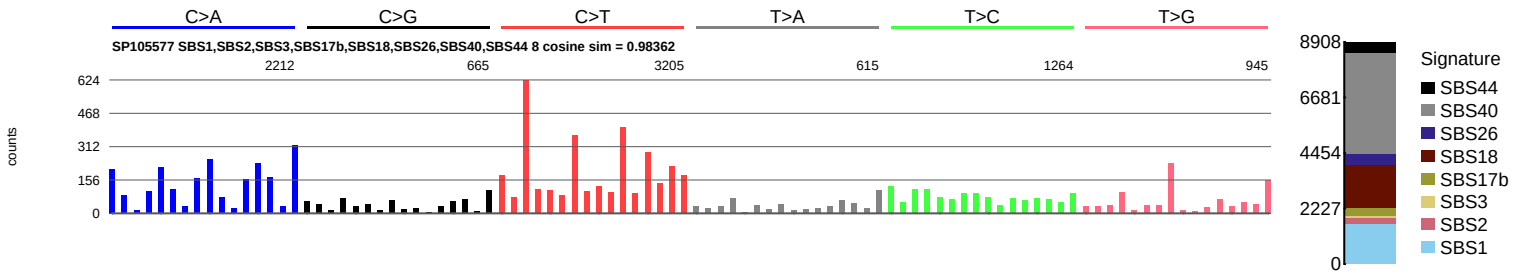

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98374
2212
664
3208
614
1265
945
counts
624
468
312
156
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98372
2202
659
3191
635
1267
952
counts
620
465
310
155
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98371
2202
661
3186
640
1267
950
counts
620
465
310
155
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98369
2154
662
3306
619
1350
821
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98362
2213
662
3207
612
1265
946
counts
624
468
312
156
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1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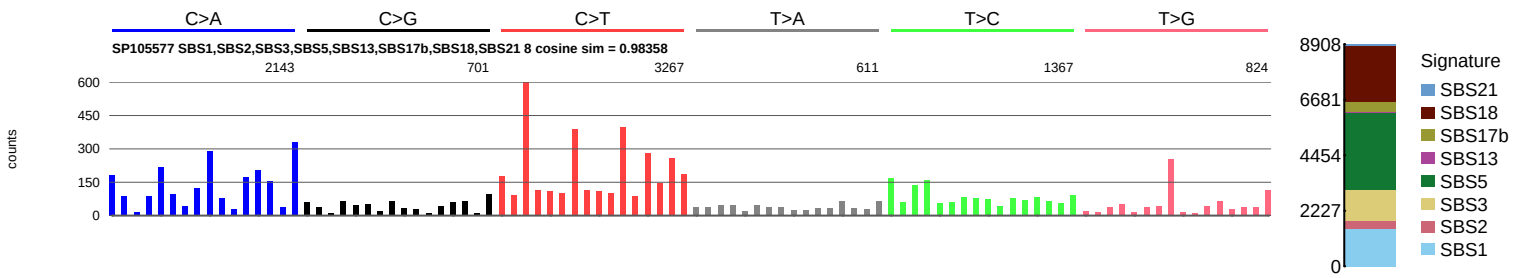
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98362
2212
665
3205
615
1264
945
counts
624
468
312
156
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1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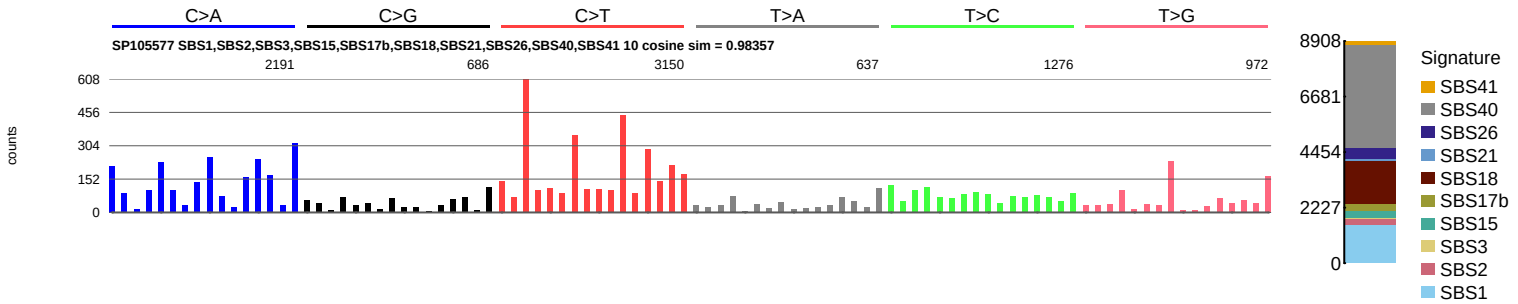
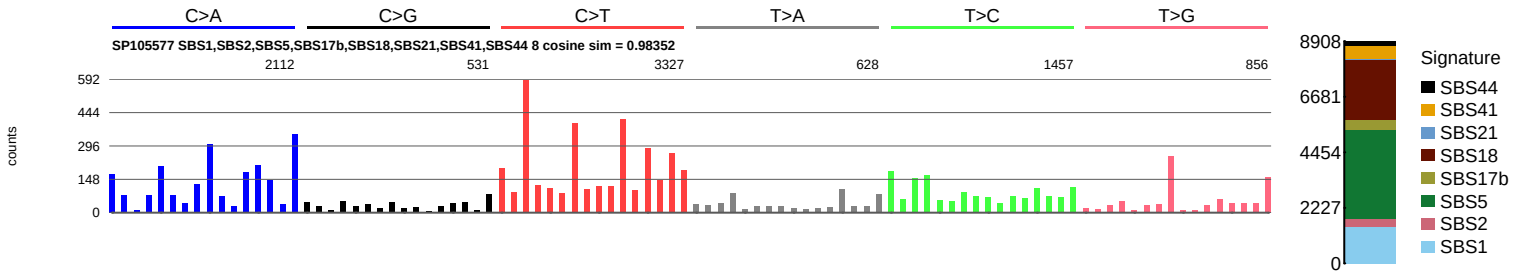
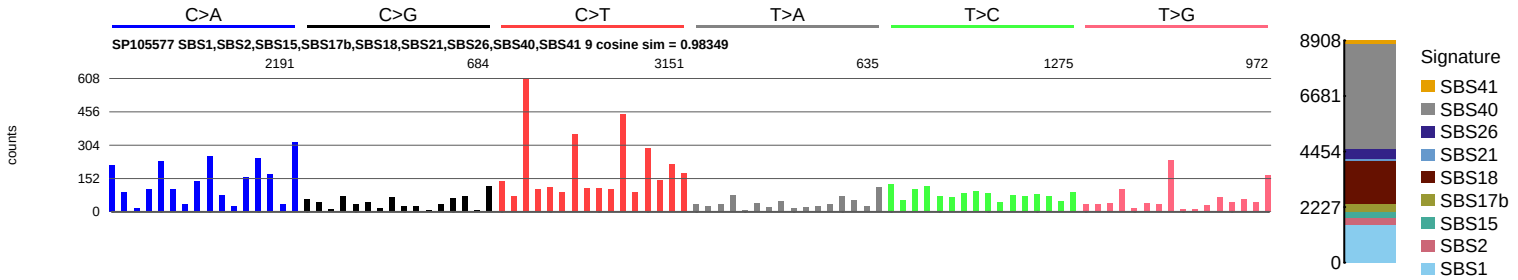
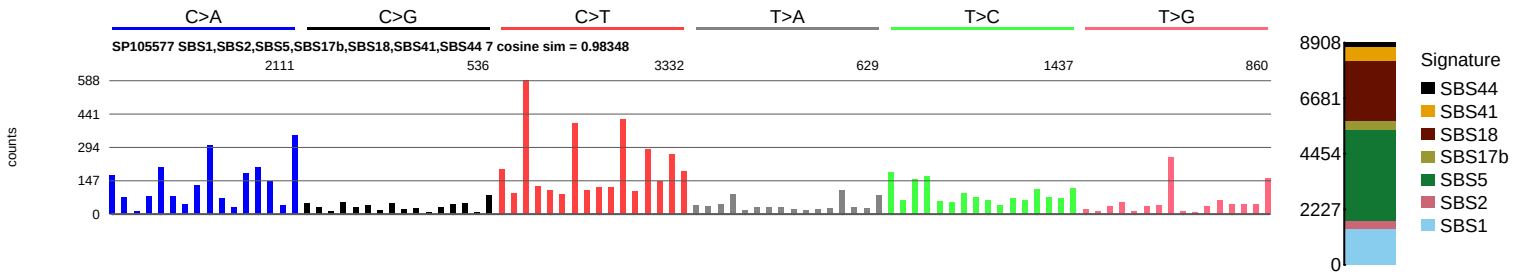
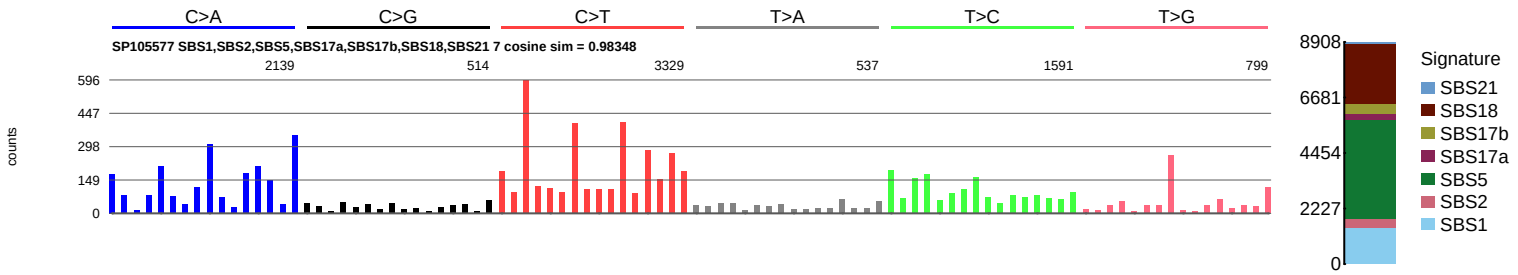
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21 8 cosine sim = 0.98358
2143
701
3267
611
1367
824
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

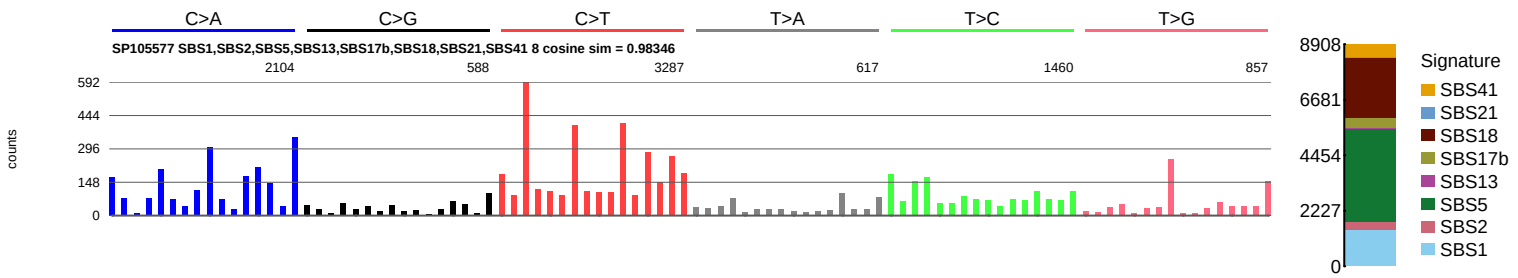

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98357
2191
686
3150
637
1276
972
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

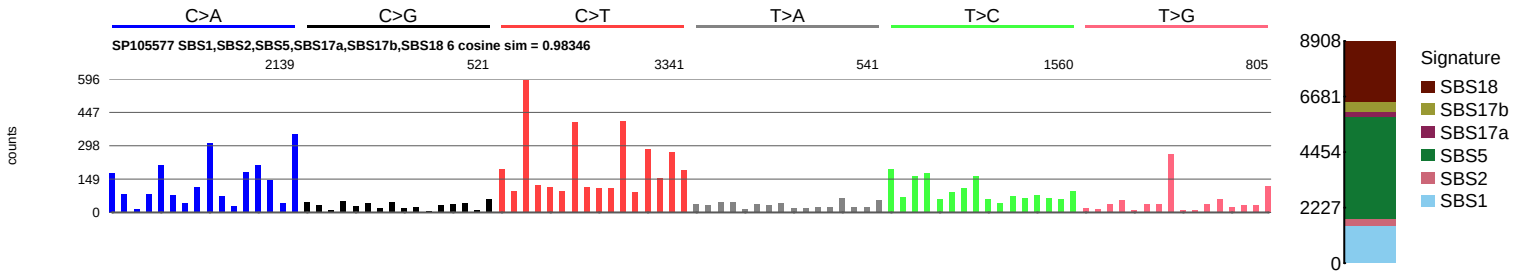

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98352
2112
531
3327
628
1457
856
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

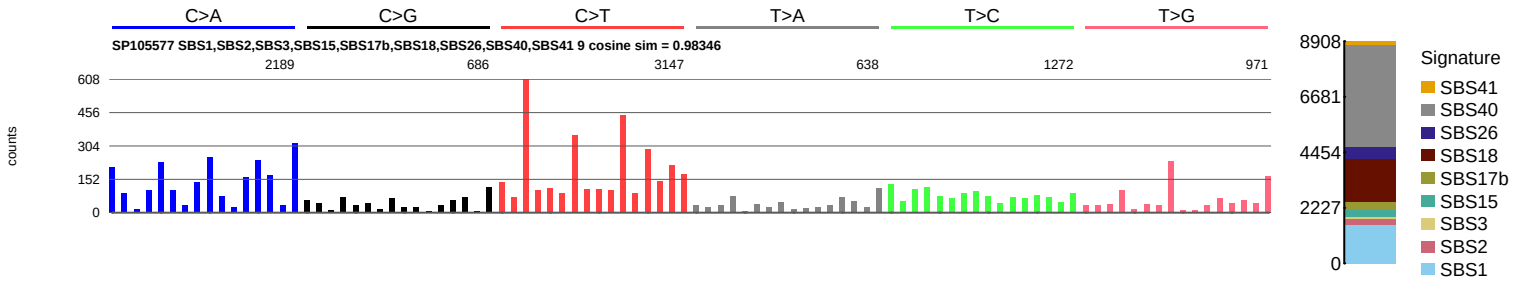

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98349
2191
684
3151
635
1275
972
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1

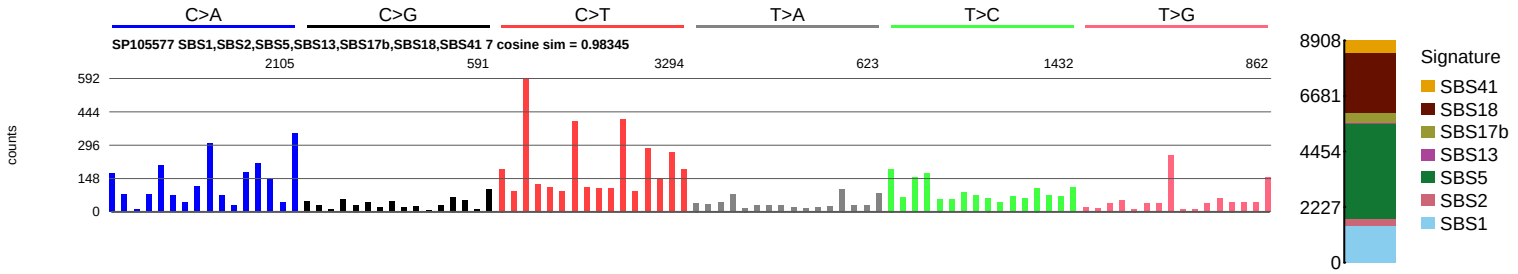

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98348
2111
536
3332
629
1437
860
588
441
294
147
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS5
SBS2
SBS1

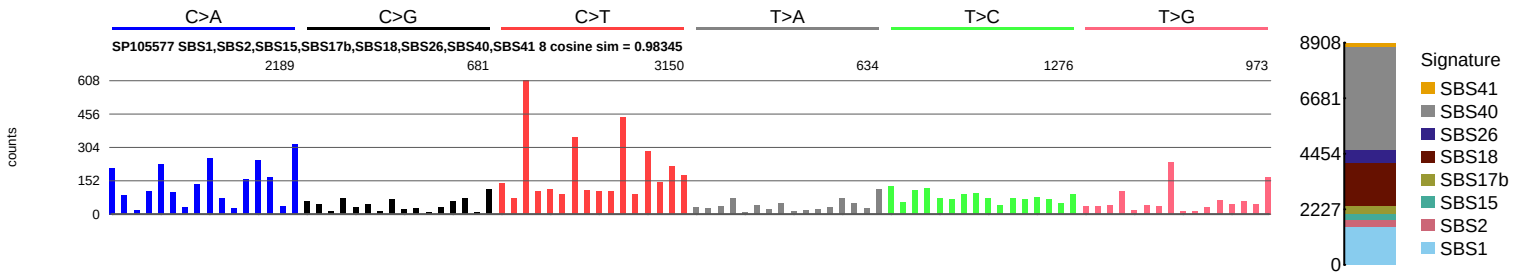

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98348
2139
514
3329
537
1591
799
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98346
2104
588
3287
617
1460
857
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98346
2139
521
3341
541
1560
805
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98346
2189
686
3147
638
1272
971
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41 7 cosine sim = 0.98345
2105
591
3294
623
1432
862
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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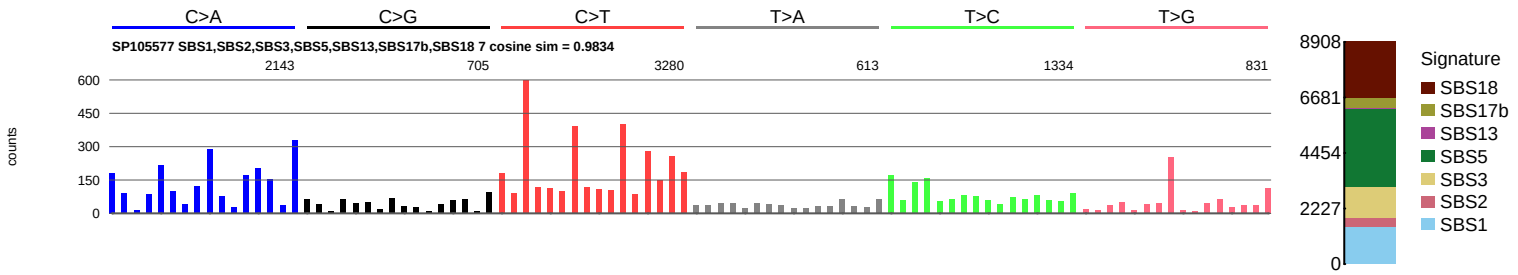
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98345
2189
681
3150
634
1276
973
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1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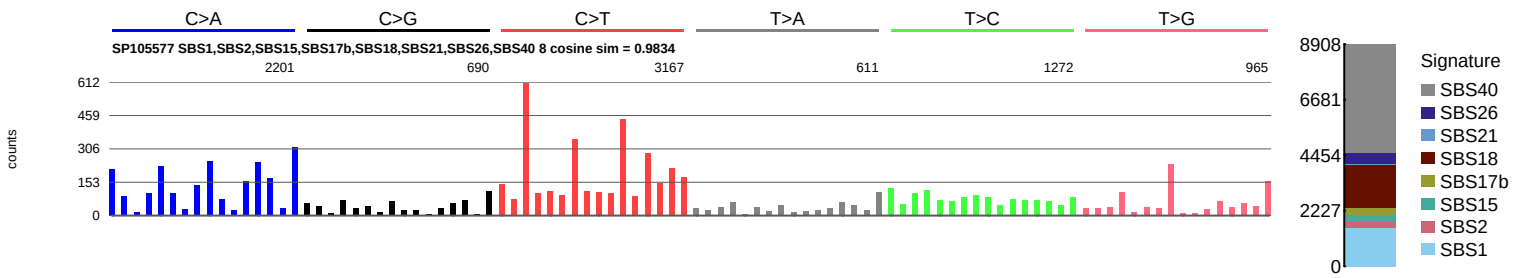
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18 7 cosine sim = 0.9834
2143
705
3280
613
1334
831
600
450
300
150
0
counts
8908
6681
4454
2227
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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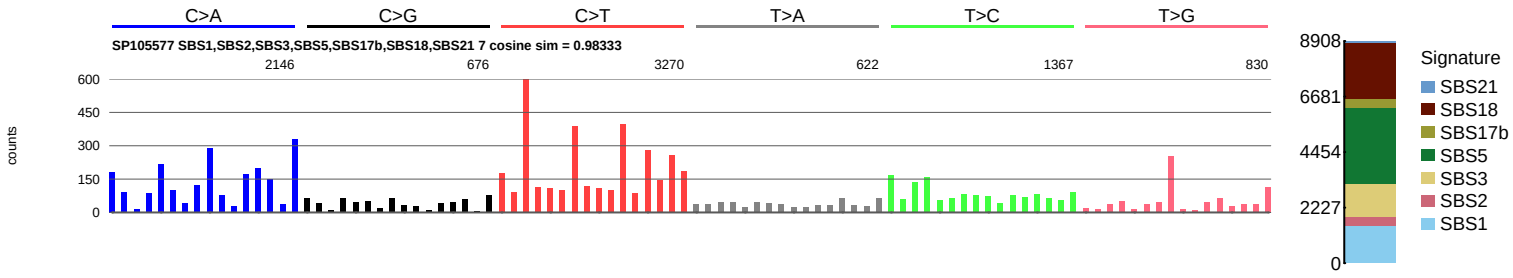
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.9834
2201
690
3167
611
1272
965
612
459
306
153
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1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C>G
C>T
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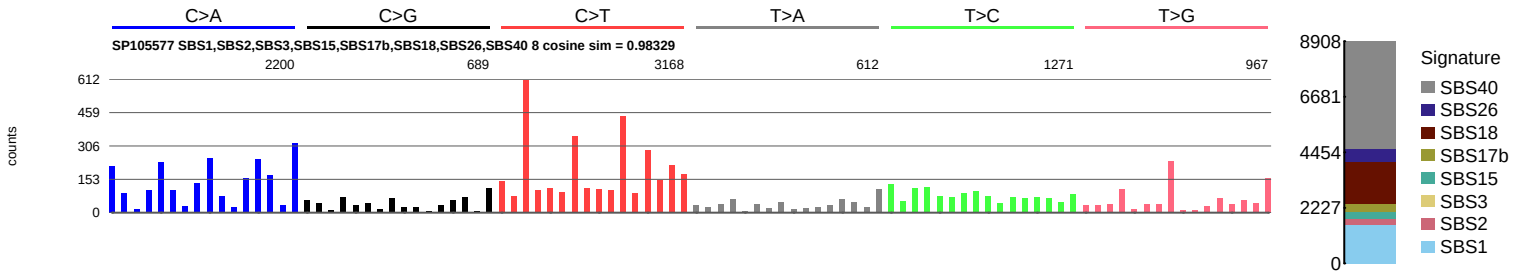
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21 7 cosine sim = 0.98333
2146
676
3270
622
1367
830
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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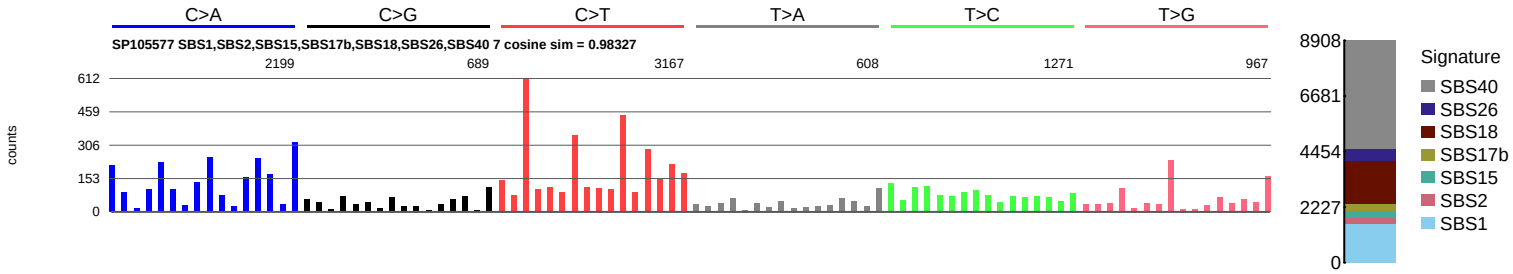
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98329
2200
689
3168
612
1271
967
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
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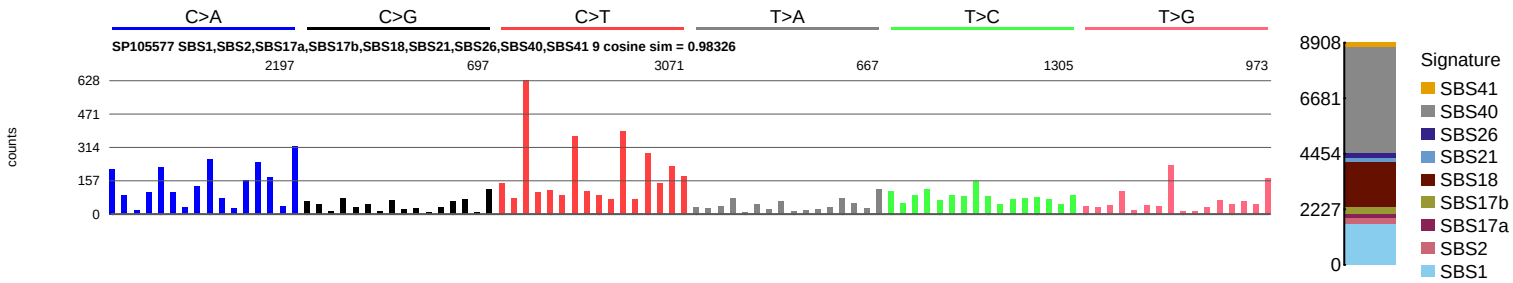
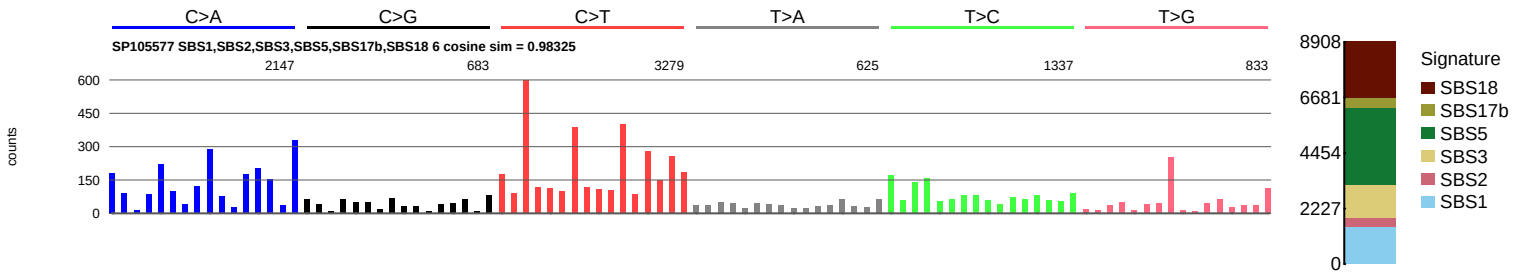
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98327
2199
689
3167
608
1271
967
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98326
2197
697
3071
667
1305
973
counts
628
471
314
157
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18 6 cosine sim = 0.98325
2147
683
3279
625
1337
833
counts
600
450
300
150
0
8908
6681
4454
2227
0
Signature
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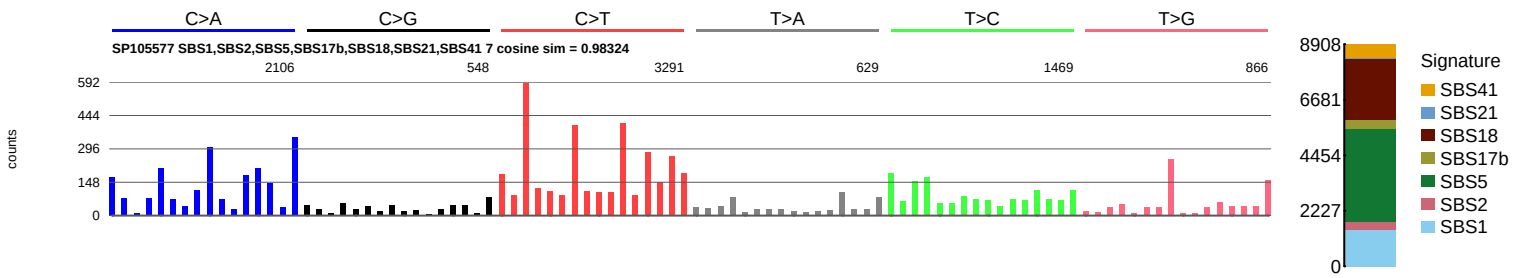
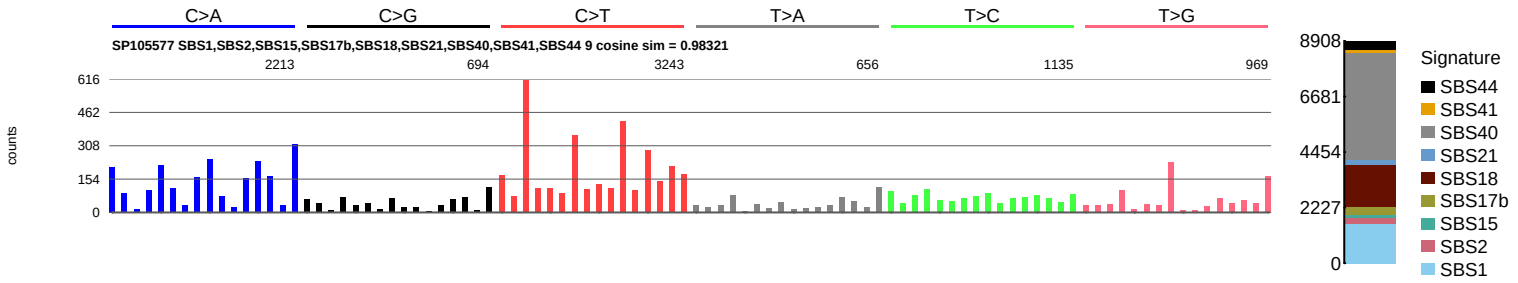
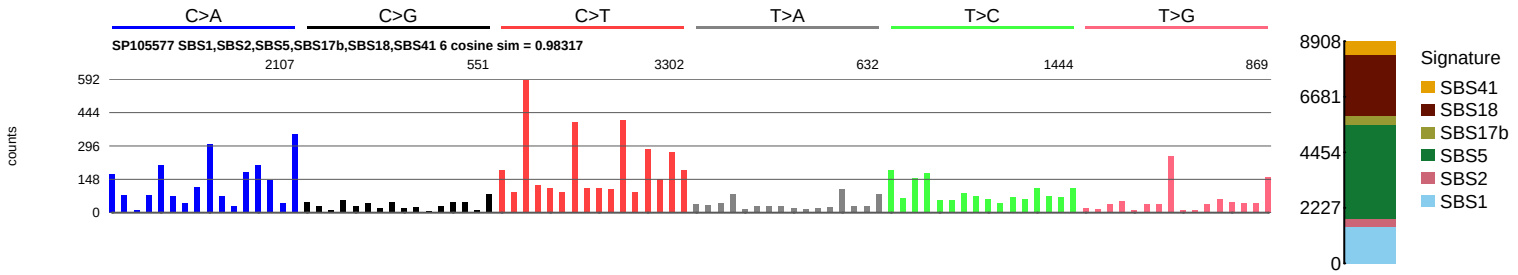
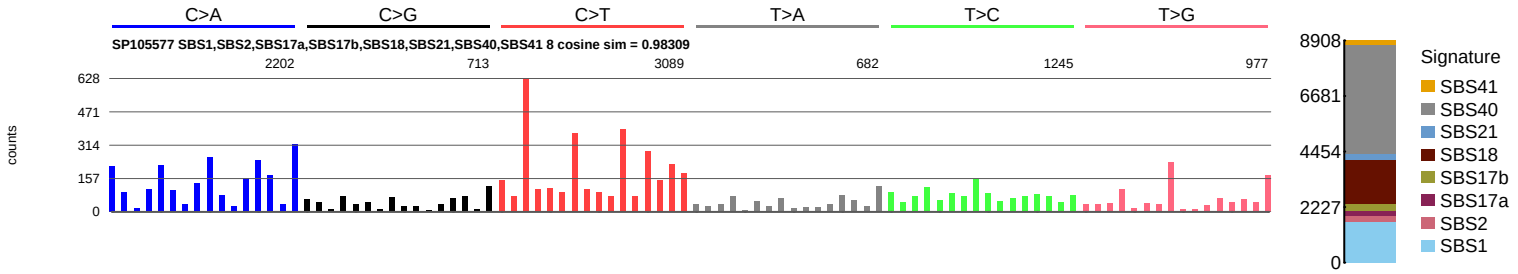
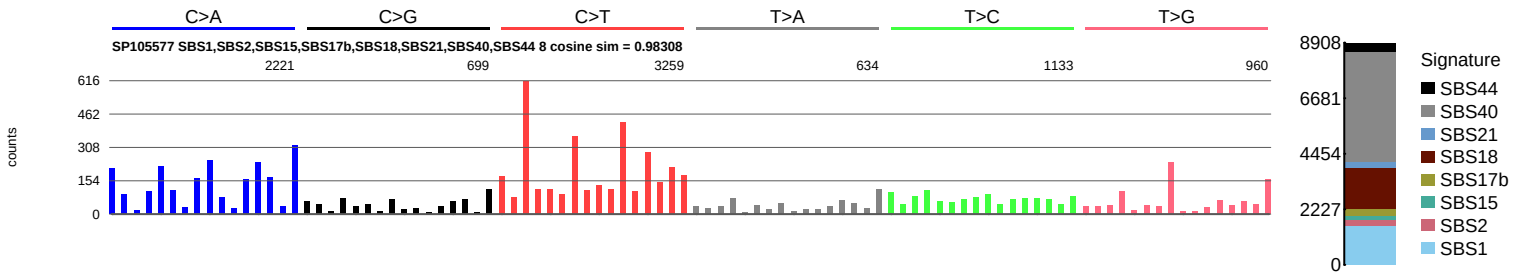
SBS2
SBS1

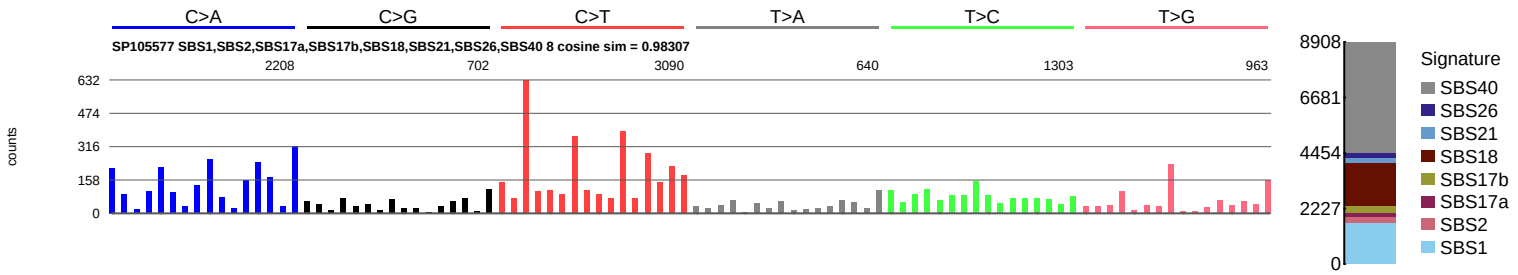

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98324
2106
548
3291
629
1469
866
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

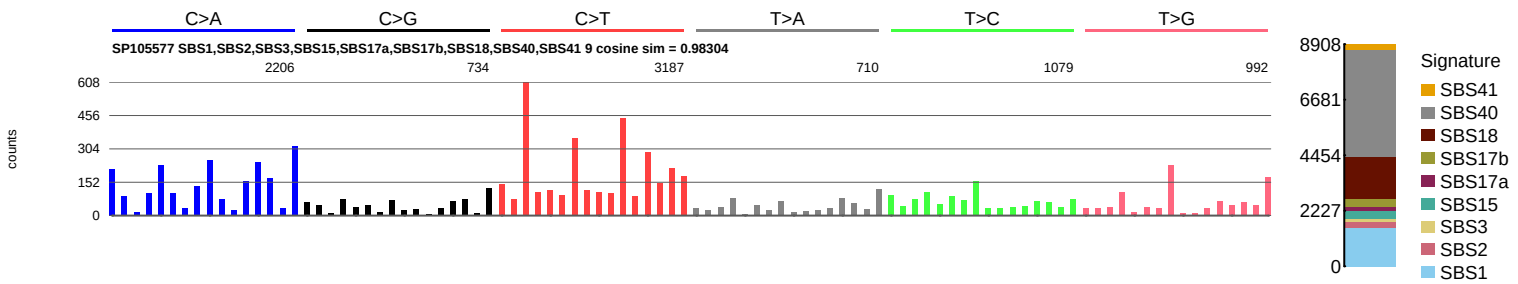

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98321
2213
694
3243
656
1135
969
counts
616
462
308
154
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS41 6 cosine sim = 0.98317
2107
551
3302
632
1444
869
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98309
2202
713
3089
682
1245
977
counts
628
471
314
157
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98308
2221
699
3259
634
1133
960
counts
616
462
308
154
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98307
2208
702
3090
640
1303
963
counts
632
474
316
158
0
8908
6681
4454
2227
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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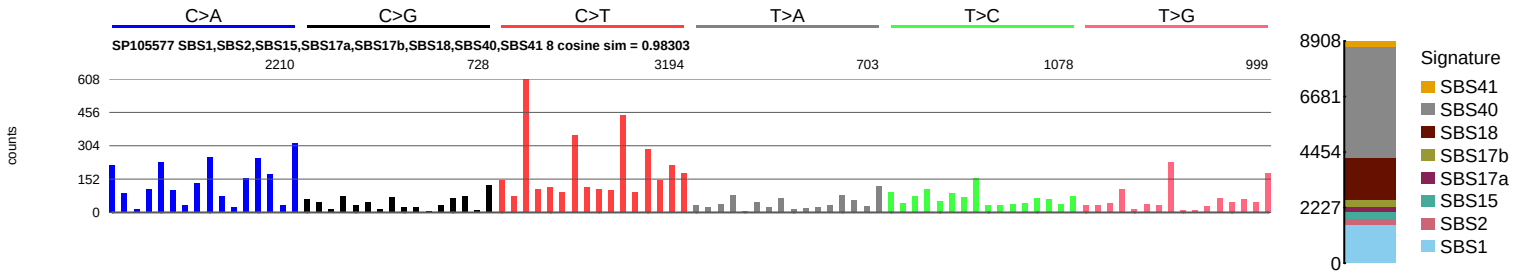
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98304
2206
734
3187
710
1079
992
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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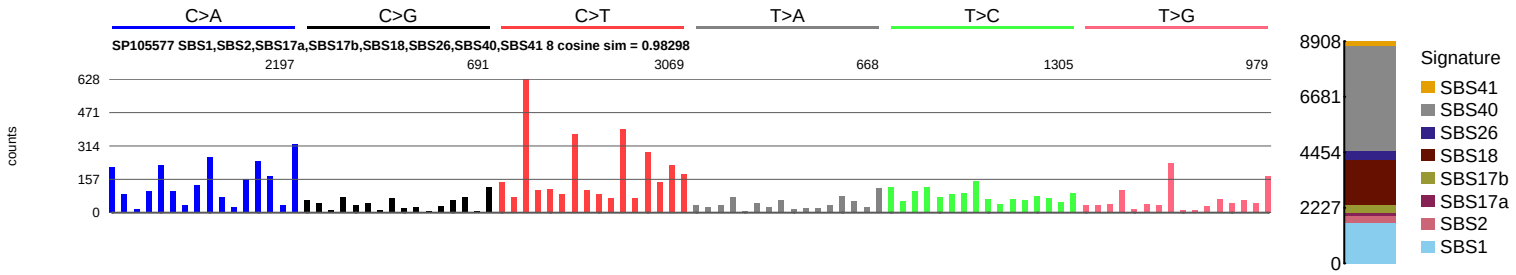
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98303
2210
728
3194
703
1078
999
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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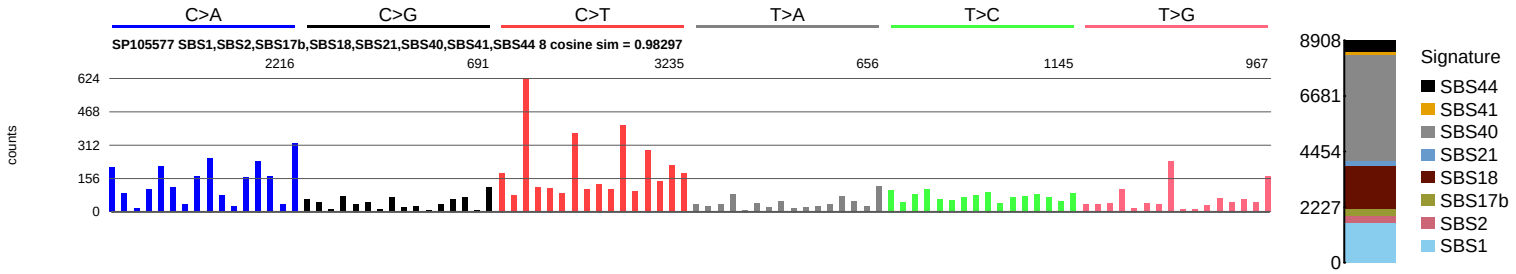
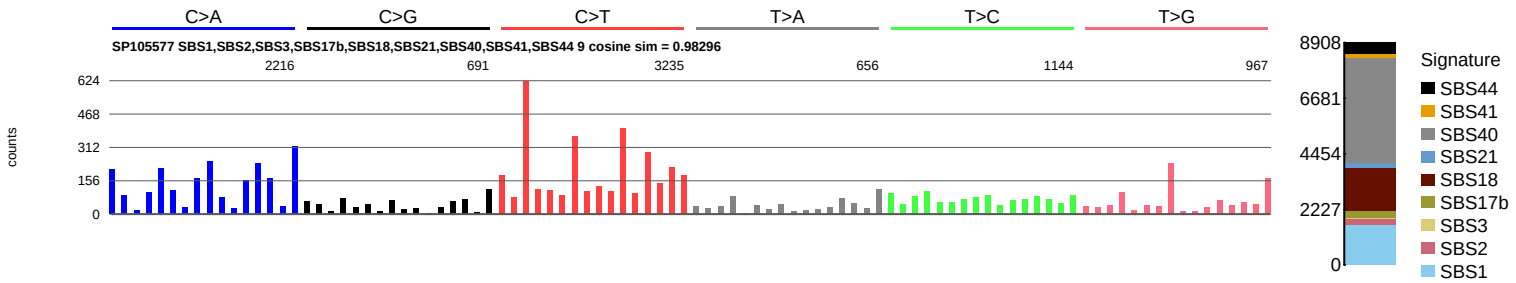
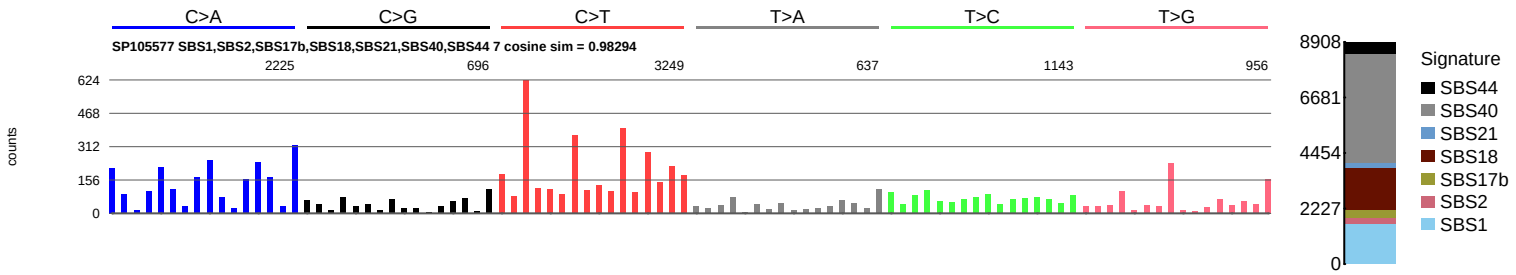
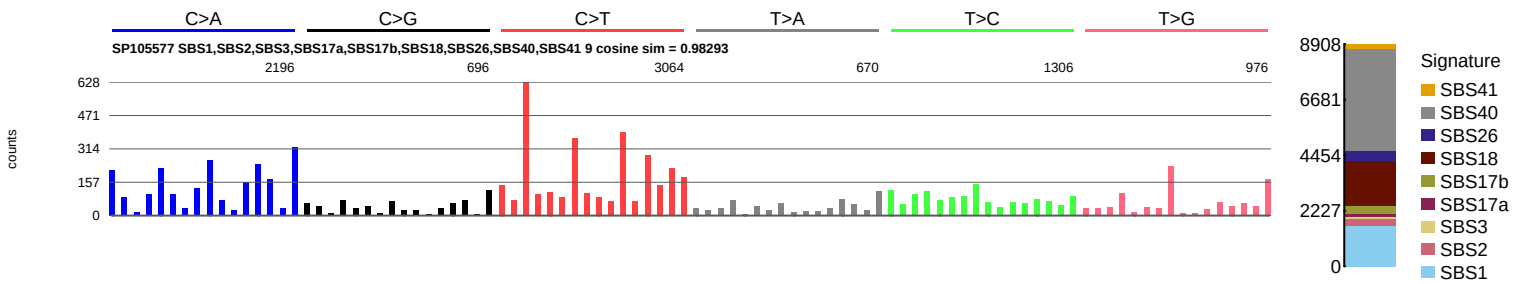
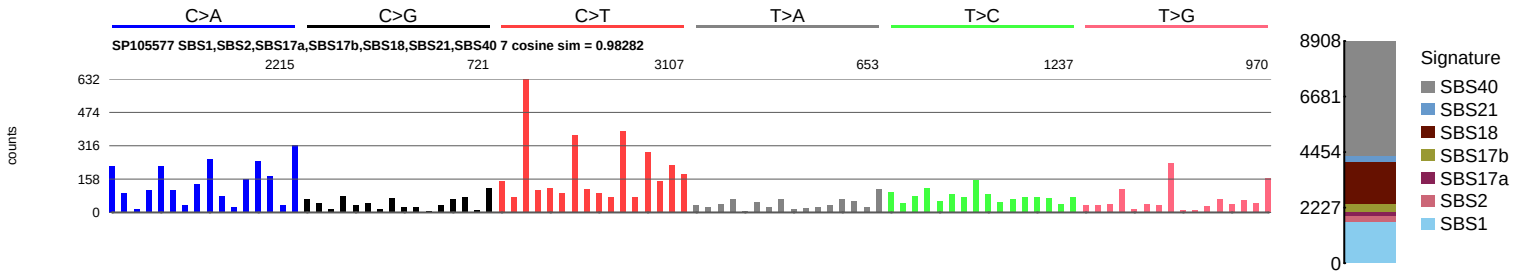
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98298
2197
691
3069
668
1305
979
628
471
314
157
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

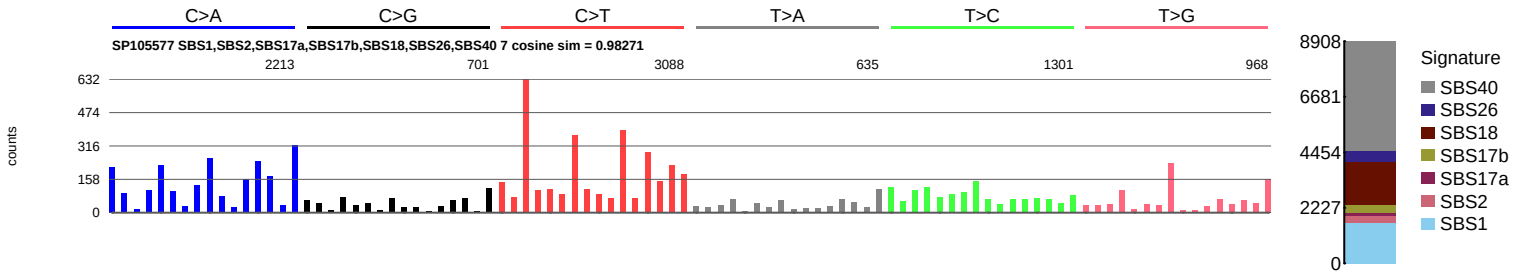

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98297
2216
691
3235
656
1145
967
624
468
312
156
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98296
2216
691
3235
656
1144
967
624
468
312
156
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98294
2225
696
3249
637
1143
956
624
468
312
156
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98293
2196
696
3064
670
1306
976
628
471
314
157
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98282
2215
721
3107
653
1237
970
counts
632
474
316
158
0
8908
6681
4454
2227
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98271
2213
701
3088
635
1301
968
counts
632
474
316
158
0
8908
6681
4454
2227
0
Signature
SBS40
SBS26
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SBS17b
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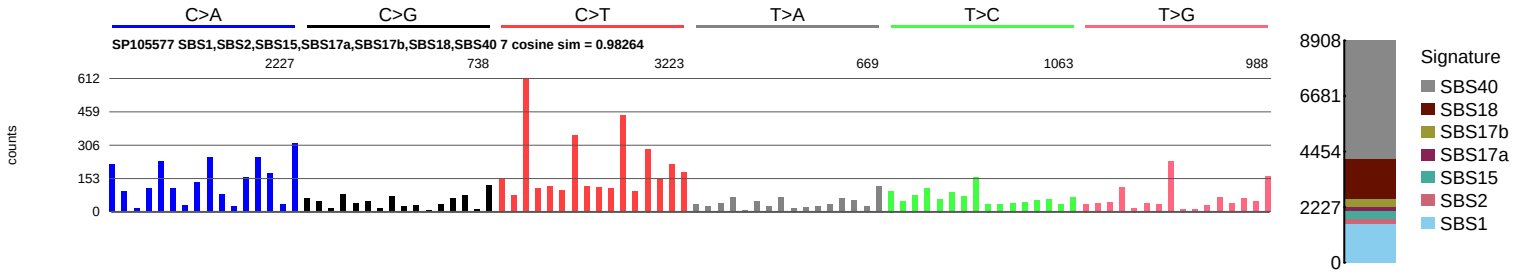
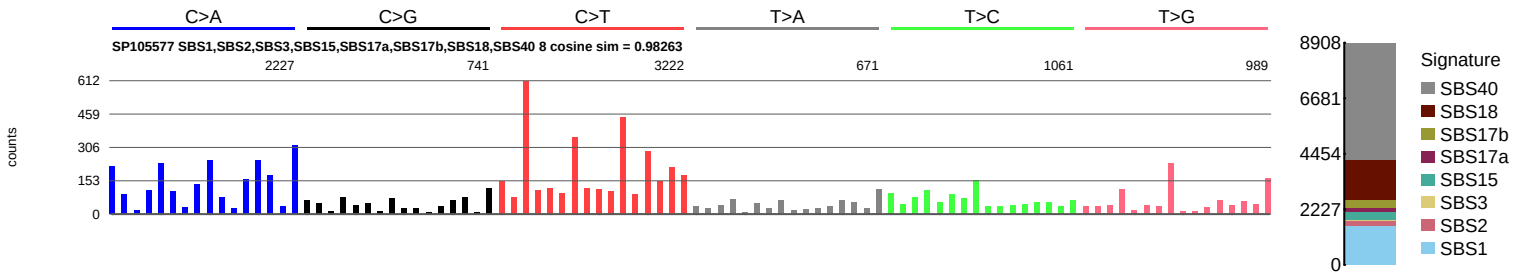
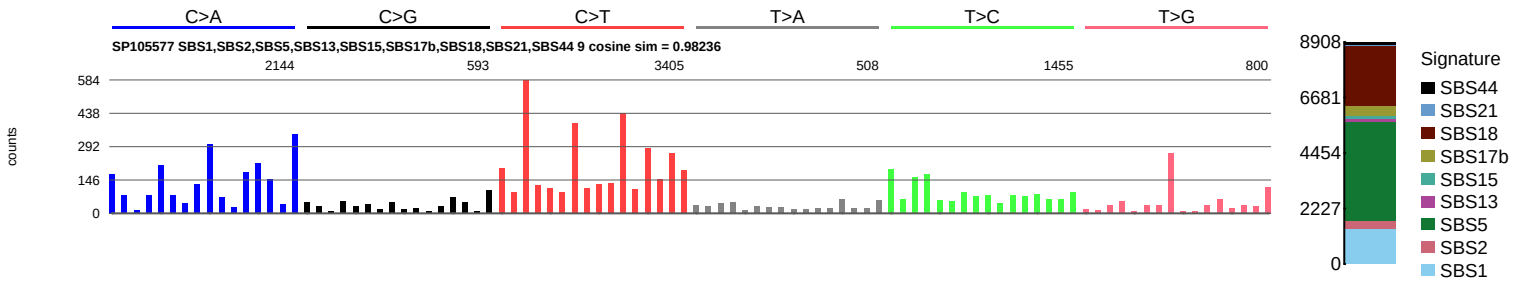
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98264
2227
738
3223
669
1063
988
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98263
2227
741
3222
671
1061
989
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98236
2144
593
3405
508
1455
800
counts
584
438
292
146
0
8908
6681
4454
2227
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Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
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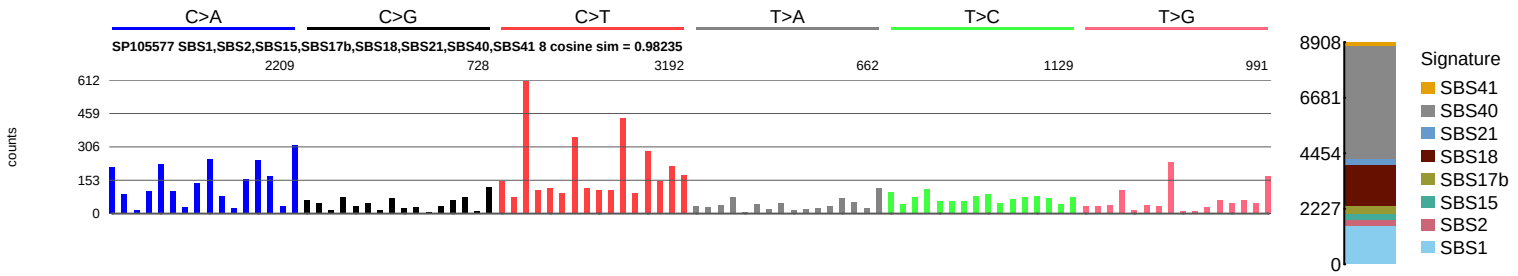
SBS2
SBS1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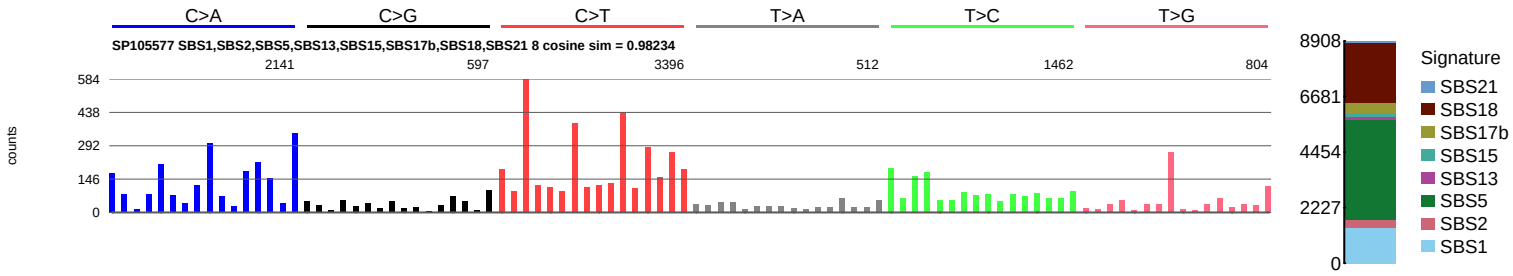
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98235
2209
728
3192
662
1129
991
counts
612
459
306
153
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1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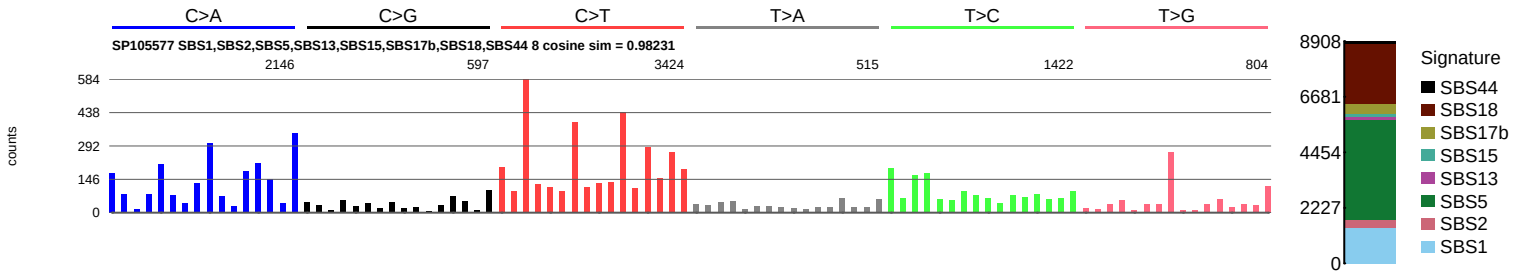
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98234
2141
597
3396
512
1462
804
584
438
292
146
0
counts
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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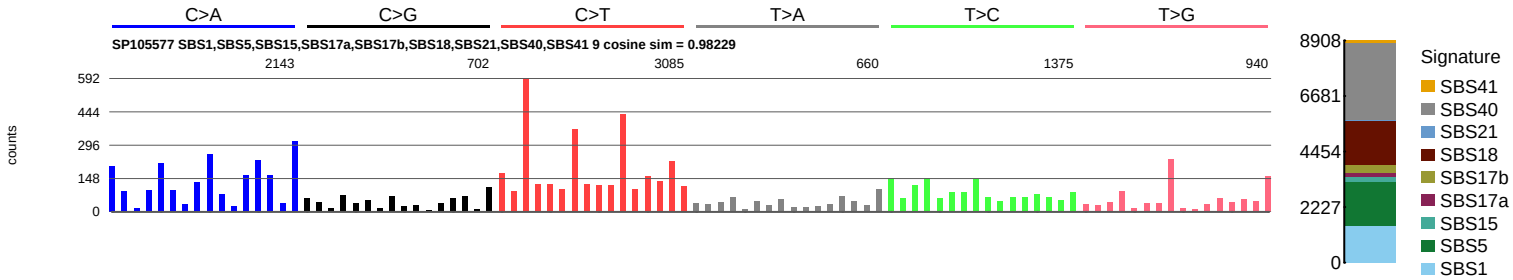
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 8 cosine sim = 0.98231
2146
597
3424
515
1422
804
584
438
292
146
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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C>T
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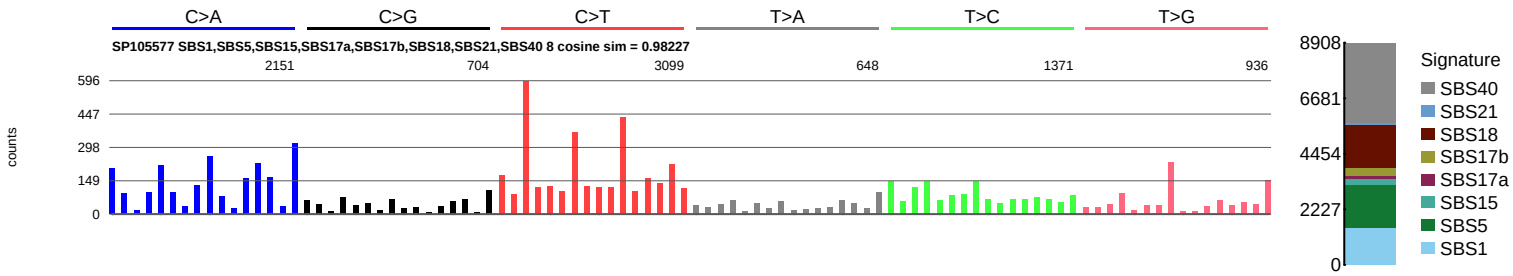
T>G
SP105577 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98229
2143
702
3085
660
1375
940
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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T>G
SP105577 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98227
2151
704
3099
648
1371
936
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS40
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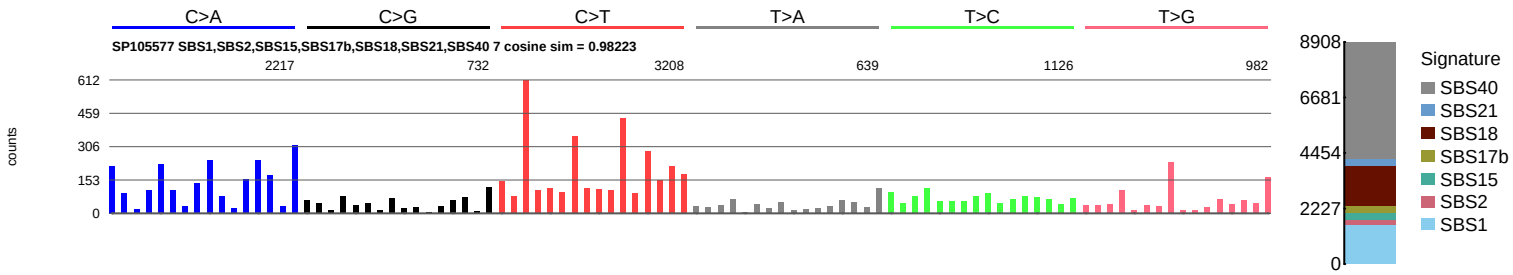
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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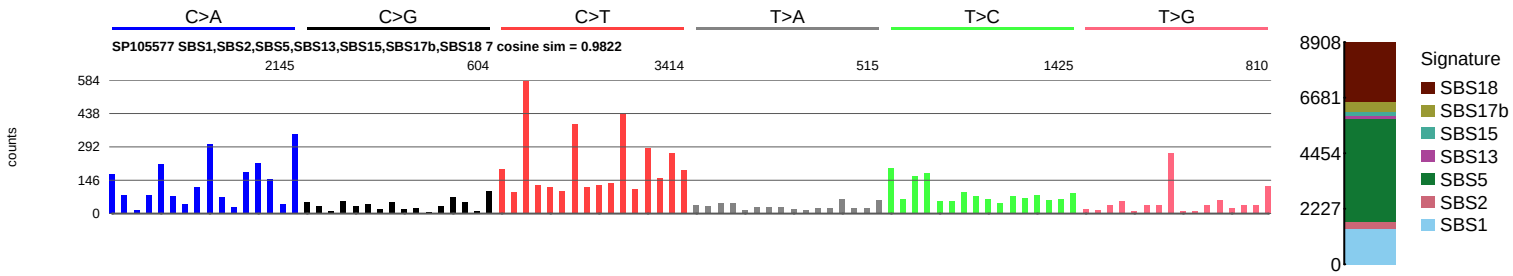
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98223
2217
732
3208
639
1126
982
612
459
306
153
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18 7 cosine sim = 0.9822
2145
604
3414
515
1425
810
584
438
292
146
0
counts
8908
6681
4454
2227
0
Signature
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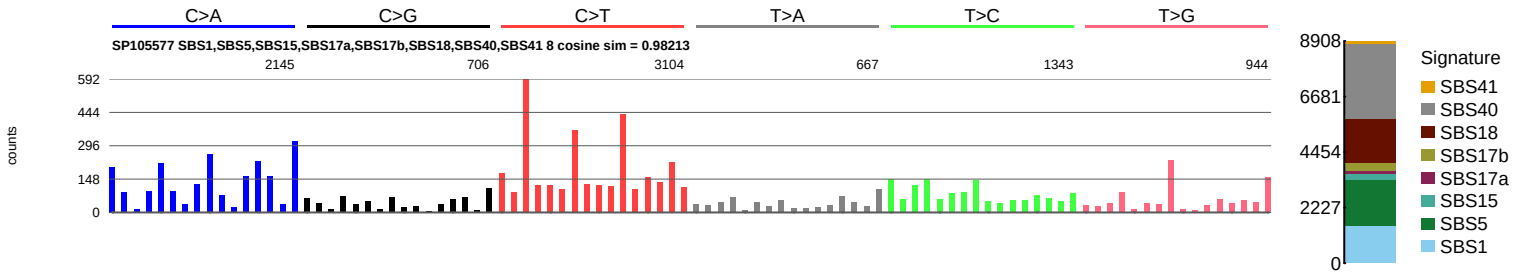
SBS2
SBS1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C>G
C>T
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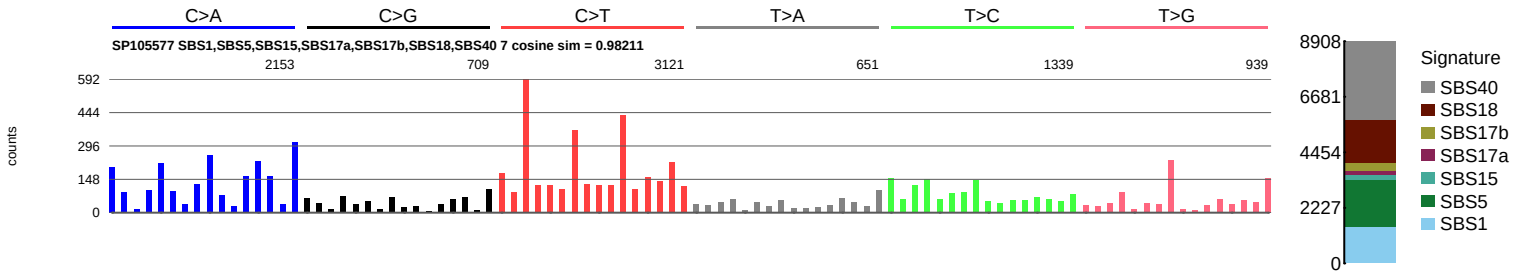
T>C
T>G
SP105577 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98213
2145
706
3104
667
1343
944
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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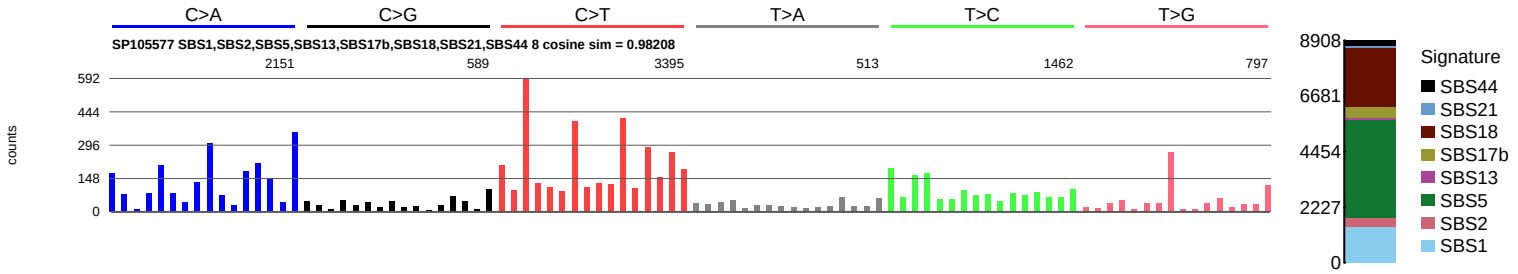
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98211
2153
709
3121
651
1339
939
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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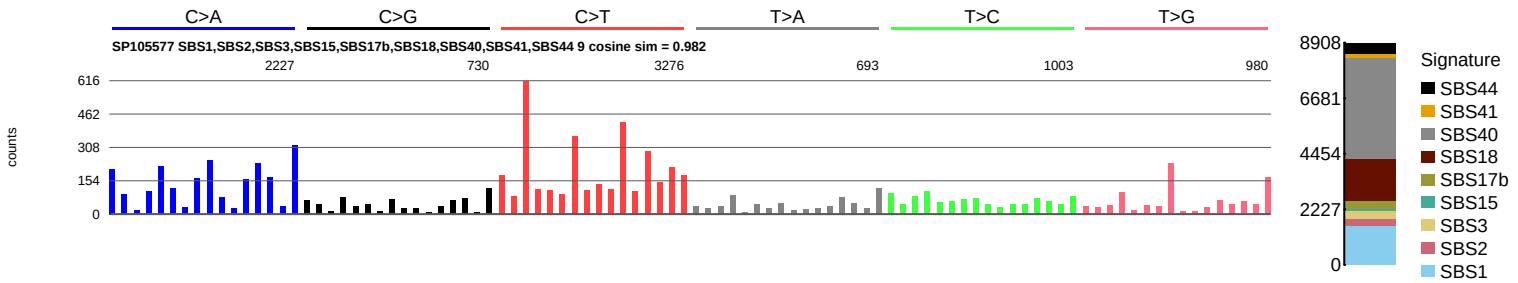
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98208
2151
589
3395
513
1462
797
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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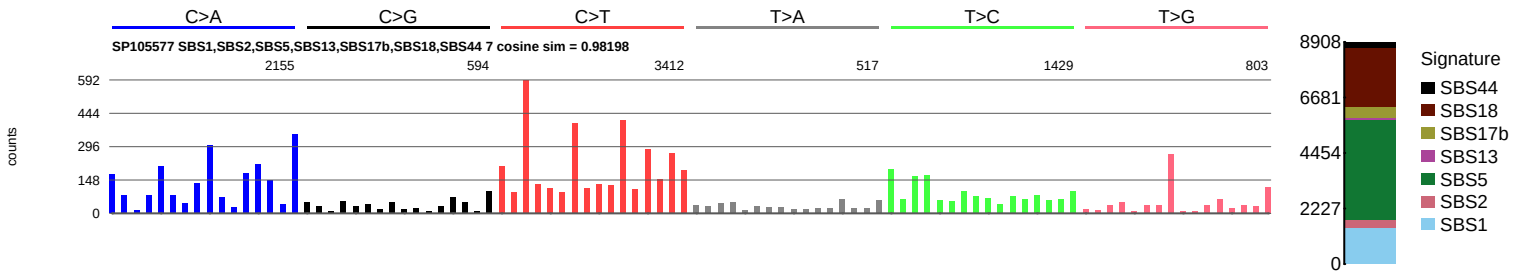
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.982
2227
730
3276
693
1003
980
616
462
308
154
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS44 7 cosine sim = 0.98198
2155
594
3412
517
1429
803
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
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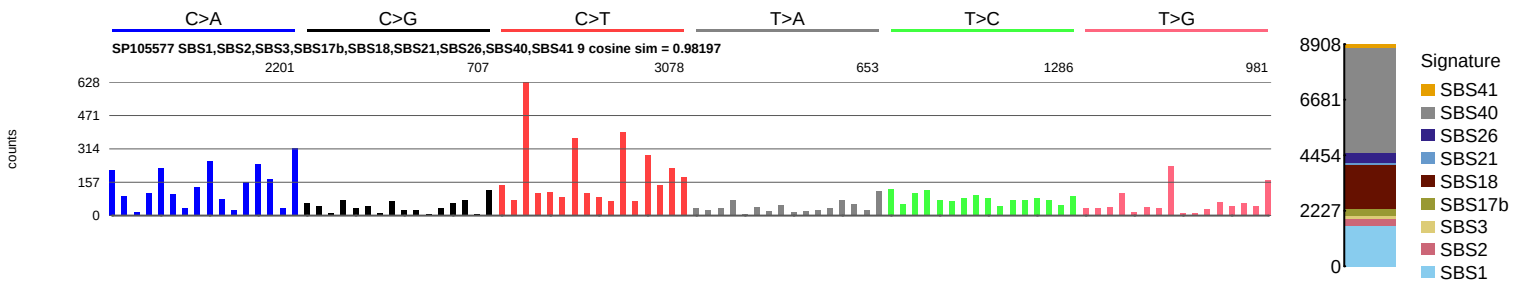
SBS2
SBS1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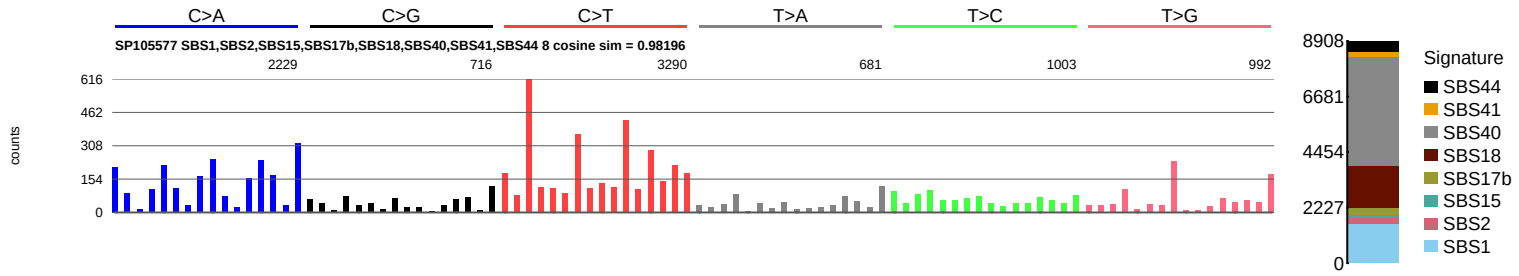
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98197
2201
707
3078
653
1286
981
628
471
314
157
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1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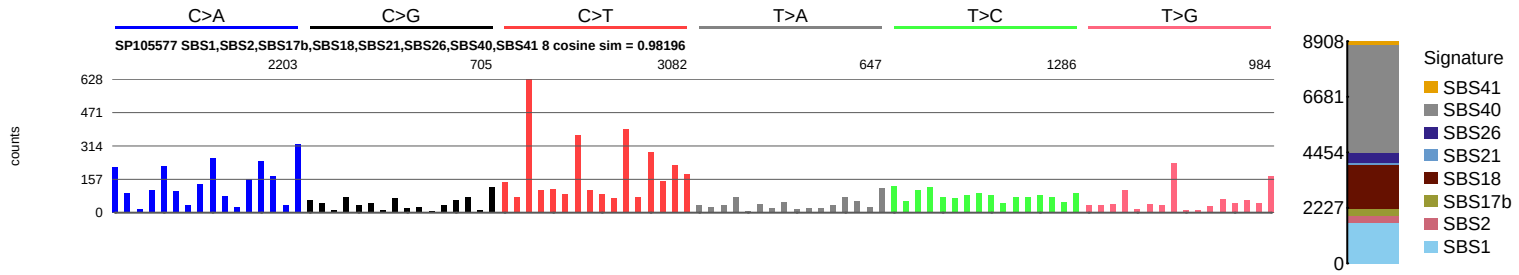
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98196
2229
716
3290
681
1003
992
counts
616
462
308
154
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1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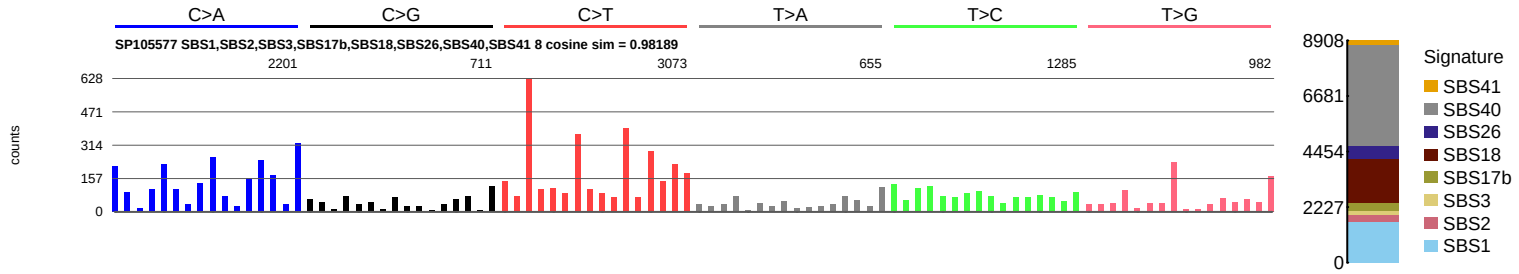
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98196
2203
705
3082
647
1286
984
counts
628
471
314
157
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1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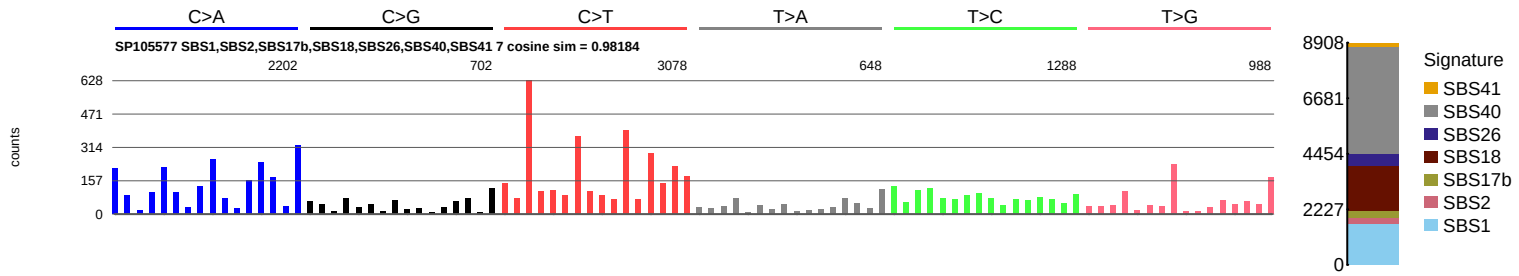
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98189
2201
711
3073
655
1285
982
counts
628
471
314
157
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1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C>G
C>T
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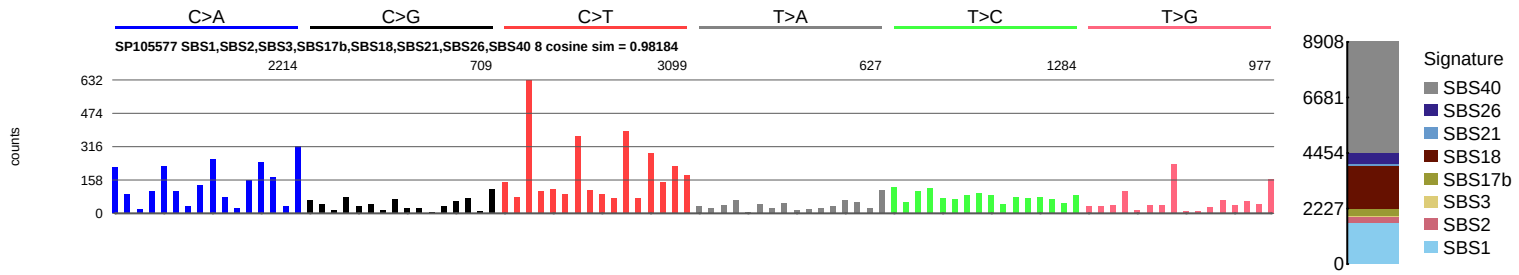
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98184
2202
702
3078
648
1288
988
counts
628
471
314
157
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1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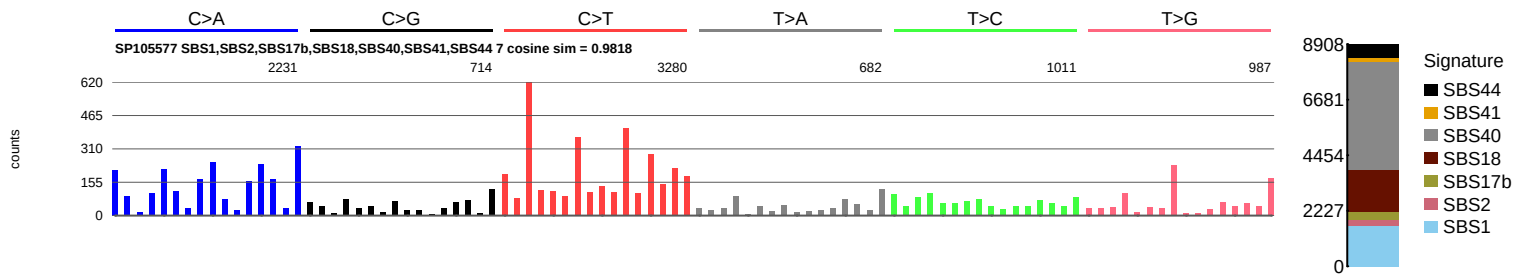
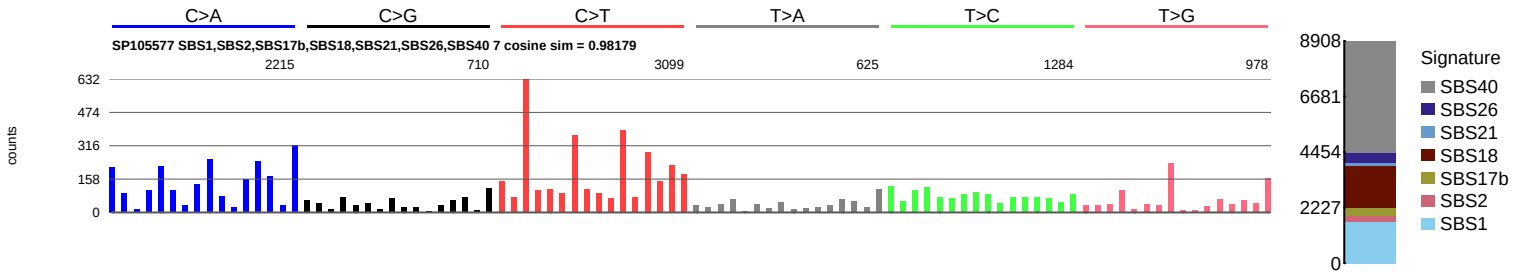
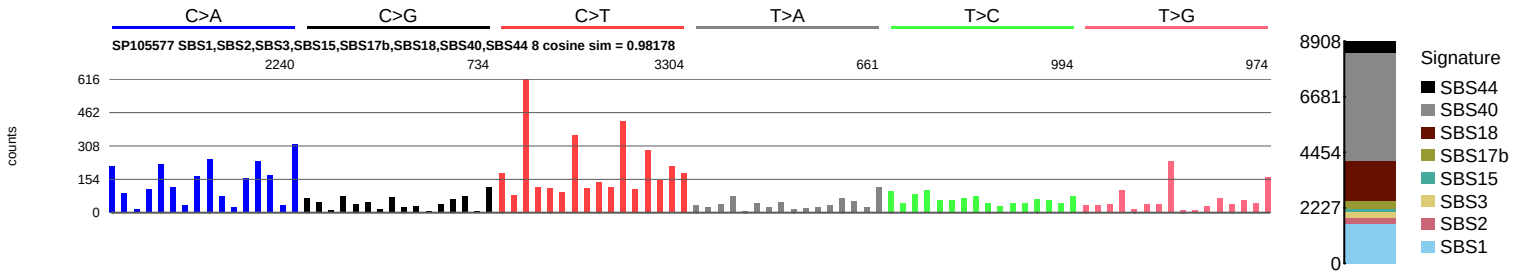
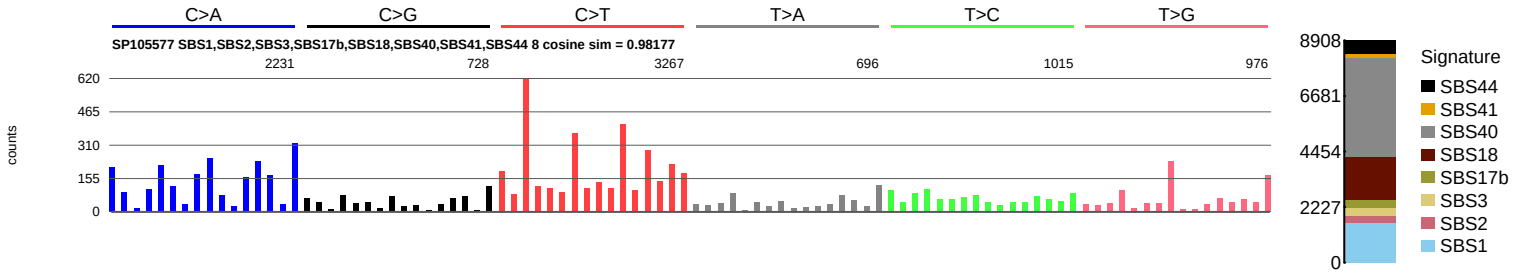
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98184
2214
709
3099
627
1284
977
counts
632
474
316
158
0
8908
6681
4454
2227
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.9818
2231
714
3280
682
1011
987
counts
620
465
310
155
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98179
2215
710
3099
625
1284
978
counts
632
474
316
158
0
8908
6681
4454
2227
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98178
2240
734
3304
661
994
974
counts
616
462
308
154
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98177
2231
728
3267
696
1015
976
counts
620
465
310
155
0
8908
6681
4454
2227
0
Signature
SBS44
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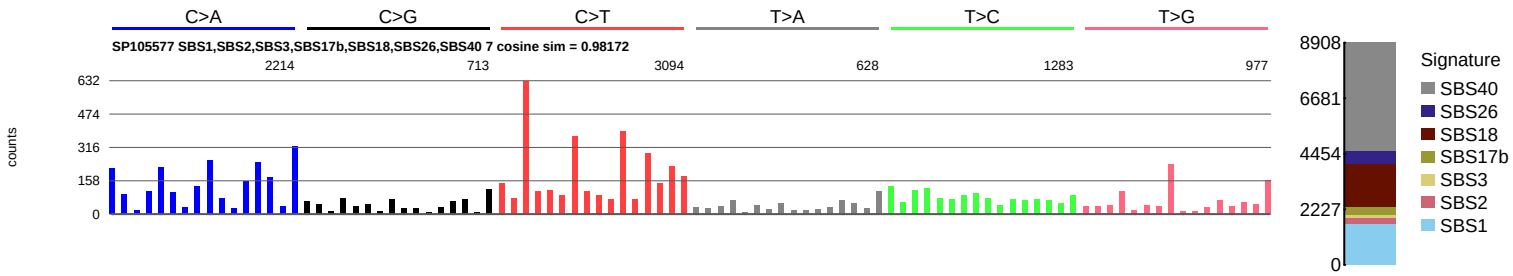
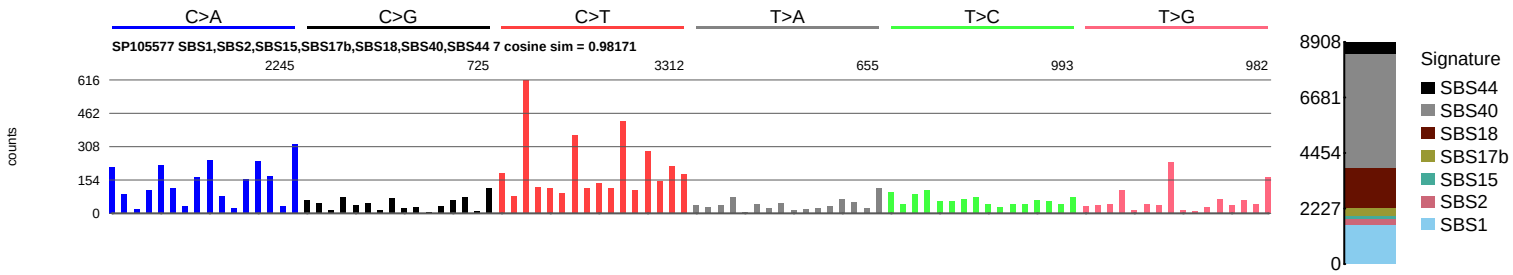
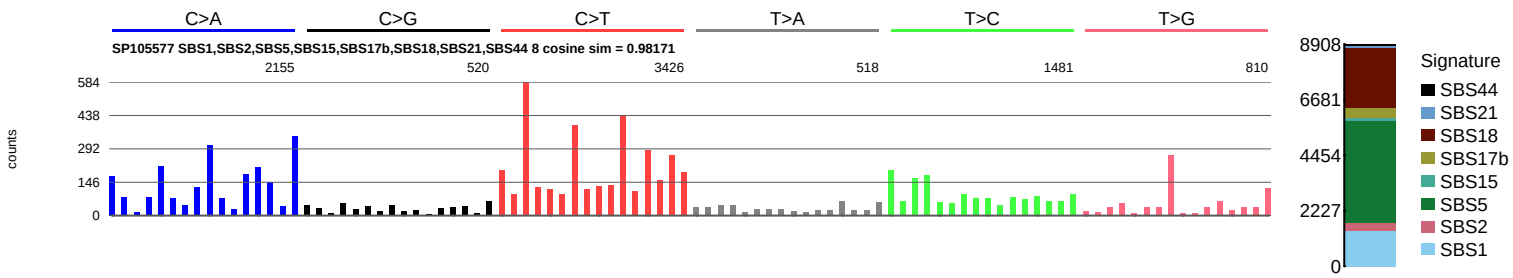
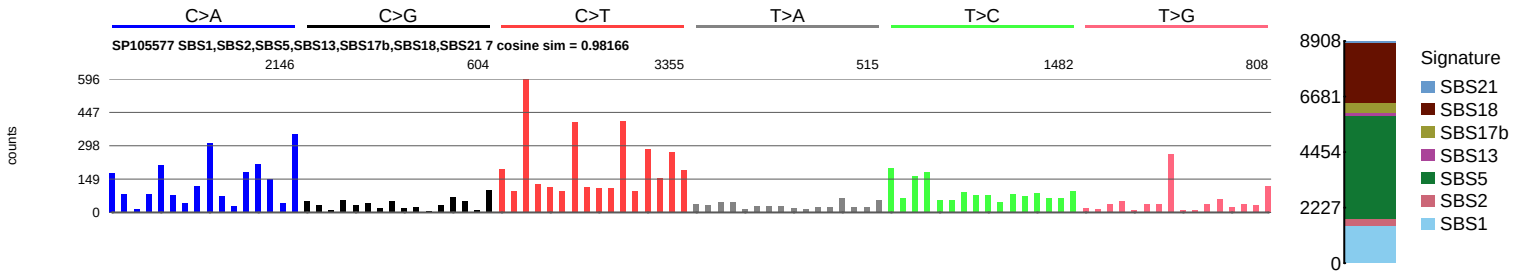
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98172
2214
713
3094
628
1283
977
counts
632
474
316
158
0
8908
6681
4454
2227
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98171
2245
725
3312
655
993
982
counts
616
462
308
154
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98171
2155
520
3426
518
1481
810
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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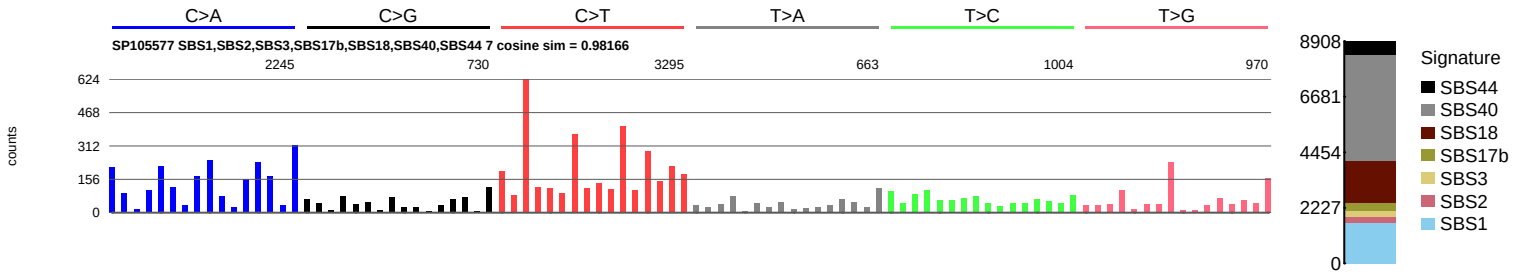
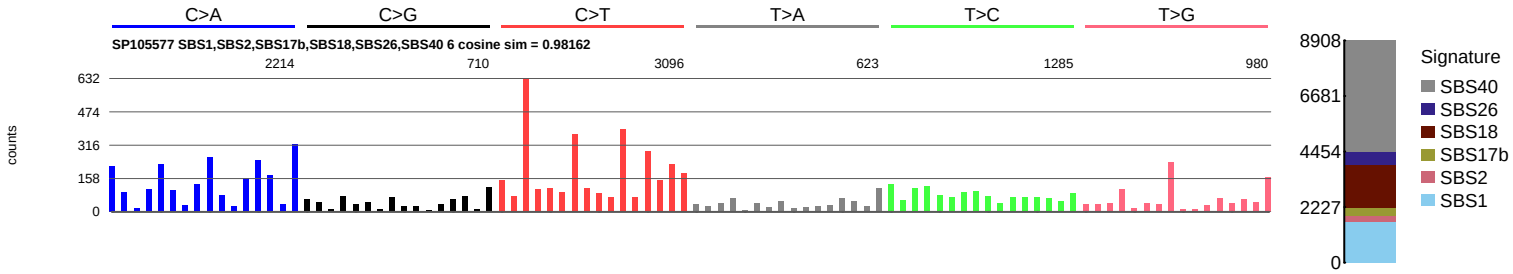
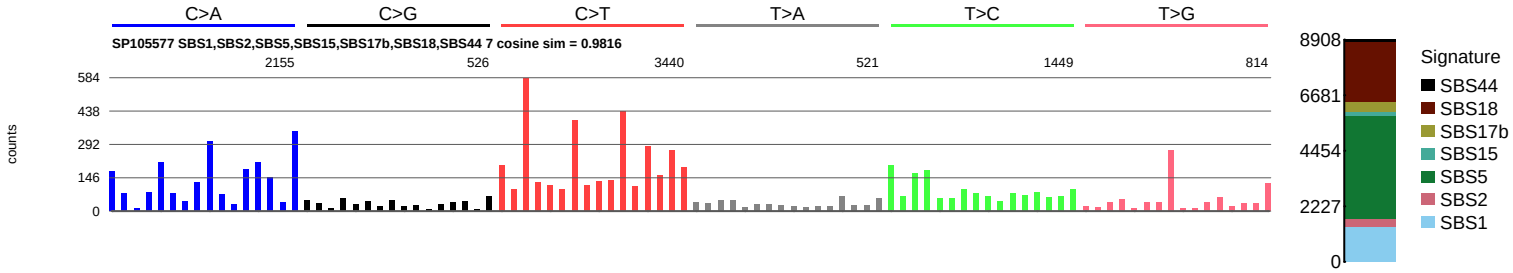
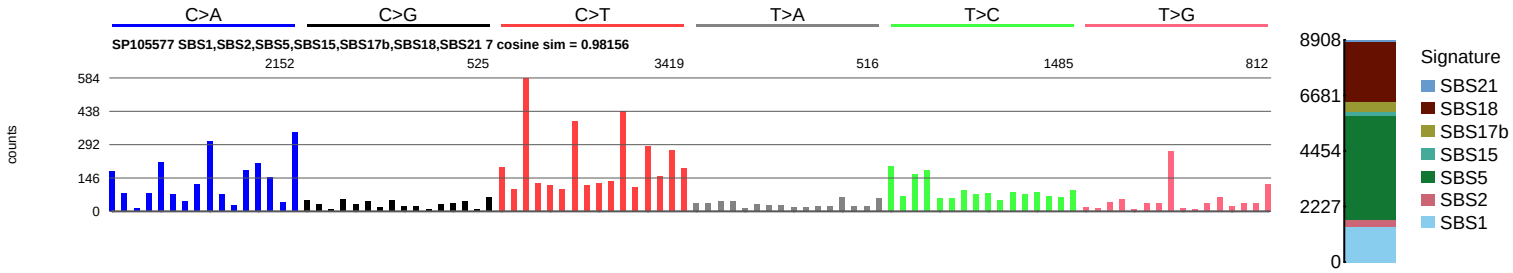
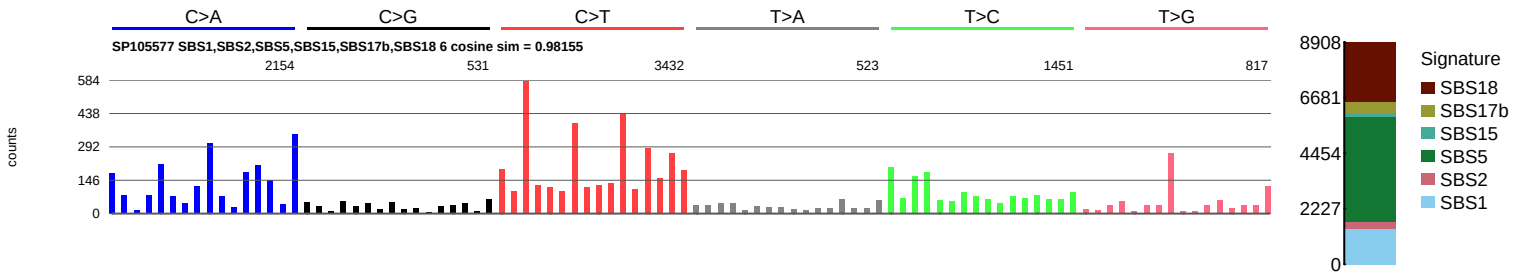
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21 7 cosine sim = 0.98166
2146
604
3355
515
1482
808
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98166
2245
730
3295
663
1004
970
counts
624
468
312
156
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98162
2214
710
3096
623
1285
980
counts
632
474
316
158
0
8908
6681
4454
2227
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS44 7 cosine sim = 0.9816
2155
526
3440
521
1449
814
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21 7 cosine sim = 0.98156
2152
525
3419
516
1485
812
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18 6 cosine sim = 0.98155
2154
531
3432
523
1451
817
counts
584
438
292
146
0
8908
6681
4454
2227
0
Signature
SBS18
SBS17b
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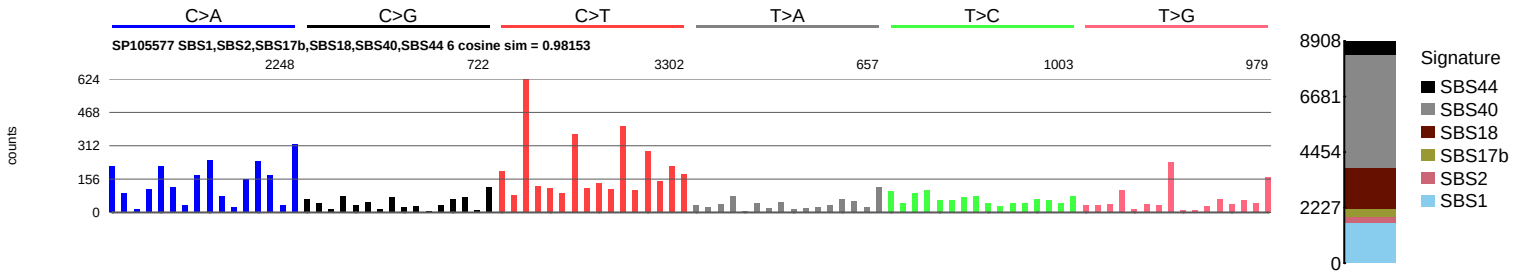
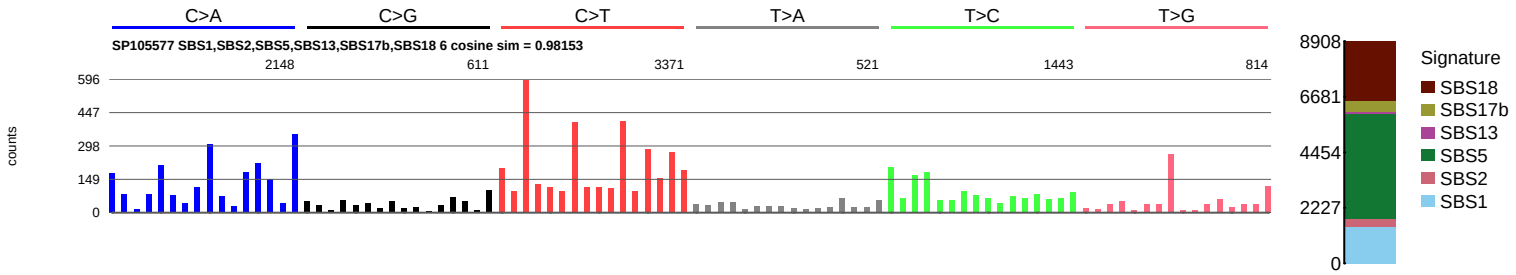
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98153
2248
722
3302
657
1003
979
counts
624
468
312
156
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18 6 cosine sim = 0.98153
2148
611
3371
521
1443
814
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
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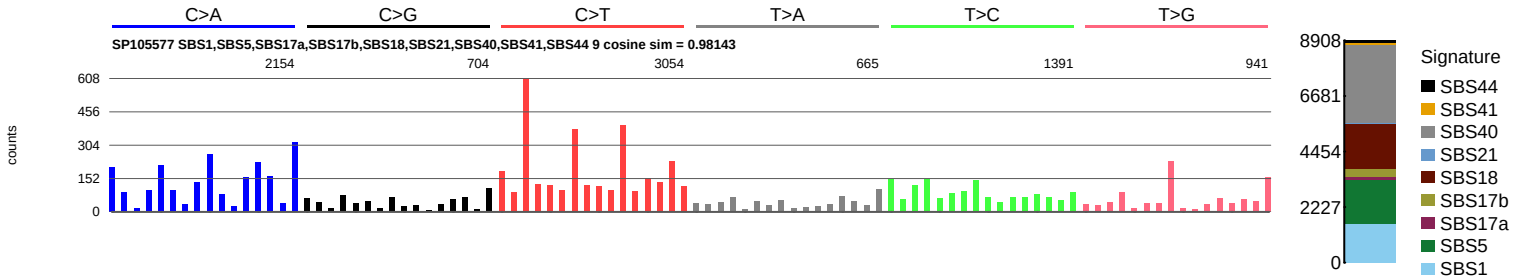
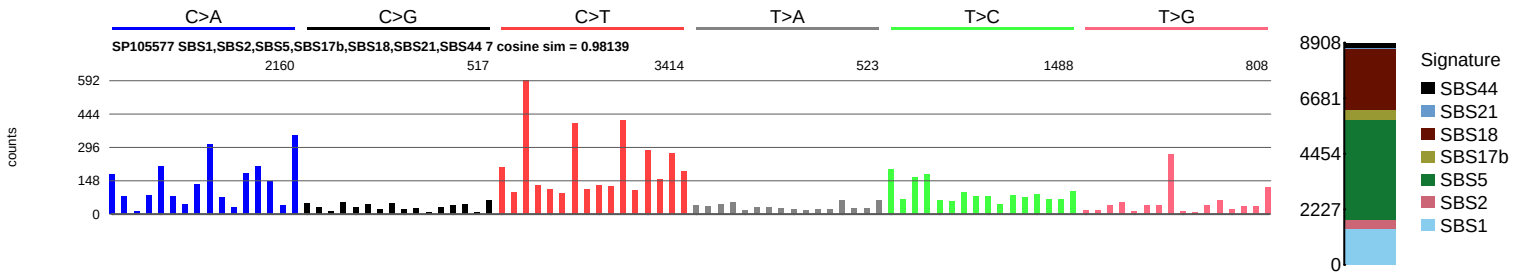
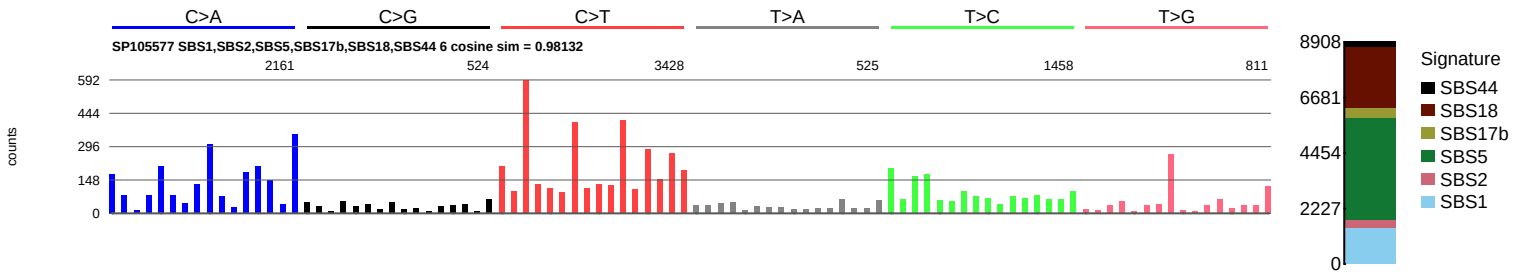
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98143
2154
704
3054
665
1391
941
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98139
2160
517
3414
523
1488
808
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS44 6 cosine sim = 0.98132
2161
524
3428
525
1458
811
counts
592
444
296
148
0
8908
6681
4454
2227
0
Signature
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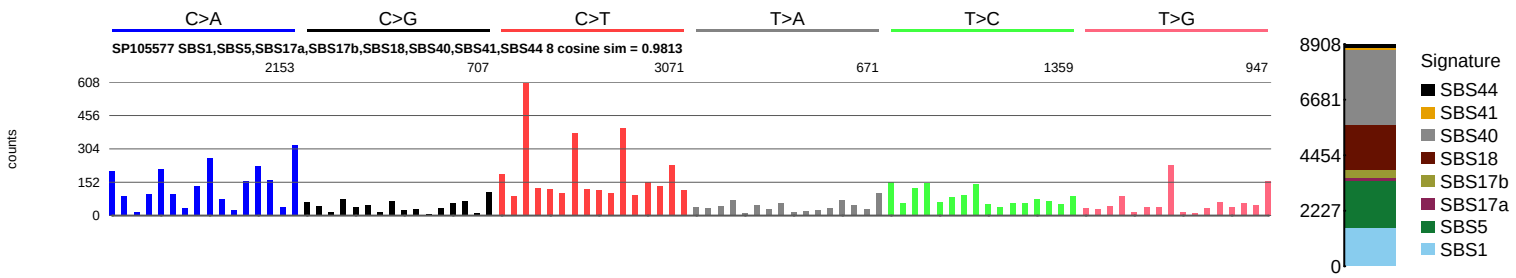
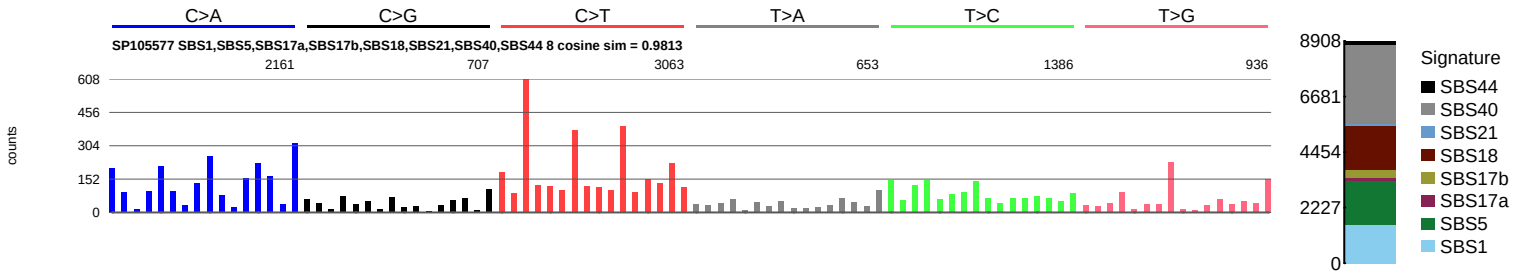
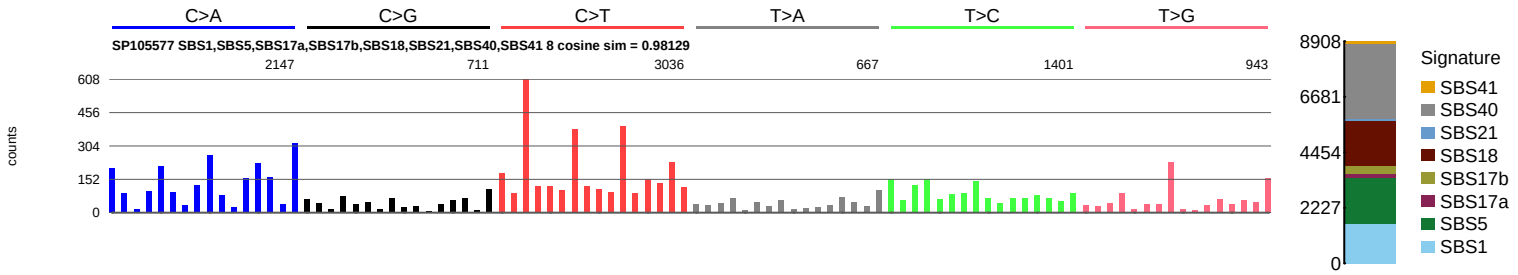
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9813
2153
707
3071
671
1359
947
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.9813
2161
707
3063
653
1386
936
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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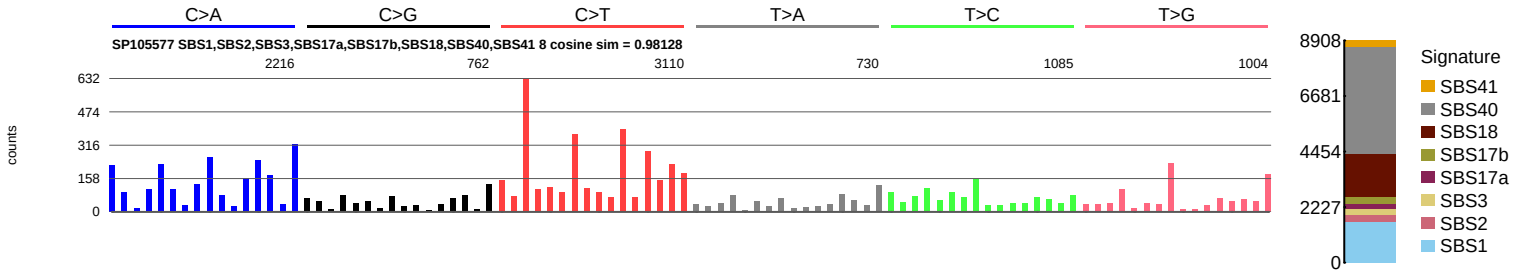
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98129
2147
711
3036
667
1401
943
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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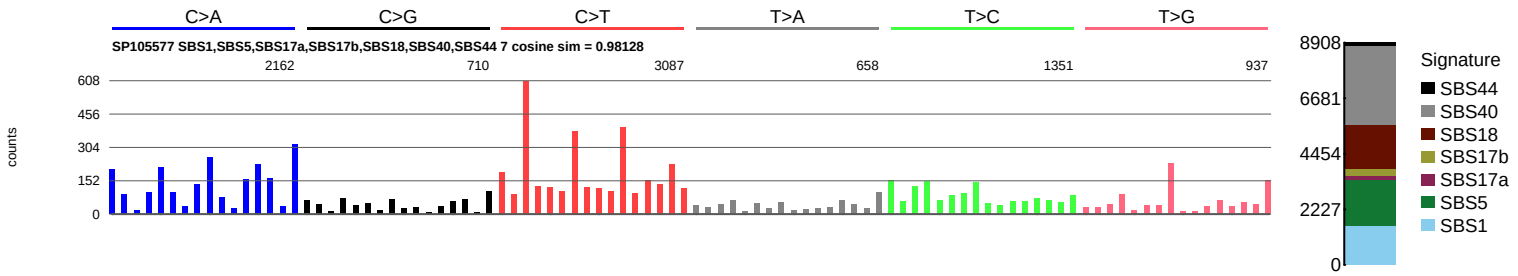
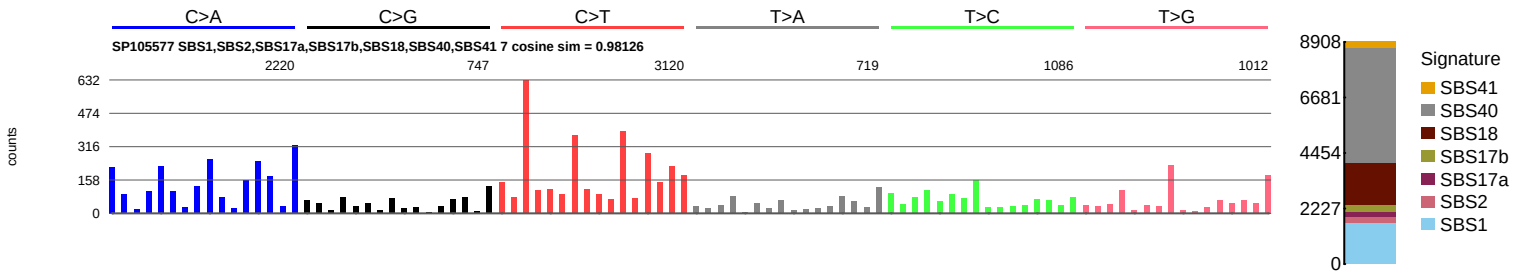
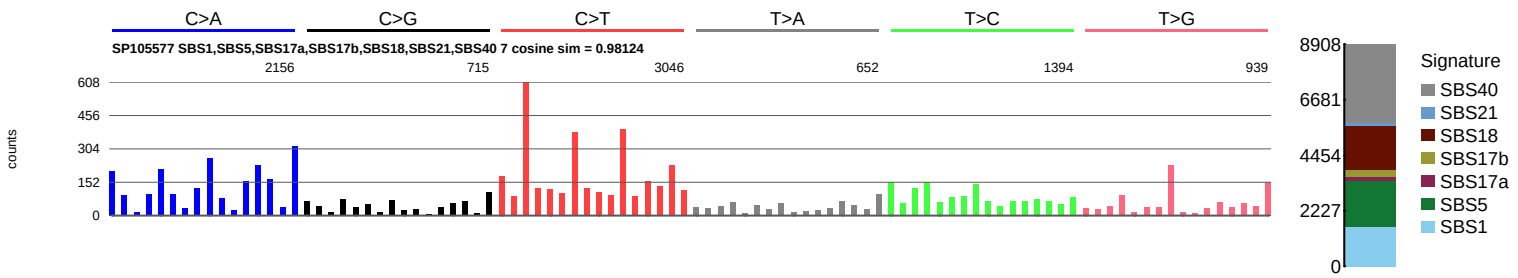
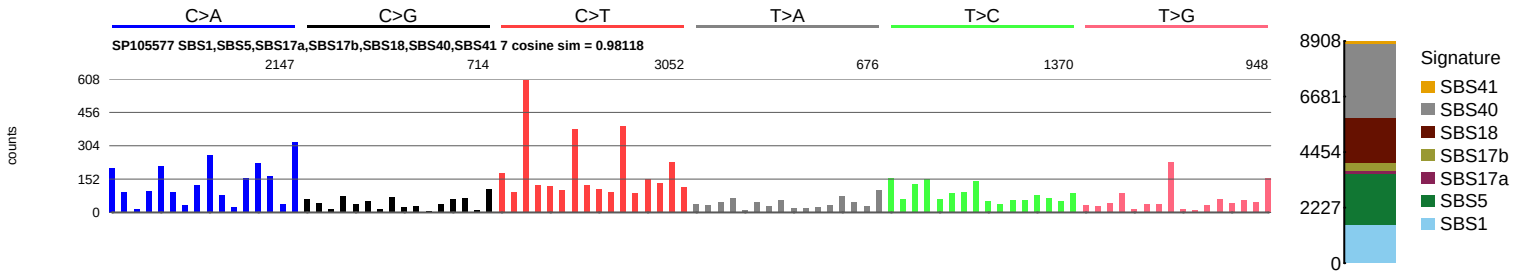
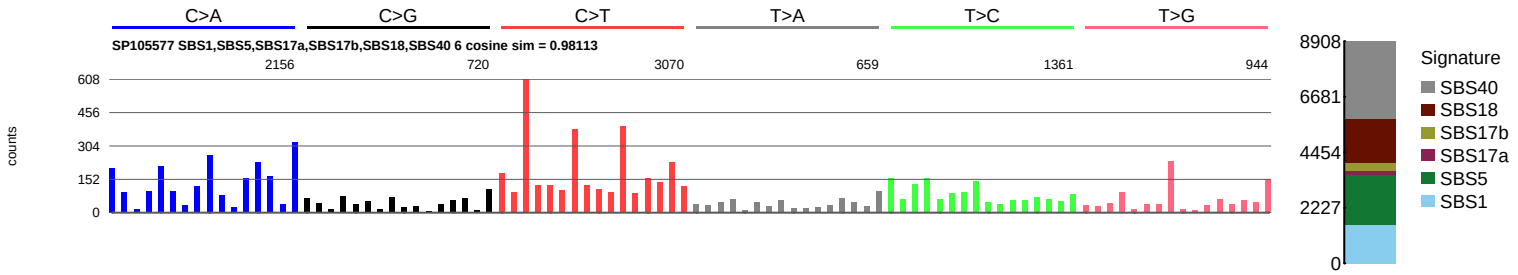
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98128
2216
762
3110
730
1085
1004
counts
632
474
316
158
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

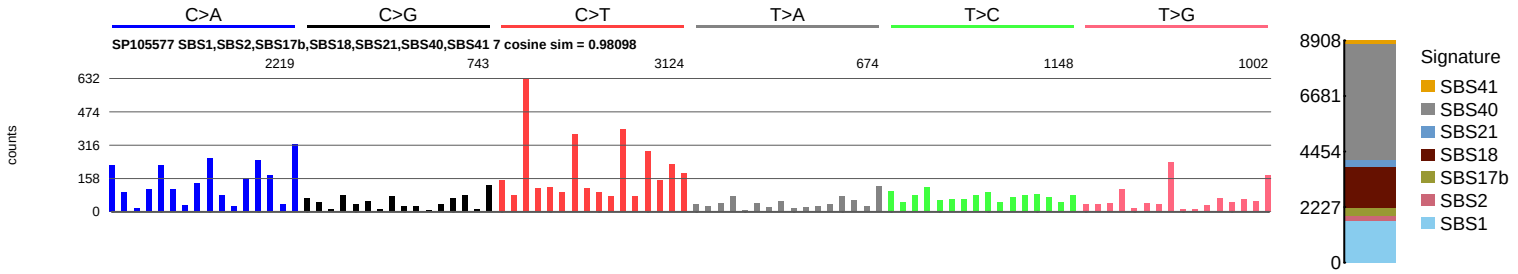

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98128
2162
710
3087
658
1351
937
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

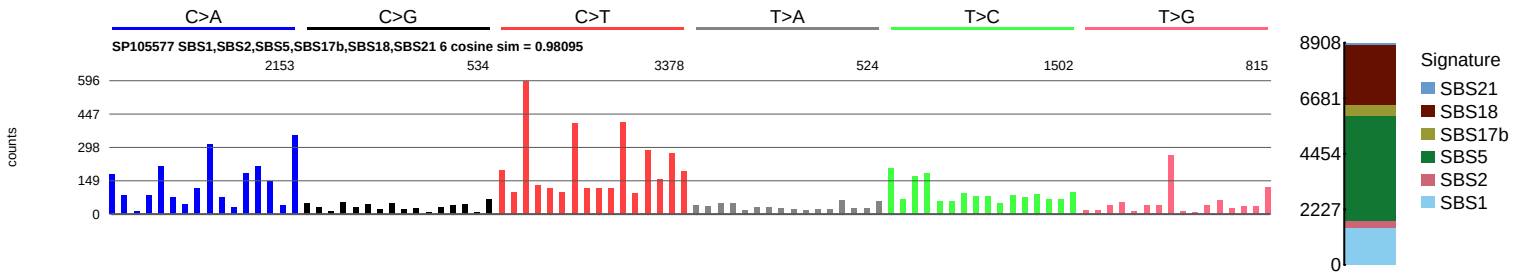

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98126
2220
747
3120
719
1086
1012
counts
632
474
316
158
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98124
2156
715
3046
652
1394
939
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98118
2147
714
3052
676
1370
948
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98113
2156
720
3070
659
1361
944
counts
608
456
304
152
0
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98098
2219
743
3124
674
1148
1002
counts
632
474
316
158
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21 6 cosine sim = 0.98095
2153
534
3378
524
1502
815
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
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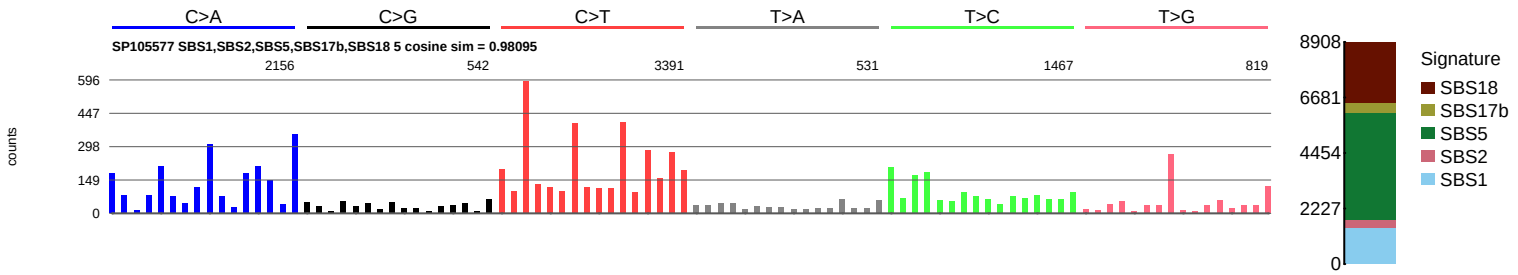
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS5,SBS17b,SBS18 5 cosine sim = 0.98095
2156
542
3391
531
1467
819
counts
596
447
298
149
0
8908
6681
4454
2227
0
Signature
SBS18
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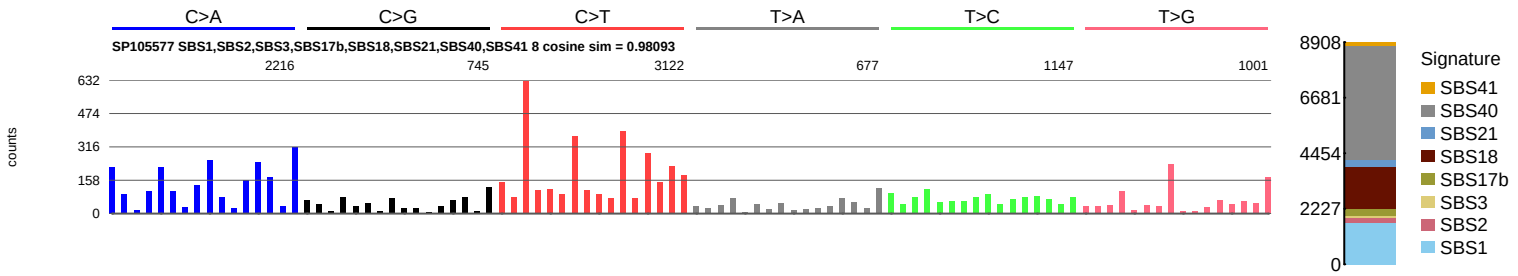
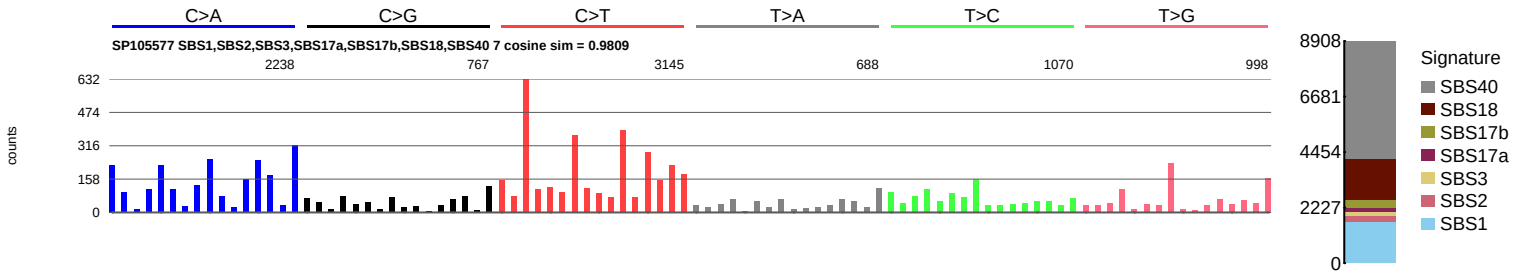
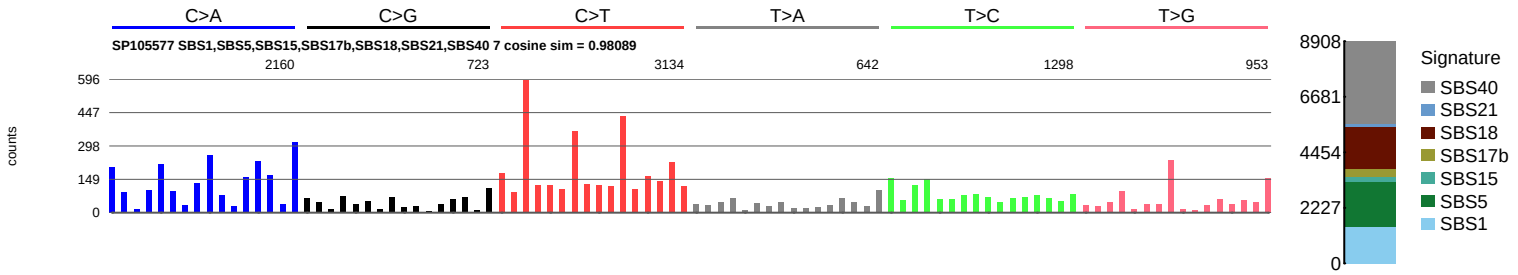
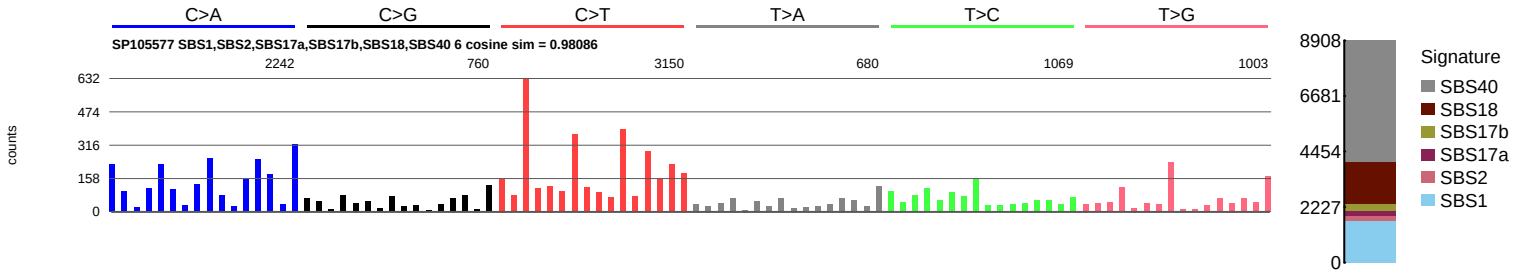
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98093
2216
745
3122
677
1147
1001
counts
632
474
316
158
0
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.9809
2238
767
3145
688
1070
998
632
474
316
158
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98089
2160
723
3134
642
1298
953
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS1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C>G
C>T
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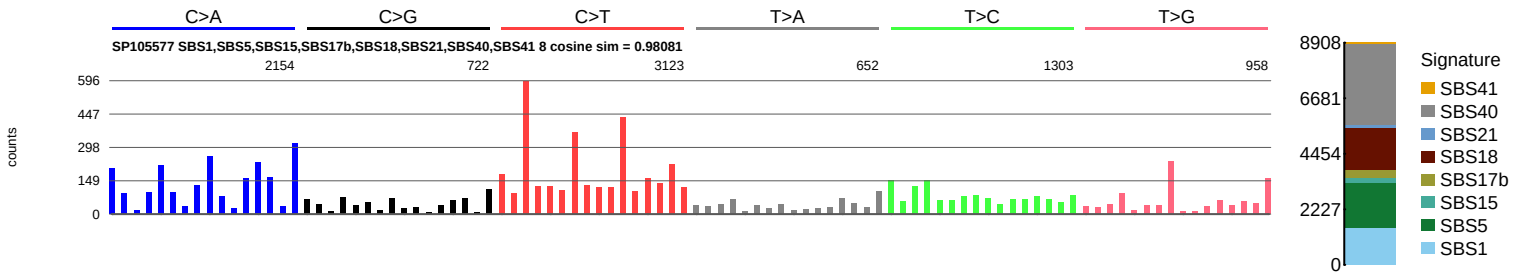
T>C
T>G
SP105577 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98086
2242
760
3150
680
1069
1003
632
474
316
158
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
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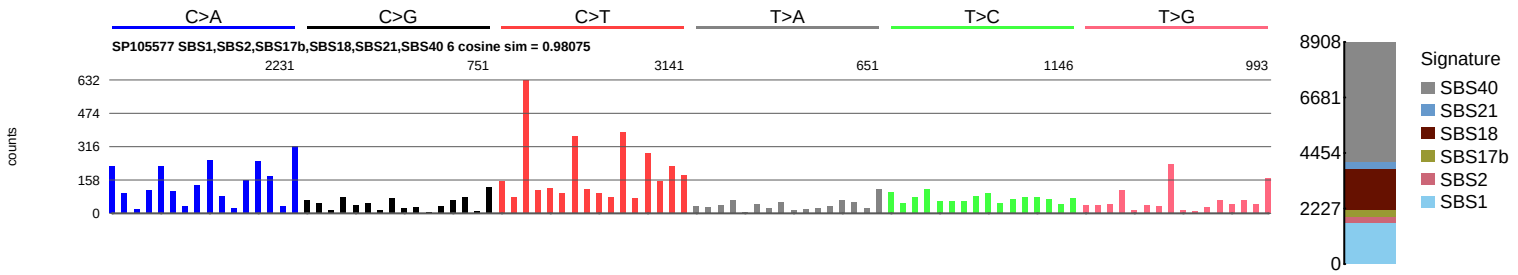
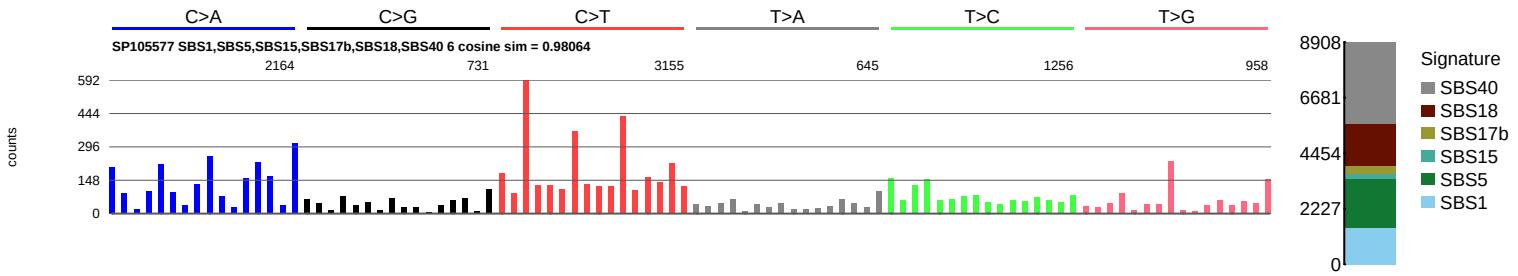
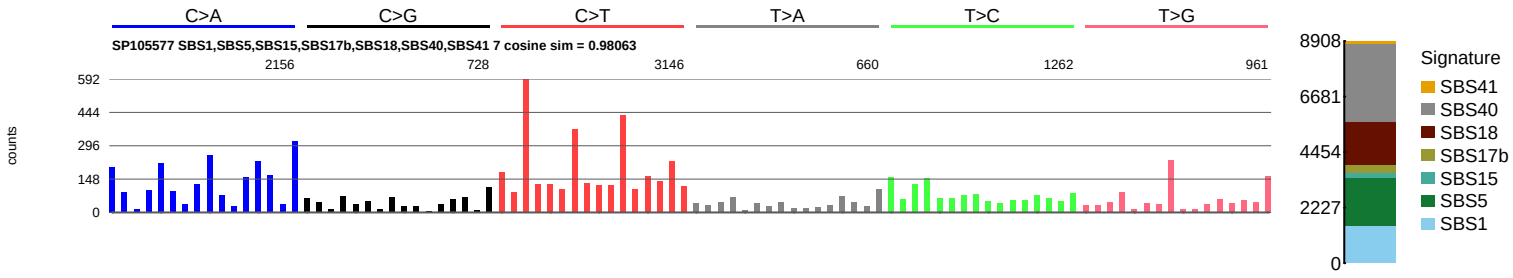
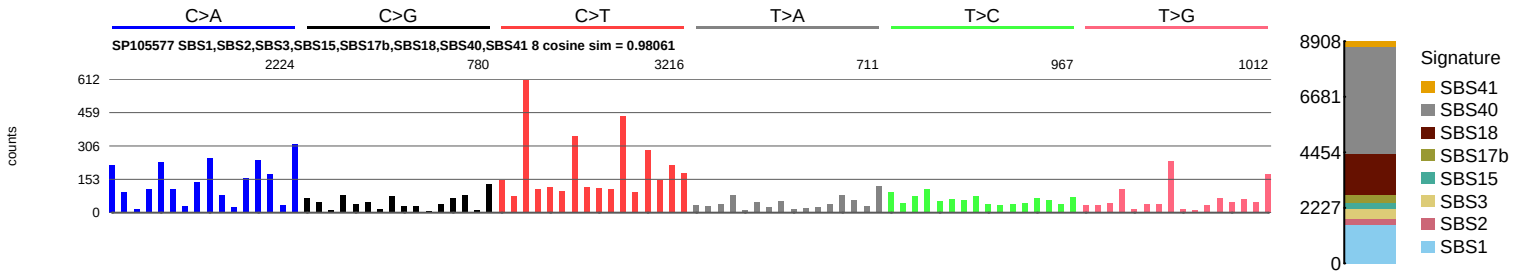
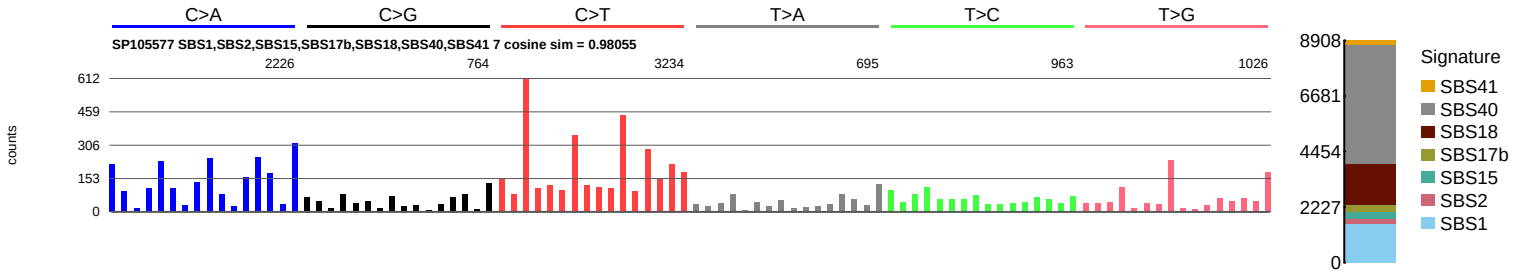
T>C
T>G
SP105577 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98081
2154
722
3123
652
1303
958
596
447
298
149
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS1

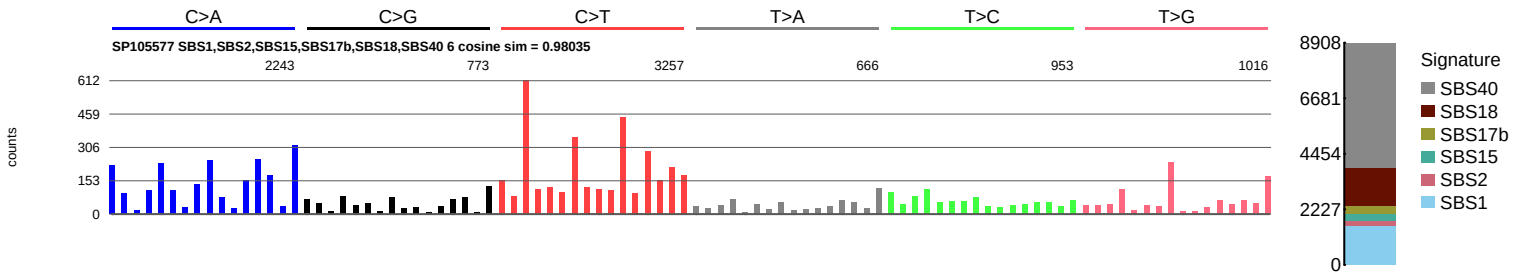

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98075
2231
751
3141
651
1146
993
632
474
316
158
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1

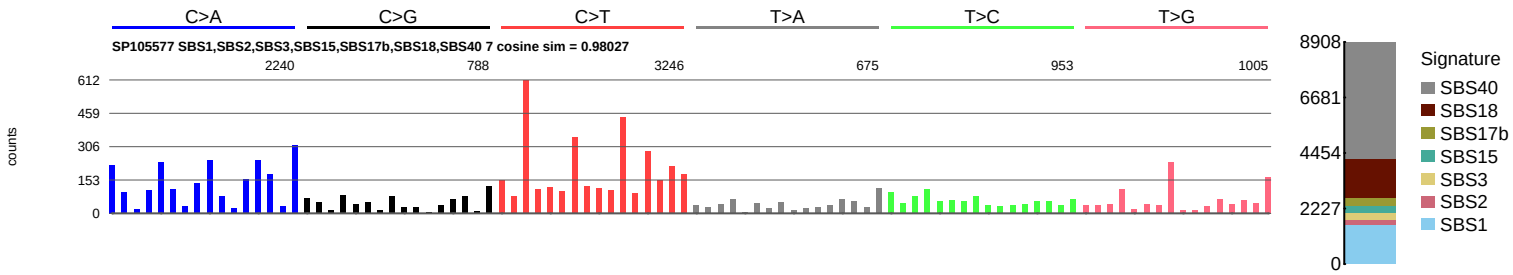

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS40 6 cosine sim = 0.98064
2164
731
3155
645
1256
958
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS1

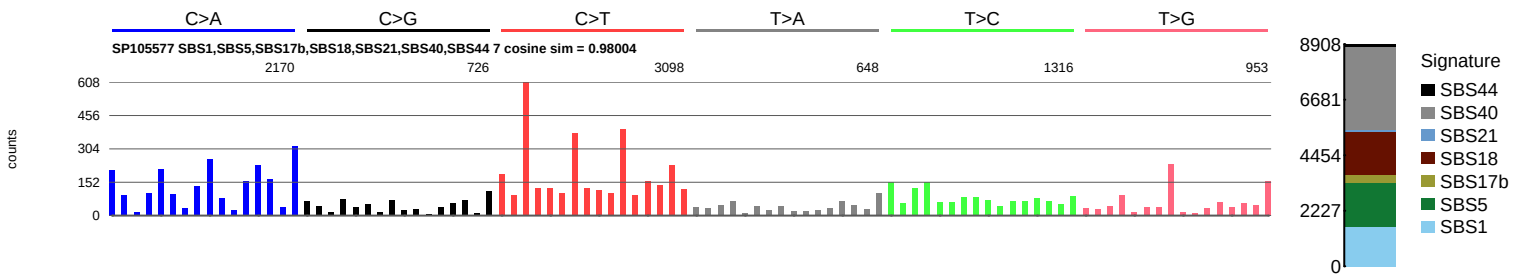

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98063
2156
728
3146
660
1262
961
592
444
296
148
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS1

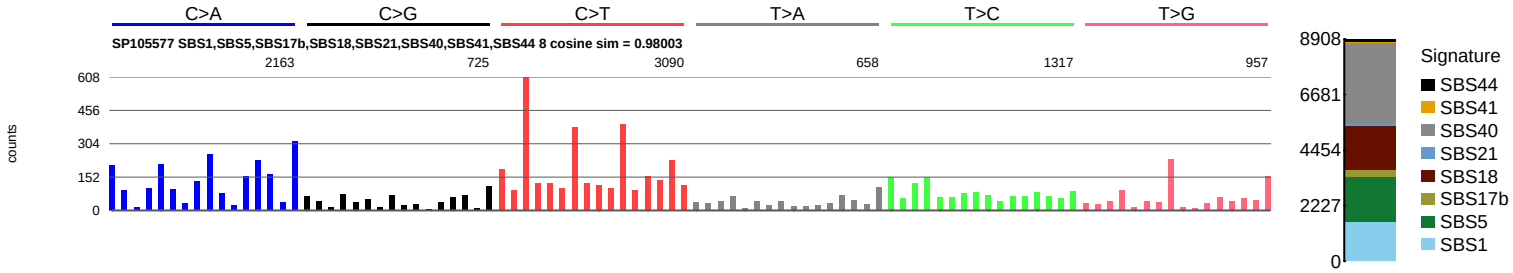

C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98061
2224
780
3216
711
967
1012
612
459
306
153
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98055
2226
764
3234
695
963
1026
612
459
306
153
0
counts
8908
6681
4454
2227
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40 6 cosine sim = 0.98035
2243
773
3257
666
953
1016
612
459
306
153
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98027
2240
788
3246
675
953
1005
612
459
306
153
0
counts
8908
6681
4454
2227
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98004
2170
726
3098
648
1316
953
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105577 SBS1,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98003
2163
725
3090
658
1317
957
608
456
304
152
0
counts
8908
6681
4454
2227
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS1